Appendix 3B: Model Building and Bridging Exercise for the Sablefish Single Region Models (Bridging from Model *25.12* to *26.1*)

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Executive Summary

The model building exercise focused on three general areas of refinement to the sablefish assessment (Table 3B.1): model parametrization (primarily aimed at improving stability), biology (focused on updating biological inputs and sex-specific modeling), and data usage (emphasizing reducing model tension, removing uncertain data, and developing a consistent approach for constructing the NOAA longline survey abundance index due to survey design changes). In addition, a number of sensitivity runs were also undertaken (Table 3B.4), but were not included in the final single region model. Sensitivity runs primarily focused on improving fits to compositional data (via time-varying selectivity), using self-weighting likelihoods (Dirichlet-multinomial) to avoid potentially overfitting the indices, and exploring time-varying growth to better represent recent declines in size and weight.

Despite minor rescaling of spawning stock biomass (SSB) across model building runs, the suite of models retained strong agreement in population trends, particularly recruitment (Figures 3B.4 – 3B.5). The differences in SSB scale can be attributed to changes in natural mortality (M) estimates (Figure 3B.6), because small changes in M have long-term impacts on SSB for a long-lived species, such as sablefish. Estimating sex-specific M values reduced estimates and helped stabilize estimation, likely reducing correlation among other scaling parameters (e.g., R_0). Updating the growth curve and length-weight parameter estimates also had a downward impact on scaling, as expected, given the decline in size- and weight-at-age of younger fish in recent years (Figure 3B.7). The changes to the input data had important impacts on scale, as well. The removal of length compositions for data sets that had age compositions (i.e., the NOAA Longline Survey and fixed gear fishery) reduced M and SSB scale, while switching to a 'carryover' approach for NOAA Longline Survey index construction had the opposite effect, particularly leading to a more rapid increase in SSB in recent years. Removal of extrapolated NOAA Longline Survey stations for the western Aleutian Islands, which are not surveyed and assume a fixed extrapolation proportion, scaled the survey and population downward at the end of the time series. The final survey index appears biologically reasonable with more gradual and consistent increases during the recent (since 2016) population growth and slightly lower overall scale (Figure 3B.3). The combination of the new index construction and removal of the fixed gear fishery Catch-per-Unit Effort (CPUE) index led to minor improvements in fitting the NOAA Longline Survey index. Similarly, dropping the length compositions led to very minor improvements in the age composition fits, but greatly improved model stability.

Despite the tranche of model changes, the general consistency across models in parameter estimates, population trends, and catch advice as well as the lack of any strong retrospective patterns within each model suggests a strong degree of viscosity towards alternative interpretations of the available data (even as different data types were added or removed). However, the continued

mediocre fits to the composition data (exhibiting age and cohort residual patterns) highlights a fundamental tension between fitting the indices and the compositions. Part of the consistency across models along with mediocre fits to the composition data can be attributed to the use of the Francis reweighting approach (Francis, 2011), which aims to 'right-weight' the data by prioritizing the abundance indices and accounting for correlations in composition data, typically resulting in lower weights for the compositions. The sensitivity runs explored alternative approaches for dealing with inherent process error when fitting composition data, including self-weighting likelihoods and/or time-varying selectivity. Although these approaches modestly improved the fit to the composition data, it often occurred at the expense of poorer fits to the NOAA Longline Survey index, and led to moderate retrospective patterns in SSB (due to overfitting compositions leading to consistent overestimation of recruitment relative to the terminal assessment). The proposed changes, as implemented in Model *26.1_1Reg*, balance fit across data sources (e.g., age compositions and indices) and model complexity, while providing a much more stable model with reduced tension caused by uncertain and potentially uninformative data (e.g., by dropping length compositions and fixed gear fishery CPUE). The sensitivity models provided potentially useful functionality that might be considered in future sablefish assessments, but diagnostics and performance did not indicate that any of the explorations led to substantially improved models at this time.

The following document summarizes model performance and results of select model runs from the stepwise model building process, primarily in comparison to the continuity model (*25.12_1Reg_Cont*). However, full model results and diagnostics are available at the Alaska sablefish assessment CIE review GitHub site, where individual folders are provided for every model run produced for the CIE Review. Folders contain SPoRC input and output R files along with PDFs of results (folders for single region model runs are available here). Full model outputs for the continuity model (*25.12_1Reg_Cont*) and the final single region model (*26.1_1Reg*) are provided in the Supplementary Material (SM3B).

1. Model Building

Introduction

In 2025, the custom sablefish assessment model implemented in AD Model Builder (ADMB) was transitioned into the R-TMB generalized Spatial Populations over Regional Components (SPoRC) stock assessment R package (Cheng et al., 2026; https://chengmatt.github.io/SPoRC/index.html). After demonstrating an exact match to the ADMB model (Figures 3B.1 – 3B.2), major modeling changes to the stock assessment were undertaken, which aimed to address model stability, better align parametrizations with stock assessment good practices, and refine data inputs (Goethel and Cheng, 2025). The final model for 2025 (Model *25.12*) was reviewed and approved at the September North Pacific Fisheries Management Council (NPFMC) Groundfish Plan Team (GPT) and October Science and Statistical Committee (SSC) meetings. However, an operational assessment was never undertaken due to the US government shutdown in the fall of 2025 (the last operational assessment was in 2024; Goethel and Cheng, 2024). With the goal of providing the 'best scientific information available' for review by the CIE and to further address GPT and SSC recommendations provided in 2025, the authors underwent another round of model building with the sablefish assessment for the CIE review resulting in the final single region model for review, Model *26.1_1Reg*.

The model building exercise focused on three general areas of refinement to the sablefish assessment (Table 3B.1): model parametrization (primarily aimed at improving stability), biology (focused on updating biological inputs and sex-specific modeling), and data usage (emphasizing reducing model tension, removing uncertain data, and developing a consistent approach for constructing the NOAA longline survey abundance index). Sablefish exhibit sexually dimorphic growth, which manifests as differences in selectivity-at-age by sex in the age-structured assessment model. A key source of model instability is the estimation of selectivity, given the increase in model parameters due to integrating a sex partition. The first step in model building was to remove selectivity parameter linkages and implement more realistic priors on the selectivity parameters, which aimed to improve model stability.

The next set of big changes aimed to improve the general portrayal of sablefish biology, which involved updating length and weight estimates as well as estimating sex-specific *M*. Given that sablefish appear to be undergoing density-dependent growth and reaching smaller sizes and weights, especially at younger ages (Cheng et al., 2024), the growth curves and length-weight parameters were reestimated (externally) with all available data through 2025 to better reflect recent changes. Then, sex-specific natural mortality was estimated to better reflect good practices for sex-specific assessments (Cheng et al., 2025a).

Next, the data inputs were revised to better address model tension and improve data inputs (e.g., the construction of the NOAA Longline Survey index). Historically, the sablefish assessment has fit both age and length composition data despite inherent tension and potential misspecification (e.g., due to lack of time-varying growth parameters when fitting length compositions). The decision to fit both types of data likely stemmed from the limited age composition data available when the sablefish assessment began using an age-structured integrated analysis paradigm in the 2000s. Moreover, certain data sources (e.g., the trawl fishery) only have length composition data available. In Model *25.12_1Reg_Cont*, length compositions were removed in years when age

compositions were available for a given data source. However, extensive tension remained between fitting age and length compositions for a given data source (as observed in the likelihood profiles; Goethel and Cheng, 2025), which could be due to continued model misspecification and/or varying signals in the data. In particular, the general lack of strong signals of recruitment in length compositions due to multiple cohorts contributing to a given length bin is likely to cause tension across fitting age and length compositions simultaneously, which is common for long-lived species. Generally, age compositions provide more direct indications of year class strength and cohort mortality compared to length data. The benefit of simultaneously fitting age and length data in an integrated assessment is an open area of research (Ono et al., 2015), but fitting length data is not recommended when growth is misspecified (i.e., if time-varying growth cannot be adequately replicated in the model; Minte-Vera et al., 2017). Given that sablefish demonstrate density-dependent growth that is not captured in the assessment (Cheng et al., 2024), there likely is model tension between age and length data, particularly for recent cohorts. Thus, the remaining length data from the fixed gear fishery (1990 – 1998) and NOAA Longline Survey (1990 – 1995) were deemed likely to be providing low signal to noise ratio and causing tension with fitting the more reliable age composition data.

Similarly, the fixed gear fishery standardized CPUE index (Cheng et al., 2023) has greater uncertainty compared to the NOAA Longline Survey index, given that the latter is a dedicated sablefish survey that historically covers a large proportion of the sablefish distribution in Alaska. Moreover, since 2022, the CPUE index can no longer be updated, given lack of important input data streams (i.e., logbooks) for creation of the index. Ultimately, NOAA Longline Survey and fixed gear fishery length compositions and the CPUE index were removed following the general principle of data quality over quantity (Payne et al., 2009), with the aim of reducing data conflict and model tension and instability.

Finally, alternative methods for index calculation of the NOAA Longline Survey relative population numbers (RPNs) were explored to provide a more consistent and reliable index. Historically, because the survey was conducted every other year in the Bering Sea (odd years, BS) and Aleutian Islands (even years, AI), an extrapolation was required for non-surveyed region RPNs to ensure a consistent scale of the index at the Alaska-wide level. The traditional approach was to use the population growth in the GOA (surveyed annually) to extrapolate abundance (from the previous RPN value) in either the BS or AI for years where a given region was not surveyed (Siwicke and Malecha, 2026). In 2025, the longline survey design was changed due to market conditions (given it is a cooperative research cost recovery survey), where only the GOA is now surveyed in odd years and only the BS and AI are surveyed in even years. The new design precludes use of the GOA extrapolation method (since it is no longer surveyed annually), so the 'carryover' approach was introduced in 2025 wherein hold-over values for non-surveyed regions were used (i.e., the last survey RPN value for a given region was used if the region was not surveyed). Although simple analysis indicated that the 'carryover' approach was likely sufficient (Goethel and Cheng, 2025), the new survey design necessitates extrapolating ~50% of sablefish abundance in a given year.

Thus, to ensure consistency across the index time series, the 'carryover' index was calculated and fit for all years (instead of using the GOA extrapolation until 2023 and switching to the carryover approach in 2025, as was done in Model *25.12_1Reg_Cont*). As part of the update to the index,

the longline survey team noted that the historical extrapolation of western AI stations (that are no longer sampled) based on eastern AI station RPNs and a fixed extrapolation ratio (estimated from Japanese survey data) was likely not appropriate (K. Siwicke, Personal Communication). Due to the rapid increase in the AI RPNs in recent years, the extrapolation was likely leading to overestimation of population growth. Thus, the final NOAA Longline Survey index utilized the carryover approach and removed western AI stations from the index (Figure 3B.3).

The three groupings of model changes, as implemented in Model *26.1_1Reg*, are expected to provide a more robust modeling framework that improves stability and biological realism of the sablefish stock assessment.

Methods

A stepwise model development process was undertaken wherein model changes were first implemented as single, one-off changes from Model *25.12* (see the sablefish CIE project GitHub for all results), then an incremental model building approach was undertaken. Each step implemented a single primary change from the previous step in the build towards the final Model *26.1_1Reg*, and the order of steps was done within the broad groupings and generally aimed to build from smaller to larger impact changes (aside from the first update, Model *26.1a*, which aimed to greatly increase stability). For all model runs in this document, the final data through the end of 2025 were utilized as would have been available for the 2025 operational assessment. All models underwent Francis reweighting to 'right-weight' the compositional data and account for correlation among composition bins (Francis, 2011, 2017).

The choice of the final model (Model *26.1_1Reg*) was made based on a combination of whether a model led to improved data fits or diagnostics, reduced model tension among data sets, increased model stability, and/or improved biological realism. In many instances, results and diagnostics were similar across model runs, which required the use of expert judgment to identify which models were likely to be more robust to the primary uncertainties associated with sablefish population dynamics while providing consistent management advice. The results focus on stepwise runs that had large impacts in comparison to the continuity model (*25.12_1Reg_Cont*), but primarily describe impacts from the full suite of changes (Model *26.1_1Reg*).

See Table 3B.1 for a full list of models with associated rationale for each and Table 3B.2 for a summary of results. Further model explorations that were deemed mostly exploratory for 2025 are then provided as sensitivity runs (Table 3B.4; see Section 2: Sensitivity Runs). A full mathematical description of the underlying SPoRC assessment framework is provided in Appendix 3A, while additional figures comparing the continuity model (*25.12_1Reg_Cont*) with the 2026 final single region model (*26.1_1Reg*) is provided in the supplement to this document (SM3B).

Primary model updates included:

- *25.12_1Reg_Cont*:
    - The final 2025 operational model with no changes except updated data through 2025.

- *26.1a_1Reg_Cont_Upd*:
    - Remove linkages among selectivity parameters (mainly sharing of parameters across sexes or time blocks, such that each selectivity parameter is estimated independently for each sex and time block).
    - Update the selectivity parameter to be more realistic (estimated in log-space) priors to ~$N(\ln(4),1.5^2)$ for the fishery and ~$N(\ln(3),1.5^2)$ for the survey.
    - Reduce the assumed survey standard error from 2*Obs_SE to 1.5*Obs_SE to reflect relative information content of the index.

- *26.1b_1Reg_Cont_Upd_ISS*:
    - Implement scaled input sample sizes (ISS) across years within a gear to better represent interannual variability in sampling, which are scaled to fleet/composition specific minimum and maximum values (e.g., age data are provided a higher max ISS than length data; formerly a time-invariant sample size of 20 was assumed for all composition data).

- *26.1c_1Reg_Cont_Upd_TF_log*:
    - Assume logistic selectivity for the trawl fishery (formerly domed selectivity was assumed estimated with a gamma function) to better reflect catch of bigger female fish and to improve model stability.

- *26.1d_1Reg_Cont_Upd_Upd_Bio*:
    - Externally re-estimate and update the growth curve and length-weight parameters using the methods outlined in Echave (2021) using all available NOAA Longline Survey age-length-weight data (1996 – 2025) and input the new size-age transition matrix and weight-at-age curve into the assessment model.

- *26.1e_1Reg_Cont_Upd_Sex_M*:
    - Estimate sex-specific *M* as independent parameters, but sharing the same prior as in Model *25.12_1Reg_Cont*.

- *26.1f_1Reg_Cont_Upd_No_CPUE:*
    - Remove the fixed gear fishery CPUE index from the assessment, including the historical Japanese CPUE (1964 – 1981) and the domestic CPUE (1995 – 2022), noting that these data are unavailable starting from 2023 onwards.

- *26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F:*
    - Remove the length data for the fixed gear fishery (1990 – 1998) and the NOAA Longline Survey (1990 – 1995) and remove the early fixed gear fishery selectivity time block (given that there is no longer composition data available to estimate it).

- *26.1h_1Reg_Cont_Upd_LLS_Carry:*
    - Implement the 'carryover' method for constructing the NOAA Longline Survey RPN index for all years.

- *26.1i_1Reg_Cont_Upd_No_wAI_LLS:*
    - Remove the extrapolated western AI stations from the NOAA Longline Survey RPN index for all years.

Results

Generally, none of the models in the model building exercise led to significant changes to the data fits or interpretation of population dynamics (Figures 3B4 – 3B.5; Table 3B.3). Moderate scaling changes in SSB resulted across runs due to changes in the biological inputs (Figure 3B.6), estimates of *M* (Figure 3B.7), NOAA Longline Survey index construction (Figure 3B.8), and selectivity assumptions (i.e., removing the dome for the trawl fishery selectivity; Figures 3B.8 – 3B.9). Although relative stock status and catch advice (Acceptable Biological Catch, based on fishing at $F_{40\%}$) remained consistent from Model *25.12_1Reg_Cont* to Model *26.1_1Reg* (same as Model *26.1i_1Reg_Cont_Upd_No_wAI_LLS*), the stepwise progression involved moderate upward and downward scaling across various model updates. Upward scaling of SSB and catch advice in Models *26.1a_1Reg_Cont_Upd, 26.1f_1Reg_Cont_Upd_No_CPUE,* and *26.1h_1Reg_Cont_Upd_LLS_Carry* was primarily associated with closer fits to the NOAA Longline Survey index (increased assumed standard error in the first, removal of contradictory CPUE signals in the second, and a more steady and consistent index in the third). Downward scaling was then associated with updating the biology (reduced growth and weight; Model *26.1d_1Reg_Cont_Upd_Upd_Bio*), estimating sex-specific *M* (reduced *M* estimates; Model *26.1e_1Reg_Cont_Upd_Sex_M*), removing length compositions (reduction in male *M* estimates; Model *26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F*), and removal of extrapolated survey stations (reductions in total NOAA Longline Survey RPNs; Model *26.1i_1Reg_Cont_Upd_No_wAI_LLS*).

Notable changes in model dynamics included:

- *25.12_1Reg_Cont* (full results here):
    - Adding a new year of data led to important changes in the shape of the trawl fishery selectivity, highlighting the lack of stability in the gamma function parameters (comparison here).
    - *M* decreased moderately (likely correlated to the change in the shape of the trawl fishery dome selectivity), leading to a slight decrease in SSB across the time series.
    - Catch advice (Acceptable Biological Catch, ABC, based on the $F_{40\%}$ harvest control rule) did not change given similar relative stock status due to the increase in $B_{40\%}$.

- *26.1a_1Reg_Cont_Upd* (full results here):
    - Assuming a lower standard error for the NOAA Longline Survey index led to minor improvements to the index fit, but no appreciable changes to fits in other data.
    - *M* increased moderately (Figure 3B.7) leading to an increase in overall scale of SSB and moderate increase in catch advice (Table 3B.4), which is due to the higher terminal SSB (Figure 3B.3) caused by a closer fit to the terminal NOAA Longline Survey index value.
    - Selectivity only changes slightly despite removing parameter linkages and updating priors, but estimates are more stable demonstrating lower correlation.

- *26.1b_1Reg_Cont_Upd_ISS* (full results here):
    - Despite higher mean Input Sample Size (ISS), final mean Effective Sample Size (ESS) values across data sources after Francis reweighting remain consistent with the previous model that assumed a time-invariant ISS.
    - Minor improvements in yearly compositional fits and residual patterns result due to better representation of interannual variability in sampling.
    - No appreciable changes in population trajectories or catch advice.

- *26.1c_1Reg_Cont_Upd_TF_log* (full results here):
    - Assuming logistic trawl fishery selectivity leads to subtle improvements in fit to the mean length by year and minor differences in length composition fits and residual patterns, as well as an important change in the shape of the ascending limb of the selectivity function for females.
    - There were no appreciable changes in population trajectories or catch advice, but trawl fishery selectivity parameters are much more stable.

- *26.1d_1Reg_Cont_Upd_Upd_Bio* (full results here):
    - Updating growth and weight parameters caused the mean weight of younger fish to decrease slightly (Figure 3B.6) reflecting decreases in average size and weight for recent cohorts, but there were similar increases in mean weight for older aged fish.
    - The scale of SSB and biomass are scaled down slightly while $B_{40\%}$ increases slightly (due to an increase in weight for older fish) resulting in a moderate decline in catch advice (Table 3B.3).

- *26.1e_1Reg_Cont_Upd_Sex_M* (full results here):
    - Allowing sex-specific *M* to be estimated leads to a downward scaling of *M* for both sexes (Figure 3B.7), though estimates remain similar (likely given the use of the same prior).
    - Reductions in *M* resulted in an upward shift in *F* and $B_{40\%}$, and a concomitant downward shift in SSB across the time series, all of which resulted in a moderate decrease in catch advice (Table 3B.3).

- *26.1f_1Reg_Cont_Upd_No_CPUE* (full results here):
    - Removing the fixed gear fishery CPUE index allows the NOAA Longline Survey index to be better fit, especially in recent years.
    - SSB demonstrates a more rapid decrease historically (due to removal of the historical Japanese CPUE) and a more rapid increase since ~2016 (more in line with the NOAA Longline Survey index trends).

- *26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F* (full results here):
    - Removing the remaining fixed gear fishery and NOAA Longline Survey length compositions led to minor improvements in the fit and residual patterning in the associated age compositions (including improved fits to mean age).

- Removal of potentially conflicting signals in the data resulted in a reduction in male *M* (Figure 3B.7), average recruitment (Figure 3B.5), and the scale of SSB (Figure 3B.4).
- The combined effect of changes in *M* and recruitment was to reduce $B_{40\%}$ and associated catch advice moderately (Table 3B.3).
- Model stability improved with slightly lower maximum gradients and reduced parameter correlation.

- *26.1h_1Reg_Cont_Upd_LLS_Carry* (full results here):
  - Using the 'carryover' method to calculate the NOAA Longline Survey index led to a more gradual and consistent increase in RPNs over the last decade, which allowed the model to better fit the index (Figure 3B.8).
  - By fitting the index more closely, the population scale increases similar to Model *26.1f_1Reg_Cont_Upd_No_CPUE* including catch advice (Table 3B.3).

- *26.1i_1Reg_Cont_Upd_No_wAI_LLS* (a.k.a. *26.1_1Reg*; full results here):
  - Dropping extrapolated stations from the western AI from the NOAA Longline Survey index reduces the scale of the index and the recent rate of increase in the index (Figure 3B.8).
  - SSB scale is reduced slightly in the recent period resulting in a small decline in catch advice (Table 3B.3).

Each model building step helped improve model stability as diagnosed via various convergence diagnostics (e.g., maximum gradient values, number and degree of parameter correlations, jitter analysis, and profile likelihoods). The resulting end model for this first segment (Model *26.1_1Reg*) has much more stable selectivity parameters, which has been an ongoing concern for the sablefish assessment, often requiring linking parameters across sexes. Moreover, improving biological realism by estimating sex-specific *M* has helped stabilize the estimates and move them towards more realistic, lower values. Similarly, removal of length compositions (for fleets where age compositions exist) has further reduced model tension, providing better alignment in scaling parameters among the age composition data and the indices (Figures SM3B.61 – SM3B.64).

Compared to *Model 25.12_ 1Reg_Cont*, *Model 26.1_1Reg* provides a slight improvement in fit to the age composition data, while also improving fit to the NOAA Longline Survey index (see Figures in SM3B). By improving model parametrization, reducing model tension from uncertain data sources, and improving biological realism, the resulting Model *26.1_1Reg* appears to provide a more robust assessment of the sablefish population. Although conflicting signals remain between the age compositions and the NOAA Longline Survey index in terms of magnitude of recent year classes and the associated rate of population growth, these conflicts appear to be reduced with the suite of model updates. Moreover, the assessment continues to produce extremely good retrospective diagnostics, showing essentially no over- or under-estimation of SSB and only moderate initial underestimation of the recent, large year classes (Figures SM3B.57 – SM3B.60).

Conclusion

Model *26.1_1Reg* provides improved parametrization and removal of uncertain and potentially contradictory data sets, greatly improves model stability, provides updates to the NOAA Longline Survey index construction that improve consistency and better reflect observed dynamics, and utilizes improved biological inputs and parametrization that likely help to better reflect sex-specific sablefish dynamics. Despite important model and data changes across the range of updates explored, SSB and recruitment trajectories are generally consistent across model runs and mostly fall within the 95% confidence intervals of the continuity Model *25.12_1Reg_Cont* (Figures 3B.5 – 3B.6). Although stock status has not changed drastically across models, important model rescaling (e.g., in SSB) due to changes in estimates of natural mortality and associated changes in survivorship have occurred, yet catch advice in Model *26.1_1Reg* remains consistent to that from Model *25.12_1Reg_Cont* (Table 3B.3).

Data fits have improved slightly, but fits to the composition data remain mediocre due to tension between the index and age composition data. The poorer fits to the composition data can partially be attributed to the Francis reweighting approach (Francis, 2011), which aims to 'right-weight' the data by prioritizing the abundance indices and accounting for correlations in composition data, typically resulting in lower weights for the compositions. Thus, the sensitivity runs were designed to explore alternative approaches to Francis reweighting for dealing with inherent process error when fitting composition data, including self-weighting likelihoods and/or time-varying selectivity (See Section 2: Sensitivity Runs). Although these approaches modestly improved the fit to the composition data, it often occurred at the expense of poorer fits to the NOAA Longline Survey index and led to moderate retrospective patterns in SSB (due to overfitting compositions leading to overestimation of recruitment). Thus, the proposed changes, as implemented in Model *26.1_1Reg*, were deemed best able to balance fit across data sources (e.g., age compositions and indices) and model complexity, while providing a much more stable model with reduced tension caused by uncertain and potentially uninformative data (e.g., by dropping length compositions and CPUE).

However, a fundamental issue for the sablefish assessment is how to handle the NOAA Longline Survey index construction and data fitting. Mainly, whether it is better to perform extrapolations for non-surveyed regions and aggregate composition data across regions (as is done in Model *26.1_1Reg*, and all associated single region models in Appendix 3B), disaggregate the indices and compositions by region (i.e., fitting the regional RPNs without any extrapolation) and use a fleets-as-areas model (see Appendix 3C), or implement a spatially explicit assessment (see Appendix 3D). All of these options maintain tradeoffs between sample size limitations, complexity, and stability. However, the impact on overall assessment bias and performance for provision of robust management advice may be relatively negligible, given the results of a sablefish-conditioned management strategy evaluation that showed essentially the same management outcomes across assessment structures (see Appendix 3E). Ultimately, as with all assessments, Model *26.1_1Reg* is not perfect and ongoing work will continue to try and better inform sex-specific dynamics and address model tension among data inputs. However, it appears to be a robust and consistent modeling framework for the provision of management advice for sablefish.

Tables

Table 3B.1. Model runs undertaken as part of the 2026 sablefish single region assessment model building process. Model development proceeded in a stepwise manner with a single change from the previous model (and including all updates from previous models). Each update was also implemented as a single, one-off change from the continuity model (*25.12*), but the results are not discussed herein.

Model Group	*Model Name*	*Major Changes*	*Rationale*
Continuity Model	*25.12_1Reg_Cont*	(*25.12*) None.	2025 continuity model.
Model Parametrization (Section 1)	*26.1a_1Reg_Cont_Upd*	(*25.12*+) Remove selectivity links, update selectivity priors, update recruit bias correction, and reduce survey CV.	Improve stability and selectivity estimation by using more realistic selectivity priors, removing selectivity parameter links (across sexes), and reducing the assumed survey CV (~7.5%) to reflect observed variance (formerly CV ~ 10%).
	26.1b_1Reg _ISS	(*26.1a*+) Implement relative scaled input sample size (ISS) across years.	Account for interannual variability in composition data (ages and lengths) by using a relative (within gears) scaling of the ISS (formerly time-invariant).
	26.1c_1Reg_TF_log	(*26.1b*+) Implement logistic selectivity for the trawl fishery.	Improve stability and account for catch of large females in the trawl fishery by using a logistic selectivity function (formerly domed gamma function).
Biology (Section 2)	*26.1d_1Reg_Upd_Bio*	(*26.1c*+) Utilize updated length and weight parameters.	Integrate recent changes in growth by updating estimates of growth and weight curves with all available data through 2025.
	26.1e_1Reg_Upd_Sex_M	(*26.1d*+) Estimate sex-specific natural mortality (*M*).	Implement best practices for sex-specific models by estimating *M* for both sexes using the same prior (formerly single *M* value shared across sexes).
Data Usage (Section 3)	*26.1f_1Reg_No_CPUE*	(*26.1e*+) Remove the fishery CPUE index.	The CPUE index is difficult to standardize given rapidly changing management and fishery practices, it is no longer able to be updated, and at times gives contradictory signals in population trends compared to the LLS; removing CPUE reduces model tension with the primary index of abundance.
	26.1g_1Reg_No_Len_2_Blks_F	(*26.1f*+) Drop fixed gear fishery and NOAA LLS lengths along with associated early fishery selectivity time block.	Historical length data for the fixed gear fishery (pre-1999) and NOAA LLS (pre-1996) is likely to be uncertain; removing this data (and pre-1995 fishery selectivity time block) reduces model tension with age comps and increases stability.
	26.1h_1Reg_LLS_Carry	(*26.1g*+) Implement 'carryover' approach for NOAA LLS calculation in all years.	The NOAA LLS extrapolation approach for non-surveyed regions in a given year (i.e., formerly using GOA growth to extrapolate for the BS or AI) is no longer feasible or defensible, so using the carryover method (i.e., for non-surveyed regions assume the same index value as the last time it was surveyed) for all years is more appropriate.
	26.1i_1Reg_No_wAI_LLS	(*26.1h*+) Drop extrapolated western AI stations from the NOAA LLS index.	Using historical extrapolations (based on the Japanese LLS) from surveyed eastern AI to non-surveyed western AI stations is no longer defensible, so remove these extrapolated station values from the index.
Final 1 Region Model	***26.1_1Reg***	(*26.1i*) None.	The 2026 final proposed single region model with all updates implemented.

Table 3B.2. Summary of results for the single region model building and bridging exercise.

Model Group	*Model Name*	*Major Changes*	*Conclusions*
Continuity Model	*25.12_1Reg_Cont*	(*25.12*) None.	The continuity model retains minor instability due to selectivity parametrizations and conflicting signals among age and length data as well as the various indices; further model updates are warranted to improve performance and increase stability.
Model Parametrization (Section 1)	*26.1a_1Reg_Cont_Upd*	(*25.12*+) Remove selectivity links, update selectivity priors, and reduce survey CV.	Improved selectivity parametrization greatly improved model stability, while decreased survey CV led to minor improvements in fit to LLS index; moderate rescaling resulted from an increase in the *M* estimate that led to increases in SSB and catch advice.
	26.1b_1Reg _ISS	(*26.1a*+) Implement relative scaled input sample size (ISS) across years.	The scaled ISS had minimal impact on estimates (e.g., selectivity) but led to minor improvements in fits to composition data and mean age.
	26.1c_1Reg_TF_log	(*26.1b*+) Implement logistic selectivity for the trawl fishery.	Implementing the logistic function for trawl fishery selectivity improved model stability and led to minimal improvements to trawl fishery length composition fits, particularly larger females.
Biology (Section 2)	*26.1d_1Reg_Upd_Bio*	(*26.1c*+) Utilize updated length and weight parameters.	Led to minor scaling changes in biomass due to slight decrease in mean weight of fish under 15 years of age.
	26.1e_1Reg_Upd_Sex_M	(*26.1d*+) Estimate sex-specific natural mortality (*M*).	Caused moderate reduction in *M* estimates across ages, which led to increases in *F* along with reductions in SSB and catch advice.
Data Usage (Section 3)	*26.1f_1Reg_No_CPUE*	(*26.1e*+) Remove the fishery CPUE index.	Resulted in minor improvements in fit to the LLS index and changes to estimates of fishery selectivity (particularly in the pre-1995 block).
	26.1g_1Reg_No_Len_2_Blks_F	(*26.1f*+) Drop fixed gear fishery and NOAA LLS lengths along with associated early fishery time block.	Improved model stability considerably and led to minor changes to fishery selectivity and male *M*, leading to increases in *F* and downward scaling of SSB and catch advice; fits to early age compositions improved slightly.
	26.1h_1Reg_LLS_Carry	(*26.1g*+) Implement 'carryover' approach for NOAA LLS calculation in all years.	The 'carryover' LLS index construction approach leads to a slightly different index trajectory that is more steadily increasing in recent years, while the model is better able to fit this trend; the lower initial index values and steady recent increases lead to an upward scaling of SSB and catch advice.
	26.1i_1Reg_No_wAI_LLS	(*26.1h*+) Drop extrapolated western AI stations from the NOAA LLS index.	Removal of the extrapolated western AI stations causes the LLS index to be scaled downwards and tampers the rate of increase in recent years (given that the increasing catch rates in the AI cause the extrapolation to have a stronger impact), which results in a minor reduction in recent SSB and catch advice.
Final 1 Region Model	***26.1_1Reg***	(*26.1i*) None.	Improved parametrization and removal of uncertain and potentially contradictory data sets greatly improved model stability; updates to the LLS index construction improve consistency and better reflect observed dynamics; and improved biological inputs and parametrization likely help to better reflect sex-specific sablefish dynamics.

Table 3B.3. Key management quantity outputs. SSB is in kt, B_Ref_Pt is the associated $B_{40\%}$ reference point in kt, and F_Ref_Pt is the associated $F_{40\%}$. The catch advice is the 2026 Acceptable Biological Catch (ABC) resulting from fishing at $F_{40\%}$. See Table 3B.1 for a summary of model changes and Table 3B.2 for a summary of results. Note that Model *26.1i_1Reg_No_wAI_LLS* represents the final single region model (Model *26.1_1Reg*).

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
25.12_1Reg_Cont	1	206.6999	357.0208	0.04784668	39.55138	131.4549	0.07875	1.57240	0.57896	0.60758
26.1a_1Reg_Cont_Upd	1	236.8887	374.7811	0.04088599	48.47894	134.3321	0.08233	1.76346	0.63207	0.49659
26.1b_1Reg_Cont_Upd_ISS	1	236.7808	372.1708	0.04010794	49.30014	134.0740	0.08287	1.76605	0.63622	0.48397
26.1c_1Reg_Cont_Upd_TF_log	1	239.0244	374.6549	0.04057036	49.03881	135.6938	0.08350	1.76150	0.63799	0.48589
26.1d_1Reg_Cont_Upd_Bio	1	236.9829	378.2417	0.04239512	46.94926	138.3195	0.08301	1.71330	0.62654	0.51072
26.1e_1Reg_Cont_Upd_Sex_M	1	219.4031	373.0349	0.04583853	41.16434	140.1182	0.07836	1.56584	0.58816	0.58497
26.1f_1Reg_Cont_Upd_No_CPUE	1	230.2295	384.2490	0.04435560	42.49986	143.0228	0.07854	1.60974	0.59917	0.56478
26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F	1	204.7788	360.0882	0.04872276	38.40136	135.5328	0.07778	1.51092	0.56869	0.62644
26.1h_1Reg_Cont_Upd_LLS_Carry	1	230.4754	382.9105	0.04375466	43.52004	142.4181	0.07947	1.61830	0.60190	0.55056
26.1i_1Reg_Cont_Upd_No_wAI_LLS	1	216.4521	369.6689	0.04642913	40.85173	139.0172	0.07903	1.55702	0.58553	0.58749

Figures

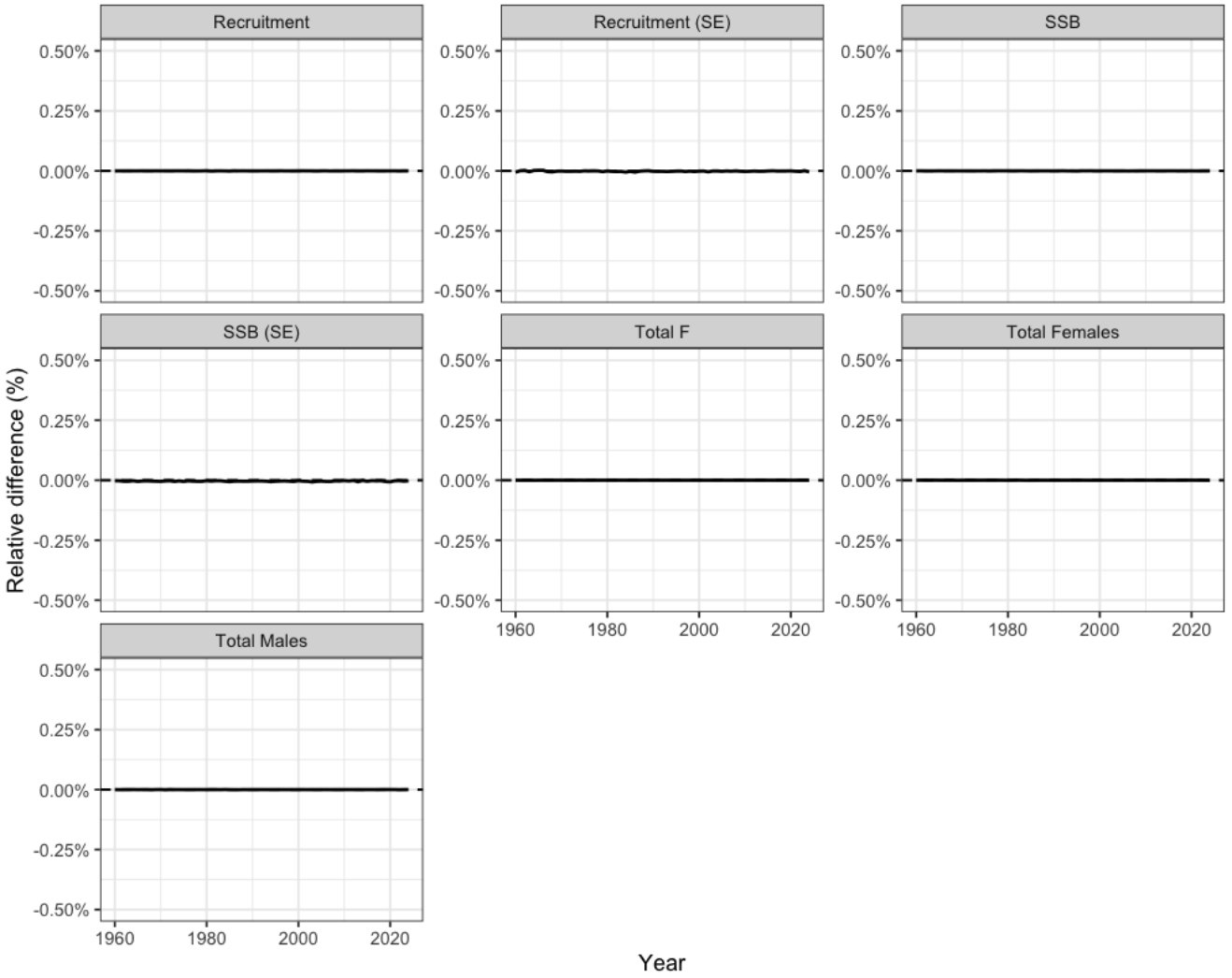

Figure 3B.1. Results of the 2025 model-bridging exercise moving the sablefish assessment from ADMB to SPoRC (R-TMB), which demonstrates the post-optimization relative percent difference ((ADMB - R-TMB) / ADMB) in time-series quantities. The black dashed horizontal line indicates 0% difference between models.

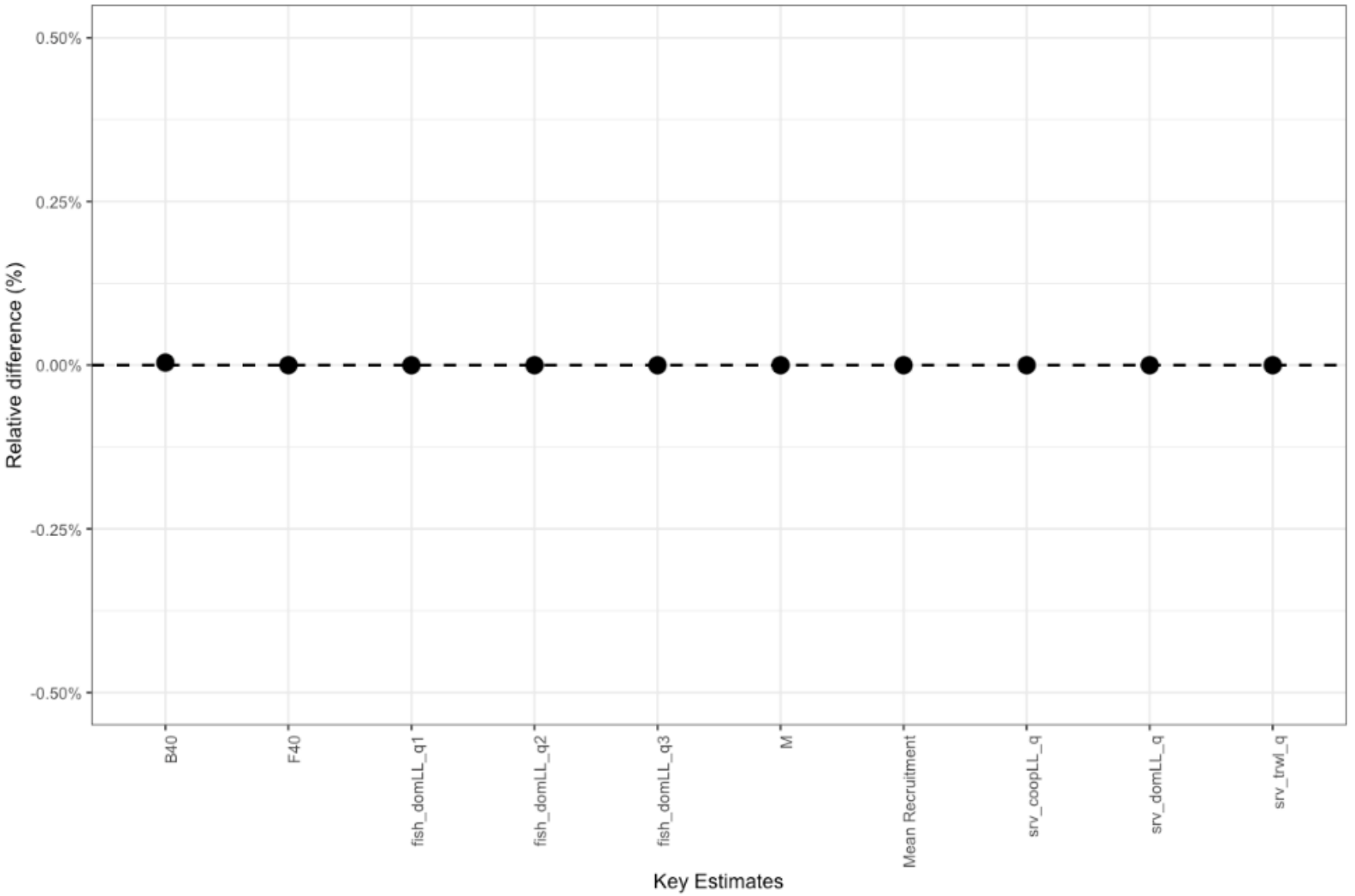

Figure 3B.2. Results of the 2025 model-bridging exercise moving the sablefish assessment from ADMB to SPoRC (R-TMB), which demonstrates the post-optimization relative percent difference ((ADMB - R-TMB) / ADMB) key parameter estimates. The black dashed horizontal line indicates 0% difference between models.

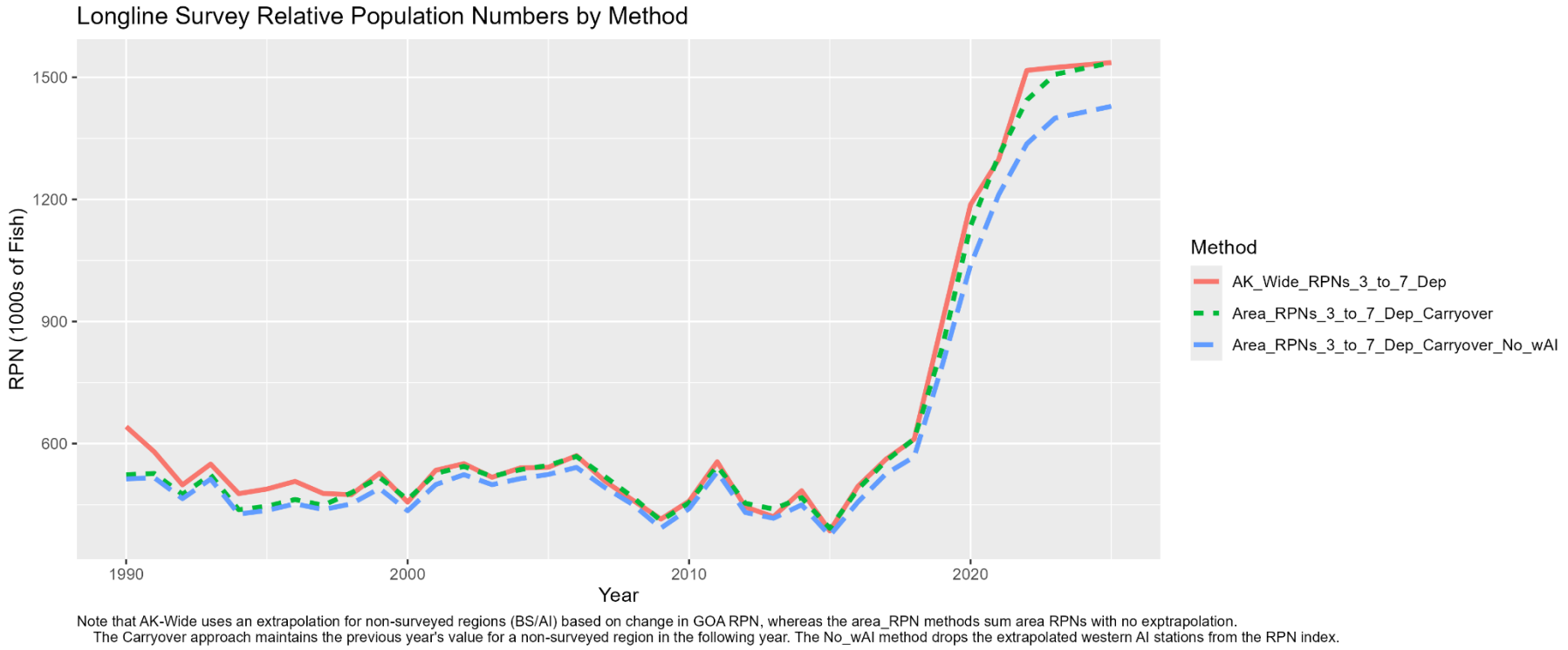

Figure 3B.3. Comparison of NOAA Longline Survey indices (relative population numbers, RPNs) resulting from different approaches for calculating the index. The 'AK_Wide_RPNs' method (red line) uses the growth in the GOA to extrapolate RPNs for non-surveyed regions (BS or AI) in off years, but switched to the 'carryover' method in 2025 (since it cannot be continued in the future given the new survey design). The 'Area_RPNs_Carryover' method (green line) uses the last survey RPN value for a non-surveyed region in off years as a carryover value. The 'Area_RPN_Carryover_No_wAI' method (blue line) uses the carryover approach, but drops the extrapolated (not surveyed in all years) western AI stations from the index. All models through Model *26.1g* (including Model *25.12_1Reg_Cont*) utilize the 'AK_Wide' method, Model *26.1h* uses the 'Area_RPNs_Carryover' method, and Model *26.1i* (including Model *26.1_1Reg*) uses the 'Area_RPN_Carryover_No_wAI' method.

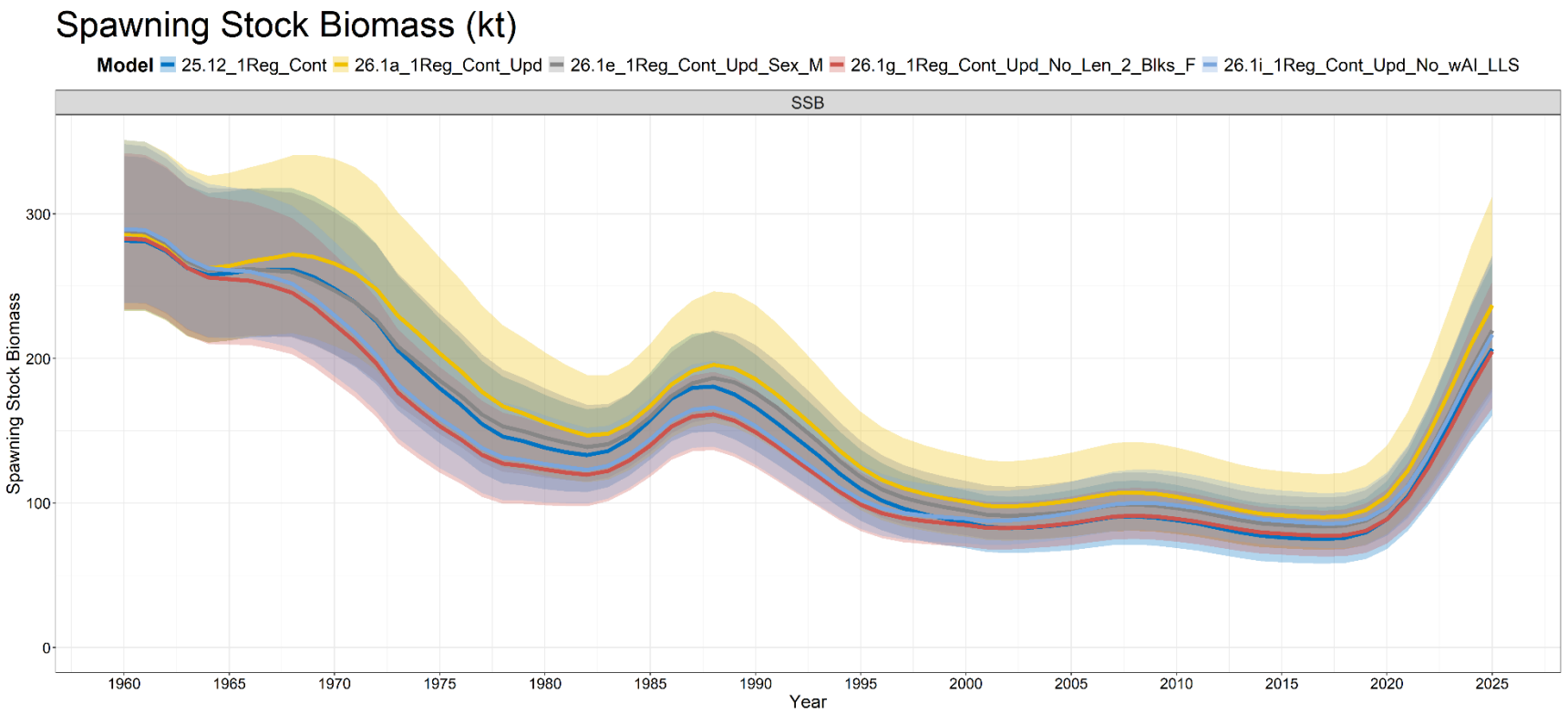

Figure 3B.4. Comparison of spawning stock biomass (SSB, in kilotons) across select model runs representing each grouping, where shaded regions represent the 95% confidence intervals.

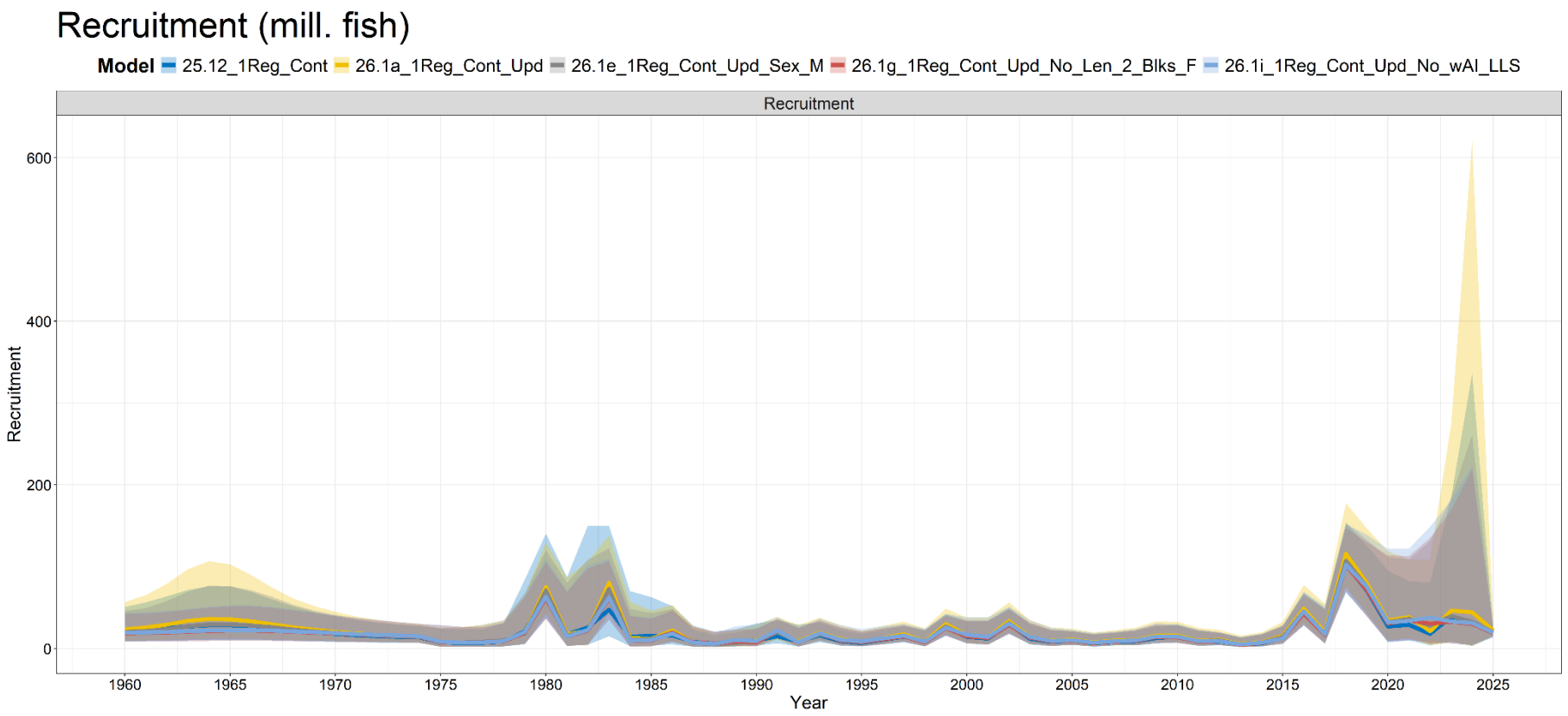

Figure 3B.5. Comparison of recruitment (in millions of fish) across select model runs representing each grouping, where shaded regions represent the 95% confidence intervals.

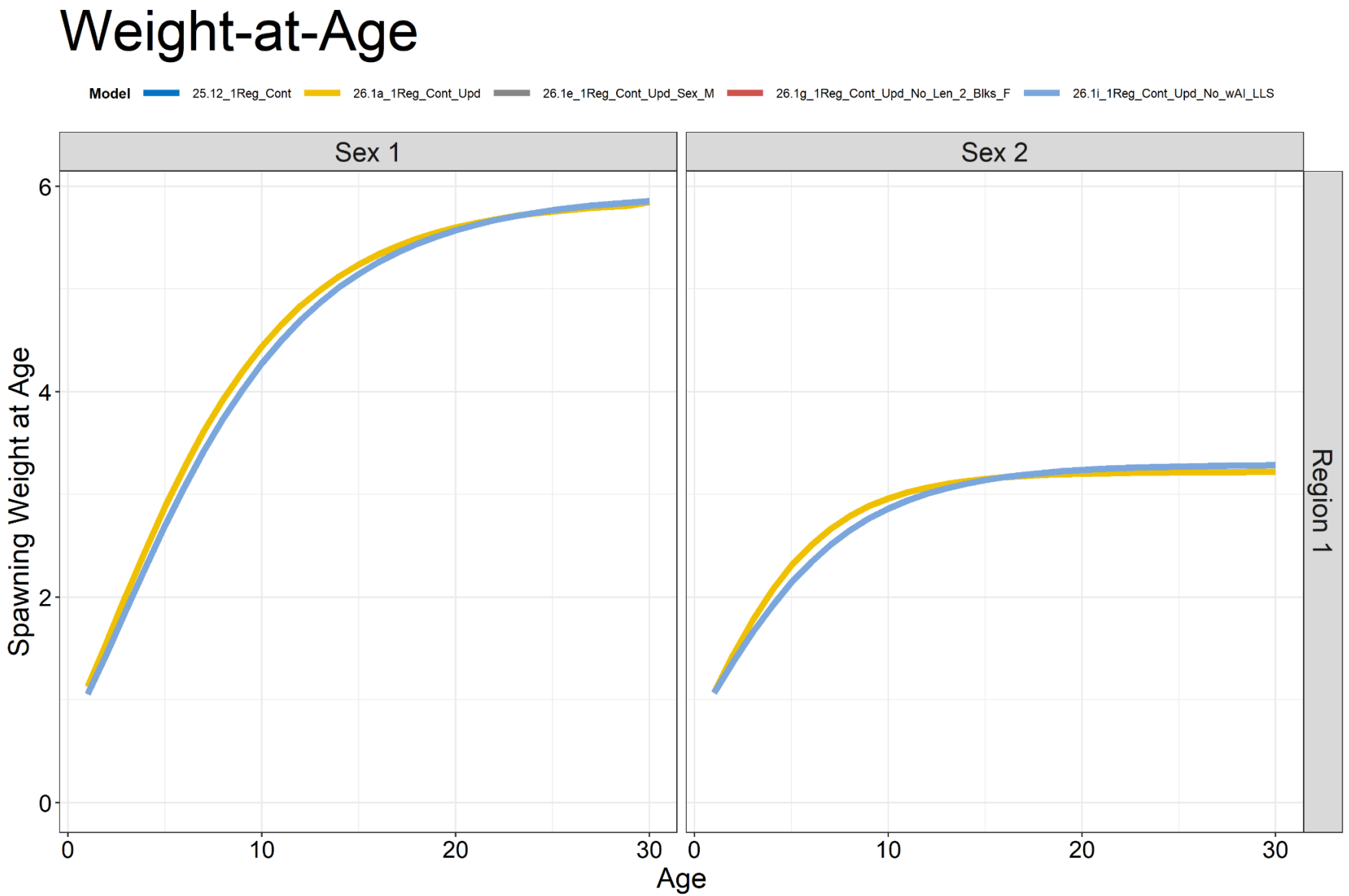

Figure 3B.6. Comparison of weight-at-age across select model runs representing each grouping with yellow line indicating original growth parameters and blue-line represents updated biological data. All models from Model *26.1d* onward use updated growth and weight-at-age (blue line). Sex 1 is females and sex 2 is males.

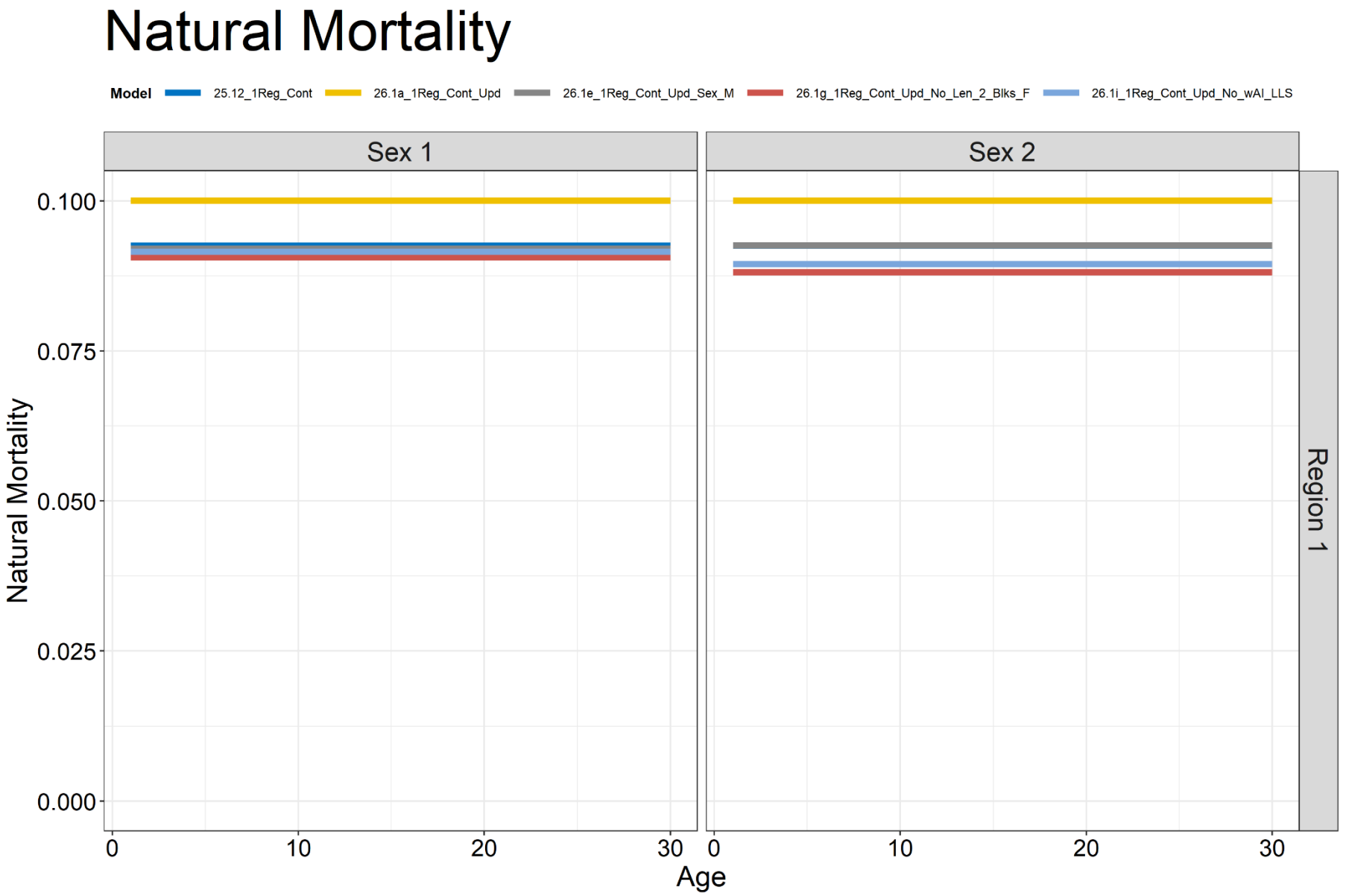

Figure 3B.7. Comparison of natural mortality estimates across select model runs representing each grouping. All models from Model *26.1e* onward use estimate sex-specific *M*. Sex 1 is females and sex 2 is males.

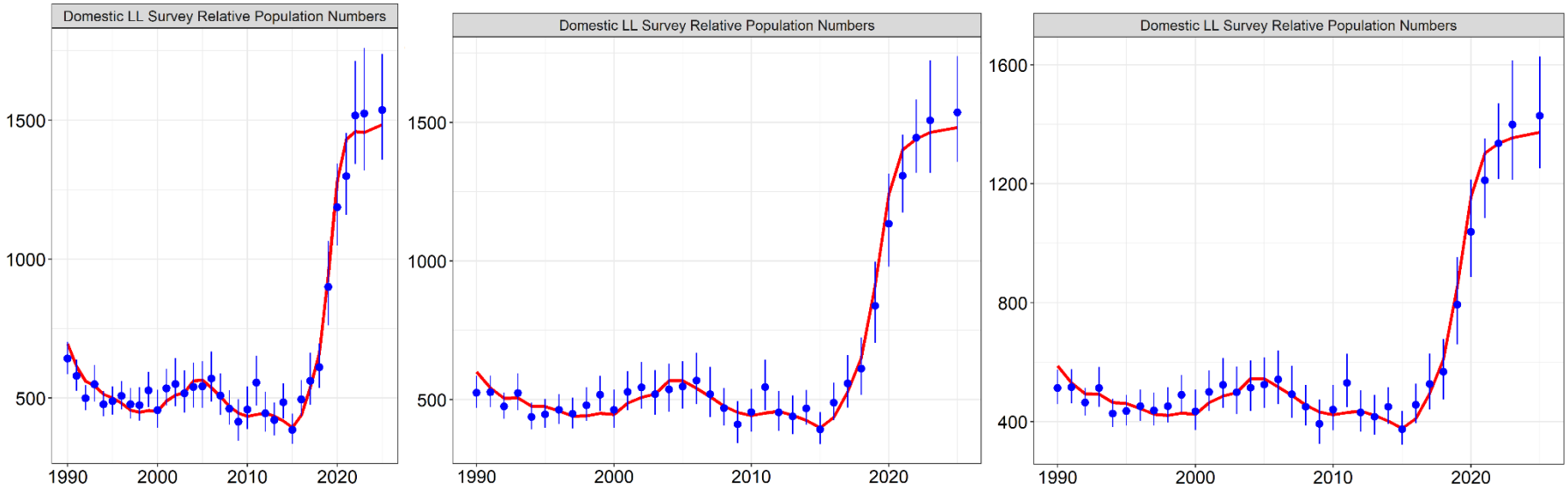

Figure 3B.8. Fits (red lines) to the NOAA Longline Survey abundance indices in relative population numbers (RPNs – 1000s of fish; blue dots with vertical lines represent 95% confidence intervals) for models with various methods of construction (see text and Figure 3B.3). The left panel is Model *26.1e*, which uses the traditional 'AK_Wide' index using an extrapolation for non-surveyed regions based on GOA growth, and still fits to the fixed gear fishery CPUE index (see text). The center panel is Model *26.1h*, which uses the 'carryover' index using an extrapolation for non-surveyed regions based on a carryover value from the last time that region was surveyed, and does not fit to the fixed gear fishery CPUE index. The right panel is Model *26.1i*, which is the same as Model *26.1h* except that extrapolations for non-surveyed western AI stations are removed from the index. Note the difference in y-axis (RPN) scaling across panels.

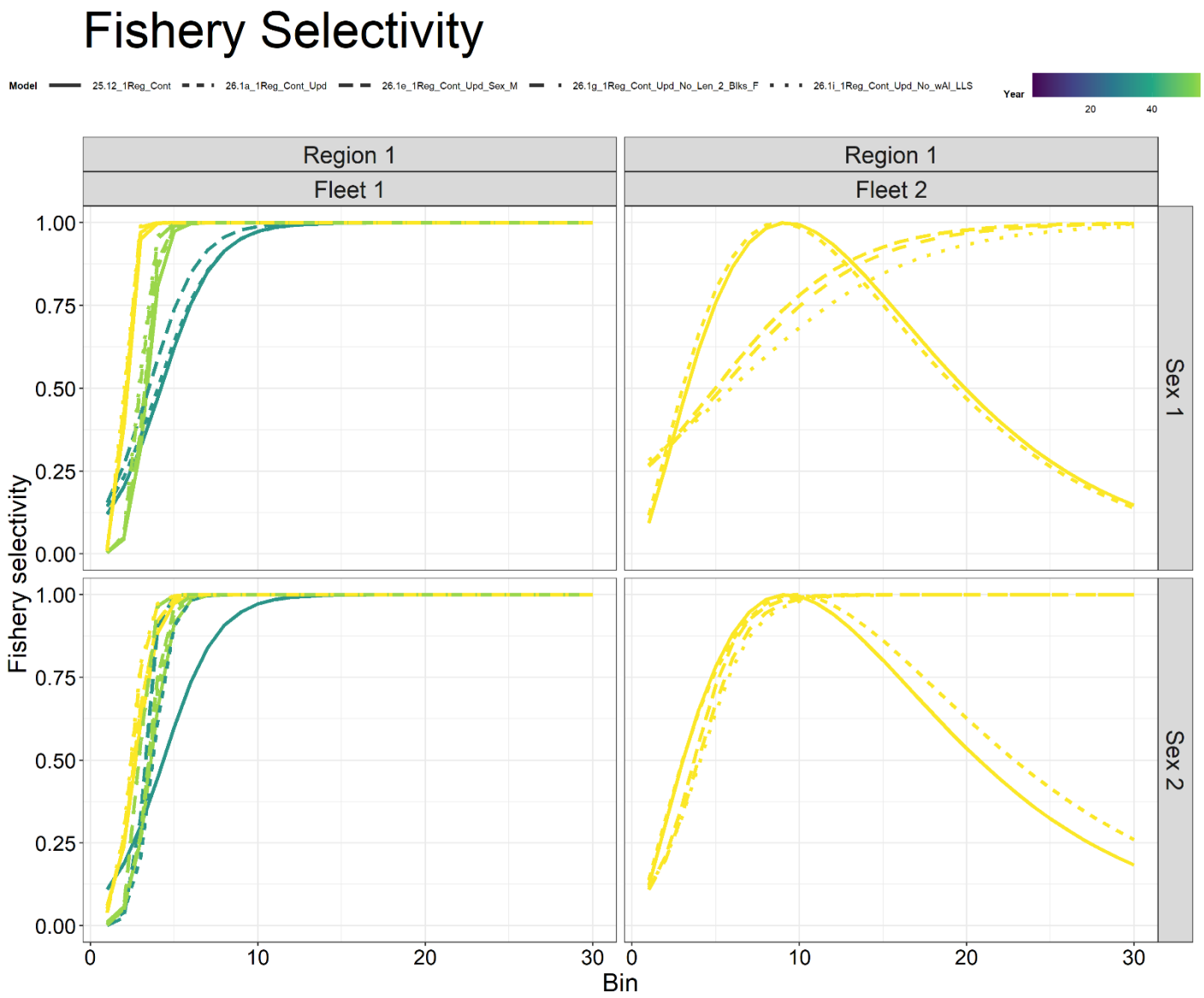

Figure 3B.9. Comparison of fishery selectivity across for select model runs representing each grouping. All models from Model *26.1c* onward assume a logistic (instead of dome) selectivity for the trawl fishery. Sex 1 is females and sex 2 is males. Fleet 1 is the fixed gear fishery and Fleet 2 is the trawl fishery.

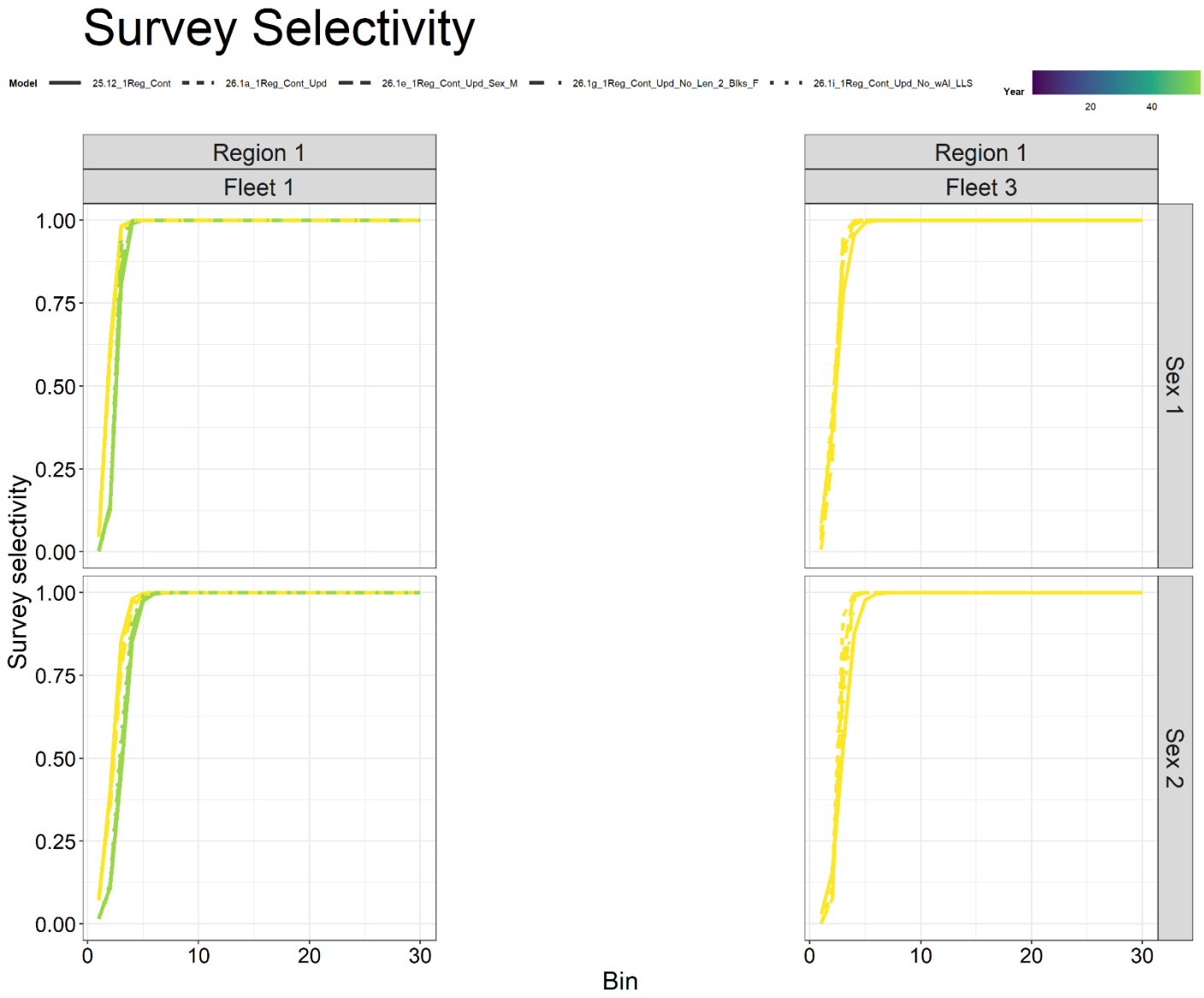

Figure 3B.10. Comparison of survey selectivity across for select model runs representing each grouping. Sex 1 is females and sex 2 is males. Fleet 1 is the NOAA Longline Survey and Fleet 3 is the Japanese Longline Survey.

2. Sensitivity Runs

Introduction

Based on previous research on sablefish dynamics, as well as insights gained from the model-building exercise conducted for the single-region model, several sensitivity runs were developed. The first sensitivity model explored the use of time-varying, empirical weight-at-age (EWAA) to account for density-dependent growth associated with recent large year classes (*26.1_1Reg_TVWAA*; Cheng et al. 2024), with the objective of better reflecting the biological characteristics of Alaska sablefish. A second sensitivity model implemented continuous semi-parametric time-varying survey selectivity for the NOAA longline survey to evaluate whether a more flexible selectivity formulation could better capture temporal changes in survey availability (*26.1_1Reg_ContTVSel*). In particular, this approach was intended to represent the spatial progression of large recruitment events originating in western regions and subsequent ontogenetic movement toward eastern regions (see Appendix 3D), while also reducing tension between composition data and abundance indices. Although a similar approach could also be considered for the longline fishery, additional exploration was not pursued because: 1) previous analyses indicated relatively limited evidence for substantial temporal variation in fishery selectivity, and 2) the fixed-gear fishery is primarily concentrated in the GOA, where large-scale shifts in availability to the fishery are less likely to occur. For similar reasons, an additional sensitivity model implementing the Dirichlet-multinomial likelihood for composition data was explored to evaluate whether model tension arising from unaccounted-for process error could be better accommodated relative to the Francis re-weighting approach (*26.1_1Reg_DM*). In particular, the Francis method has the potential to substantially downweight composition data, whereas the Dirichlet-multinomial likelihood provides an alternative framework for accounting for overdispersion in composition data. The final sensitivity run explored the combined effects of continuous semi-parametric time-varying survey selectivity, and the Dirichlet-multinomial likelihood for composition data (*26.1_1Reg_ContTVSel_DM*). This combined sensitivity model was intended to evaluate the extent to which time-varying selectivity and the Dirichlet-multinomial jointly interacted.

Methods

The sensitivity model runs follow the parameterization of Model *26.1_1Reg* with updates as noted below, and results are compared to that model. Francis reweighting was only utilized for model *26.1_1Reg_TVWAA*. This was done because Francis reweighting is known to substantially upweight composition data when time-varying selectivity is specified (Xu *et al.* 2019), and because the Dirichlet-multinomial formulation inherently accounts for overdispersion in composition data and therefore does not require additional composition re-weighting.

The configuration of sensitivity model runs was as follows:

- *26.1_1Reg_TVWAA*:
    - Time-varying EWAA was incorporated into the model following the methods from Cheng et al. (2024). Specifically, annual weight-at-age values were estimated external to the assessment model. An initial set of EWAA estimates was first derived using bootstrapping, after which a model-based smoothing framework was applied to account for correlations among age, year, and cohort effects.

- *26.1_1Reg_ContTVSel*:
    - The NOAA Longline Survey selectivity was parameterized to exhibit continuous semi-parametric time-varying selectivity. The parametric form assumed a logistic shape, while semi-parametric lognormal deviations were specified for each age to be independent and identically distributed (iid) random variables following a normal distribution. These deviations were estimated within a penalized likelihood framework, with the standard deviation fixed at 0.5.

- *26.1_1Reg_DM*:
    - Composition likelihoods were specified assuming the linear parameterization of a Dirichlet-multinomial distribution (Thorson *et al.*, 2017). The Dirichlet-multinomial likelihood was applied to all composition datasets except those associated with the Japanese longline survey length compositions. Owing to the sparse nature of the Japanese longline survey data, and the resulting difficulty in estimating overdispersion parameters for this dataset during initial model explorations, a multinomial likelihood was instead assumed for these observations.

- *26.1_1Reg_ ContTVSel_DM*:
    - It is generally recommended to explore the use of continuous time-varying selectivity and self-weighting likelihoods in tandem (Xu *et al.* 2020). Accordingly, this model configuration combined the attributes of Models *26.1_1Reg_ContTVSel* and *26.1_1Reg_DM*, such that time-varying survey selectivity and Dirichlet-multinomial likelihoods were implemented simultaneously.

Results

- *26.1_1Reg_TVWAA* (full results here):
    - Similar to the results from Cheng et al. (2024), recent declines in weight-at-age were apparent particularly for young to intermediate-aged individuals, likely attributed to density dependent effects from recent large recruitment events (Fig. 3B.11).
    - Potentially unrealistic increases in weight-at-age were observed for older females, likely reflecting parameter correlation between the intrinsic growth rate and asymptotic length parameters, exacerbated by limited data availability for older ages (Fig. 3B.11).
    - Incorporation of EWAA into the assessment yielded intuitive results, where recent declines in weight-at-age coincided with declines in *SSB*, along with a downward revision in population scale relative to Model *26.1_1Reg* (Fig. 3B.12). Data fits remained largely unchanged (because indices were based on abundance rather than biomass) and estimates of stock status also maintained similar levels (Fig. 3B.13).

- *26.1_1Reg_ContTVSel* (full results here):
    - Specifying continuous semi-parametric time-varying survey selectivity for the NOAA longline survey did not lead to substantiative improvements in model fits to composition data nor abundance indices relative to model *26.1_1Reg* (Fig. 3B.14

– 3B.19). Moreover, aggregate fits to young individuals appeared somewhat degraded for *26.1_1Reg_ContTVSel.*

- Moderate variability in survey selectivity estimates was detected, with recent years exhibiting some degree of doming, likely reflecting the influence of large recruitment events. The magnitude of this variability was partly attributable to the subjective specification of a standard deviation of 0.5 within the penalized likelihood framework (chosen based on initial explorations attempting to balance composition and index fits). Attempts to estimate time-variation in a state-space framework were attempted, but proved computationally challenging (i.e., convergence issues were encountered) (Fig. 3B.20).
- The introduction of additional variability in survey selectivity led to degraded retrospective performance, resulting in moderate and consistent overestimation of *SSB* relative to the terminal assessment (Fig. 3B.21).
- In general, while moderate variability in survey selectivity was detected, no substantial differences in model fit, estimated stock status, or interpretation of population dynamics were observed compared to model *26.1_1Reg* (Fig. 3B.22 – 3B.23). Moreover, moderate retrospective patterns were evident, likely driven by continual revision of recruitment and survey selectivity estimates as cohorts progressively entered the observed population. These dynamics introduce instability in retrospective performance, making this model difficult to recommend as a viable option for assessment use.

- *26.1_1Reg_DM* (full results here):
  - The use of a Dirichlet-multinomial likelihood resulted in slightly degraded fits to the abundance index. Relative to Francis reweighting, composition data were generally upweighted, which contributed to the slight deterioration in fit to the abundance index (Fig. 3B.24). Despite this upweighting, improvements in fits to composition data were limited (Fig. 3B.25 – 3B.29).
  - The introduction of additional overdispersion parameters resulted in slightly degraded retrospective patterns compared to *26.1_1Reg*, particularly with respect to recruitment estimates of large year classes being often underestimated relative to the terminal assessment. *SSB* retrospective patterns remained small (Mohn's rho of ~0.02; Fig. 3B.30).
  - Recruitment estimates became more variable compared to model *26.1_1Reg*, with an upward revision in overall population scale (Fig. 3B.31 – 3B.32). Presumably, the upward revision in population scale is due to upweighting composition data relative to the Francis re-weighting approach, where the model then attempts to increase the number of recruits in the population to better fit the composition data. The increase in the number of recruits then increases population scale in order to adequately fit to catch data (i.e., to remove the "correct" amount of individuals from the population).
  - Although the Dirichlet-multinomial likelihood is more statistically rigorous than Francis reweighting, the model was generally unable to provide an adequate trade-off between fit to composition and indices simultaneously.

- *26.1_1Reg_ContTVSel_DM* (full results here):

- Combining continuous time-varying survey selectivity with a Dirichlet-multinomial likelihood for composition data generally improved model fits. However, fits to composition data still remained moderately overdispersed and showed little substantive change in interpretation relative to *26.1_1Reg* (Fig. 3B.33 – 3B.38).
- Strong retrospective patterns persisted, likely driven by constant revision of recruitment estimates as cohorts aged into the population given the more flexible survey selectivity parameterization (estimates were largely similar to *26.1_1Reg_ContTVSel*; Fig. 3B.20, 3B.39). Nonetheless, interpretation of population dynamics remained similar, although some differences in recruitment estimates were detected towards the end of the time-series (Fig. 3B.41 – 3B.42). Given the increased retrospective instability and the lack of meaningful improvement in model fit, this configuration does not appear to offer substantive improvement over Model *26.1_1Reg*.

Conclusions

The sensitivity models illustrate both the potential benefits and limitations of more complex model parameterizations. In general, none of the sensitivity runs led to substantial changes in the interpretation of population dynamics, nor did they meaningfully improve model fits as compared to Model *26.1_1Reg*. Moreover, none of the sensitivity runs appear to address the issue of extrapolating for unsampled survey regions for the NOAA Longline Survey indices. Incorporating EWAA appears to be a potentially useful option; however, several issues remain that warrant attention, particularly regarding the realism of estimates at older ages. These may be addressed through the application of smoothing penalties as well as by incorporating EWAA directly within the assessment model itself. Moreover, the inclusion of time-varying survey selectivity or self-weighting likelihoods did not consistently improve fits to the data and, in some cases, resulted in degraded retrospective performance. Collectively, the sensitivity runs provide a useful foundation for exploring alternative model parameterizations in future assessments, but further investigation is required to better understand their performance and stability before they could be recommended as the basis for operational management advice.

Tables

Table 3B.4. Summary of sensitivity model runs and associated results.

Model Group	*Model Name*	*Major Changes*	*Rationale*	*Conclusions*
Sensitivity Runs	*26.1_1Reg_TVWAA*	(*26.1_1Reg+*) Incorporate time-varying empirical weight-at-age.	Large recruitment events in recent years likely resulted in density-dependent declines in weight-at-age. Incorporating time-varying weight-at-age is expected to improve the biological realism of the model.	Declines in weight-at-age were evident in recent years. However, unrealistic increases in weight-at-age for older females were detected, likely due to sparse data at older ages. Further investigation and potential smoothing are required before operational use.
	26.1_1Reg_ContTVSel	(*26.1_1Reg +*) Specify continuous semi-parametric time-varying survey selectivity for the domestic NOAA longline survey.	Intended to better capture temporal changes in survey selectivity and associated changes in availability due to recruitment and ontogenetic movement.	No substantive improvements in fit to composition data or abundance indices were observed relative to *26.1_1Reg*. Moderate variability in selectivity was detected, but retrospective performance was degraded.
	26.1_1Reg_DM	(*26.1_1Reg +*) Specify Dirichlet-multinomial likelihoods for composition data.	Designed to account for overdispersion in composition data and provide a more flexible statistical treatment than Francis reweighting.	Fits to abundance indices were slightly degraded, and composition data were upweighted relative to indices. Retrospective performance of recruitment degraded, with an increase in recruitment variability.
	26.1_1Reg_ContTVSel_DM	(*26.1_1Reg +*) Combines *26.1_1Reg_ContTVSel* and *26.1_1Reg_DM* and specifies continuous semi-parametric time-varying survey selectivity alongside Dirichlet-multinomial likelihoods for composition data.	Time-varying selectivity and self-weighting likelihoods likely interact. This model is intended to address the concerns of both *26.1_1Reg_ContTVSel* and *26.1_1Reg_DM*.	Some reconciliation of misfit was observed (relative to *26.1_1Reg_ContTVSel* and *26.1_1Reg_DM*) when combining approaches; however, composition data remained overdispersed and overall interpretation changed little relative to *26.1_1Reg*. Strong retrospective instability persisted, and no meaningful improvement in fit was achieved.

Table 3B.5. Key management quantity outputs for sensitivity runs. SSB is in kt, B_Ref_Pt is the associated $B_{40\%}$ reference point in kt, and F_Ref_Pt is the associated $F_{40\%}$.

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
26.1i_1Reg_Cont_Upd_No_wAI_LLS	1	215.8208	369.2842	0.04654268	40.71033	139.1895	0.07907	1.55055	0.58443	0.58864
26.1i_1Reg_Cont_Upd_No_wAI_LLS_ContTVSel	1	240.1340	405.2891	0.04234058	44.32321	152.7854	0.07810	1.57171	0.59250	0.54212
26.1i_1Reg_Cont_Upd_No_wAI_LLS_ContTVSel_DM	1	221.8317	389.8287	0.04638192	44.66541	153.8605	0.08404	1.44177	0.56905	0.55192
26.1i_1Reg_Cont_Upd_No_wAI_LLS_DM	1	303.2590	441.4774	0.03691477	63.00107	155.6534	0.09728	1.94830	0.68692	0.37946
26.1i_1Reg_Cont_Upd_No_wAI_LLS_TVWAA	1	165.3681	312.4824	0.05666288	30.36136	117.3838	0.07249	1.40878	0.52921	0.78162

Figures

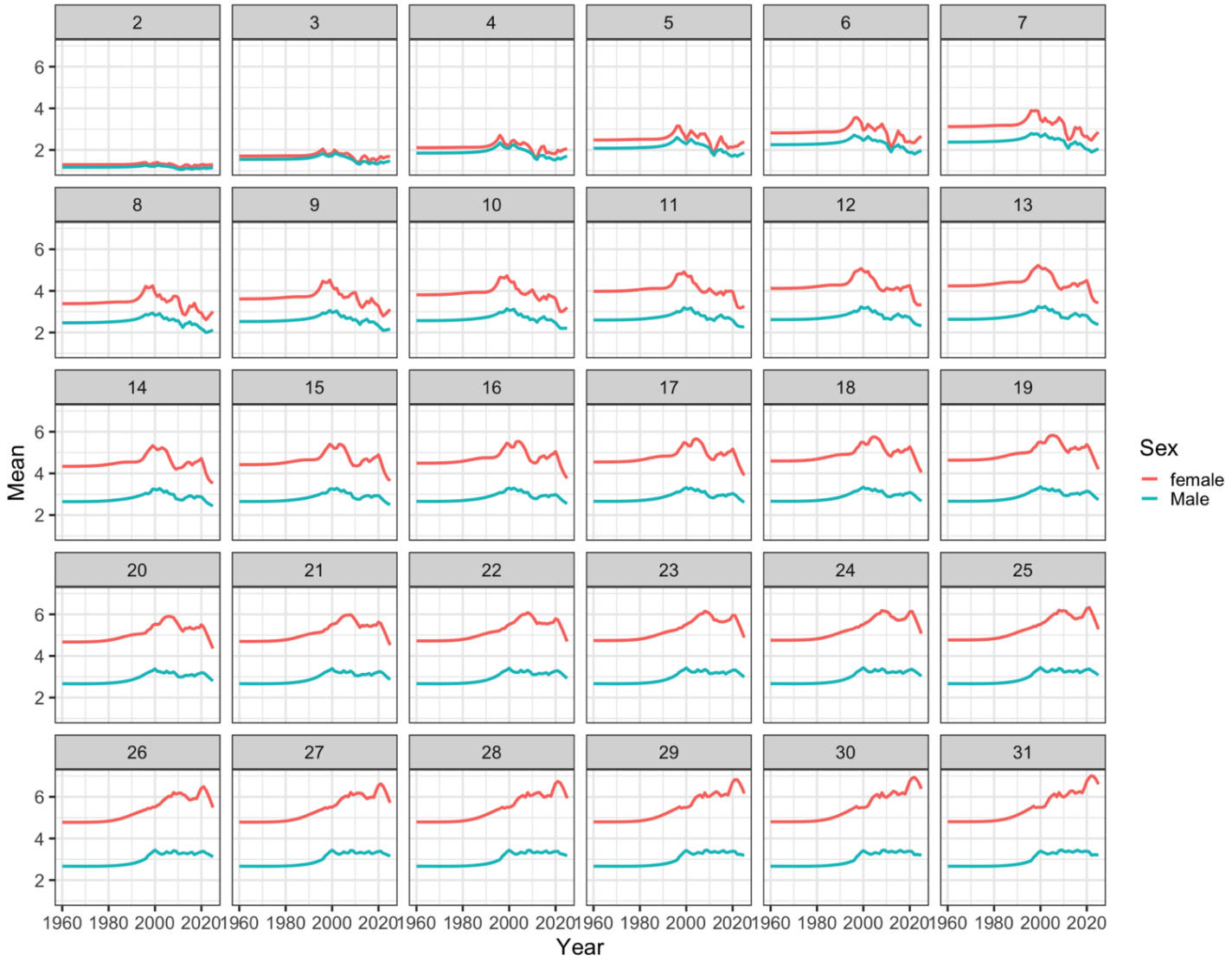

Figure 3B.11. Empirical weight-at-age estimates used in model *26.1_1Reg_TVWAA*. Panels represent different ages, while colored lines represent sexes. Estimates were derived following similar methods as Cheng et al. 2024.

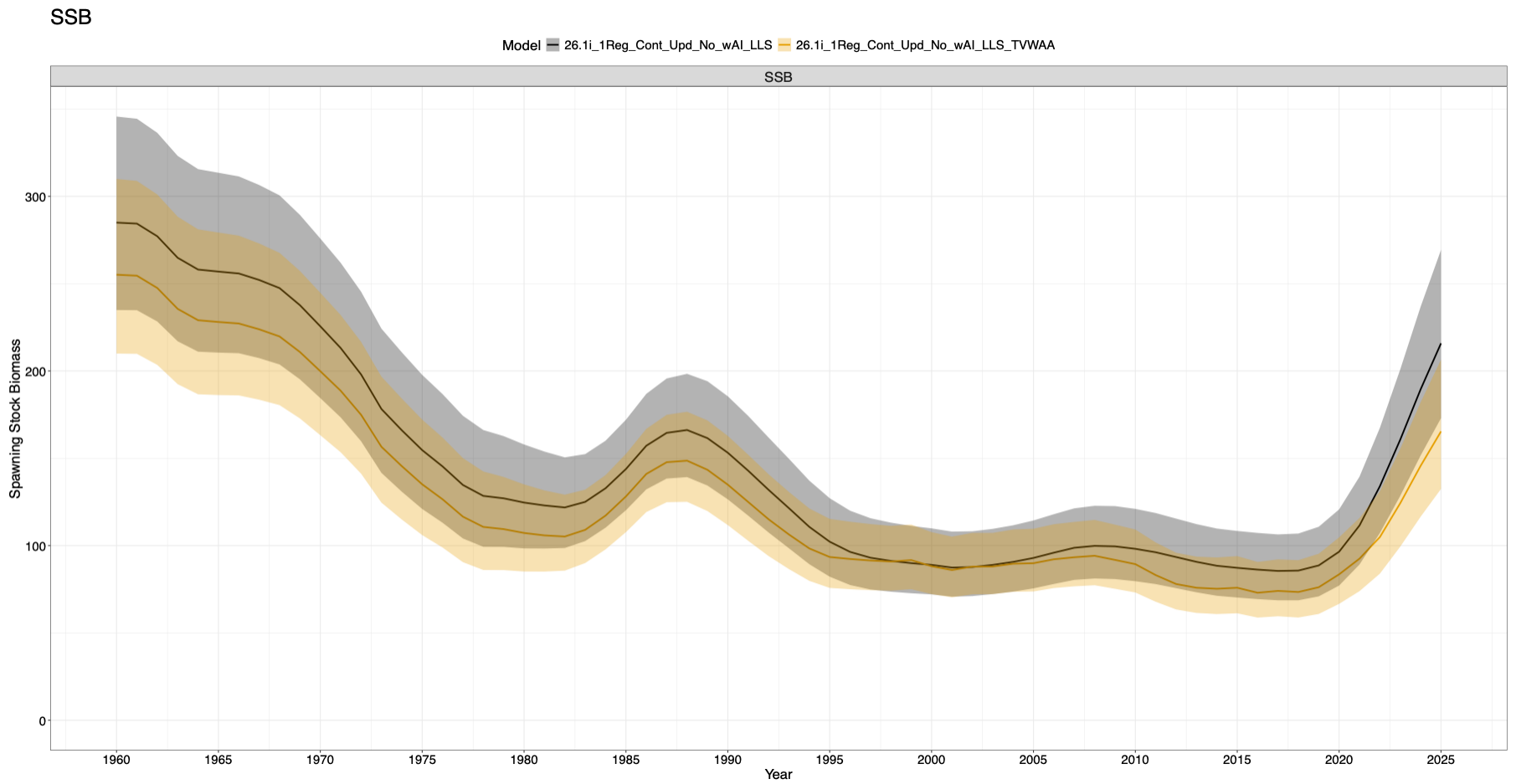

Figure 3B.12. Comparison of spawning stock biomass (SSB, in kilotons) between Model *26.1_1Reg* (also termed *26.1i_1Reg_Cont_Upd_No_wAI_LLS; black line*), along with a configuration that incorporates time-varying empirical weight-at-age (*26.1i_1Reg_Cont_Upd_No_wAI_LLS_TVWAA; yellow line*). Shaded regions represent the 95% confidence intervals.

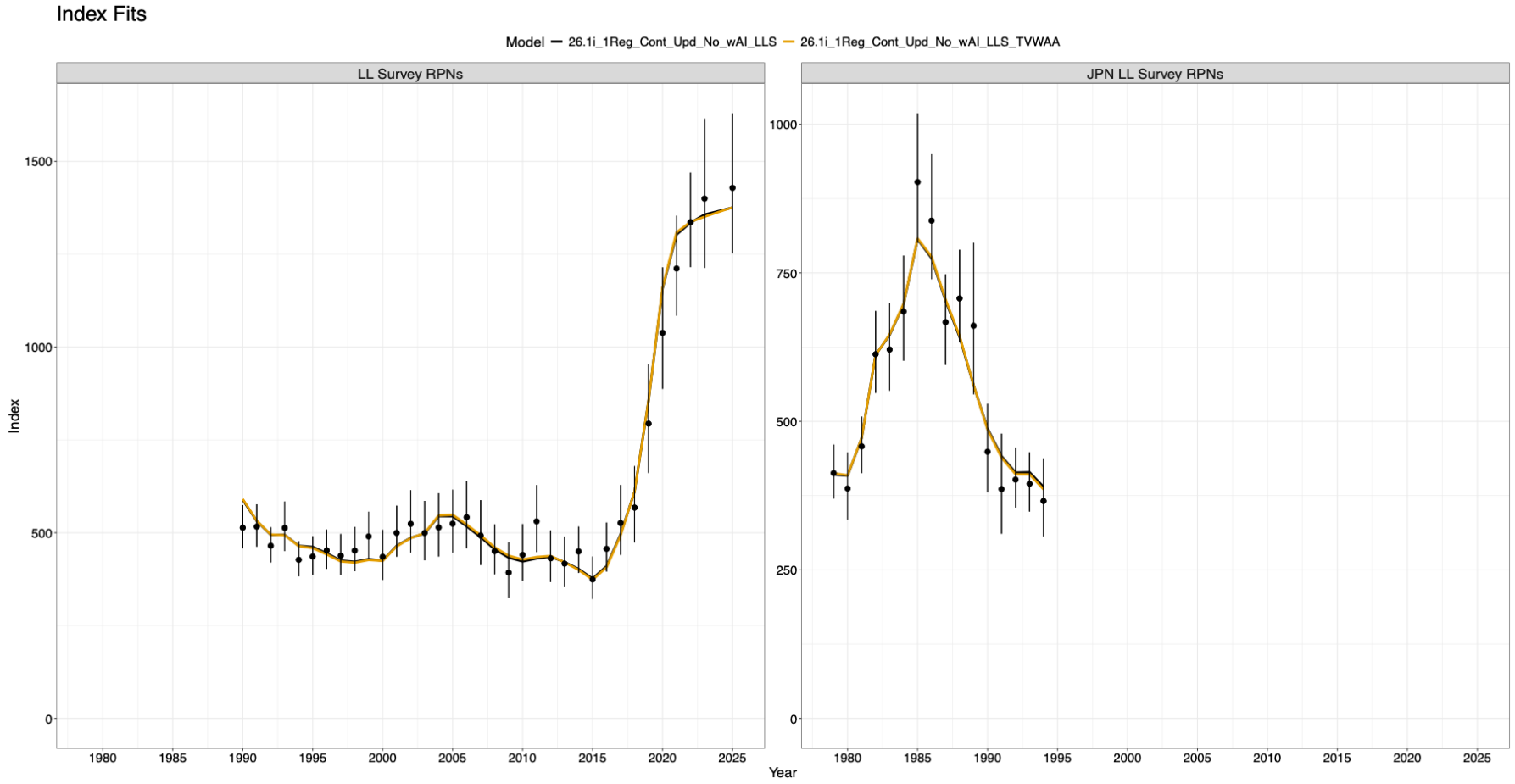

Figure 3B.13. Model fits to survey abundance indices between Model *26.1_1Reg* (also termed *26.1i_1Reg_Cont_Upd_No_wAI_LLS; black line*), along with a configuration that incorporates time-varying empirical weight-at-age (*26.1i_1Reg_Cont_Upd_No_wAI_LLS_TVWAA; yellow line*). Point ranges denote observed data and associated 95% confidence intervals. Solid colored lines represent the model expectation.

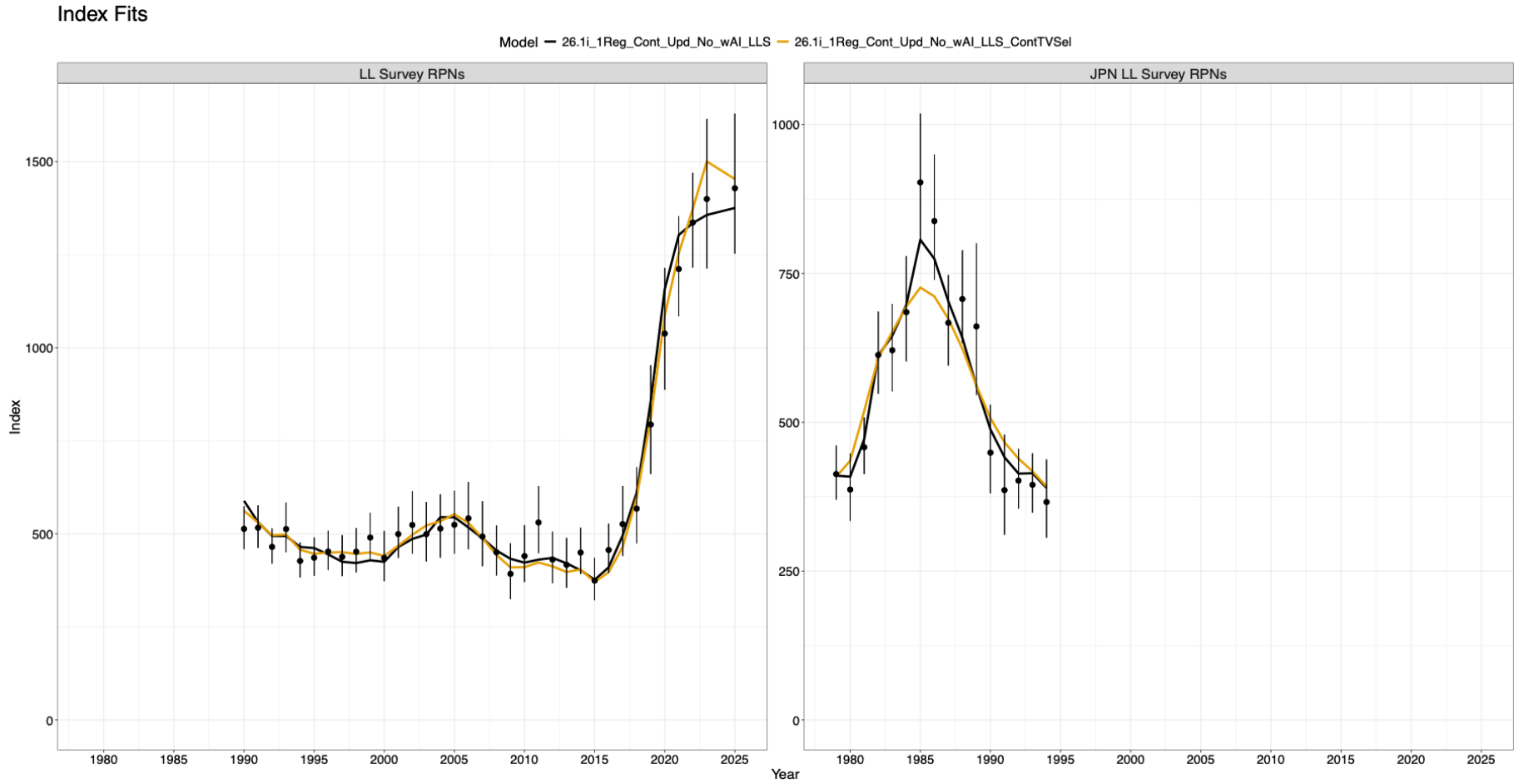

Figure 3B.14. Model fits to survey abundance indices between Model *26.1_1Reg* (also termed *26.1i_1Reg_Cont_Upd_No_wAI_LLS; black line*), along with a configuration that incorporates continuous semi-parametric time-varying survey selectivity (*26.1i_1Reg_Cont_Upd_No_wAI_LLS_ContTVSel; yellow line*). Point ranges denote observed data and associated 95% confidence intervals. Solid colored lines represent the model expectation.

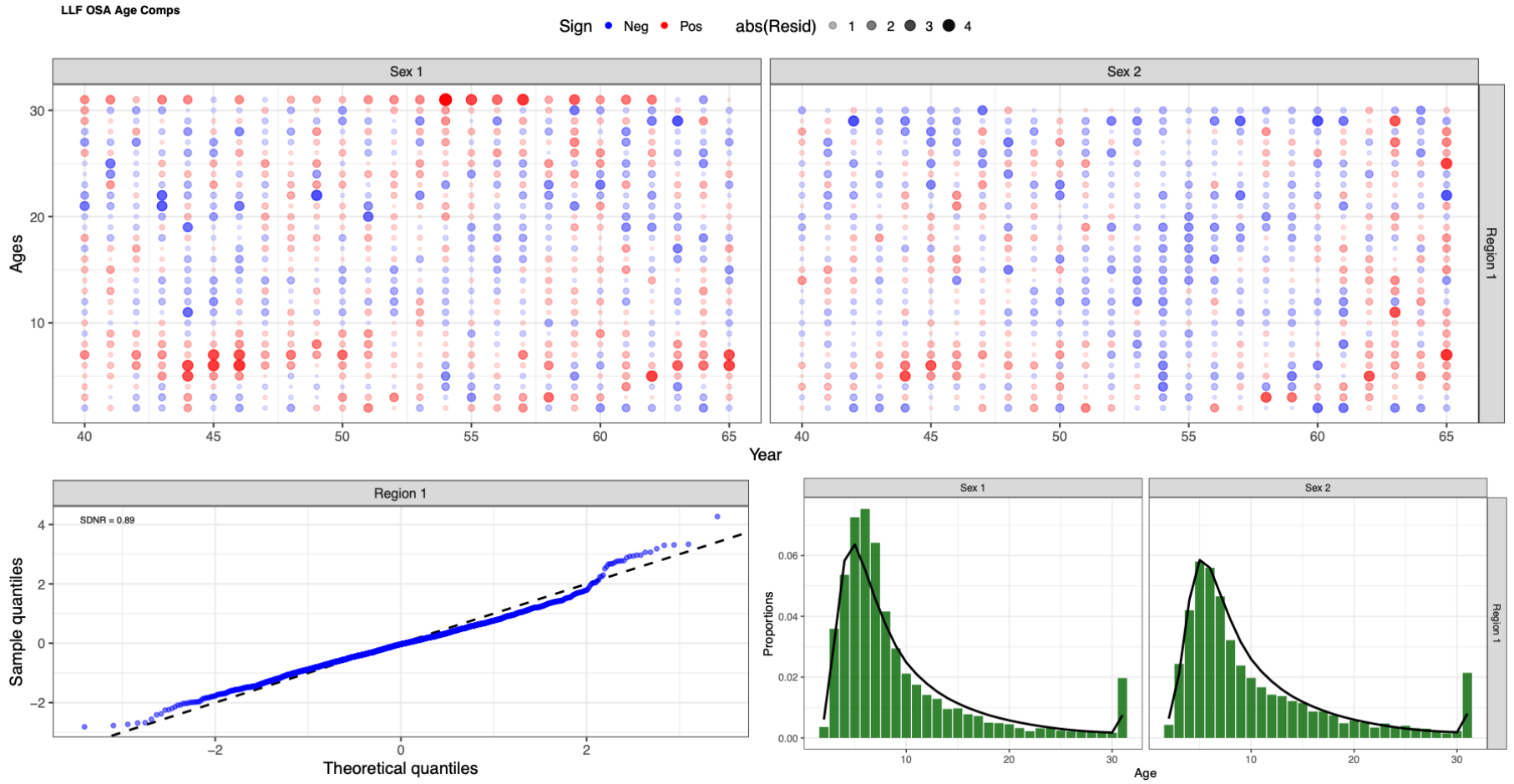

Figure 3B.15. Fits to longline fishery age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg_ContTVSel*.

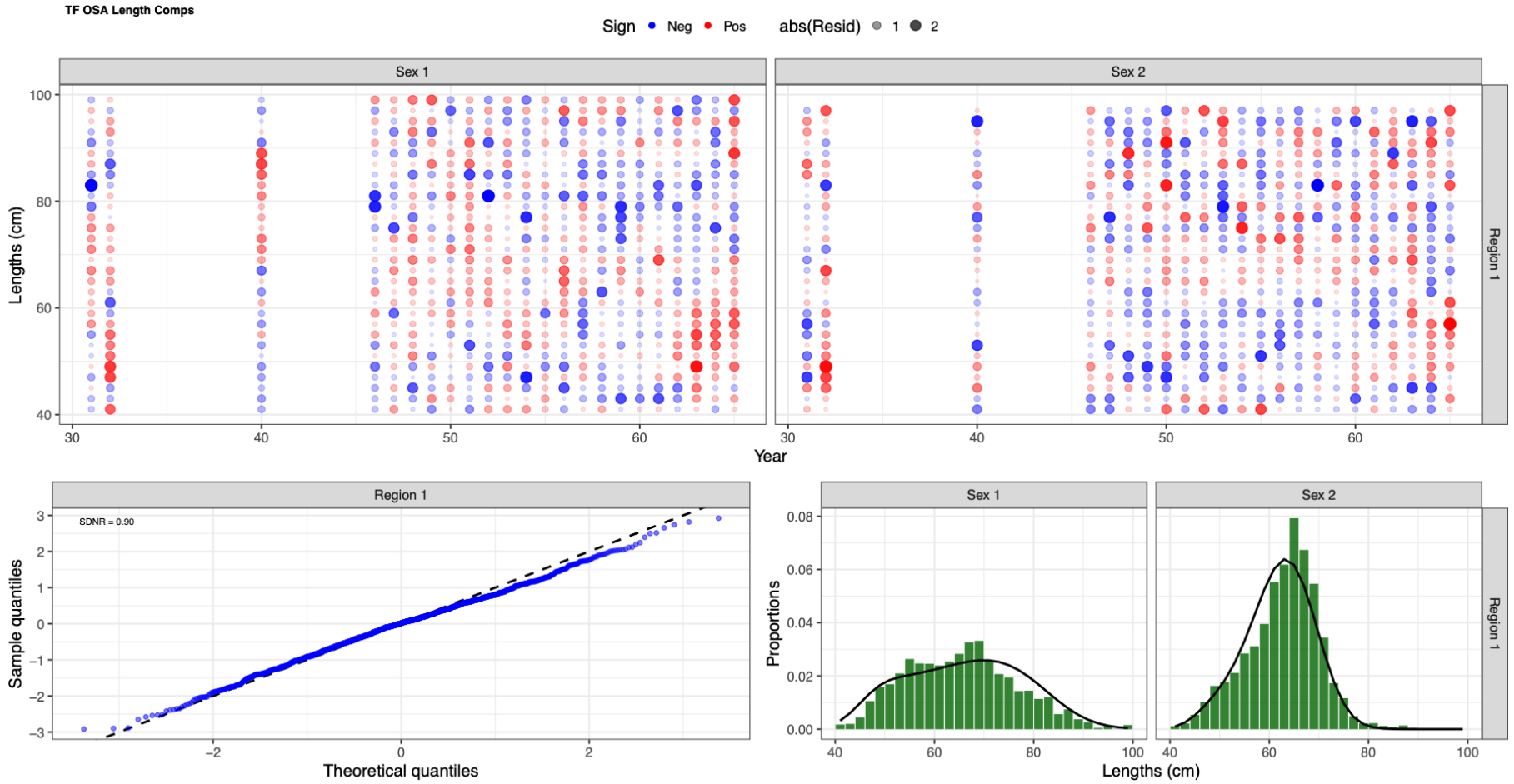

Figure 3B.16. Fits to trawl fishery length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg_ContTVSel*.

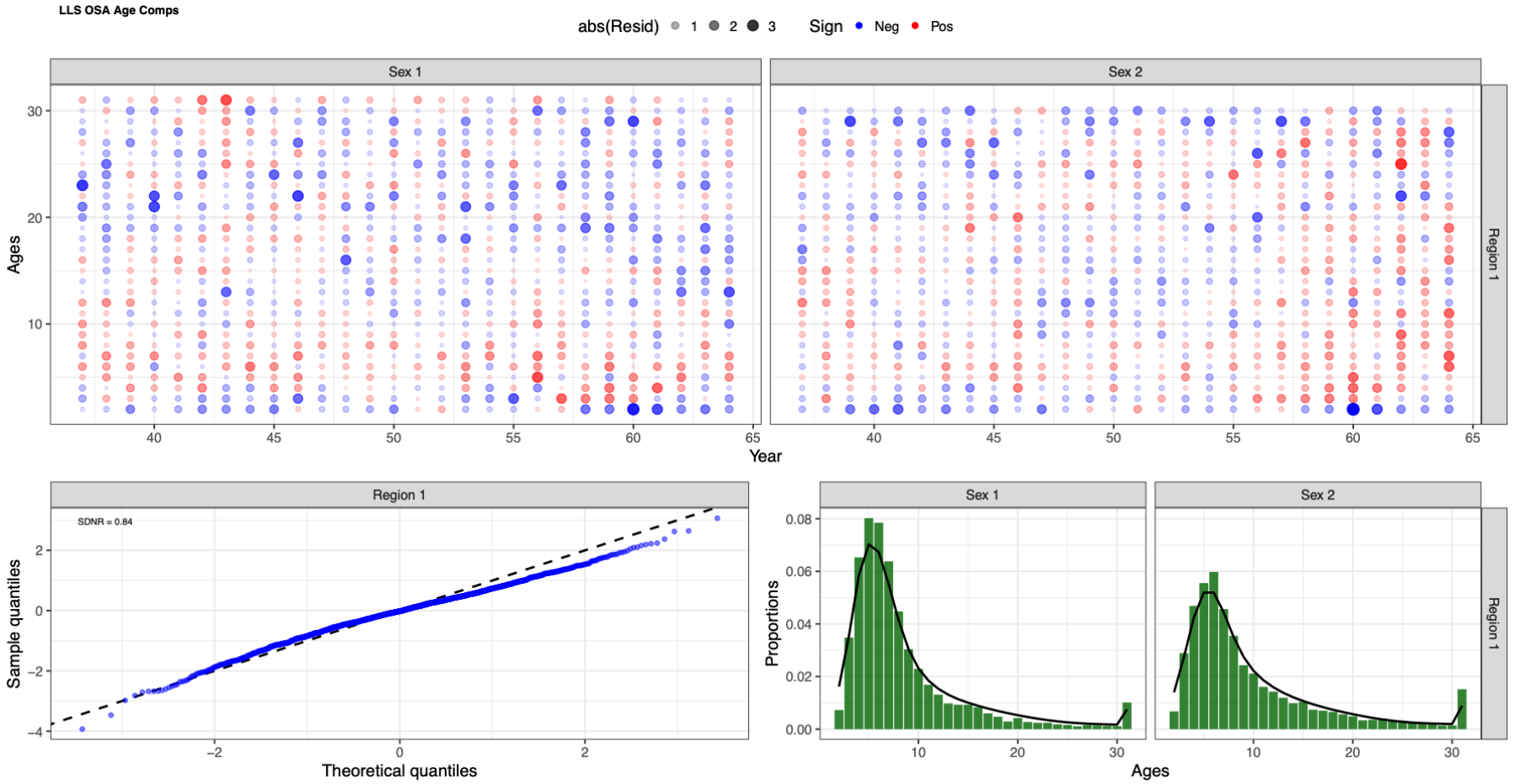

Figure 3B.17. Fits to NOAA longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg_ContTVSel*.

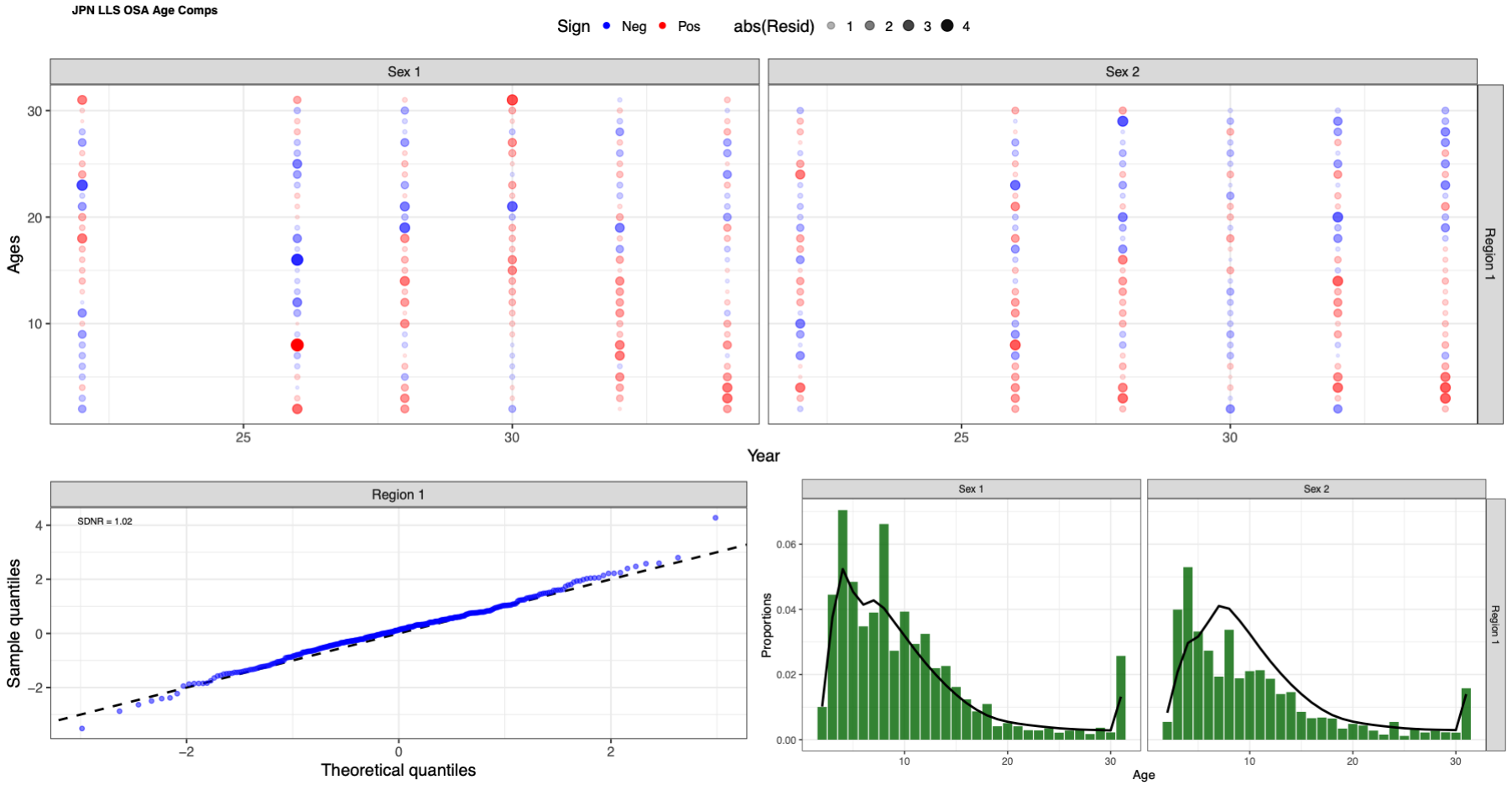

Figure 3B.18. Fits to Japanese longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg_ContTVSel*.

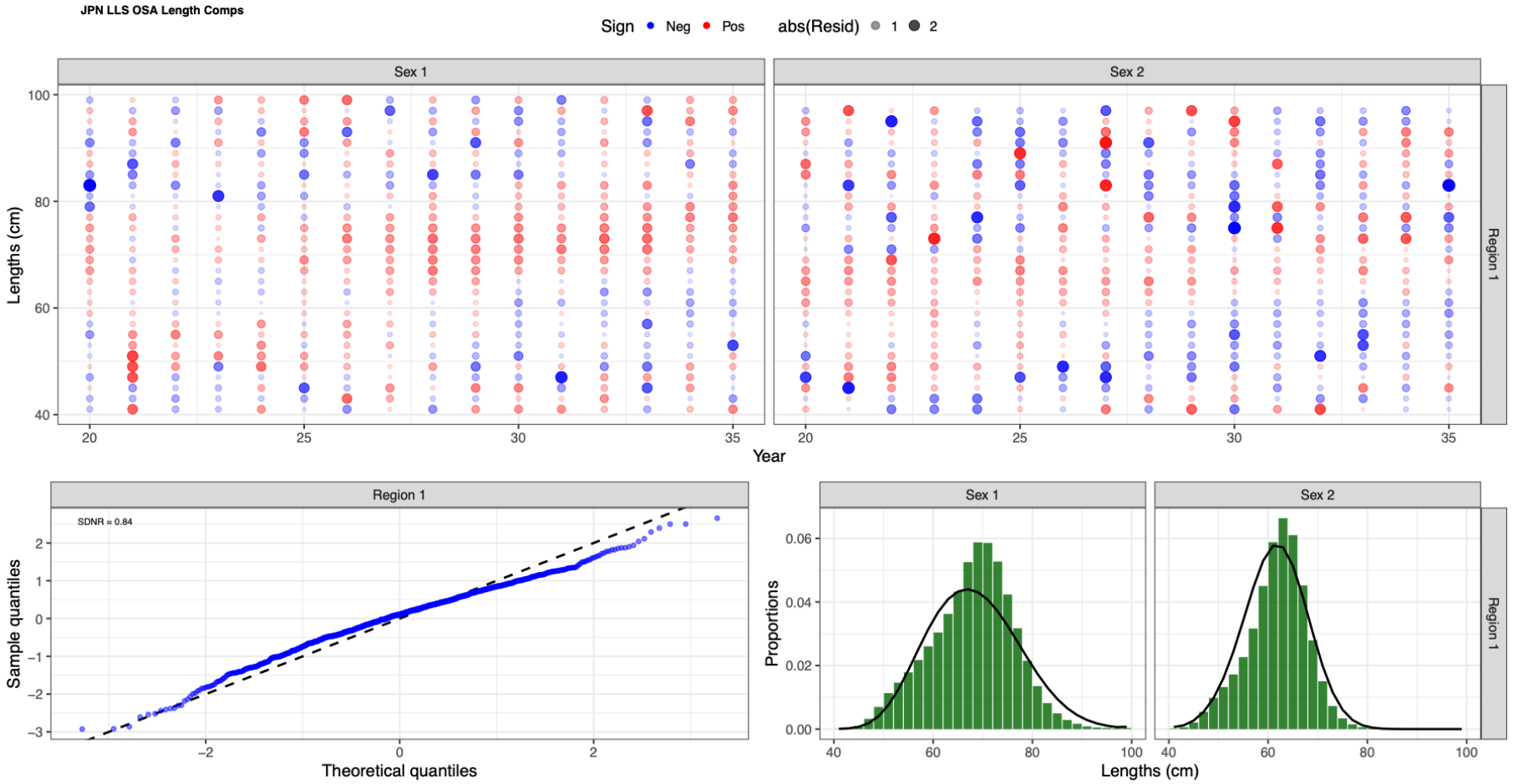

Figure 3B.19. Fits to Japanese longline survey length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg_ContTVSel*.

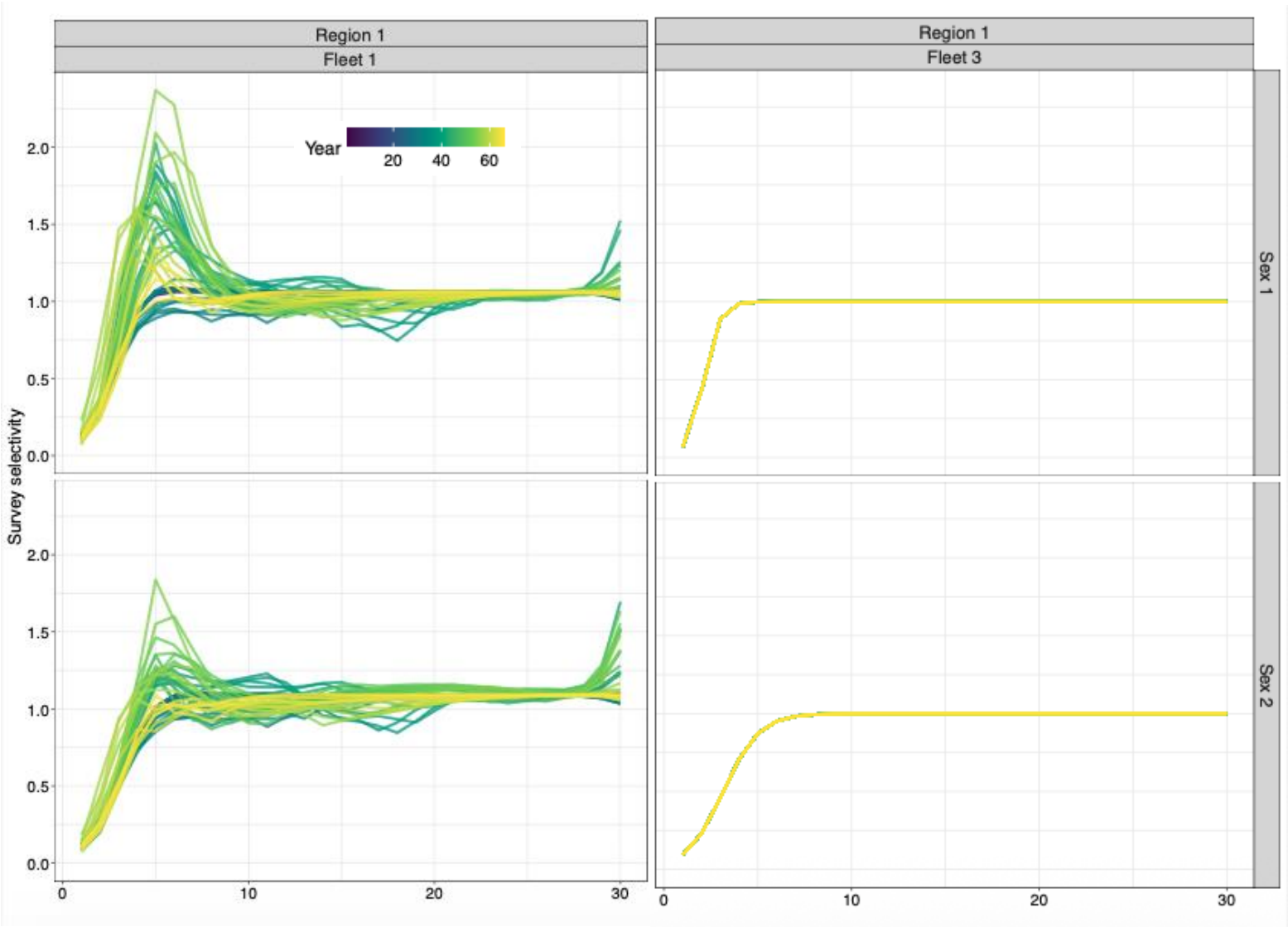

Figure 3B.20. Survey selectivity estimates from Model *26.1_1Reg_ContTVSel*. Fleet 1 is the domestic longline survey fleet and fleet 2 is the Japanese longline survey fleet. Sex 1 is females and sex 2 is males.

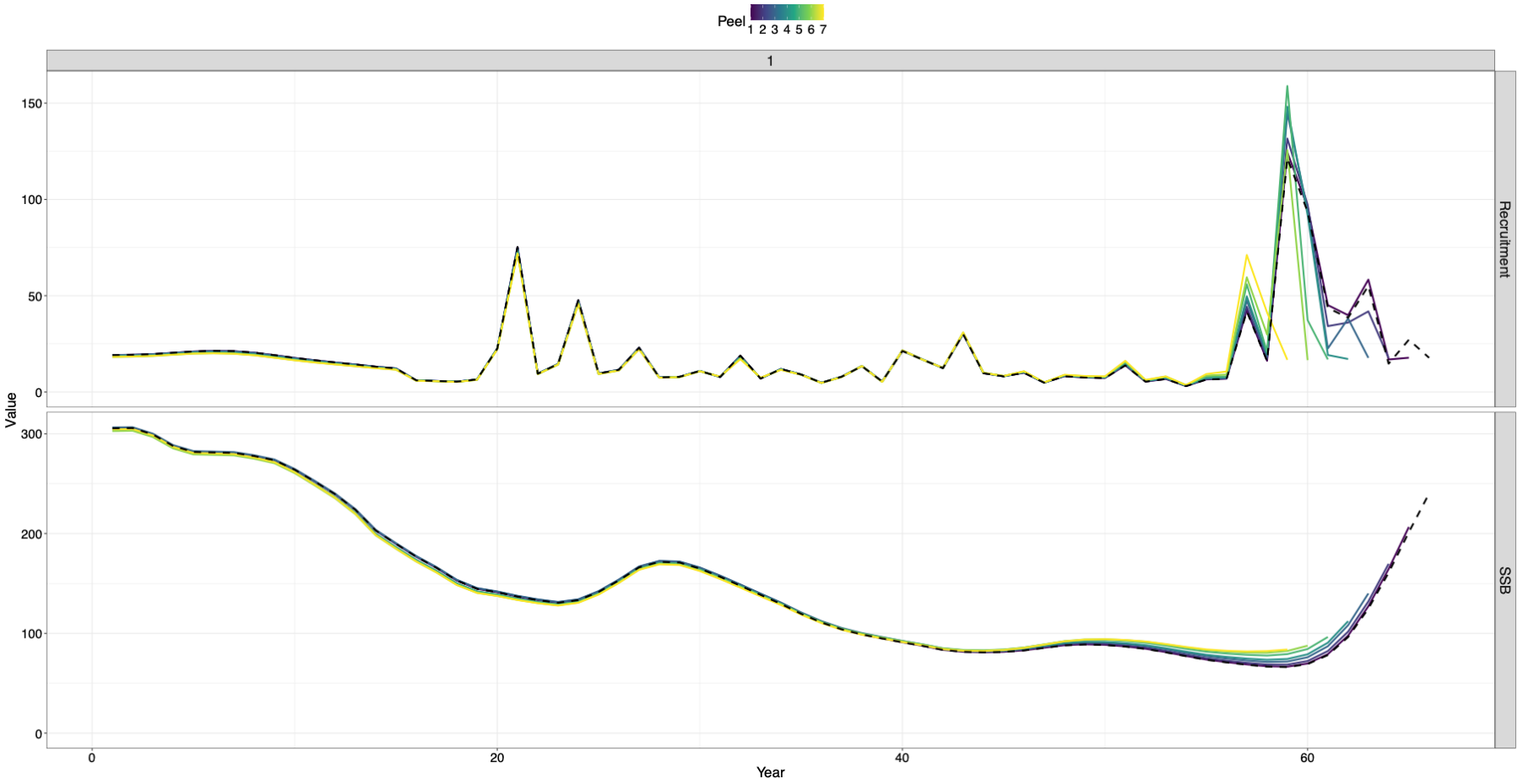

Figure 3B.21. Comparison of recruitment (millions of fish; top panel) and spawning stock biomass (SSB; bottom panel) time series estimates from a seven year retrospective analysis for Model *26.1_1Reg_ContTVSel*. Mohn's rho for SSB was 0.161.

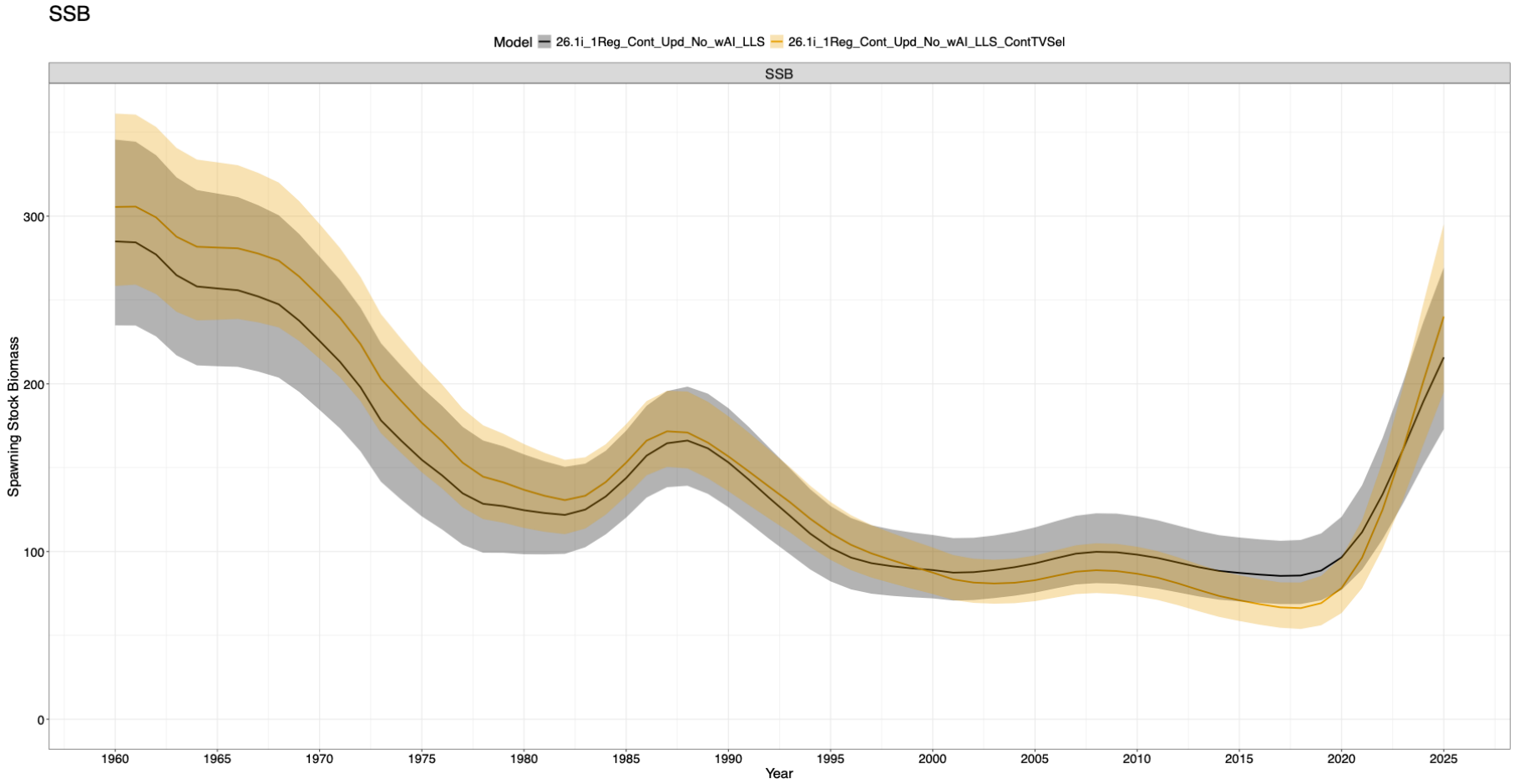

Figure 3B.22. Comparison of spawning stock biomass (SSB, in kilotons) between Model *26.1_1Reg* (also termed *26.1i_1Reg_Cont_Upd_No_wAI_LLS; black line*), along with a configuration that incorporates continuous time-varying semi-parametric survey selectivity (*26.1i_1Reg_Cont_Upd_No_wAI_LLS_ContTVSel; yellow line*). Shaded regions represent the 95% confidence intervals.

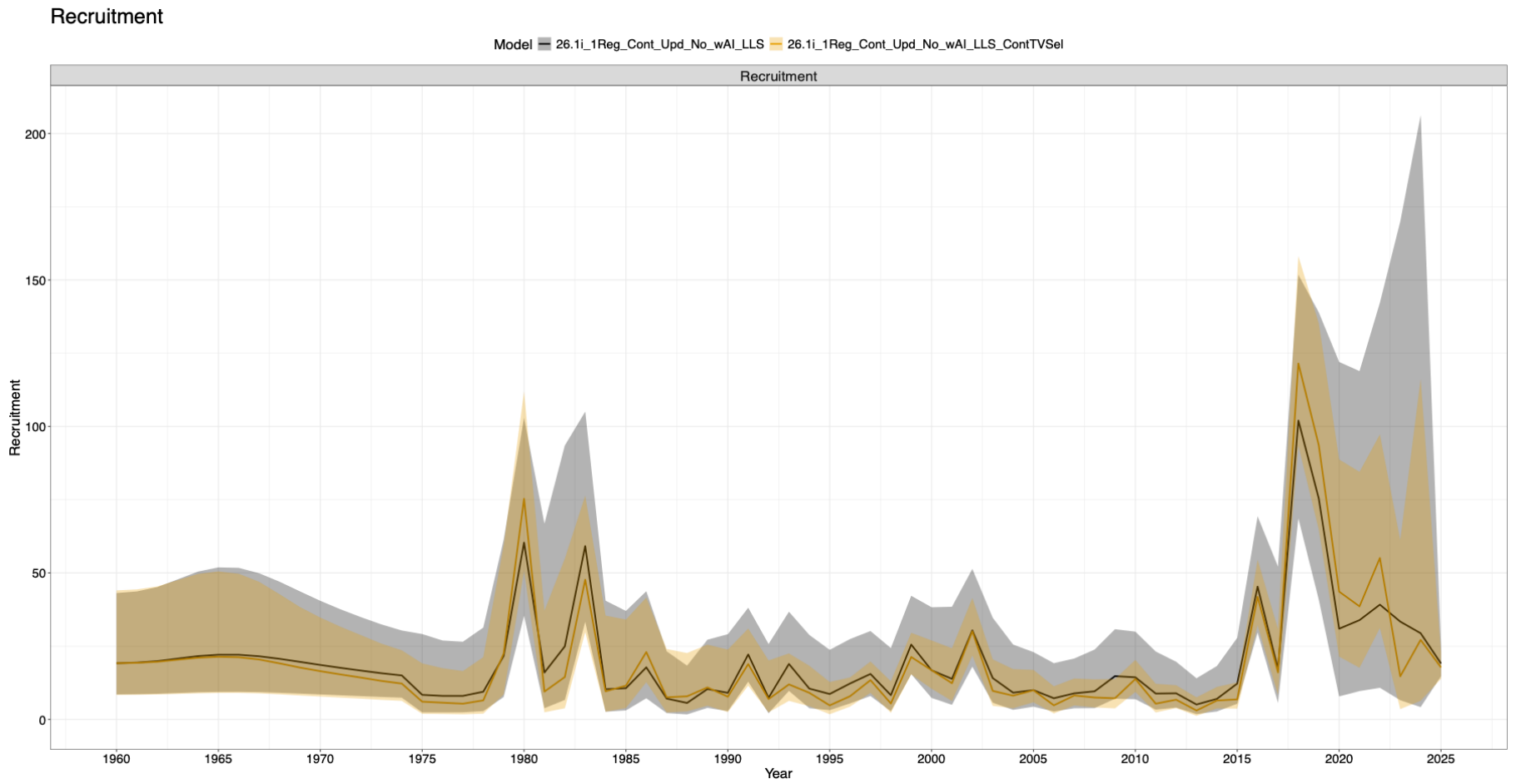

Figure 3B.23. Comparison of age-2 recruitment (in millions of individuals) between Model *26.1_1Reg* (also termed *26.1i_1Reg_Cont_Upd_No_wAI_LLS; black line*), along with a configuration that incorporates continuous time-varying semi-parametric survey selectivity (*26.1i_1Reg_Cont_Upd_No_wAI_LLS_ContTVSel; yellow line*). Shaded regions represent the 95% confidence intervals.

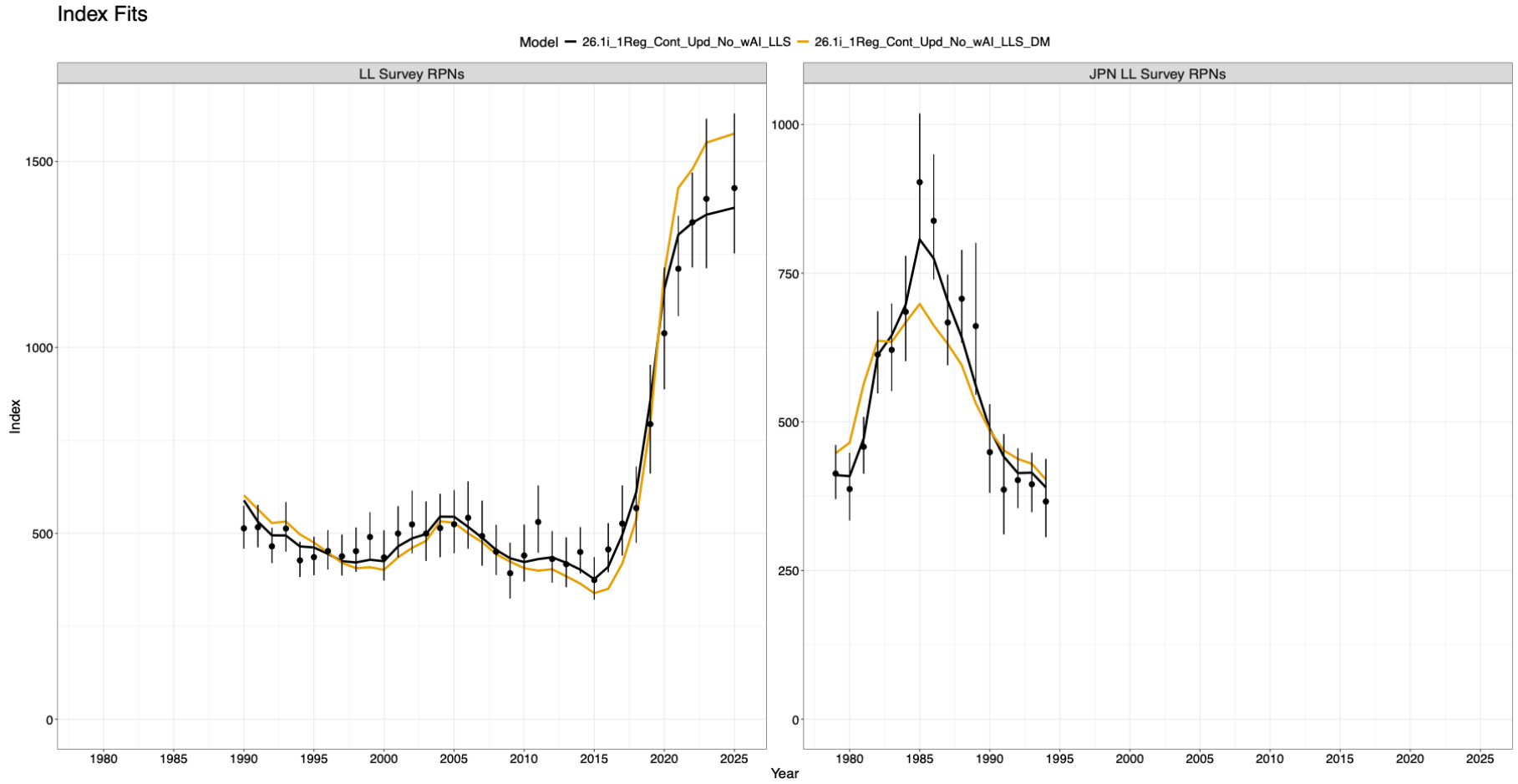

Figure 3B.24. Model fits to survey abundance indices between Model *26.1_1Reg* (also termed *26.1i_1Reg_Cont_Upd_No_wAI_LLS; black line*), along with a configuration that incorporates the Dirichlet-multinomial likelihood for composition data (*26.1i_1Reg_Cont_Upd_No_wAI_LLS_DM; yellow line*). Point ranges denote observed data and associated 95% confidence intervals. Solid colored lines represent the model expectation.

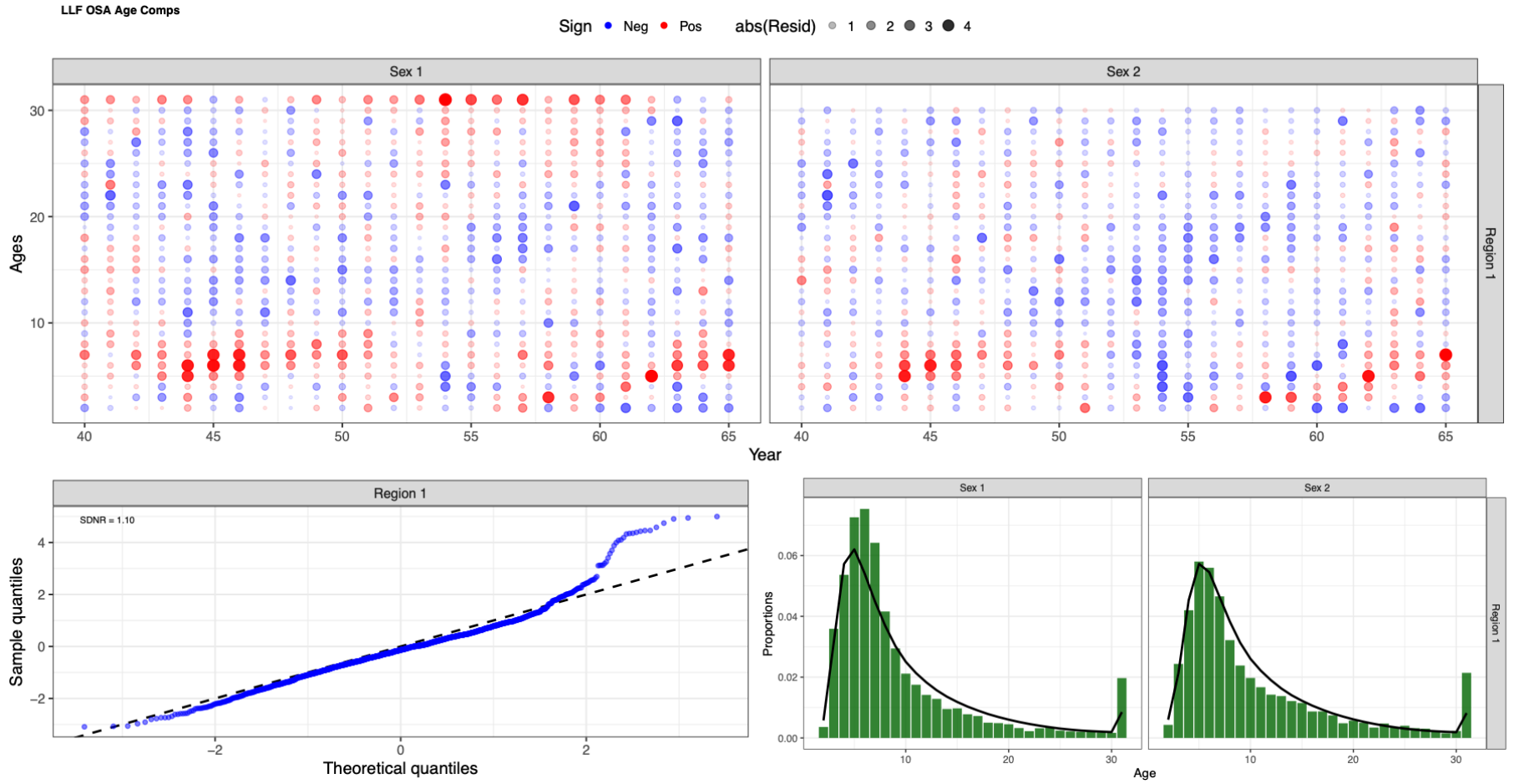

Figure 3B.25. Fits to longline fishery age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg_DM*.

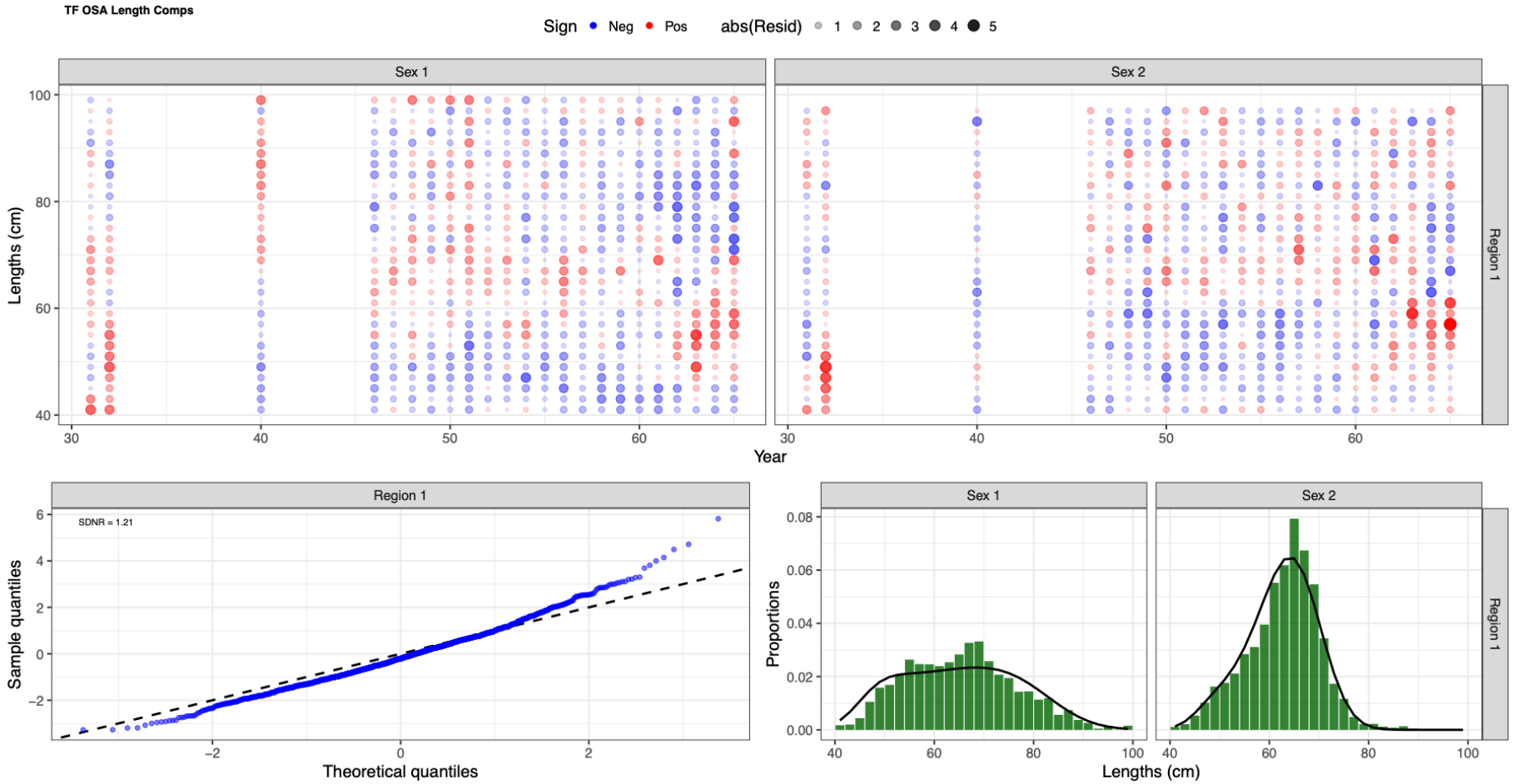

Figure 3B.26. Fits to trawl fishery length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg_DM*.

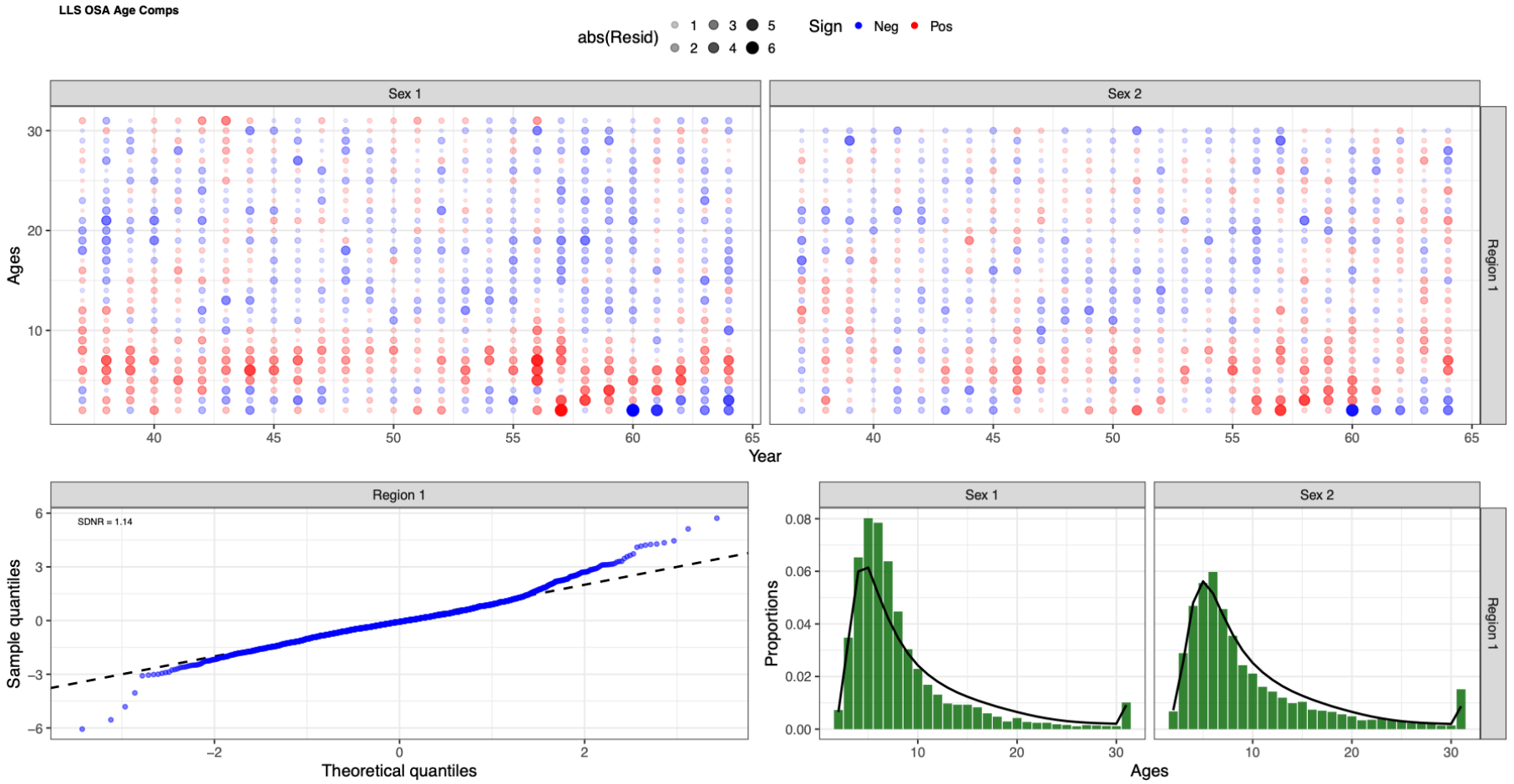

Figure 3B.27. Fits to NOAA longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg_DM*.

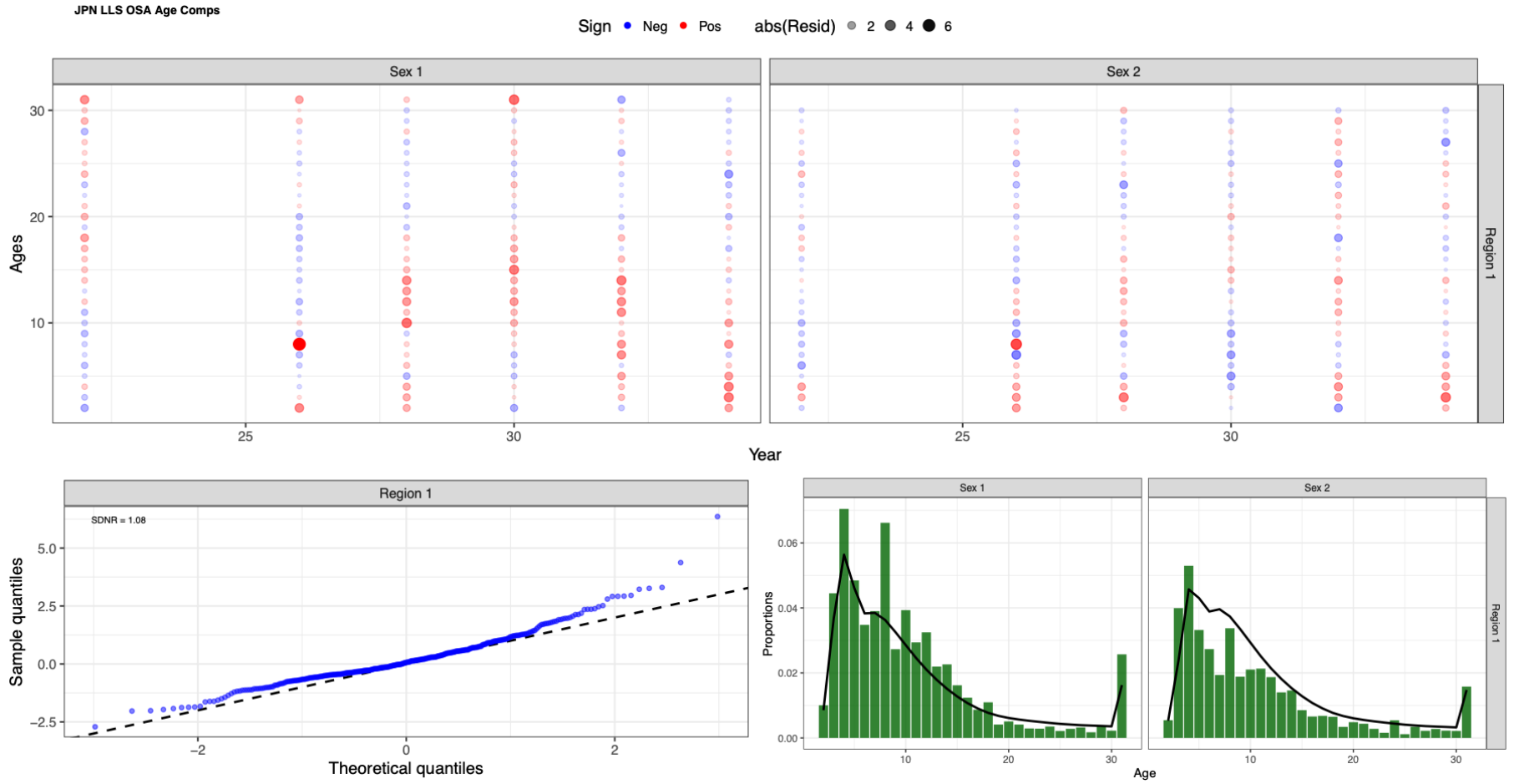

Figure 3B.28. Fits to Japanese longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg_DM*.

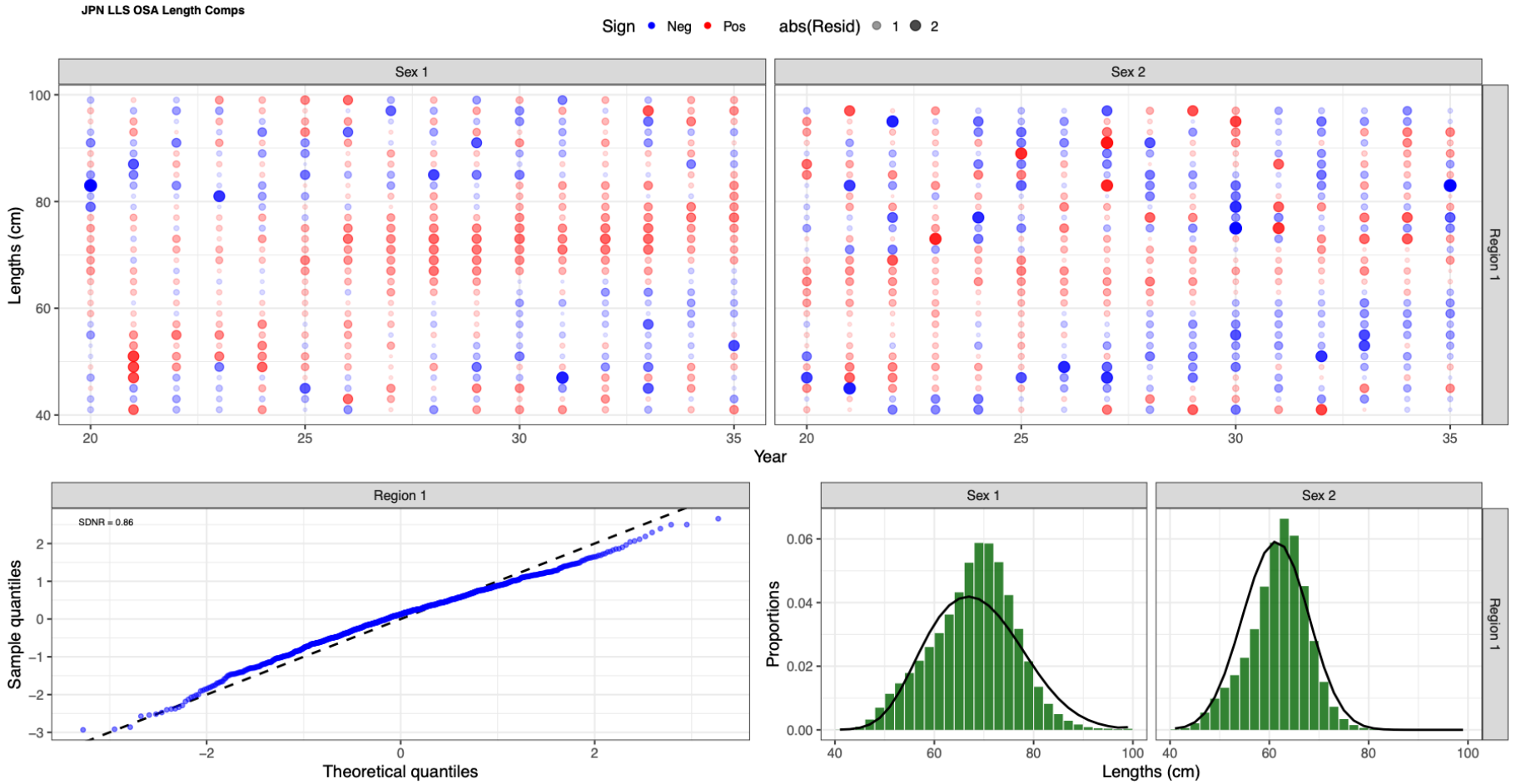

Figure 3B.29. Fits to Japanese longline survey length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg_DM*.

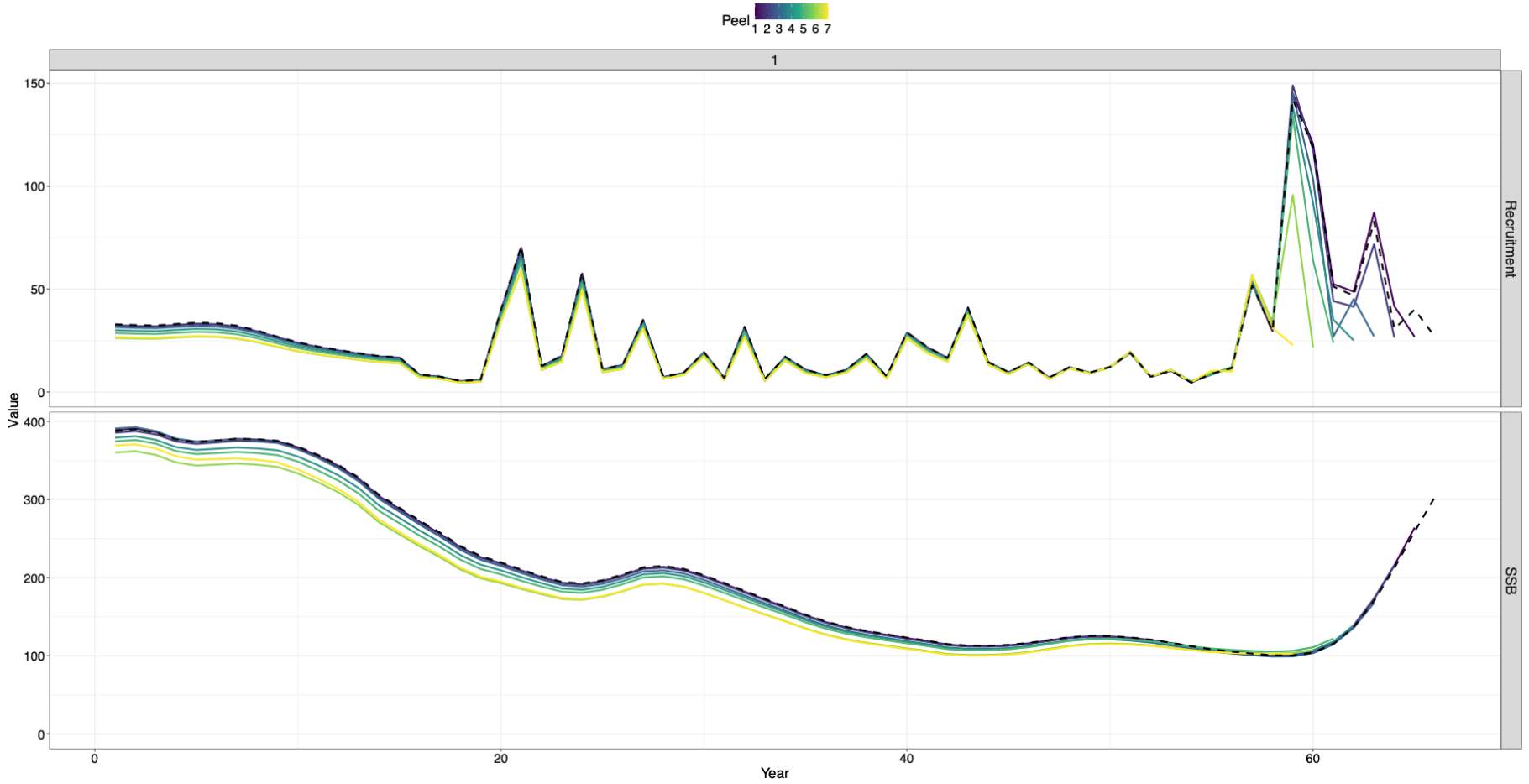

Figure 3B.30. Comparison of recruitment (millions of fish; top panel) and spawning stock biomass (SSB; bottom panel) time series estimates from a seven year retrospective analysis for Model *26.1_1Reg_DM*. Mohn's rho for SSB was 0.022.

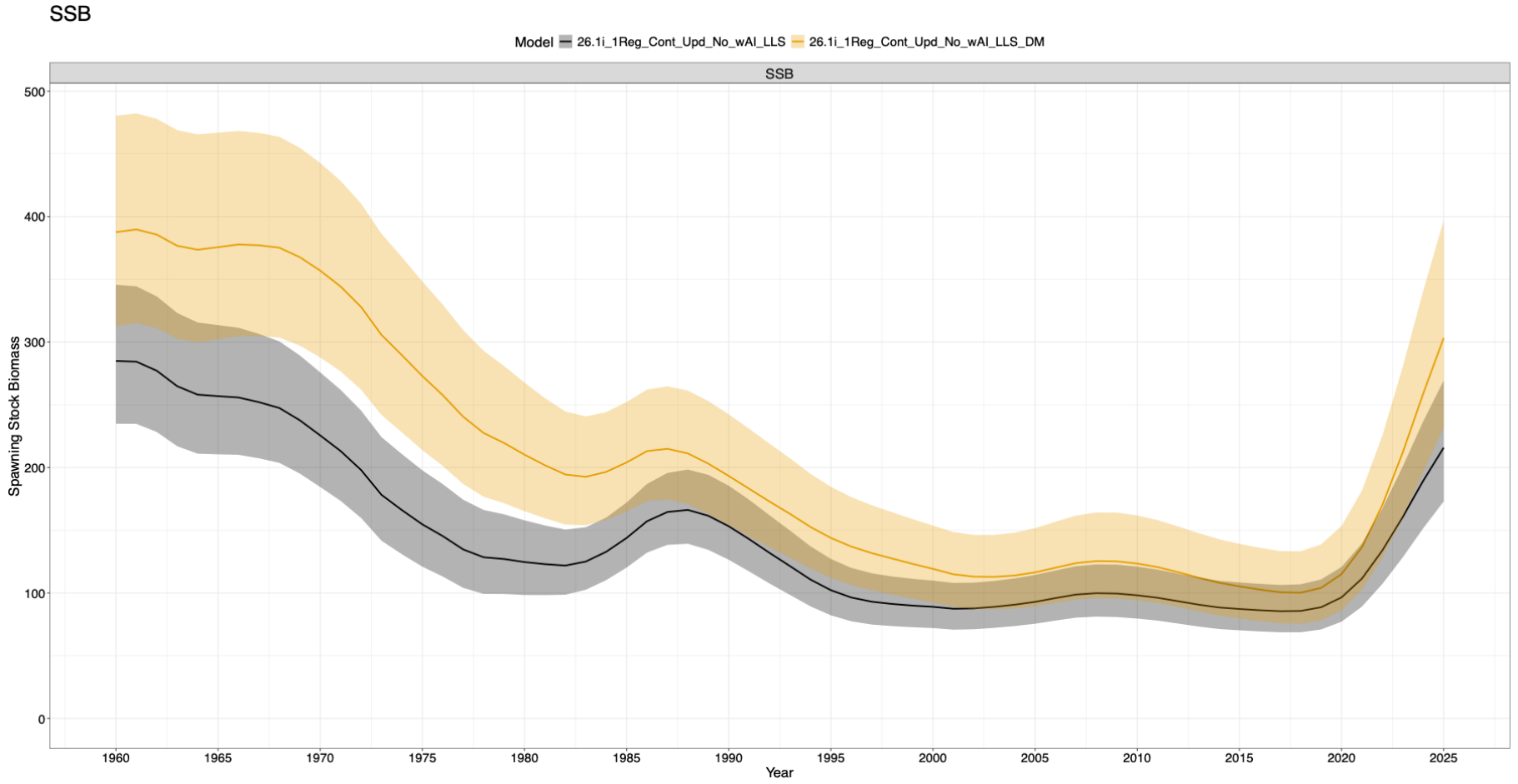

Figure 3B.31. Comparison of spawning stock biomass (SSB, in kilotons) between Model *26.1_1Reg* (also termed *26.1i_1Reg_Cont_Upd_No_wAI_LLS; black line*), along with a configuration that incorporates the Dirichlet-multinomial likelihood for composition data (*26.1i_1Reg_Cont_Upd_No_wAI_LLS_DM; yellow line*). Shaded regions represent the 95% confidence intervals.

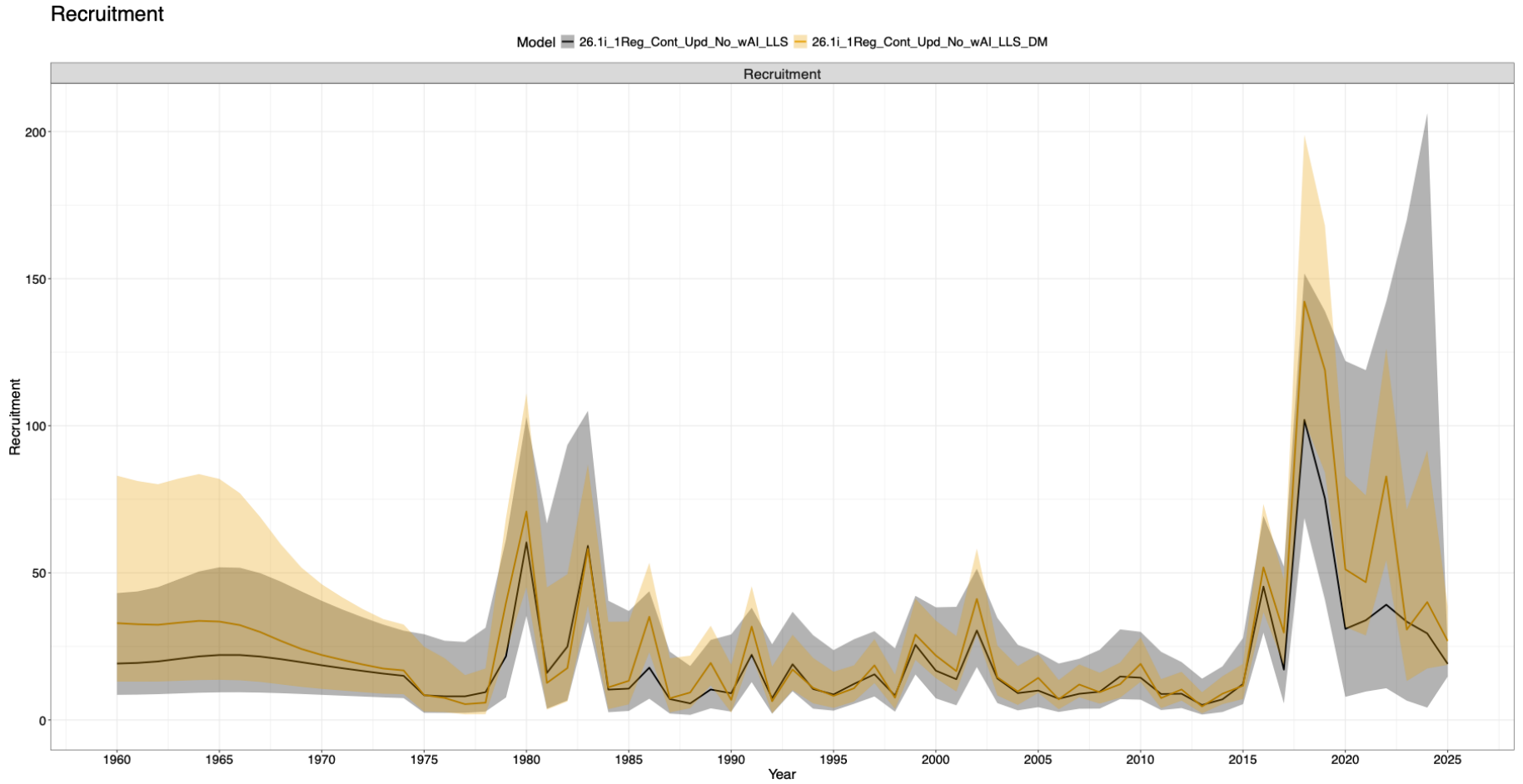

Figure 3B.32. Comparison of age-2 recruitment (millions of individuals) between Model *26.1_1Reg* (also termed *26.1i_1Reg_Cont_Upd_No_wAI_LLS; black line*), along with a configuration that incorporates the Dirichlet-multinomial likelihood for composition data (*26.1i_1Reg_Cont_Upd_No_wAI_LLS_DM; yellow line*). Shaded regions represent the 95% confidence intervals.

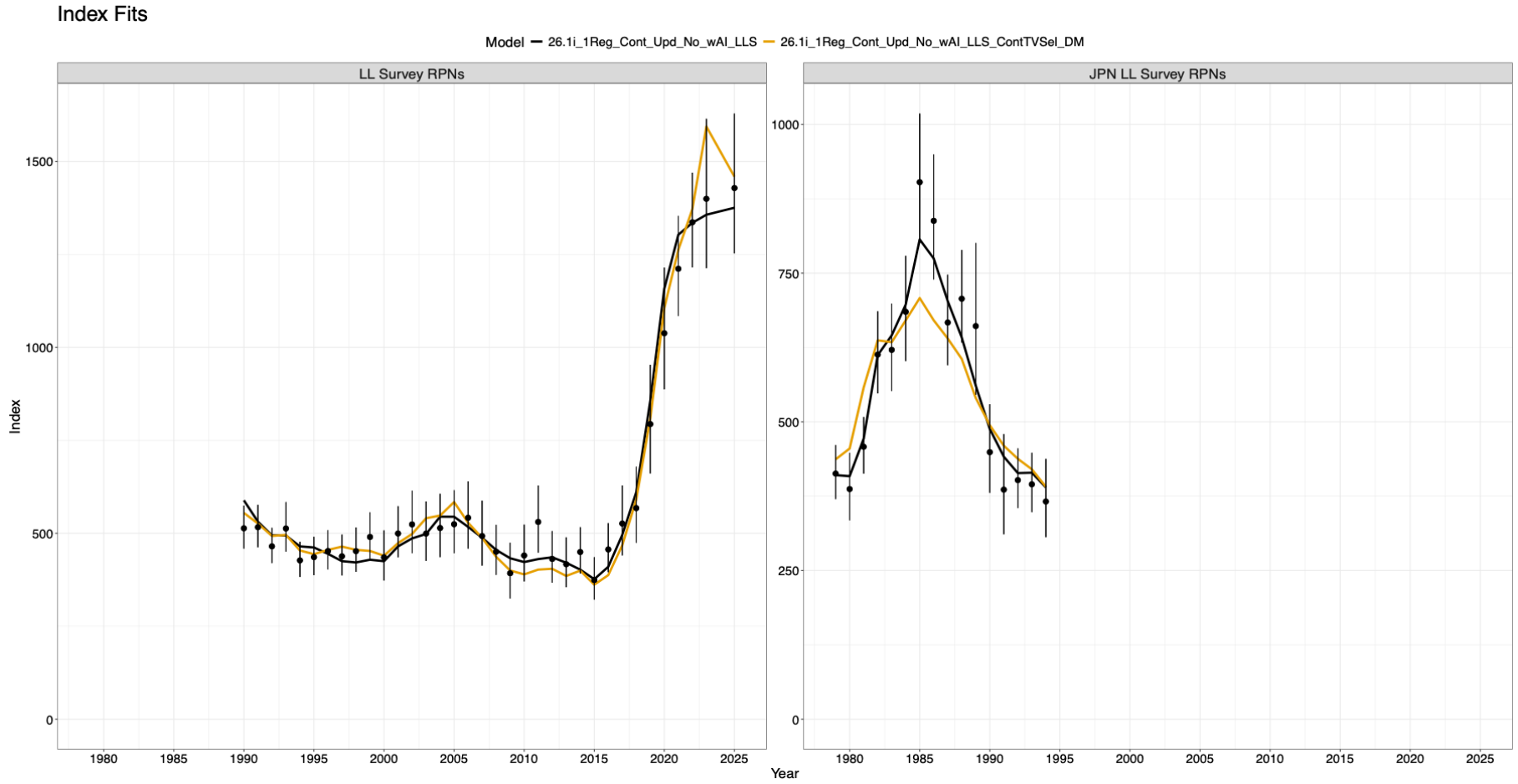

Figure 3B.33. Model fits to survey abundance indices between Model *26.1_1Reg*) also termed *26.1i_1Reg_Cont_Upd_No_wAI_LLS; black line*), along with a configuration that incorporates continuous semi-parametric time-varying survey selectivity and Dirichlet-multinomial likelihoods for composition data (*26.1i_1Reg_Cont_Upd_No_wAI_LLS_ContTVSel_DM; yellow line*). Point ranges denote observed data and associated 95% confidence intervals. Solid colored lines represent the model expectation.

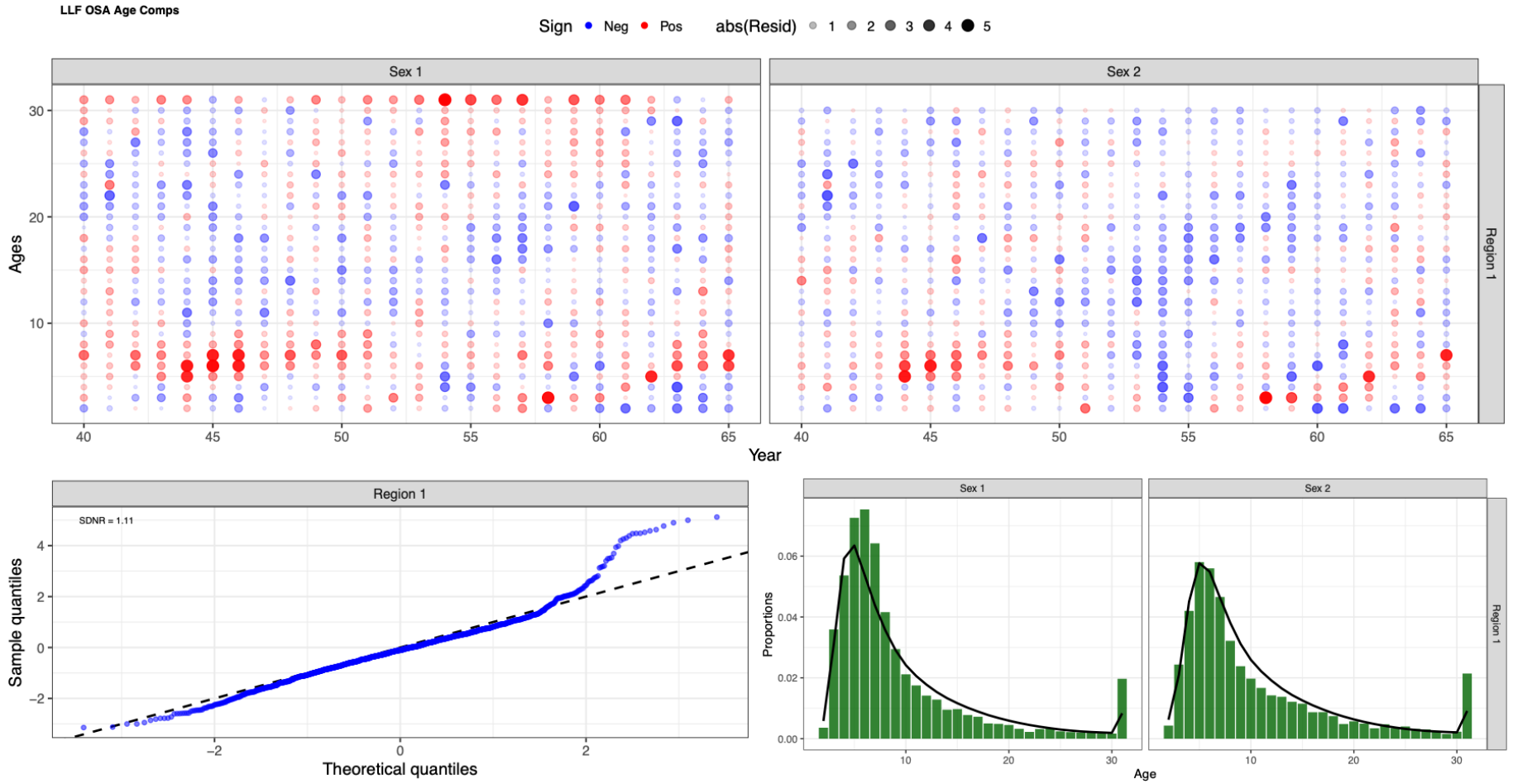

Figure 3B.34. Fits to longline fishery age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg_ContTVSel_DM*.

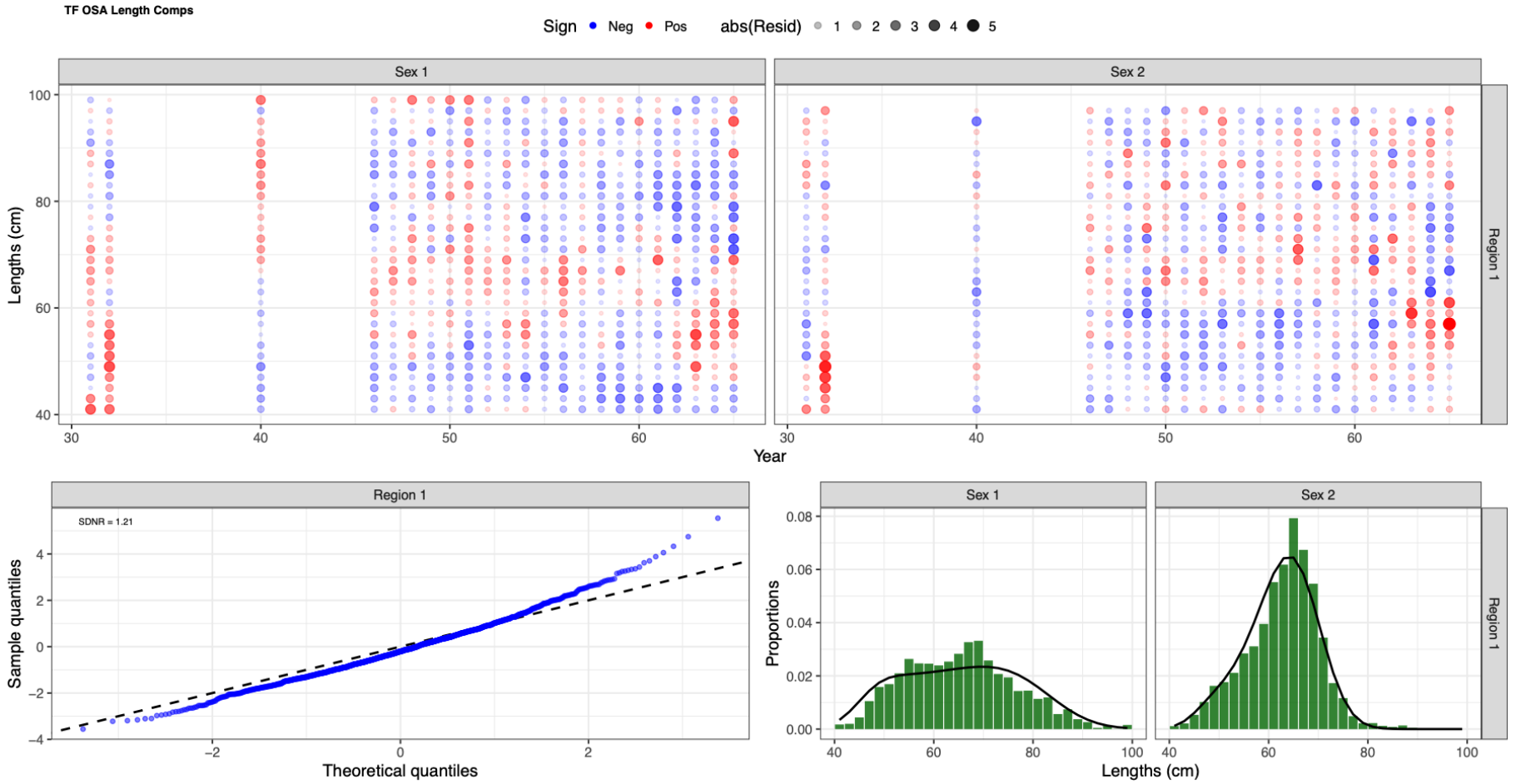

Figure 3B.35. Fits to trawl fishery length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg_DM*.

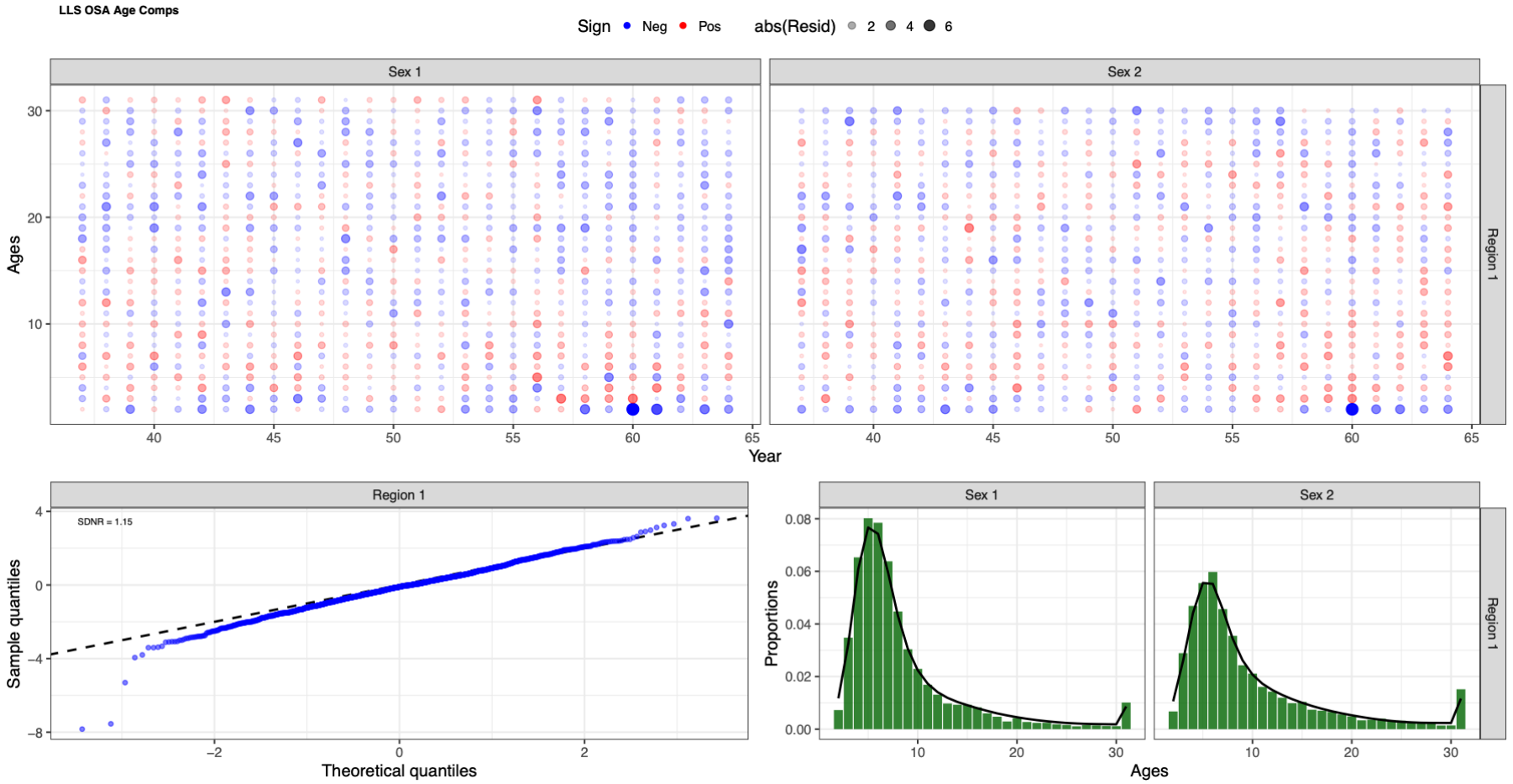

Figure 3B.36. Fits to NOAA longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg_ContTVSel_DM*.

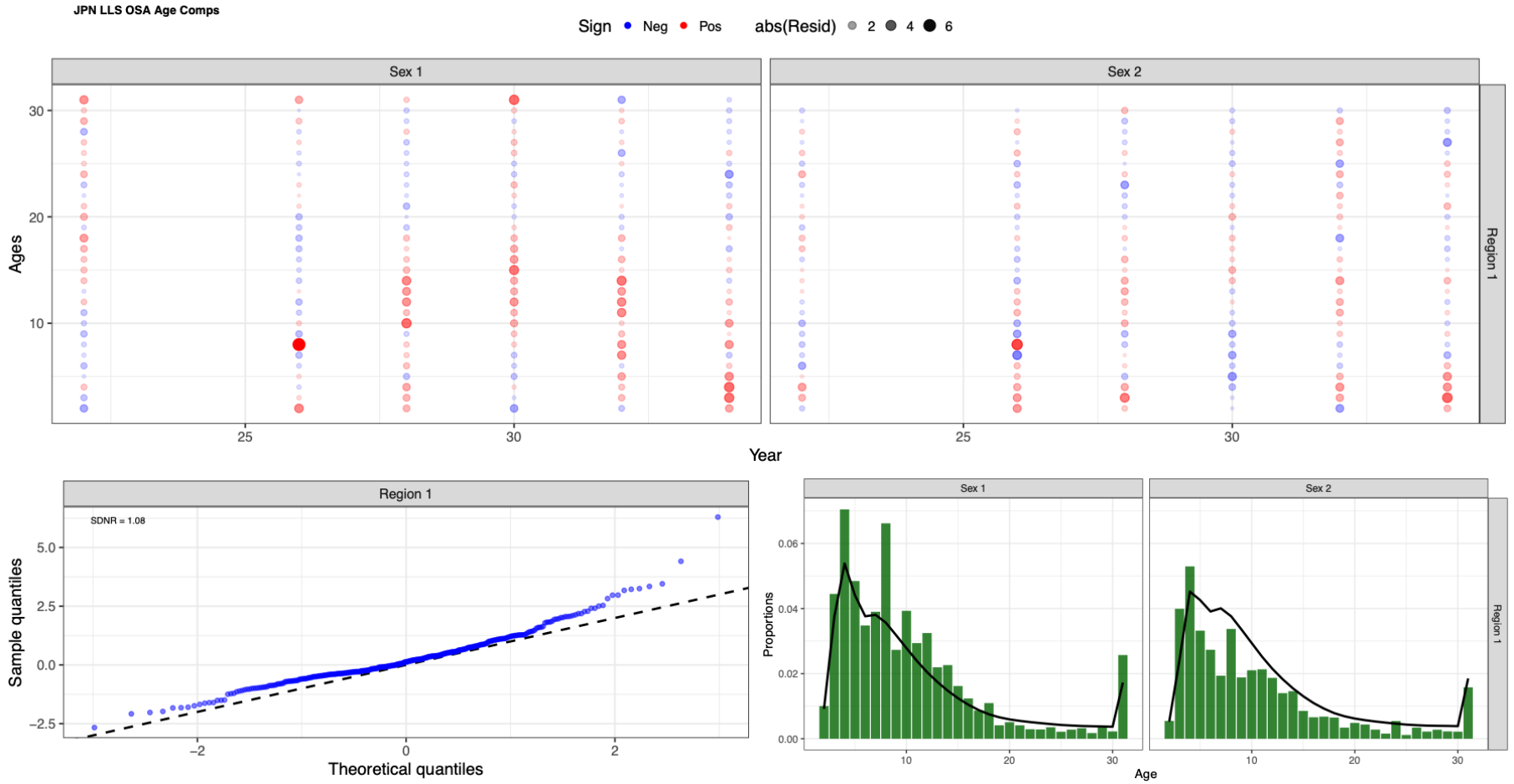

Figure 3B.37. Fits to Japanese longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg_ContTVSel_DM*.

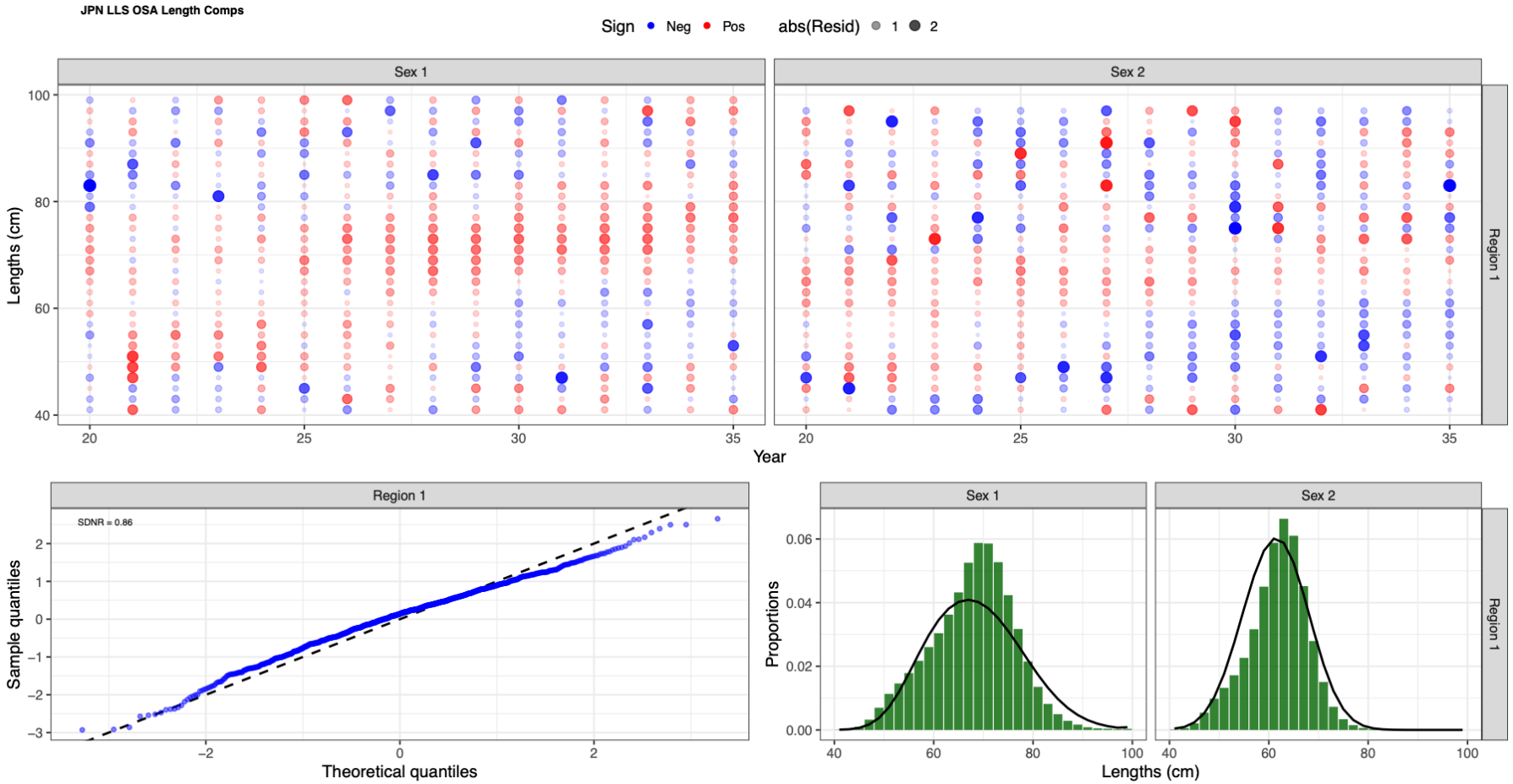

Figure 3B.38. Fits to Japanese longline survey length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg_ContTVSel_DM*.

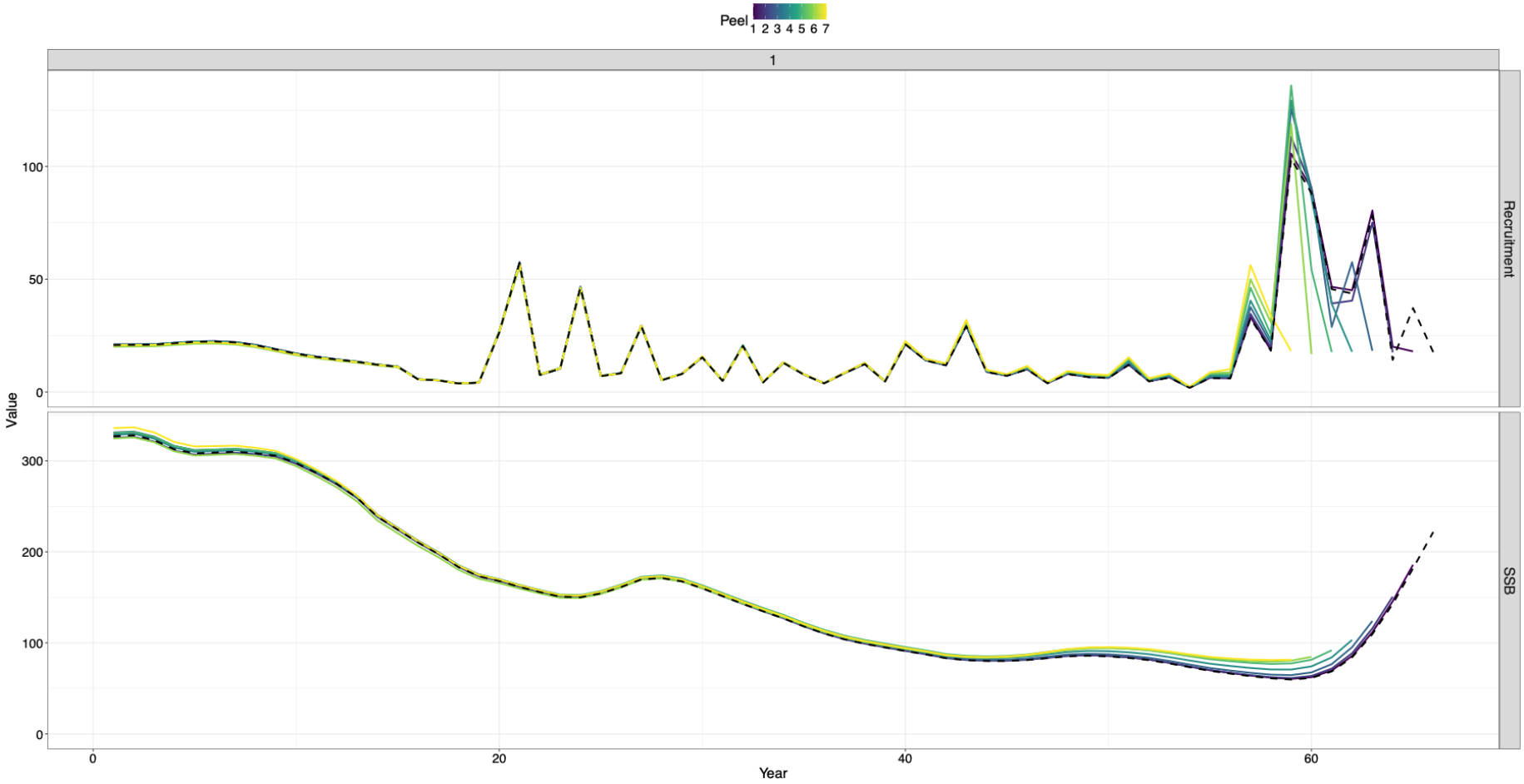

Figure 3B.39. Comparison of recruitment (millions of fish; top panel) and spawning stock biomass (SSB; bottom panel) time series estimates from a seven year retrospective analysis for Model *26.1_1Reg_ContTVSel_DM*. Mohn's rho for SSB was 0.217.

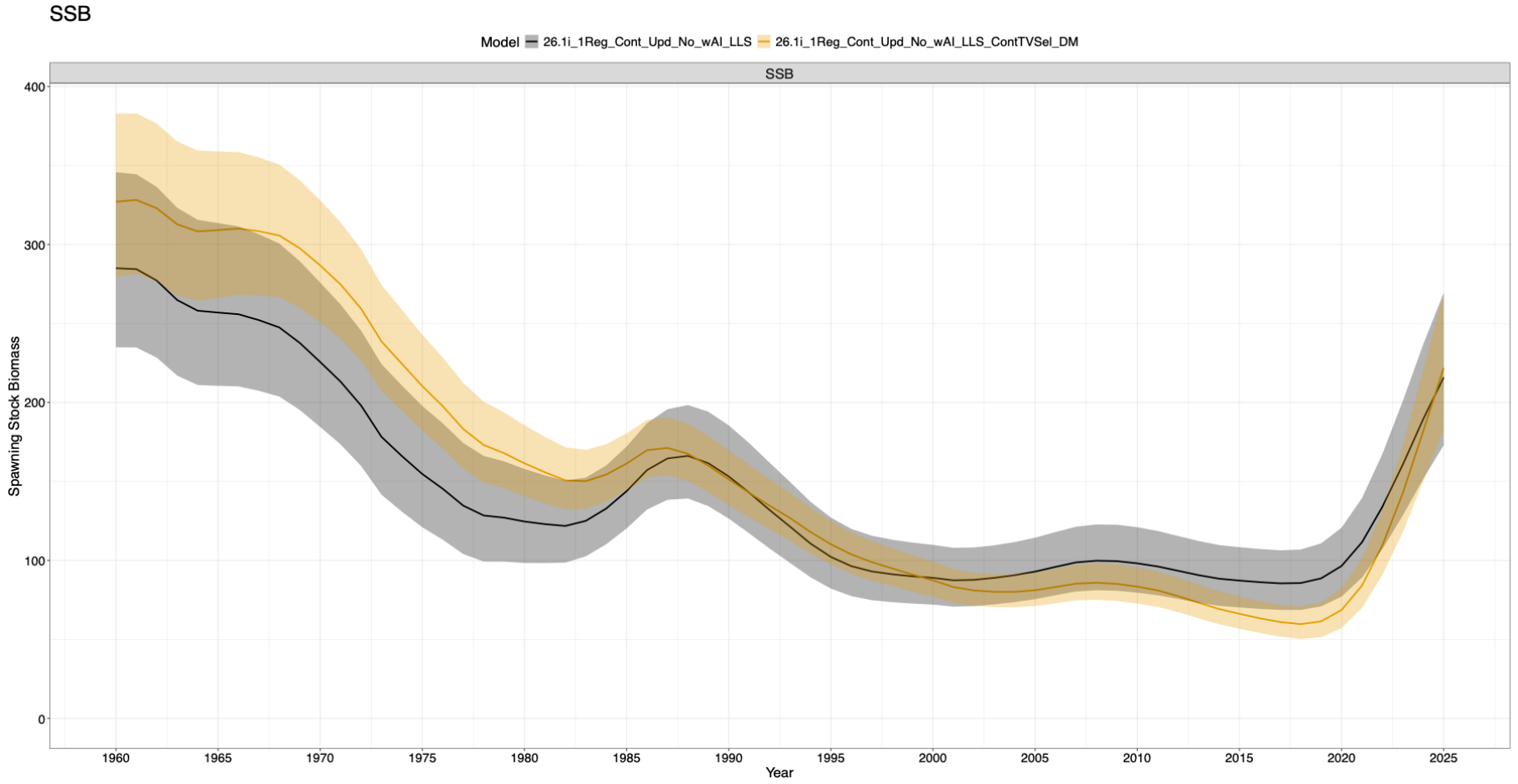

Figure 3B.40. Comparison of spawning stock biomass (SSB, in kilotons) between Model *26.1_1Reg* (also termed *26.1i_1Reg_Cont_Upd_No_wAI_LLS; black line*), along with a configuration that incorporates continuous time-varying semi-parametric survey selectivity and Dirichlet-multinomial likelihoods for composition data (*26.1i_1Reg_Cont_Upd_No_wAI_LLS_ContTVSel_DM; yellow line*). Shaded regions represent the 95% confidence intervals.

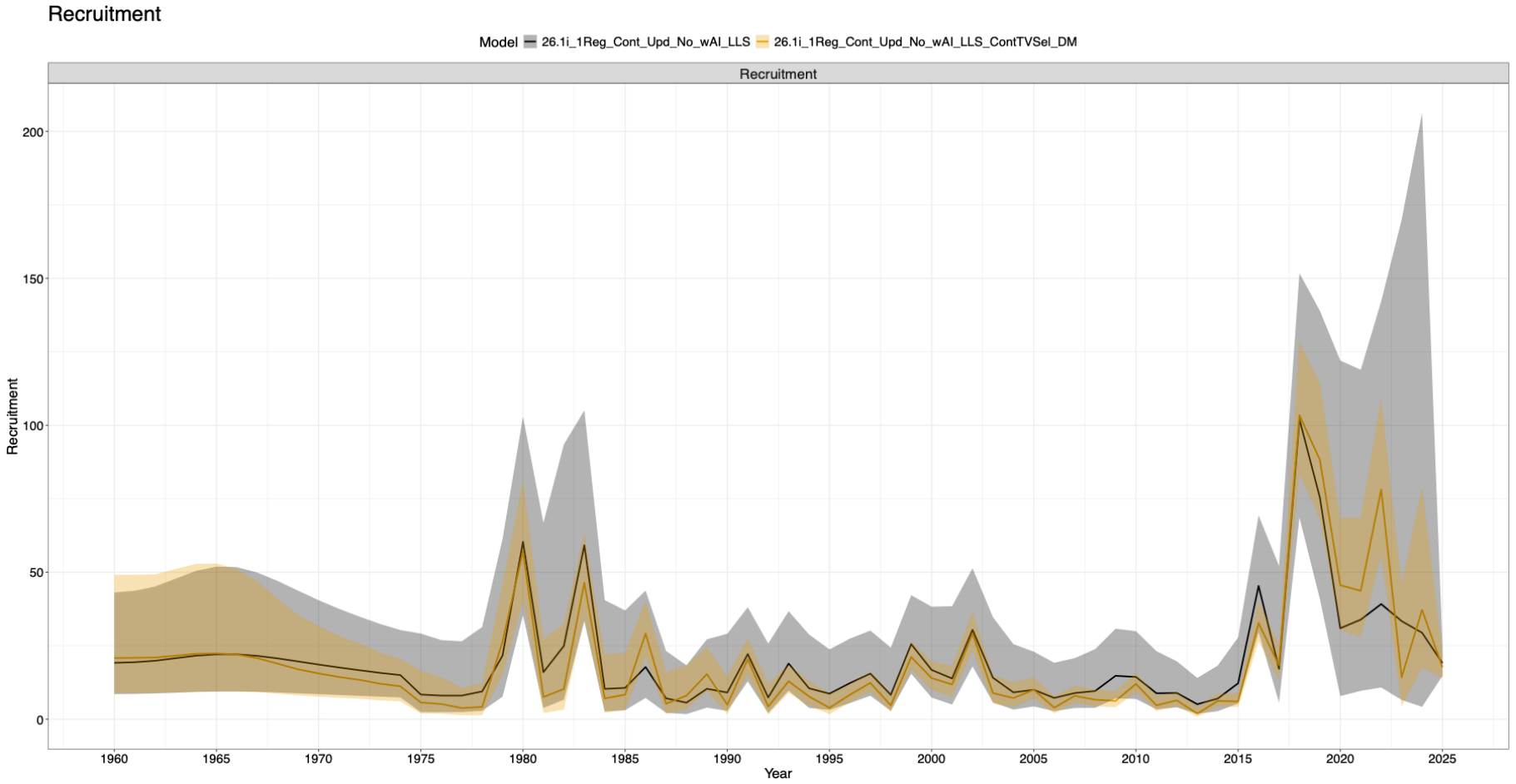

Figure 3B.41. Comparison of age-2 recruitment (millions of fish) between Model *26.1_1Reg* (also termed *26.1i_1Reg_Cont_Upd_No_wAI_LLS; black line*), along with a configuration that incorporates continuous time-varying semi-parametric survey selectivity and Dirichlet-multinomial likelihoods for composition data (*26.1i_1Reg_Cont_Upd_No_wAI_LLS_ContTVSel_DM; yellow line*). Shaded regions represent the 95% confidence intervals.

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4. Supplemental Material 3B: Additional Figures

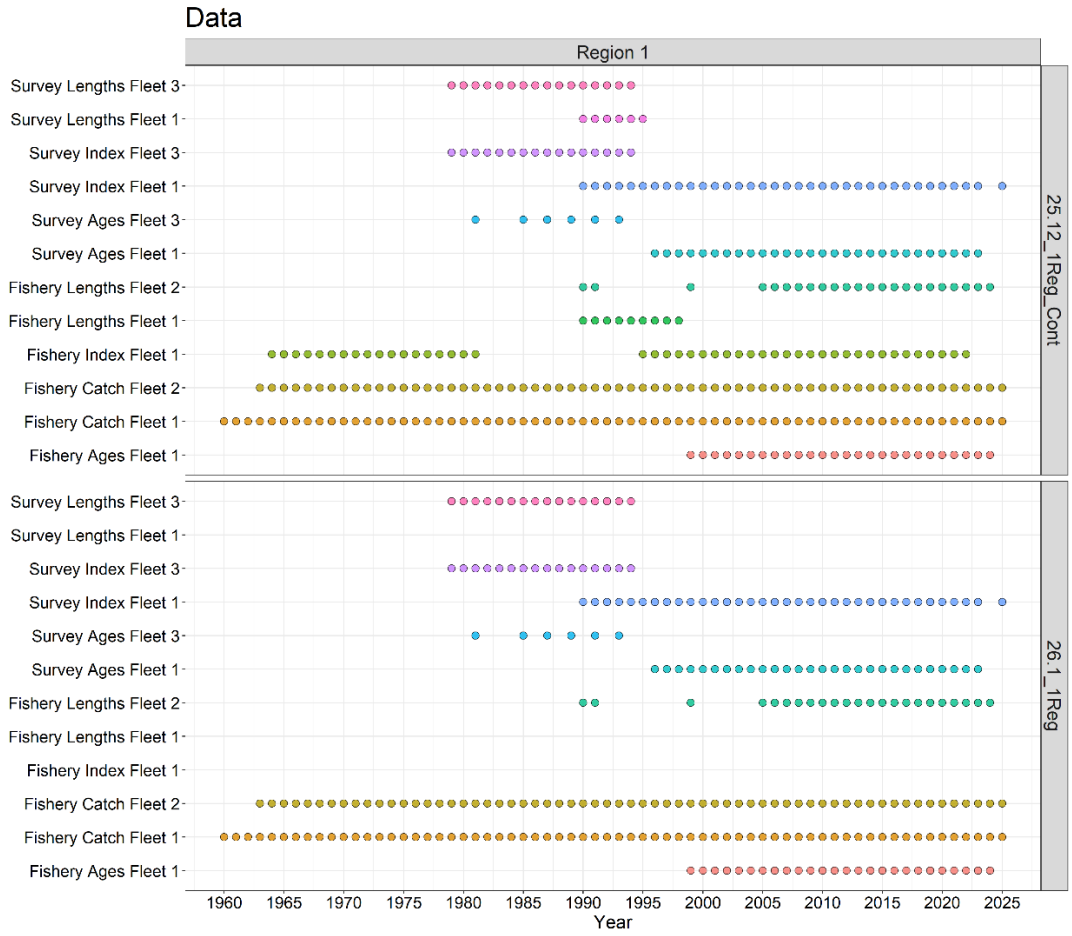

Figure SM3B.1. Data inputs used by Model *25.12_1Reg_Cont* and Model *26.1_1Reg*. Note that Fishery Fleet 1 is the fixed gear fleet, Fishery Fleet 2 is the trawl fleet, Survey Fleet 1 is the NOAA Longline Survey, and survey Fleet 3 is the Japanese Longline Survey. Primary differences in data usage are that Model *26.1_1Reg* no longer fits to fixed gear fishery lengths or the associated CPUE index (including the historical Japanese fishery CPUE and the recent domestic fishery CPUE) as well as not fitting NOAA Longline Survey lengths.

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
25.12_1Reg_Cont	1	206.6999	357.0208	0.04784668	39.55138	131.4549	0.07875	1.57240	0.57896	0.60758
26.1_1Reg	1	216.4521	369.6689	0.04642913	40.85173	139.0172	0.07903	1.55702	0.58553	0.58749

Figure SM3B.2. Key management quantity outputs from Model *25.12_1Reg_Cont* and Model *26.1_1Reg*. SSB is in kt, B_Ref_Pt is the associated $B_{40\%}$ reference point in kt, and F_Ref_Pt is the associated $F_{40\%}$.

Source	SE	Eff_SE	Data_Wt	ISS	ESS
LL Survey Index	0.09	0.09	1.00	NA	NA
JPN LL Survey Index	0.09	0.09	1.00	NA	NA
Fixed Gear Fishery CPUE	0.20	0.20	1.00	NA	NA
Fixed Gear Fish Ages	NA	NA	0.80	20	16.00
LL Survey Ages	NA	NA	3.54	20	70.75
JPN LL Survey Ages	NA	NA	0.64	20	12.87
Fixed Gear Fish Lengths	NA	NA	3.48	20	69.54
Trawl Fish Lengths	NA	NA	0.35	20	6.99
LL Survey Lengths	NA	NA	2.29	20	45.75
JPN LL Survey Lengths	NA	NA	1.03	20	20.56

Source	SE	Eff_SE	Data_Wt	ISS	ESS
LL Survey Index	0.07	0.07	1.00	NA	NA
JPN LL Survey Index	0.07	0.07	1.00	NA	NA
Fixed Gear Fish Ages	NA	NA	0.12	145.15	17.69
LL Survey Ages	NA	NA	0.28	171.43	47.31
JPN LL Survey Ages	NA	NA	0.17	83.33	14.45
Trawl Fish Lengths	NA	NA	0.25	45.39	11.51
JPN LL Survey Lengths	NA	NA	0.34	54.62	18.84

Figure SM3B.3. Assumed mean (across years) statistical distribution variance terms for each data source defined in terms of standard error (SE) for indices or input sample size (ISS) for compositional data for Model *25.12_1Reg_Cont* (left) and Model *26.1_1Reg* (right). The compositional data weights (λ, Data_Wt) are calculated using the Francis reweighting method (Francis, 2011). The effective values (Eff_SE or effective sample size, ESS) represent the input values multiplied by the data weights and are the final variance terms used to fit the data in the assessment likelihoods. All compositional data are fit using the 'joint' approach (Cheng et al., 2025b), so only a single ISS/ESS or weighting term is provided across both sexes. Note that Model *25.12_1Reg_Cont* assumes a SE for indices of 2x's the observed value, while Model *26.1_1Reg* assumes a SE for indices of 1.5x's the observed value.

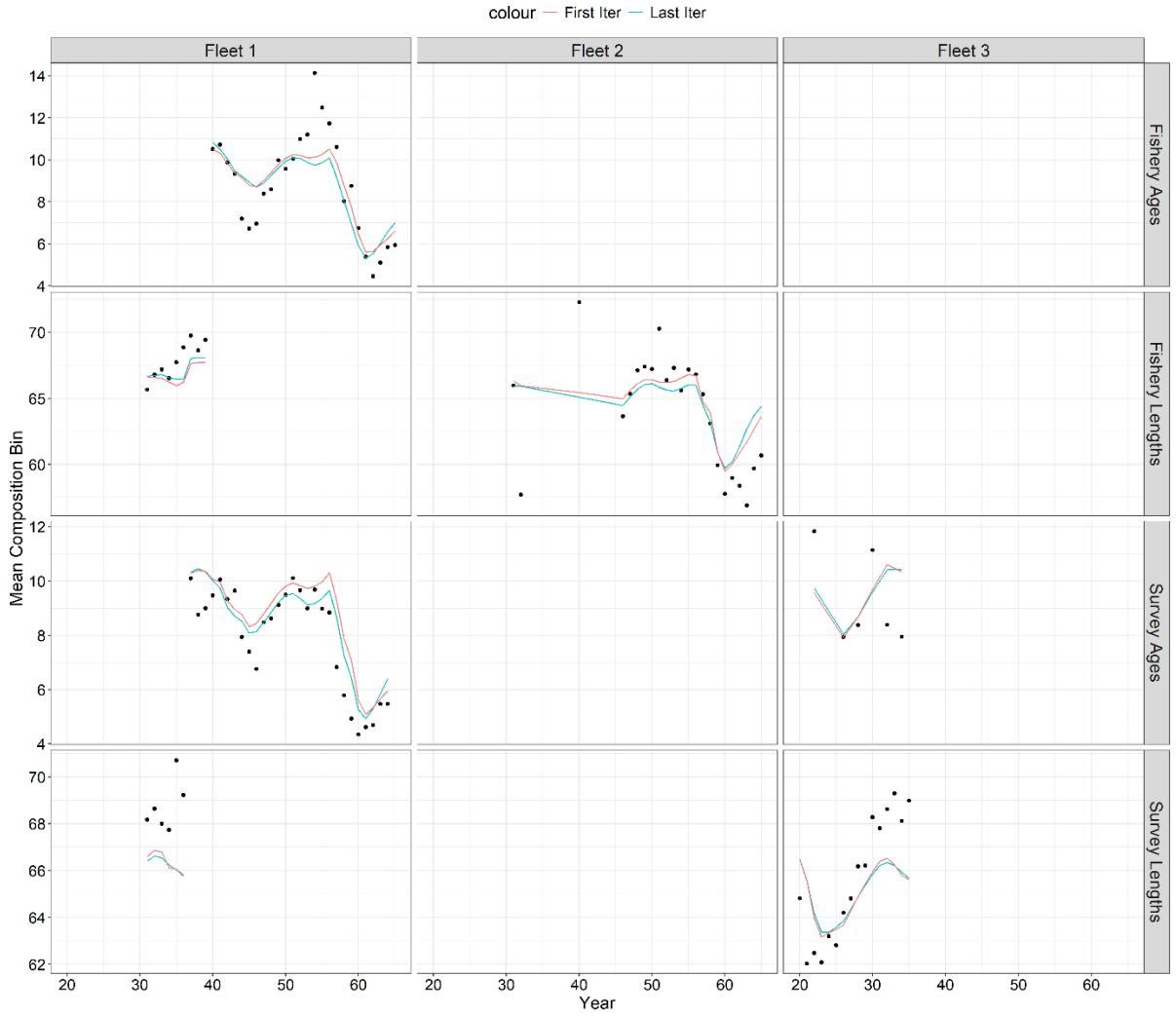

Figure SM3B.4. Fits to the mean age or mean length by year for each compositional data source at the start (red line) and end (blue lines) of the Francis reweighting method (Francis, 2011) for Model *25.12_1Reg_Cont*. Fishery fleet 1 is the fixed gear fishery and fleet 2 is the trawl fishery. Survey fleet 1 is the NOAA longline survey and fleet 3 is the Japanese longline survey.

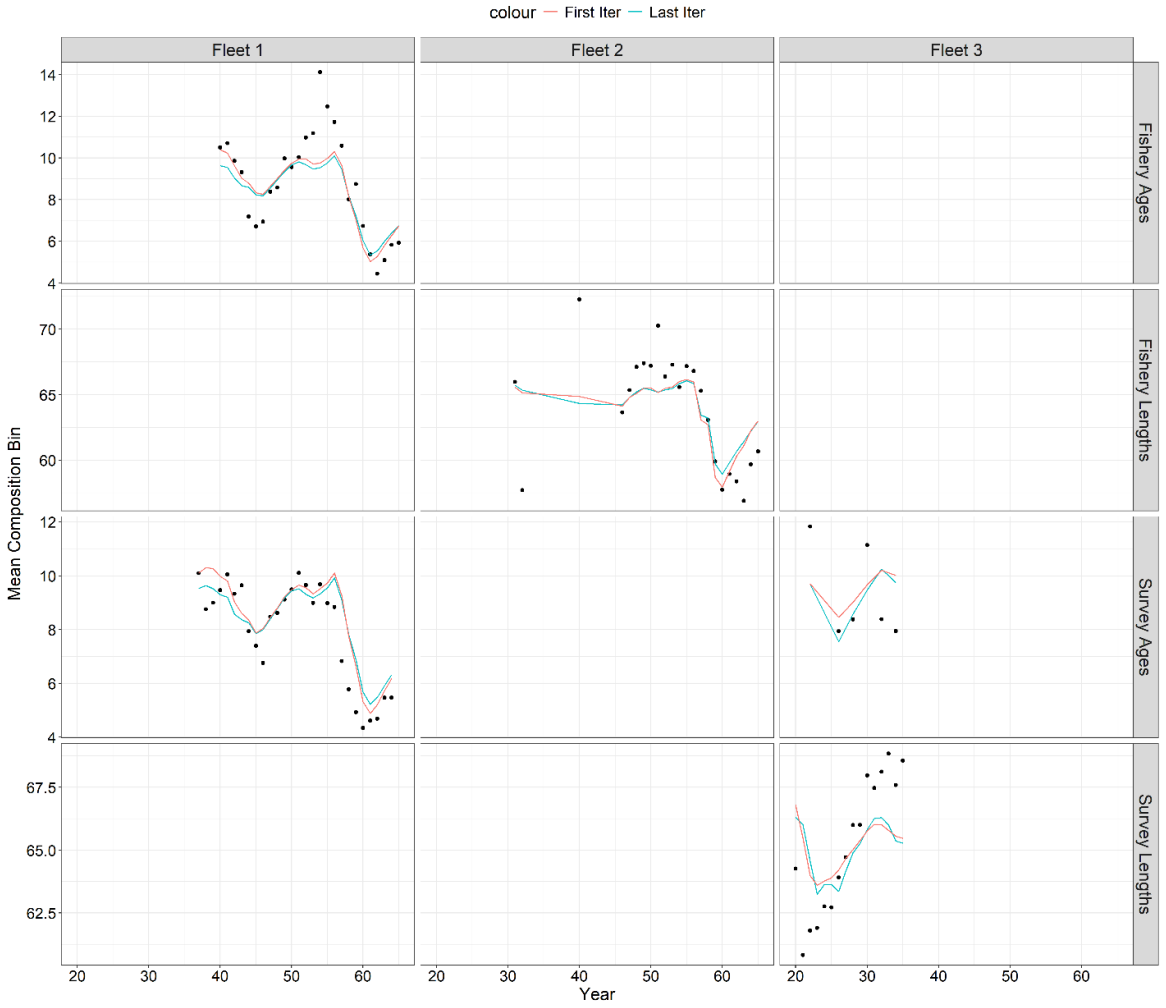

Figure SM3B.5. Fits to the mean age or mean length by year for each compositional data source at the start (red line) and end (blue lines) of the Francis reweighting method (Francis, 2011) for Model *26.1_1Reg*. Fishery fleet 1 is the fixed gear fishery and fleet 2 is the trawl fishery. Survey fleet 1 is the NOAA longline survey and fleet 3 is the Japanese longline survey.

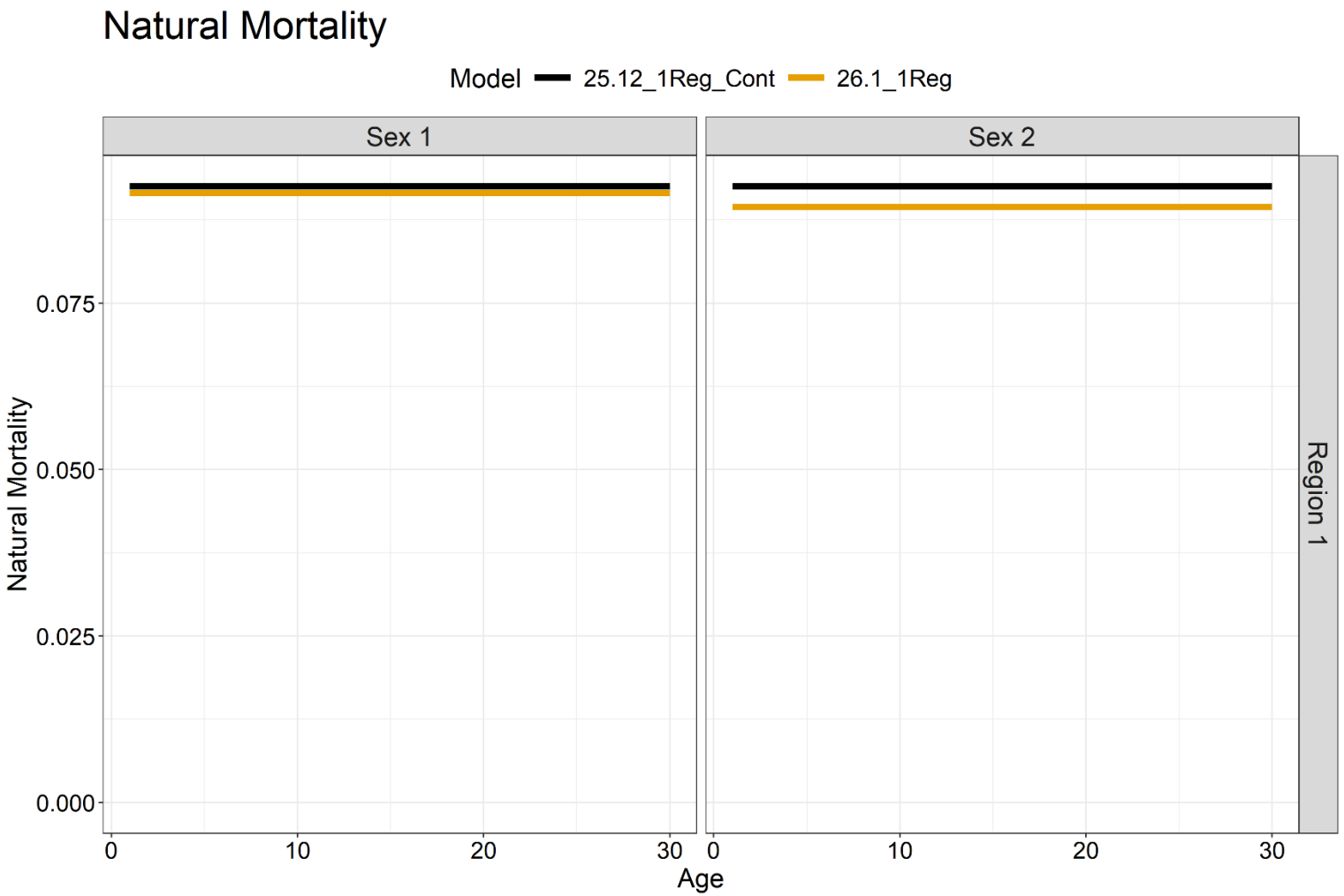

Figure SM3B.6. Natural mortality estimates from Model *25.12_1Reg_Cont* (black line) and Model *26.1_1Reg* (orange line). Sex 1 is females and sex 2 is males.

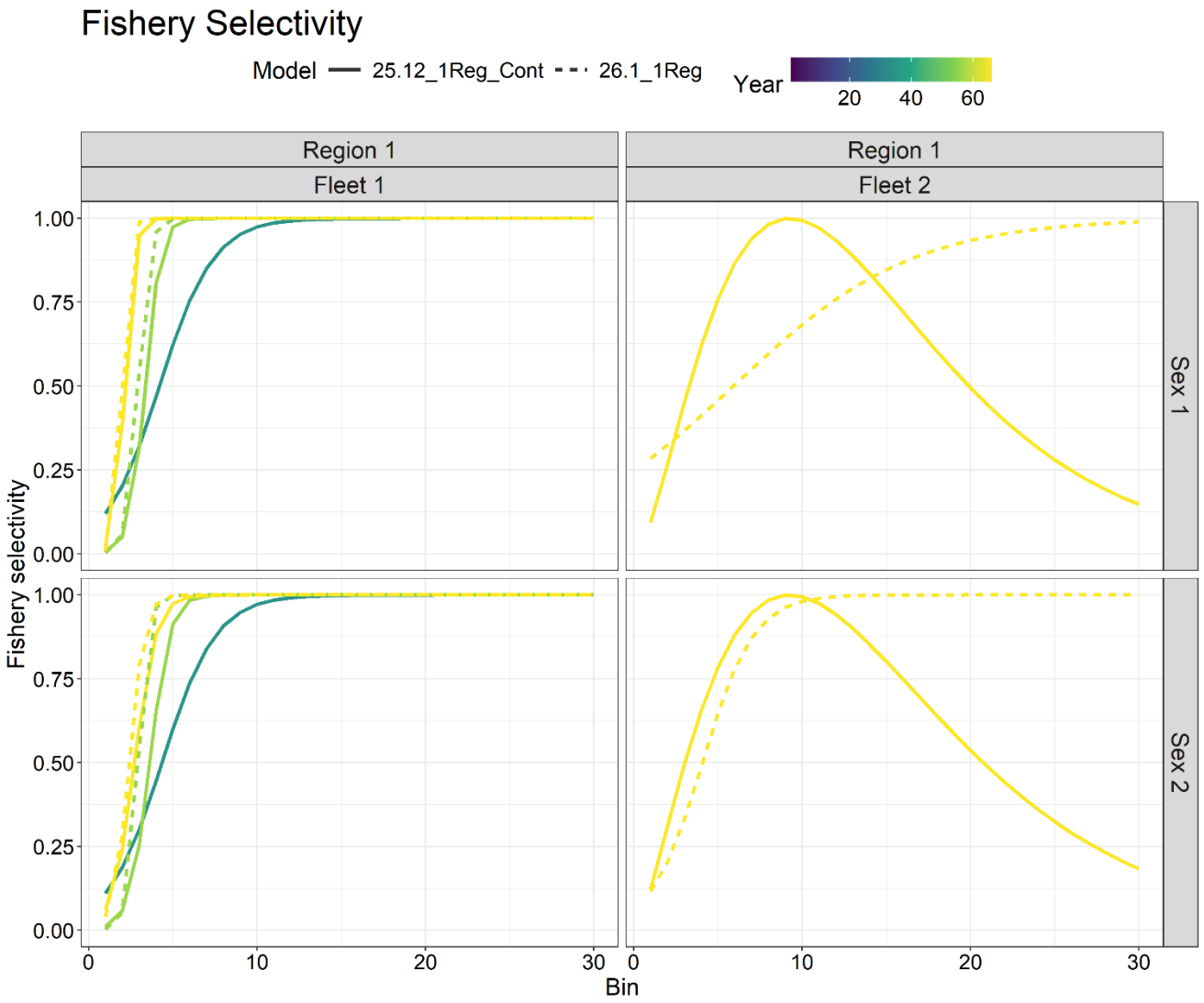

Figure SM3B.7. Fishery selectivity estimates from Model *25.12_1Reg_Cont* (solid lines) and Model *26.1_1Reg* (dashed lines) where different colors represent the different time blocks. Fleet 1 is the fixed gear fleet and fleet 2 is the trawl fleet. Sex 1 is females and sex 2 is males.

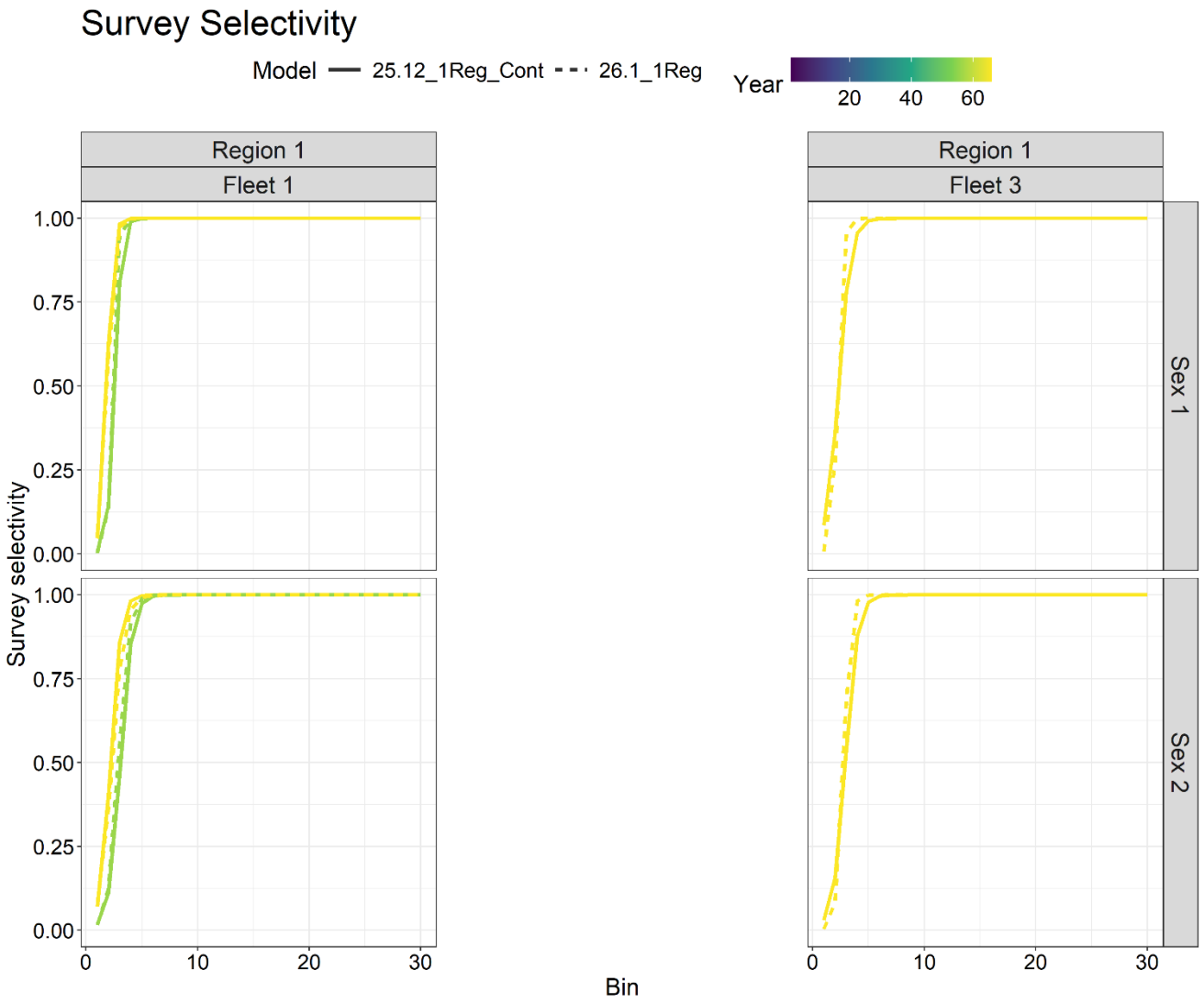

Figure SM3B.8. Survey selectivity estimates from Model *25.12_1Reg_Cont* (solid lines) and Model *26.1_1Reg* (dashed lines) where different colors represent the different time blocks. Fleet 1 is the NOAA Longline Survey and fleet 3 is the Japanese Longline Survey. Sex 1 is females and sex 2 is males.

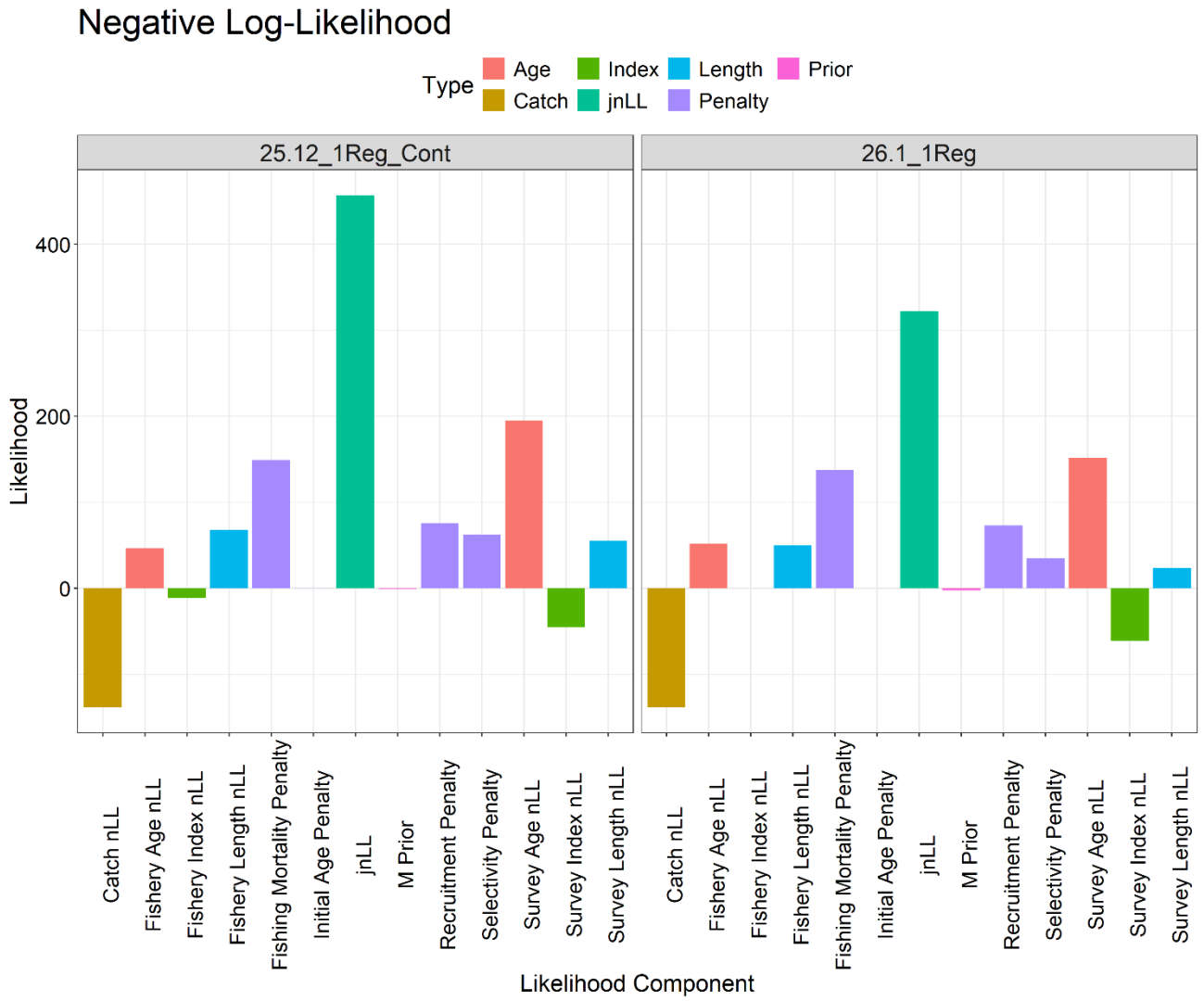

Figure SM3B.9. Joint negative log-likelihood (jnLL) components from Model *25.12_1Reg_Cont* and Model *26.1_1Reg* where different colors represent the various data sets and priors or penalty terms. Note that due to differences in data and priors or penalties used in each model, these figures are not directly comparable across models.

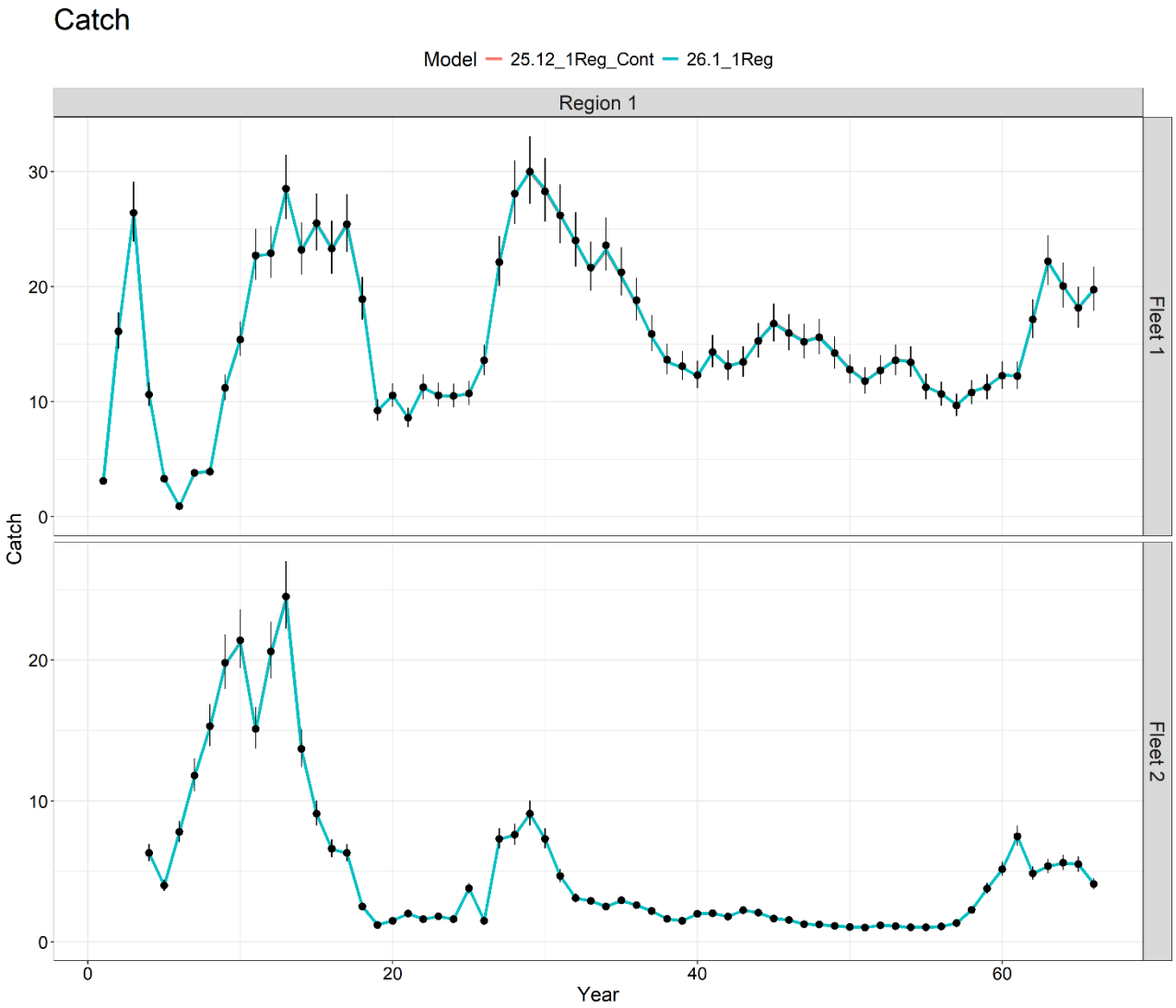

Figure SM3B.10. Fit to the catch data (black dots with 95% confidence intervals) by fleet (fleet 1 is the fixed gear fishery and fleet 2 is the trawl fleet) for Model *25.12_1Reg_Cont* (red line) and model *26.1_1Reg* (blue line).

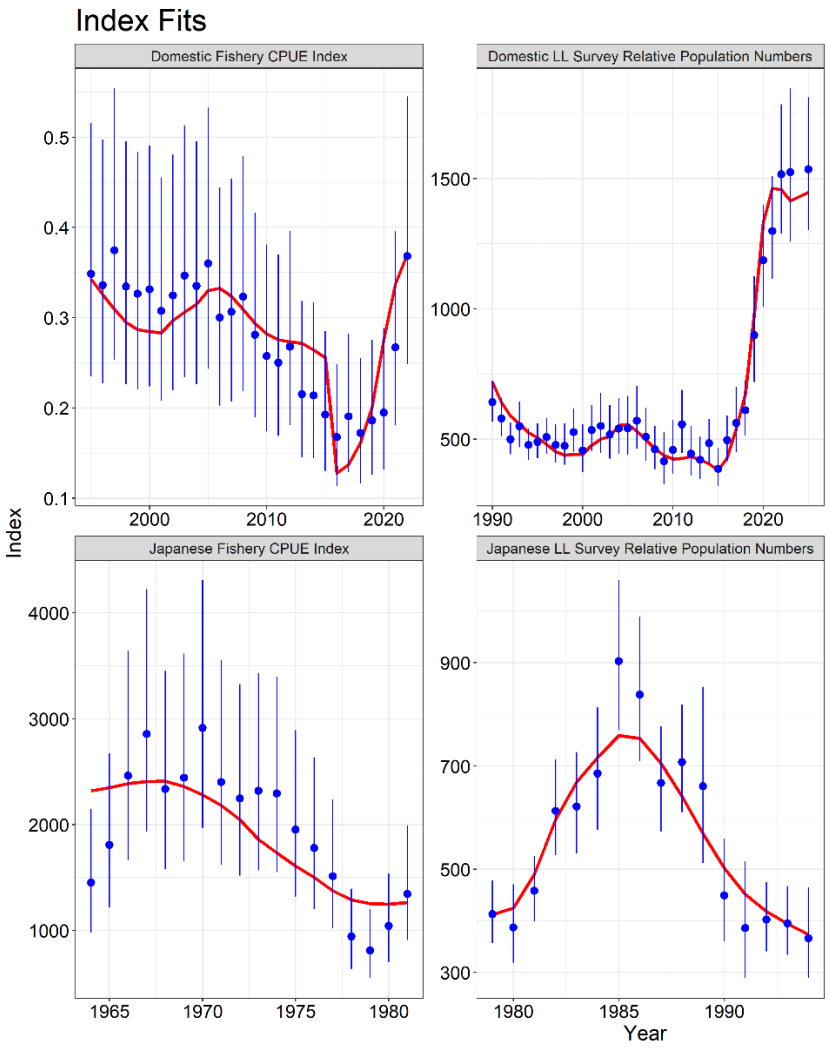

Figure SM3B.11. Model *25.12_1Reg_Cont* fit to indices, including the domestic fishery CPUE (top left), Japanese fishery CPUE (bottom left), NOAA Longline Survey RPNs (top right), and Japanese Longline Survey RPNs (bottom right). Red lines are model predicted values and blue dots are observed values (with 95% confidence intervals based on the assumed standard errors).

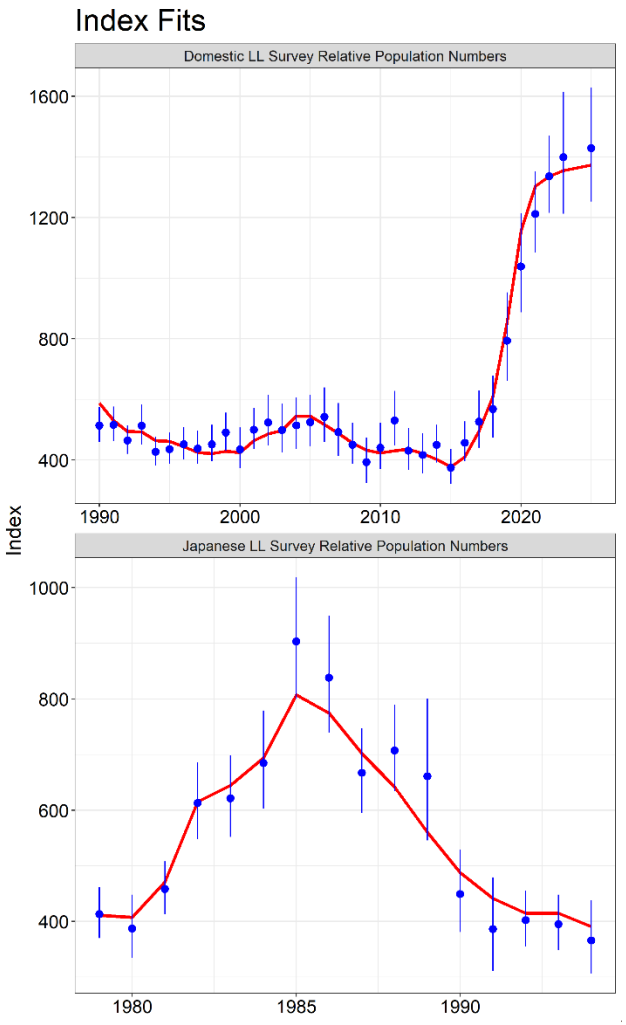

Figure SM3B.12. Model *26.1_1Reg* fit to indices, including the NOAA Longline Survey RPNs (top) and Japanese Longline Survey RPNs (bottom). Red lines are model predicted values and blue dots are observed values (with 95% confidence intervals based on the assumed standard errors). Note the changes in the NOAA Longline Survey index from Model *25.12_1Reg_Cont* due to the use of the carryover method and removal of extrapolated western AI stations (see Figure 3B.3).

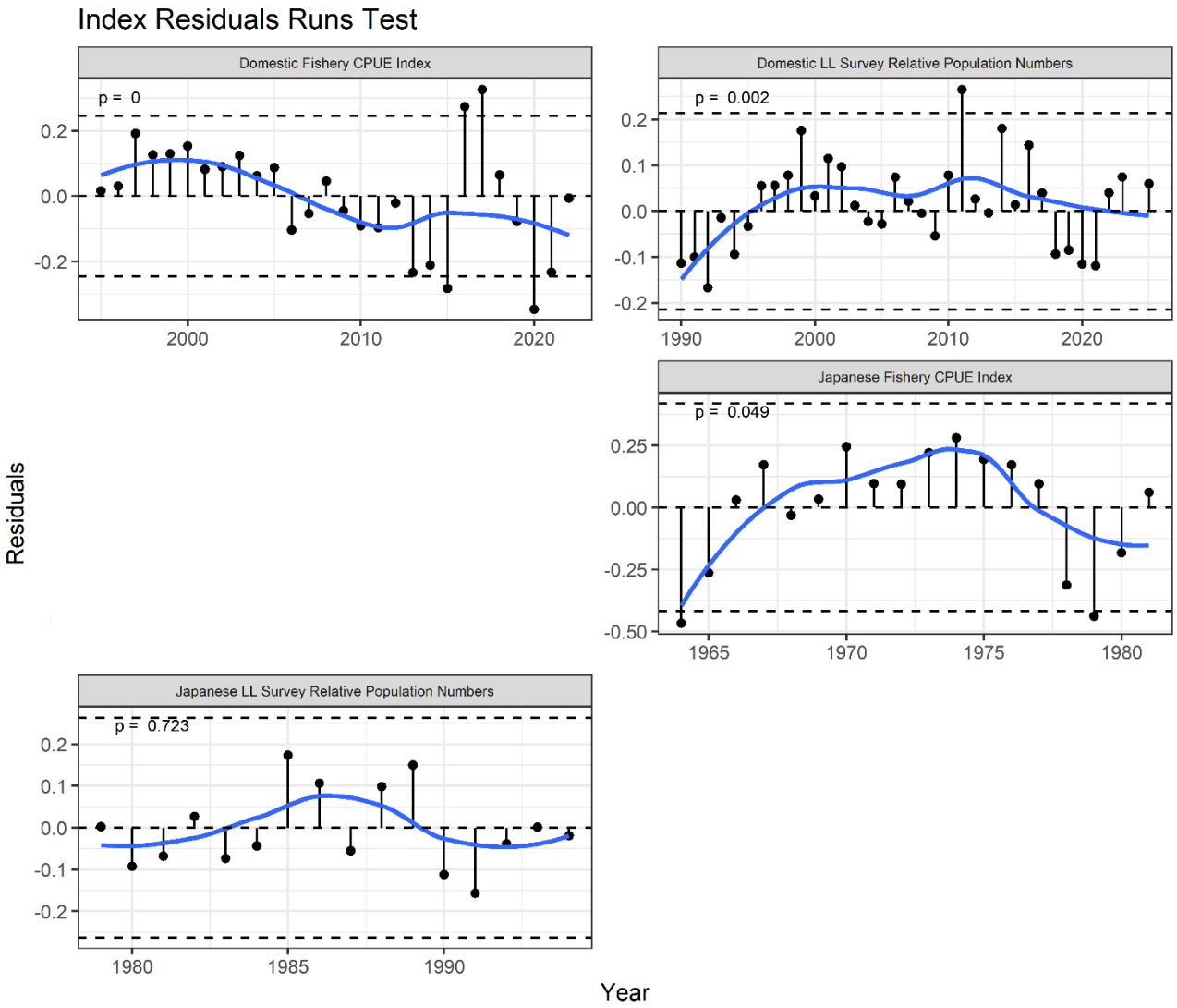

Figure SM3B.13. Results of an index residuals runs test for Model *25.12_1Reg_Cont*.

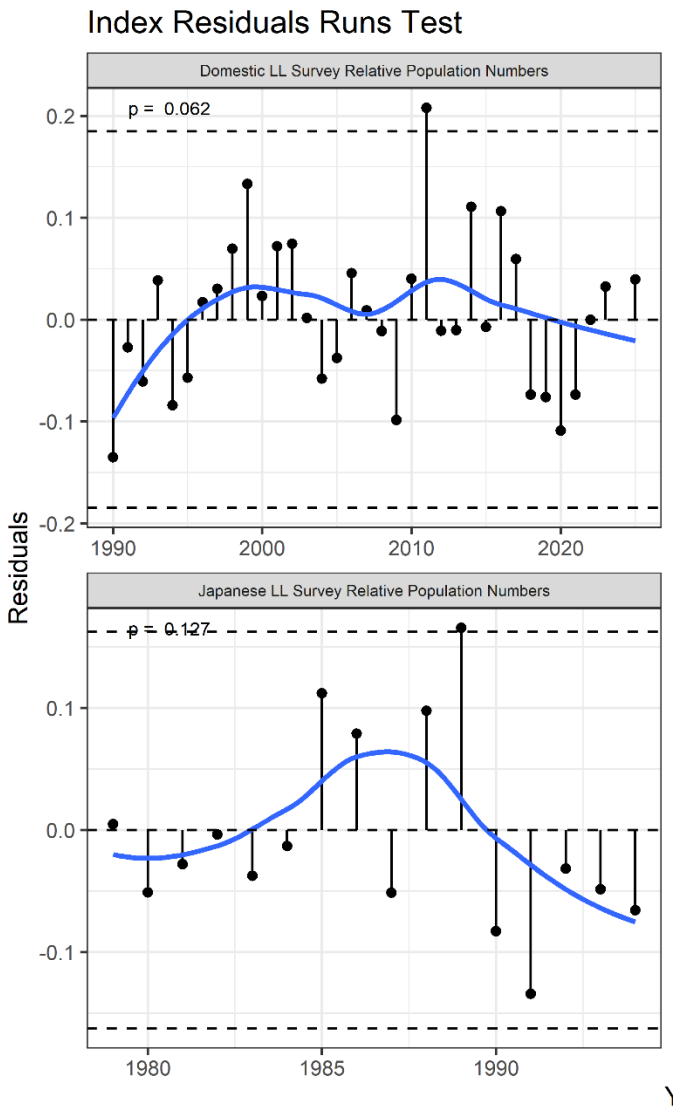

Figure SM3B.14. Results of an index residuals runs test for Model *26.1_1Reg*.

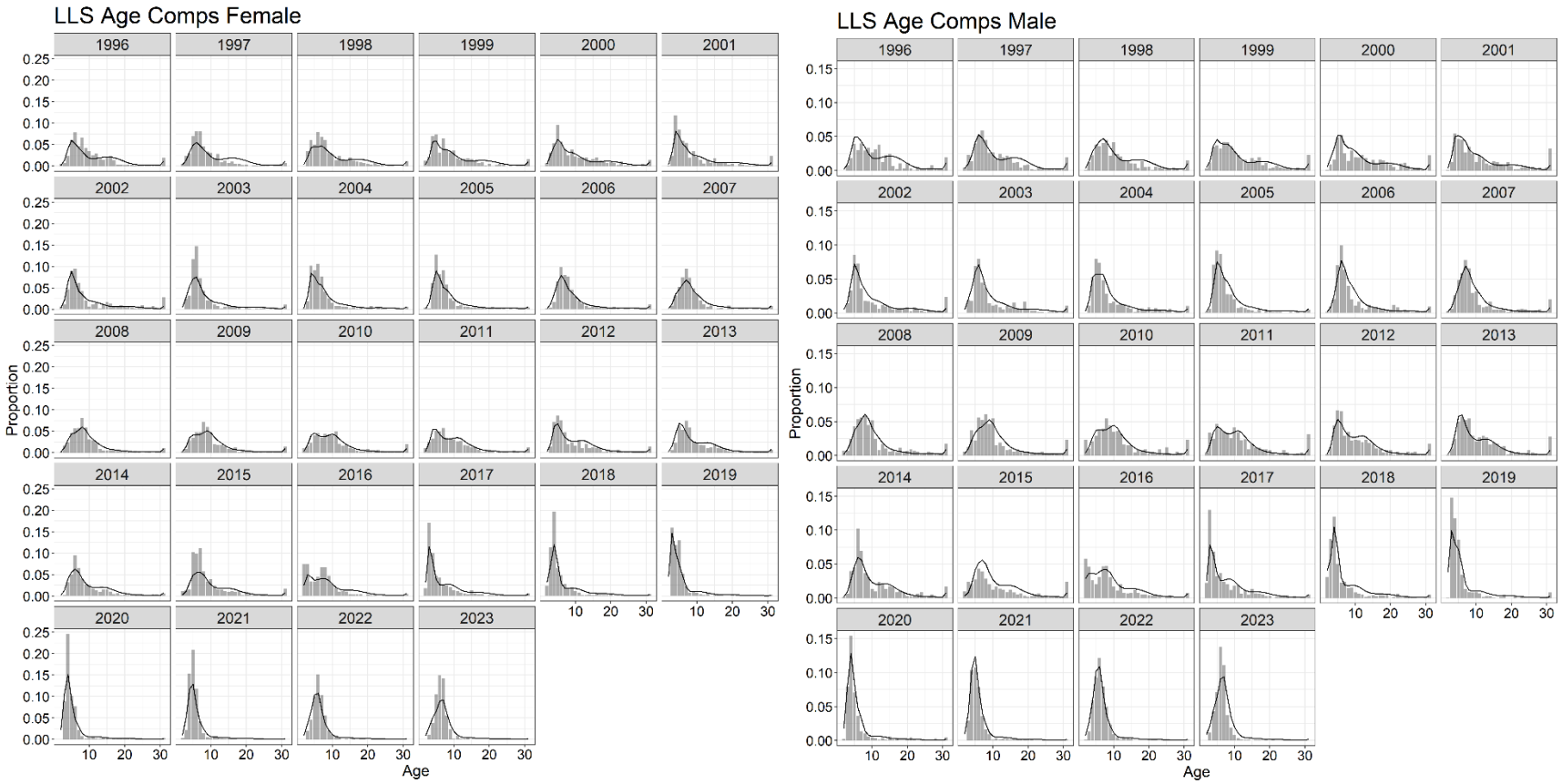

Figure SM3B.15. Fits to NOAA longline survey age composition data by year and sex (females left and males right) for model *25.12_1Reg_Cont*.

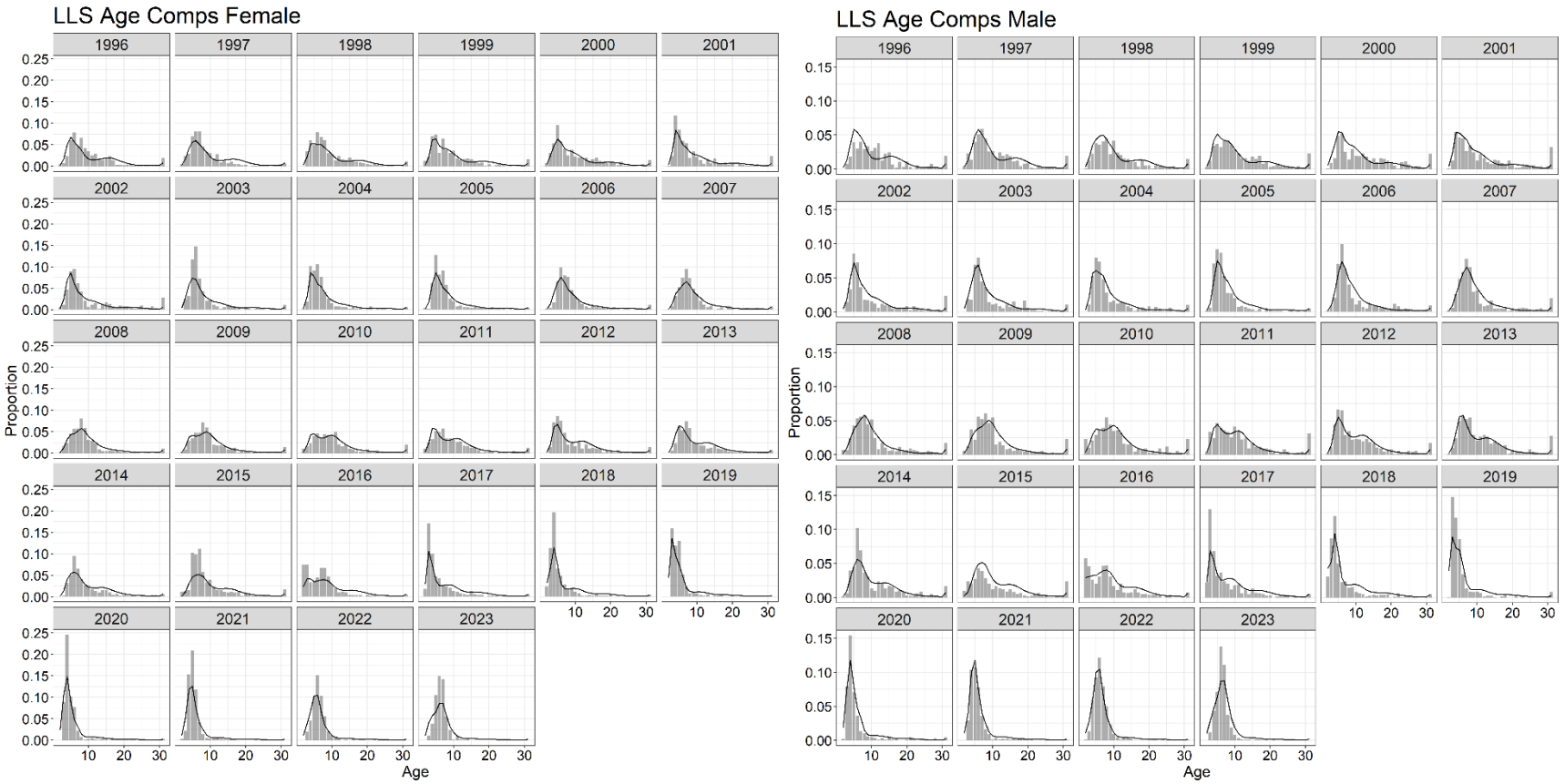

Figure SM3B.16. Fits to NOAA longline survey age composition data by year and sex (females left and males right) for model *26.16_1Reg*.

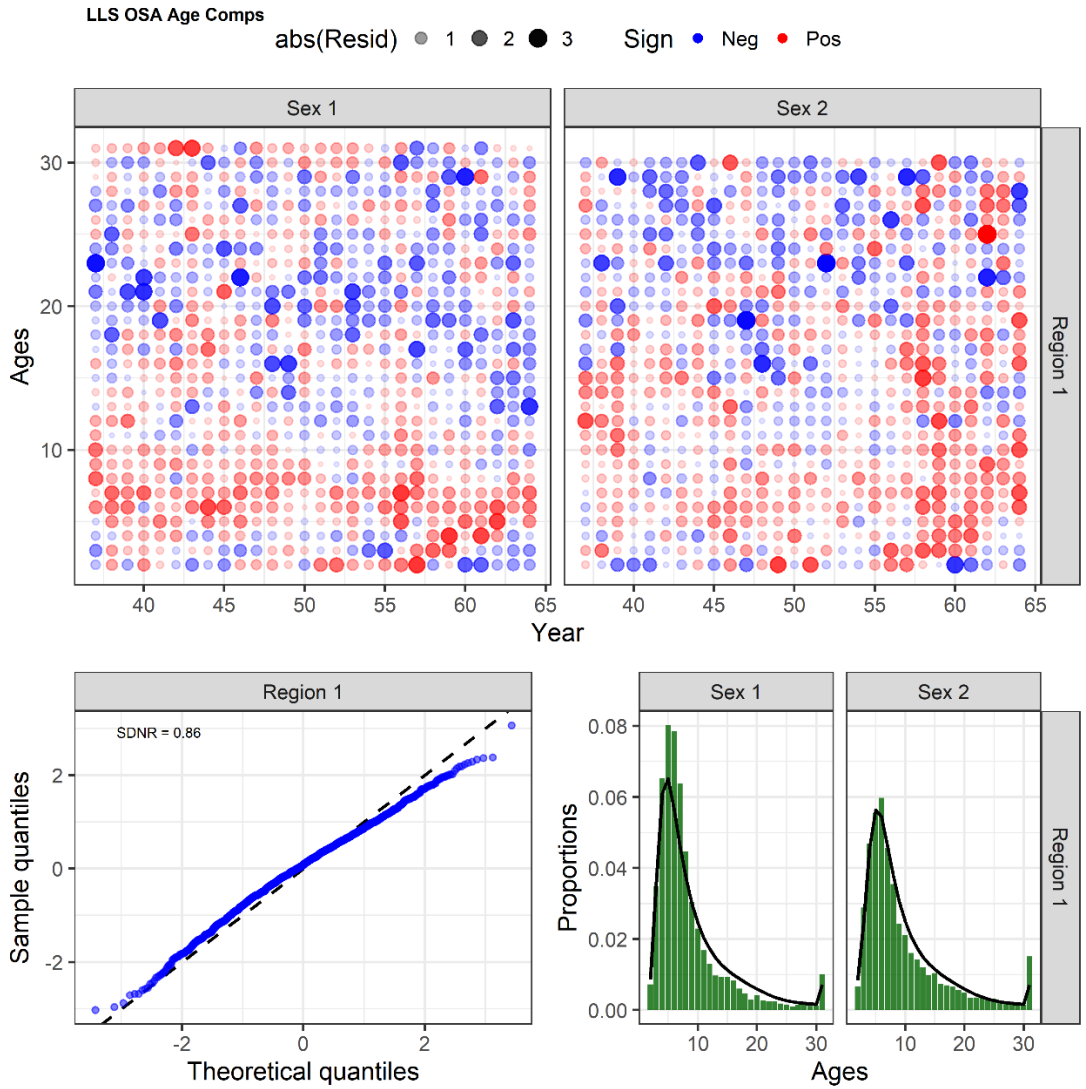

Figure SM3B.17. Fits to NOAA longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *25.12_1Reg_Cont*.

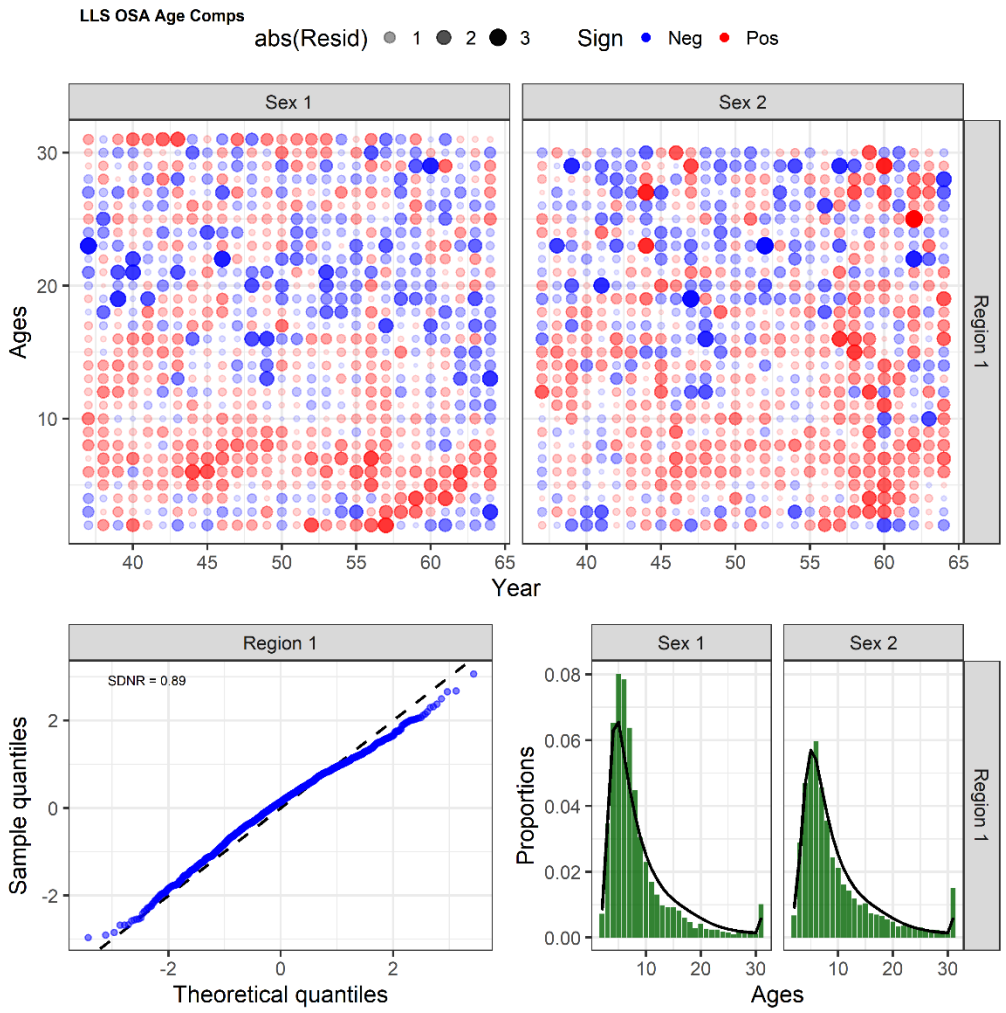

Figure SM3B.18. Fits to NOAA longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg*

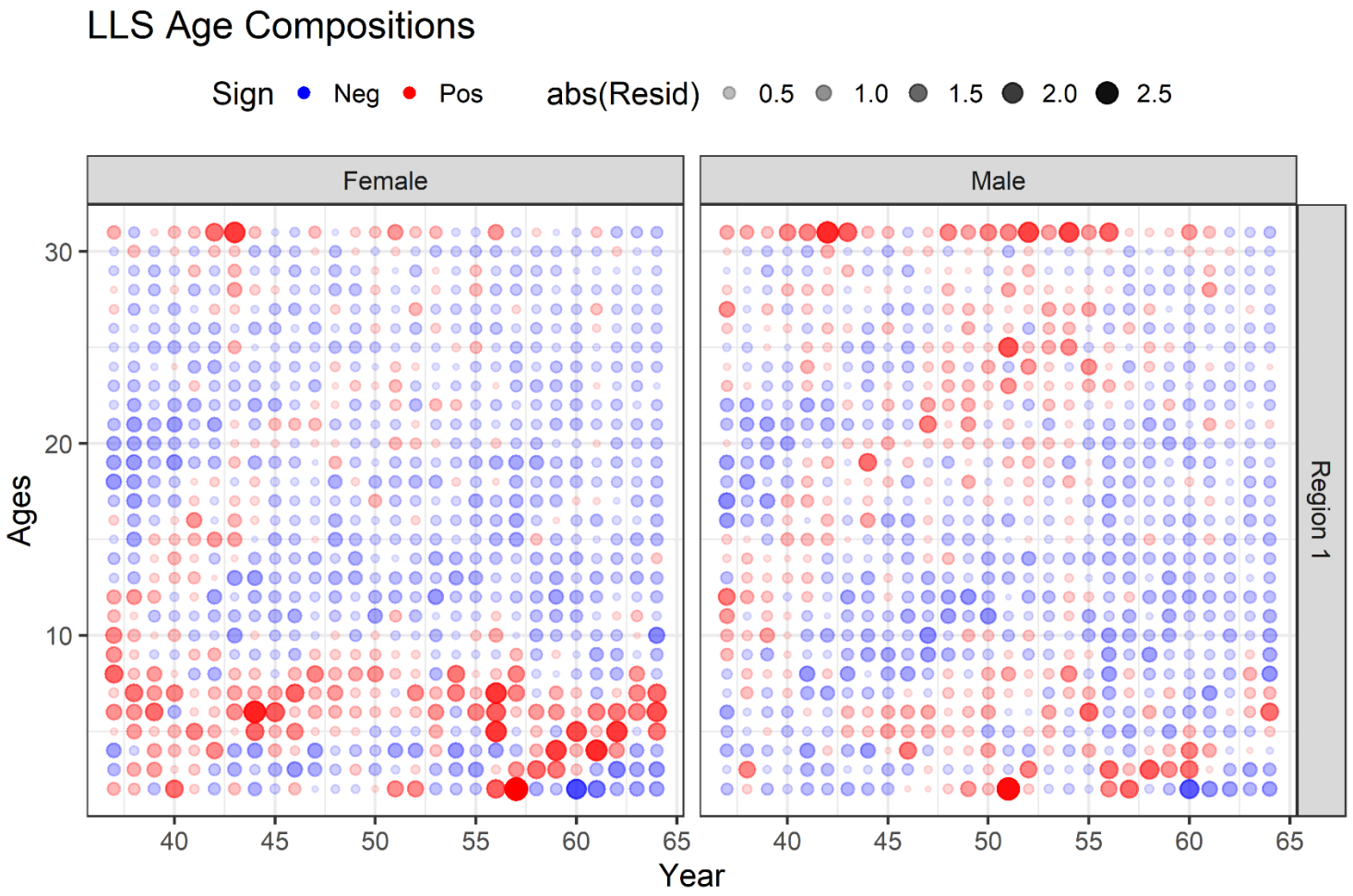

Figure SM3B.19. Fits to NOAA longline survey age composition data based on Pearson residuals for model *25.12_1Reg_Cont*.

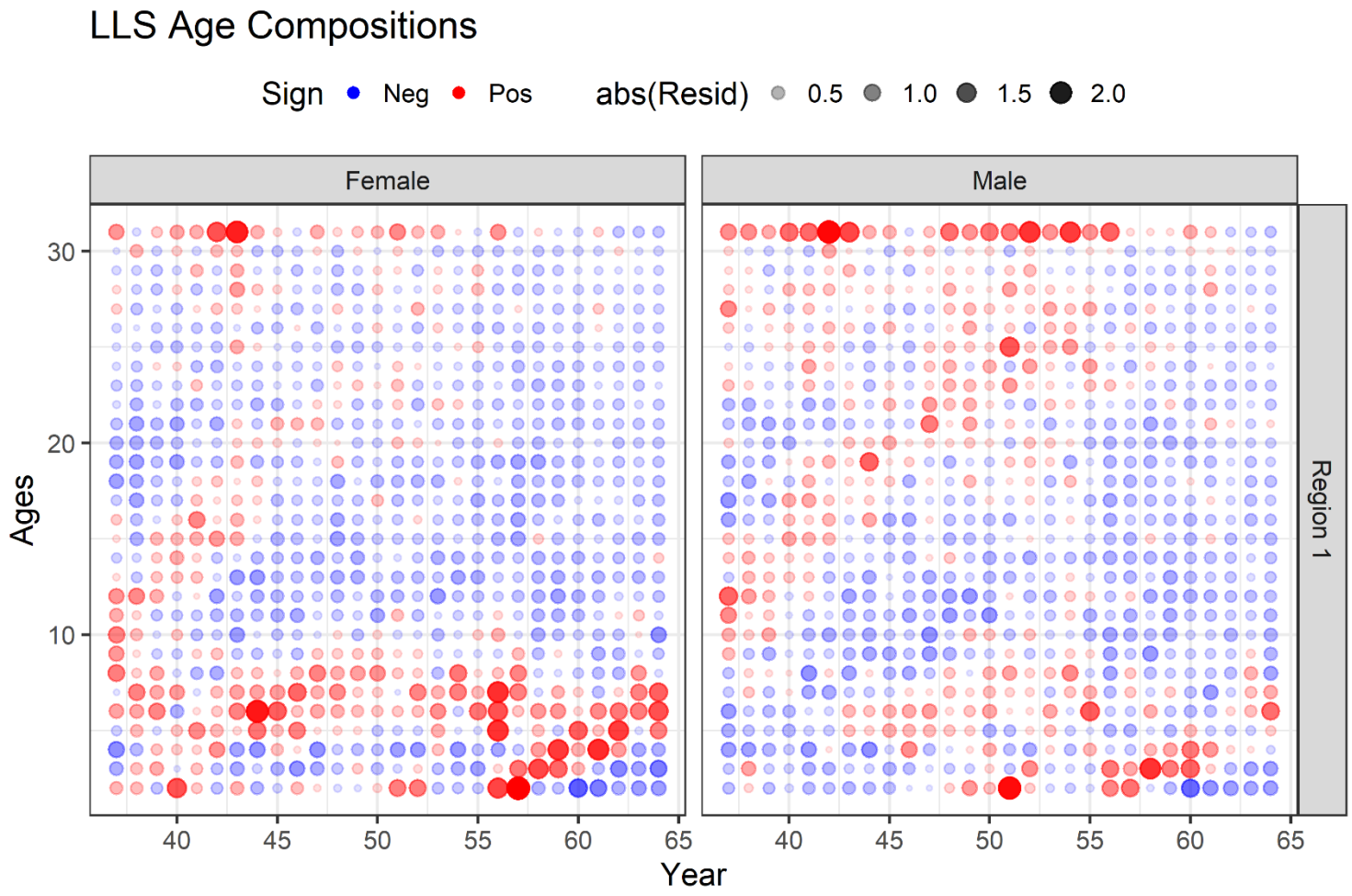

Figure SM3B.20. Fits to NOAA longline survey age composition data based on Pearson residuals for model *26.1_1Reg*.

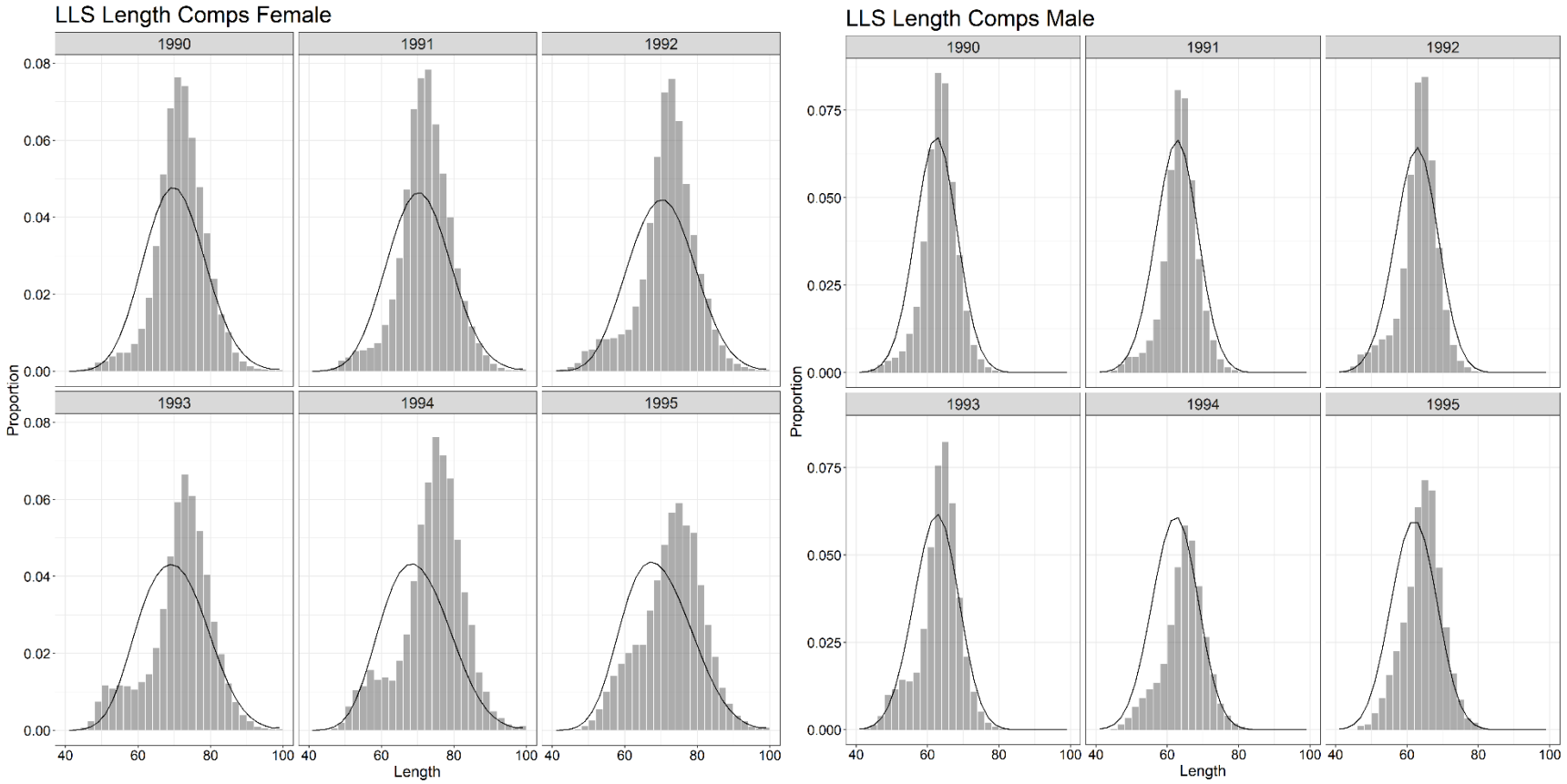

Figure SM3B.21. Fits to NOAA longline survey length composition data by year and sex (females left and males right) for Model *25.12_1Reg_Cont*.

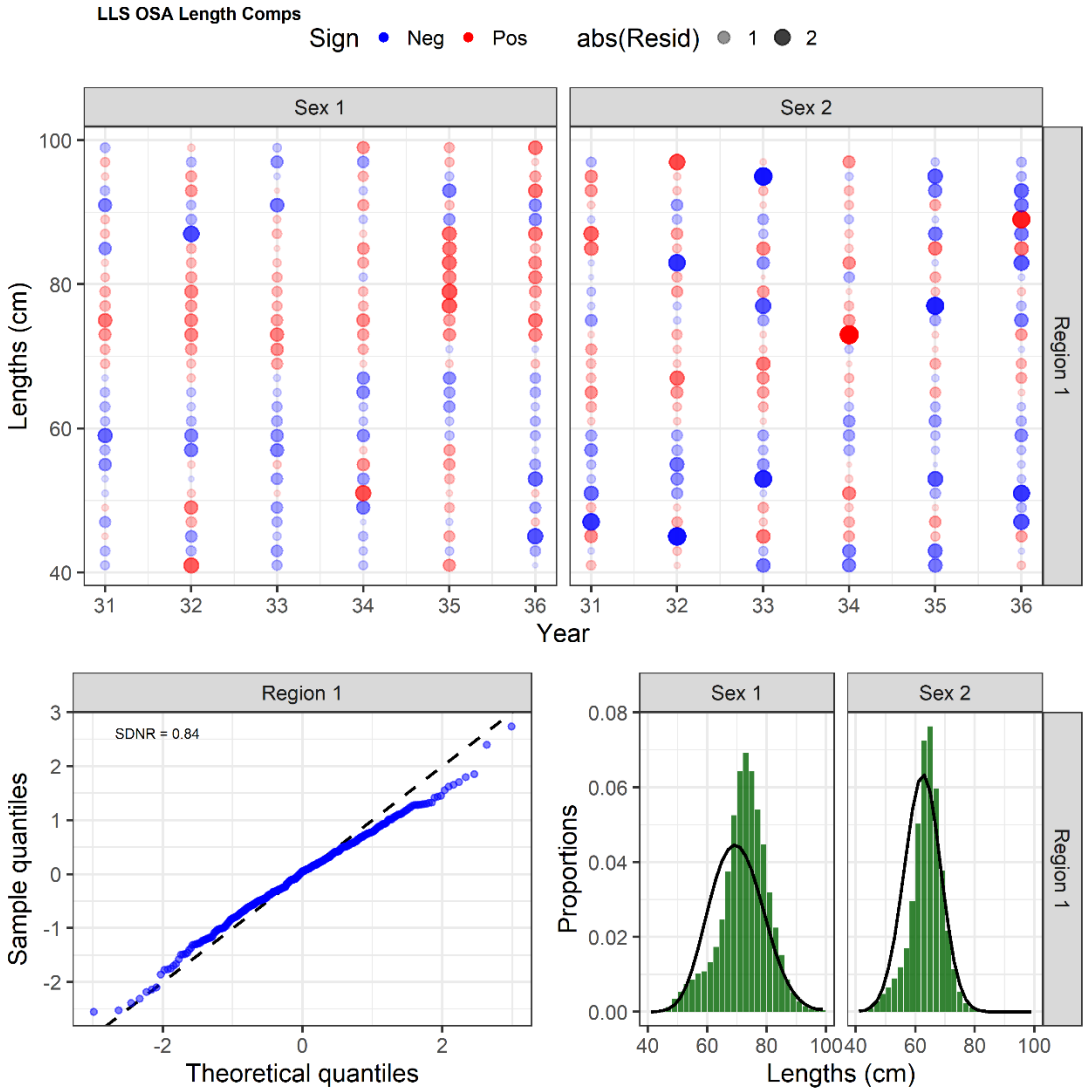

Figure SM3B.22. Fits to NOAA longline survey length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *25.12_1Reg_Cont*.

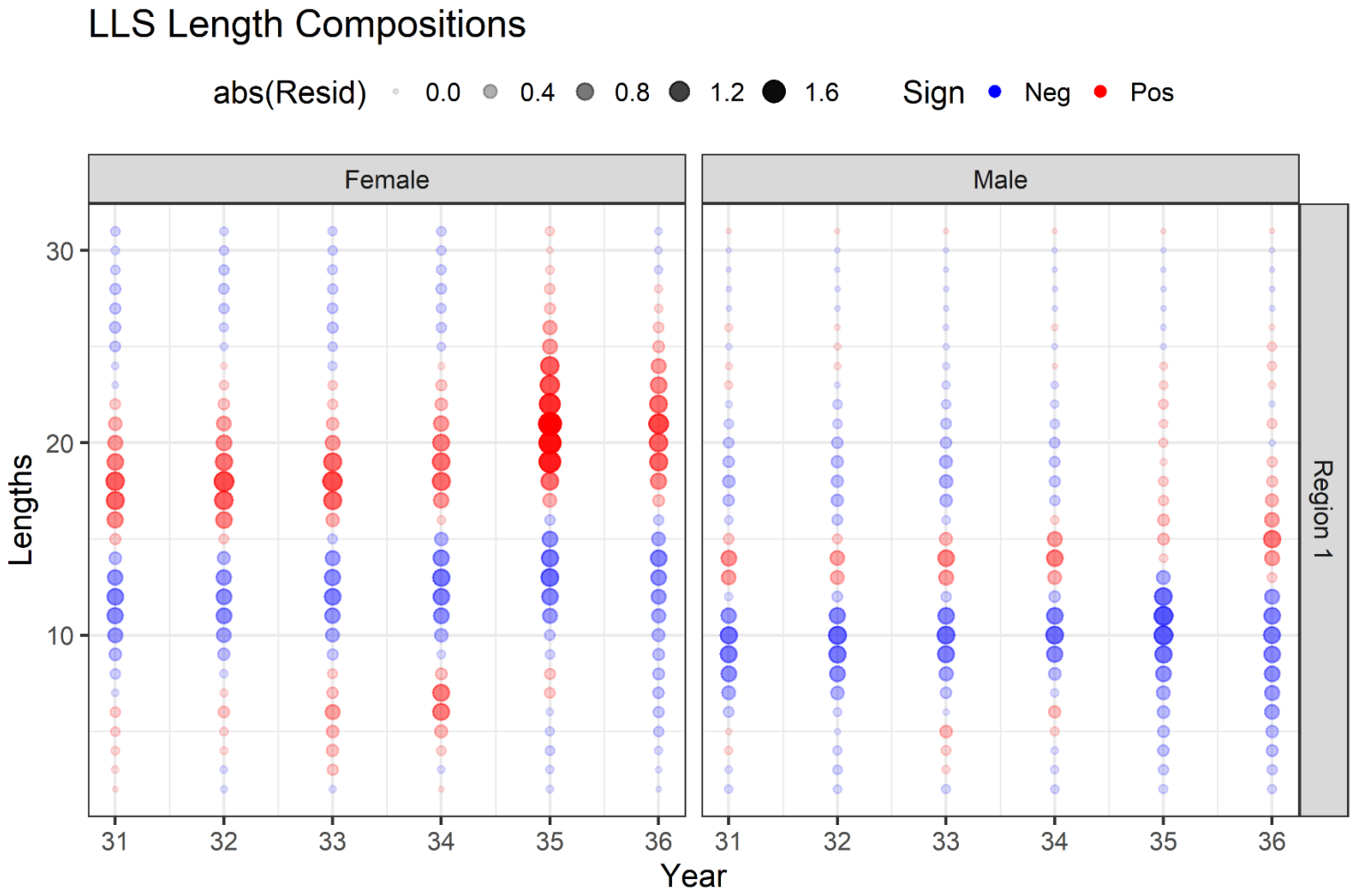

Figure SM3B.23. Fits to NOAA longline survey length composition data based on Pearson residuals for Model *25.12_1Reg_Cont*.

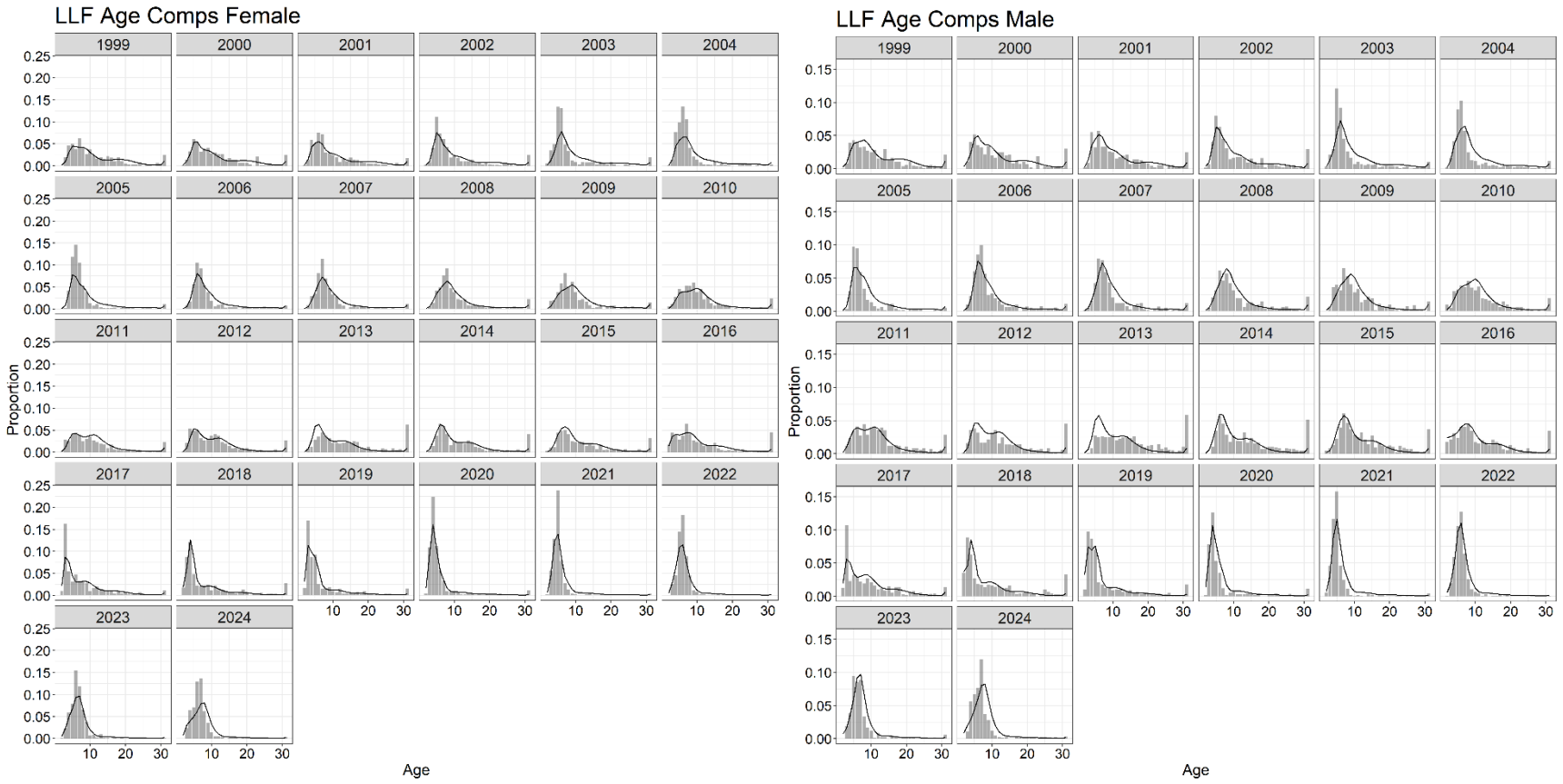

Figure SM3B.24. Fits to fixed gear fishery age composition data by year and sex (females left and males right) for model *25.12_1Reg_Cont*.

Figure SM3B.25. Fits to fixed gear fishery age composition data by year and sex (females left and males right) for model *26.1_1Reg*.

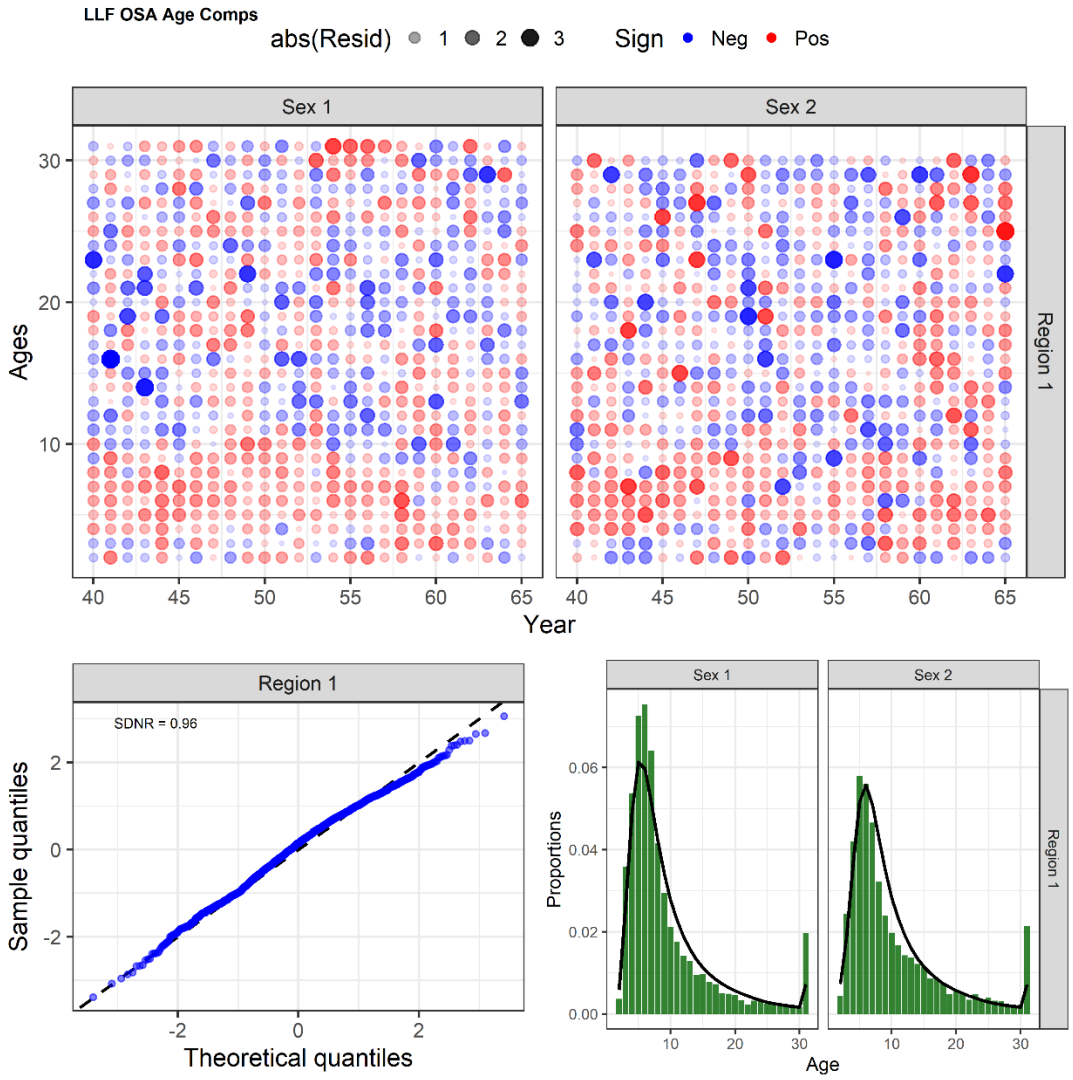

Figure SM3B.26. Fits to fixed gear fishery age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *25.12_1Reg_Cont*.

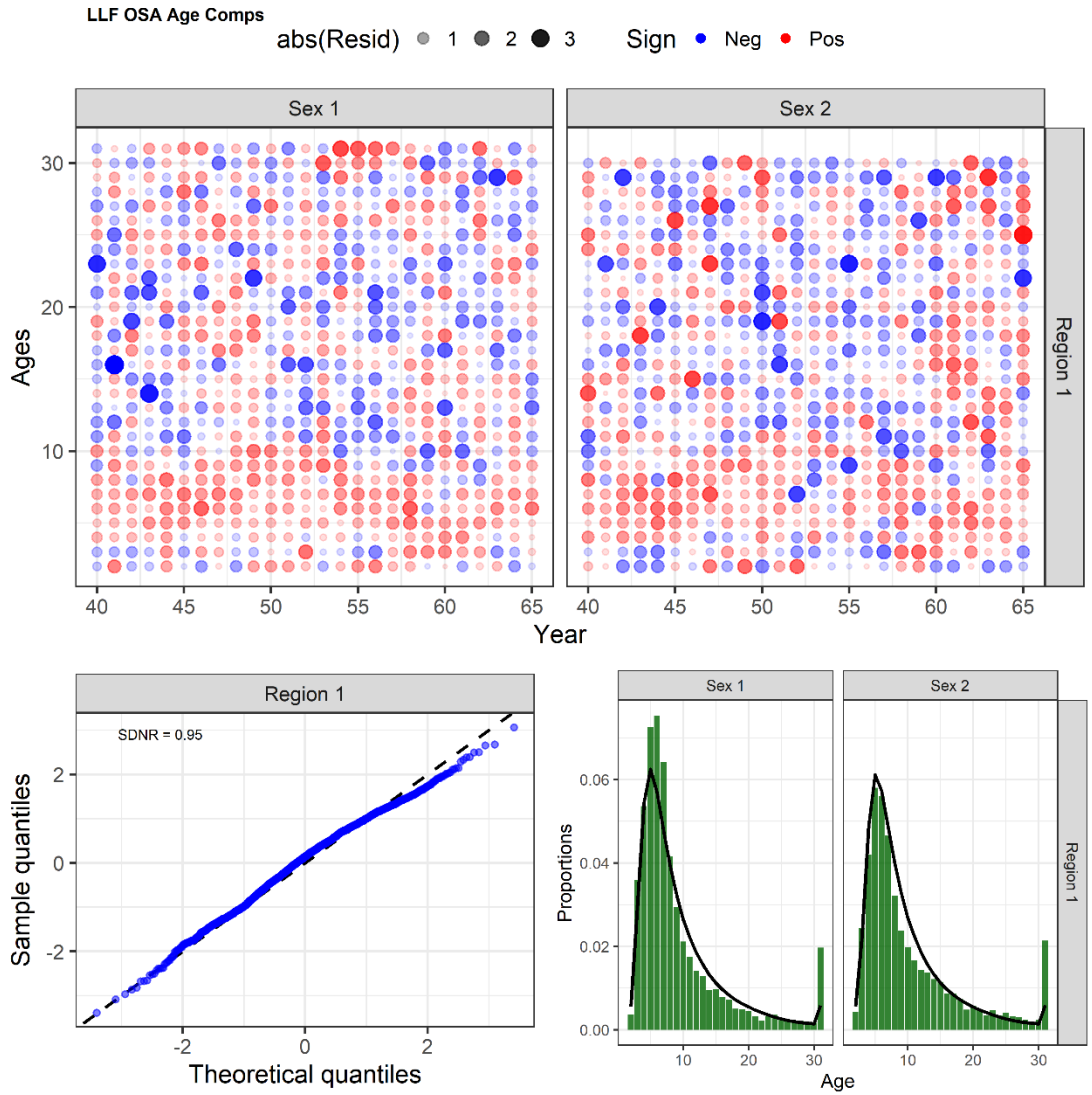

Figure SM3B.27. Fits to fixed gear fishery age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg*.

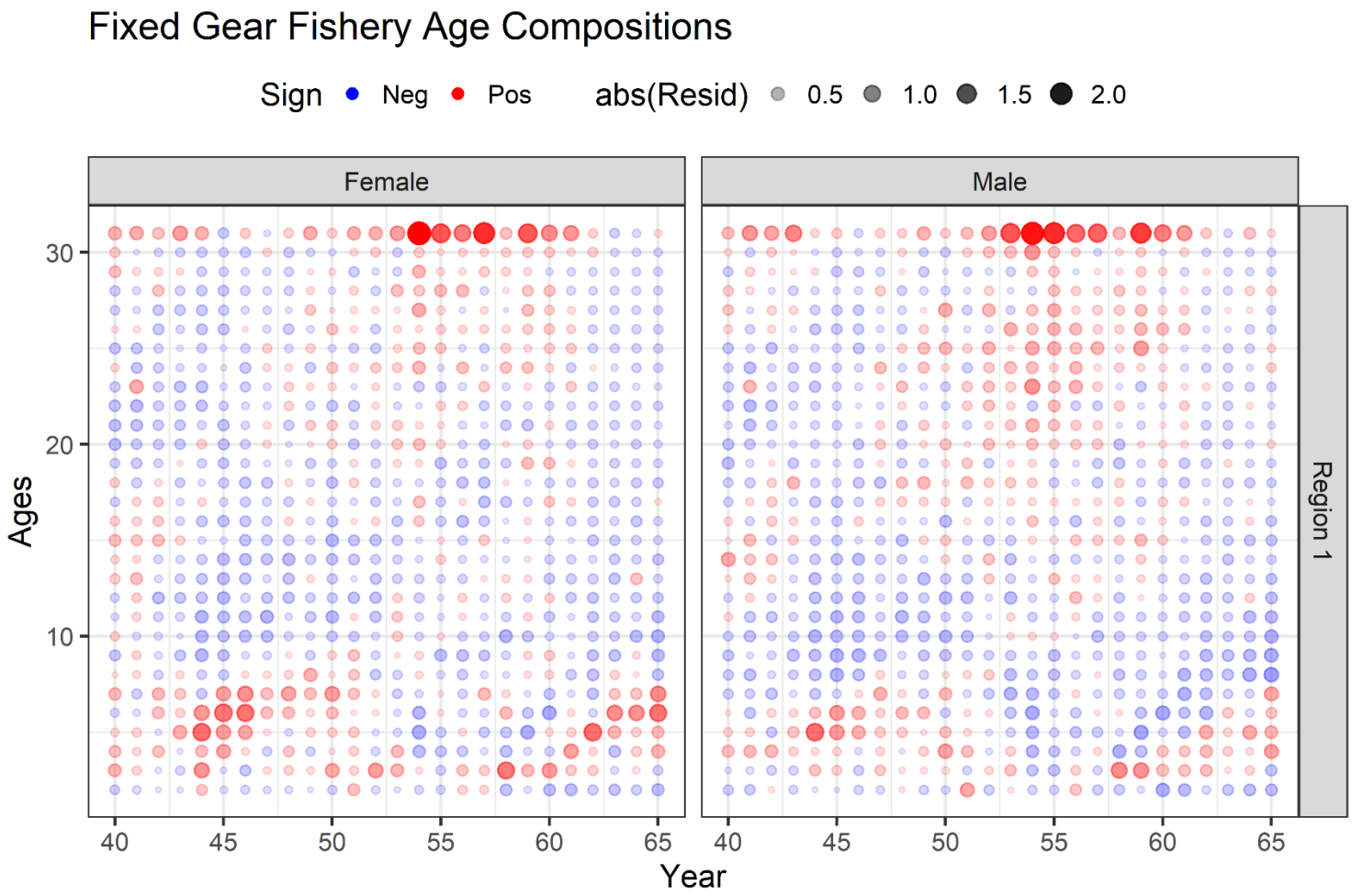

Figure SM3B.28. Fits to the fixed gear fishery age composition data based on Pearson residuals for model *25.12_1Reg_Cont*.

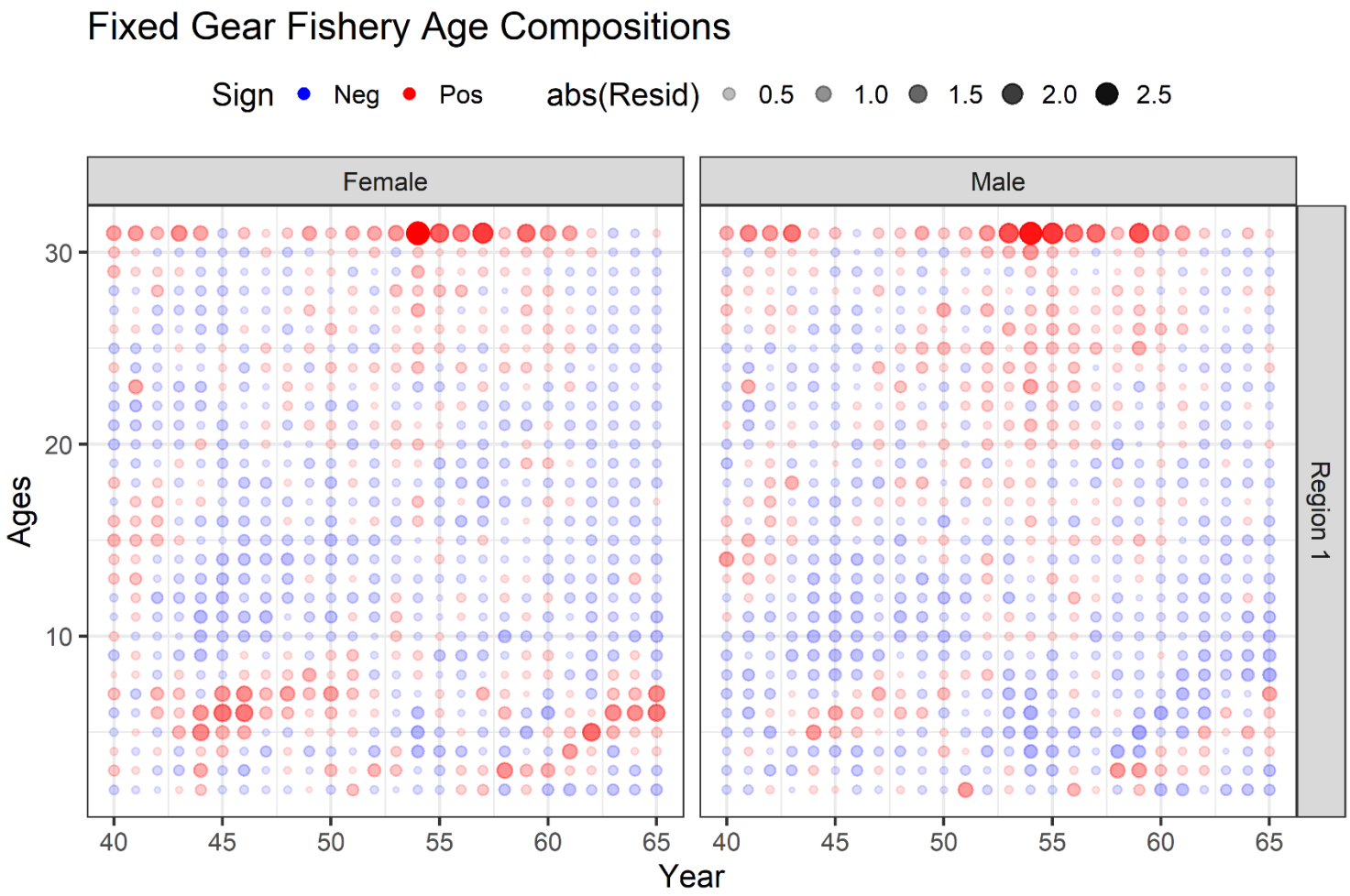

Figure SM3B.29. Fits to the fixed gear fishery age composition data based on Pearson residuals for model *26.1_1Reg*.

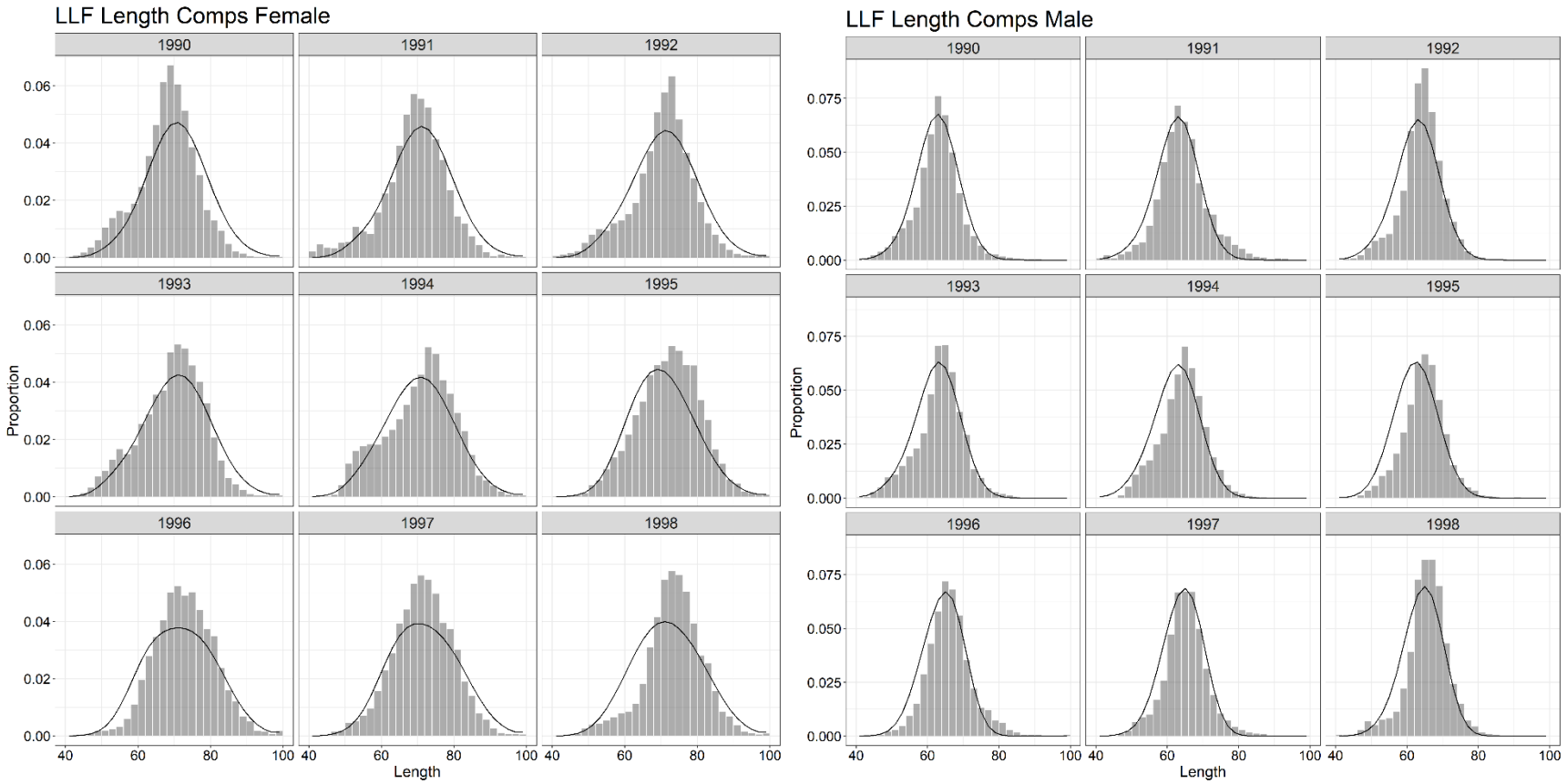

Figure SM3B.30. Fits to fixed gear fishery length composition data by year and sex (females left and males right) for model *25.12_1Reg_Cont*.

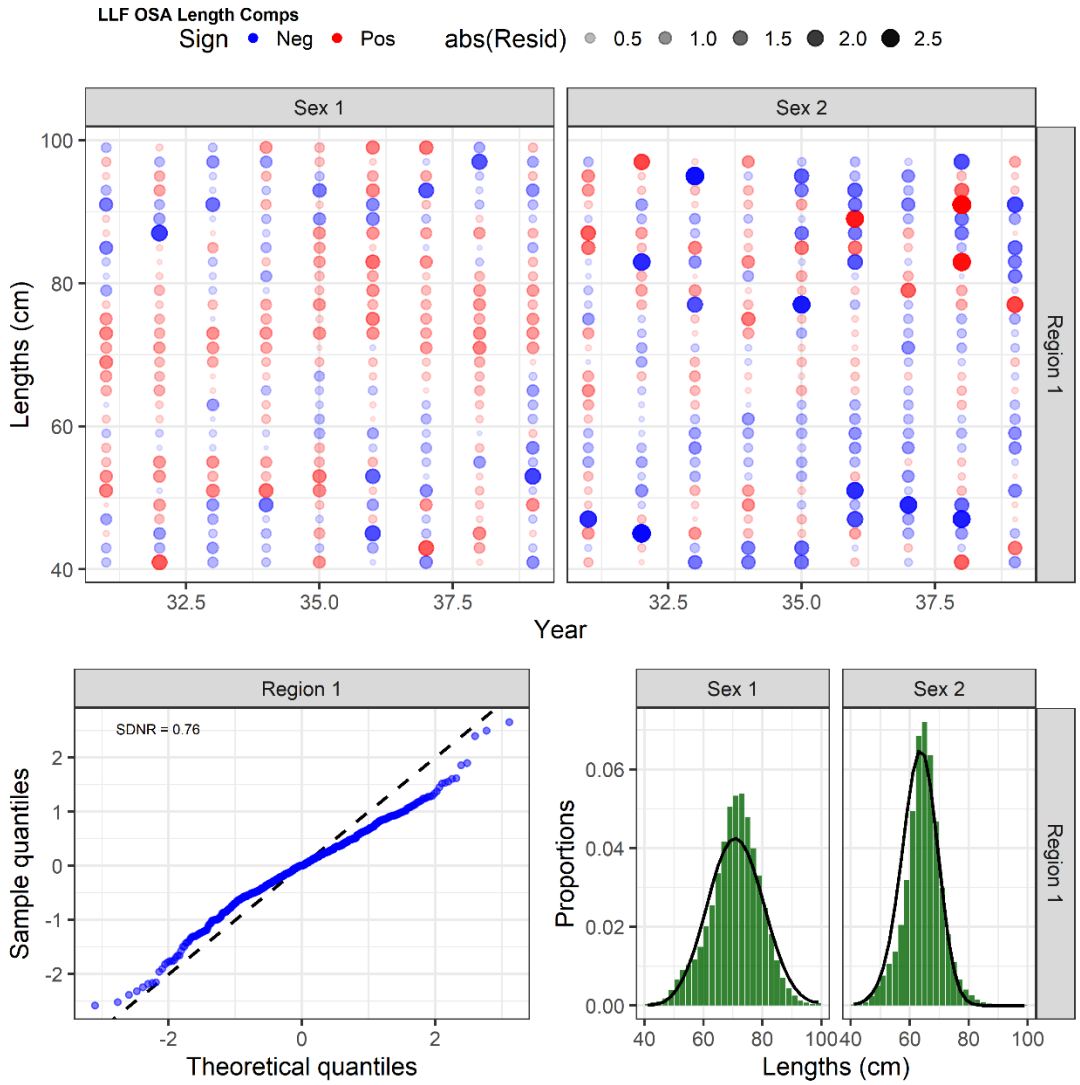

Figure SM3B.31. Fits to fixed gear fishery length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *25.12_1Reg_Cont*.

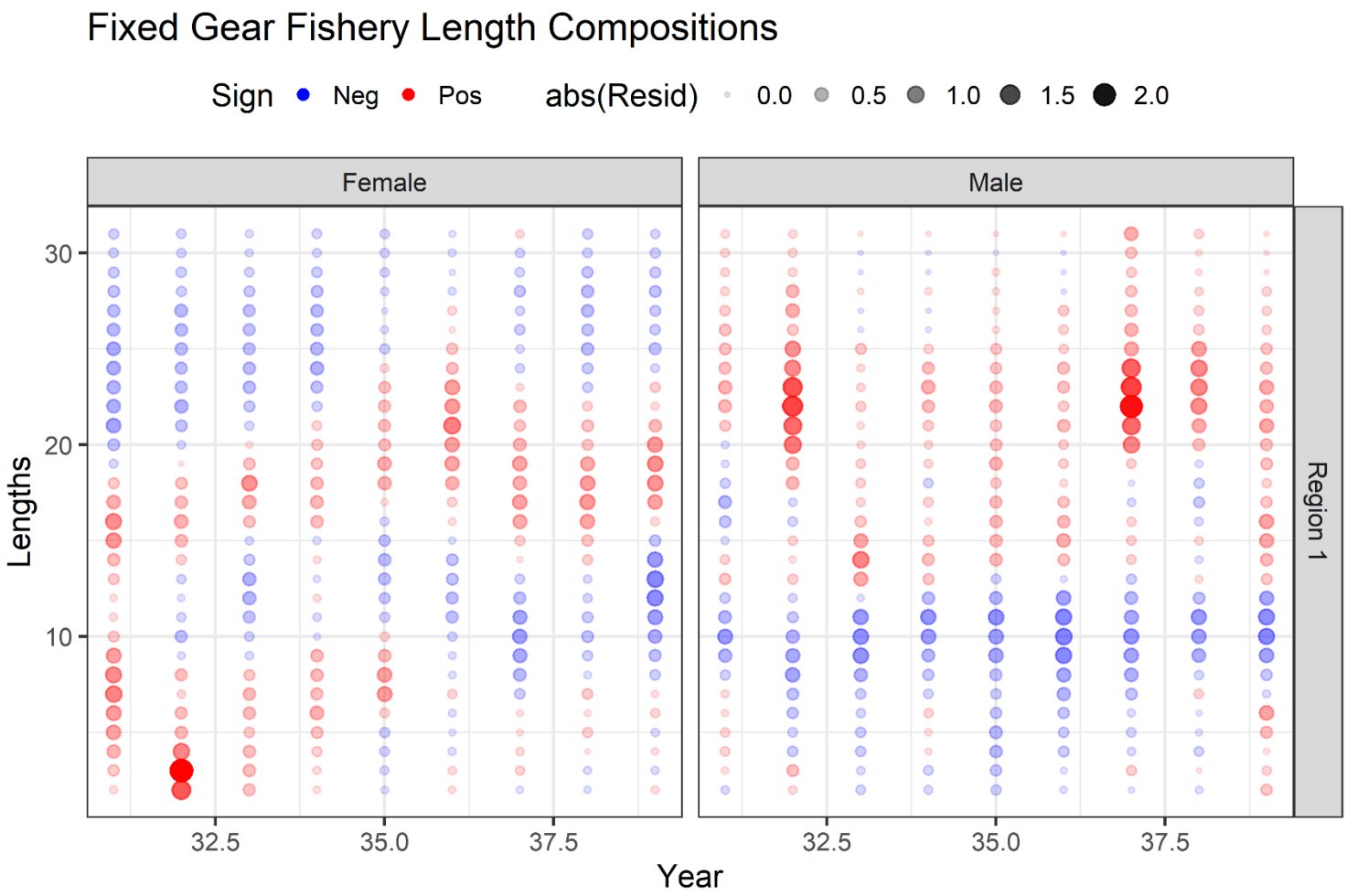

Figure SM3B.32. Fits to the fixed gear fishery length composition data based on Pearson residuals for model *25.12_1Reg_Cont*.

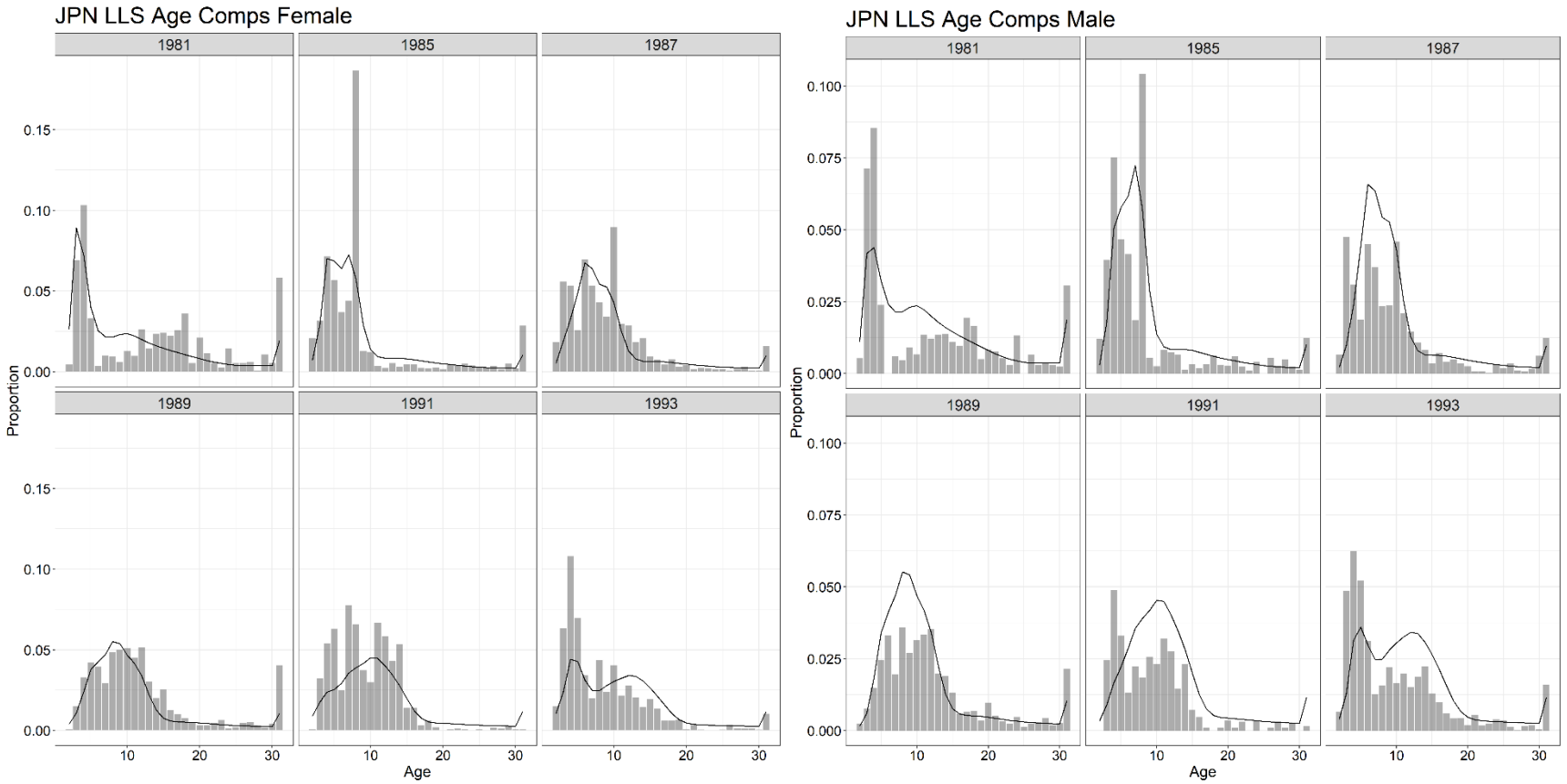

Figure SM3B.33. Fits to Japanese Longline Survey age composition data by year and sex (females left and males right) for model *25.12_1Reg_Cont*.

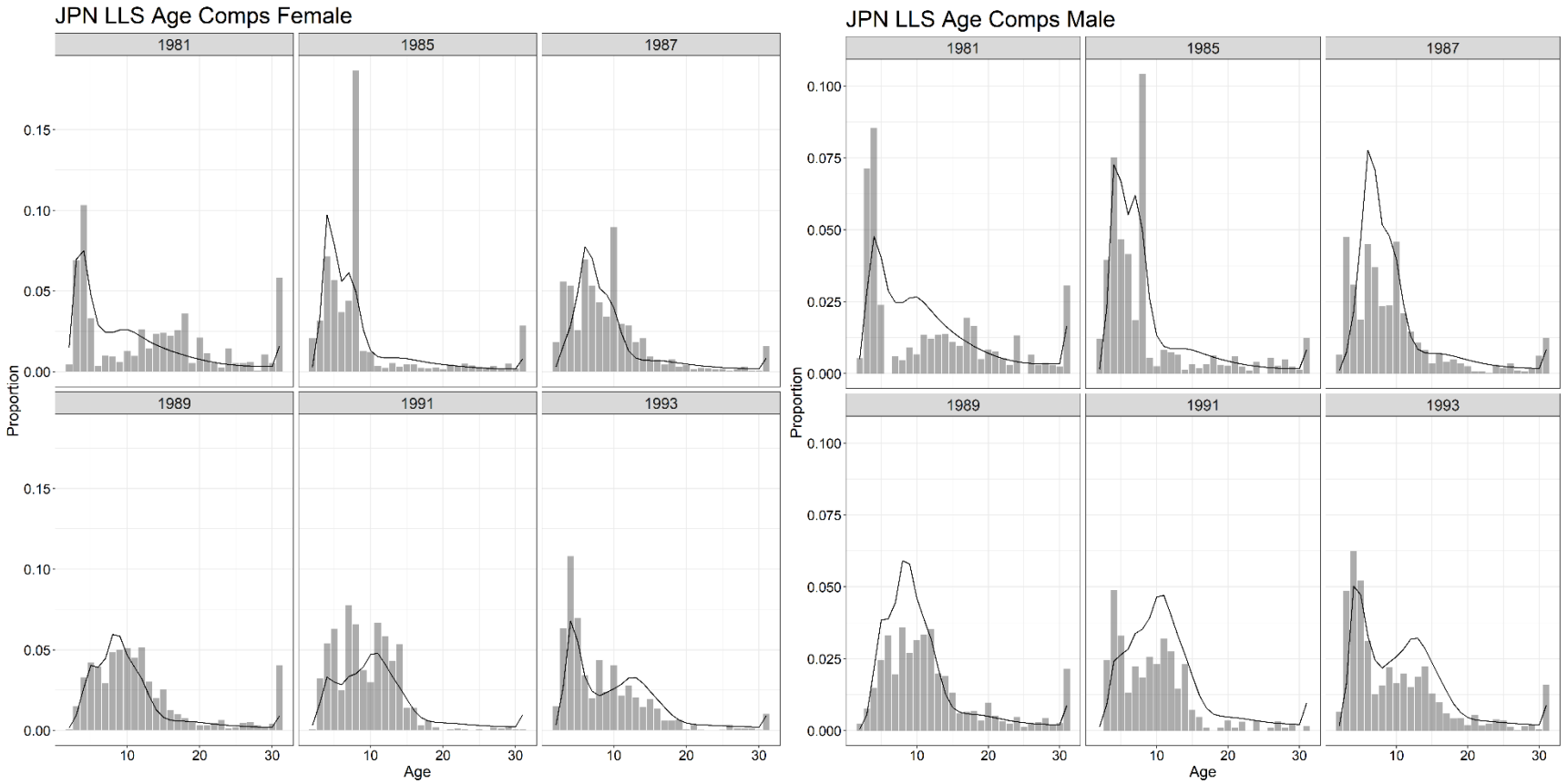

Figure SM3B.34. Fits to Japanese Longline Survey age composition data by year and sex (females left and males right) for model *26.1_1Reg*.

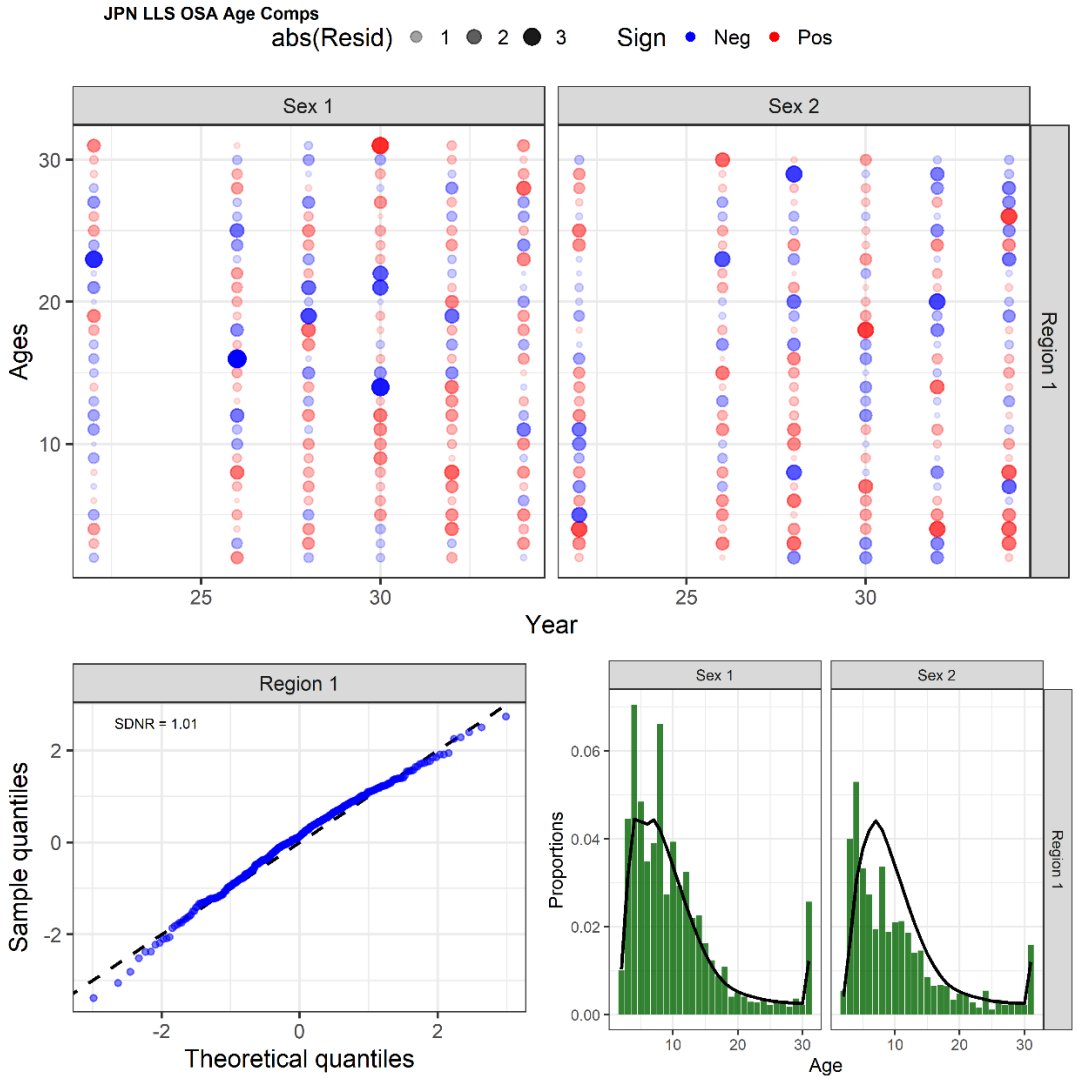

Figure SM3B.35. Fits to Japanese longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *25.12_1Reg_Cont*.

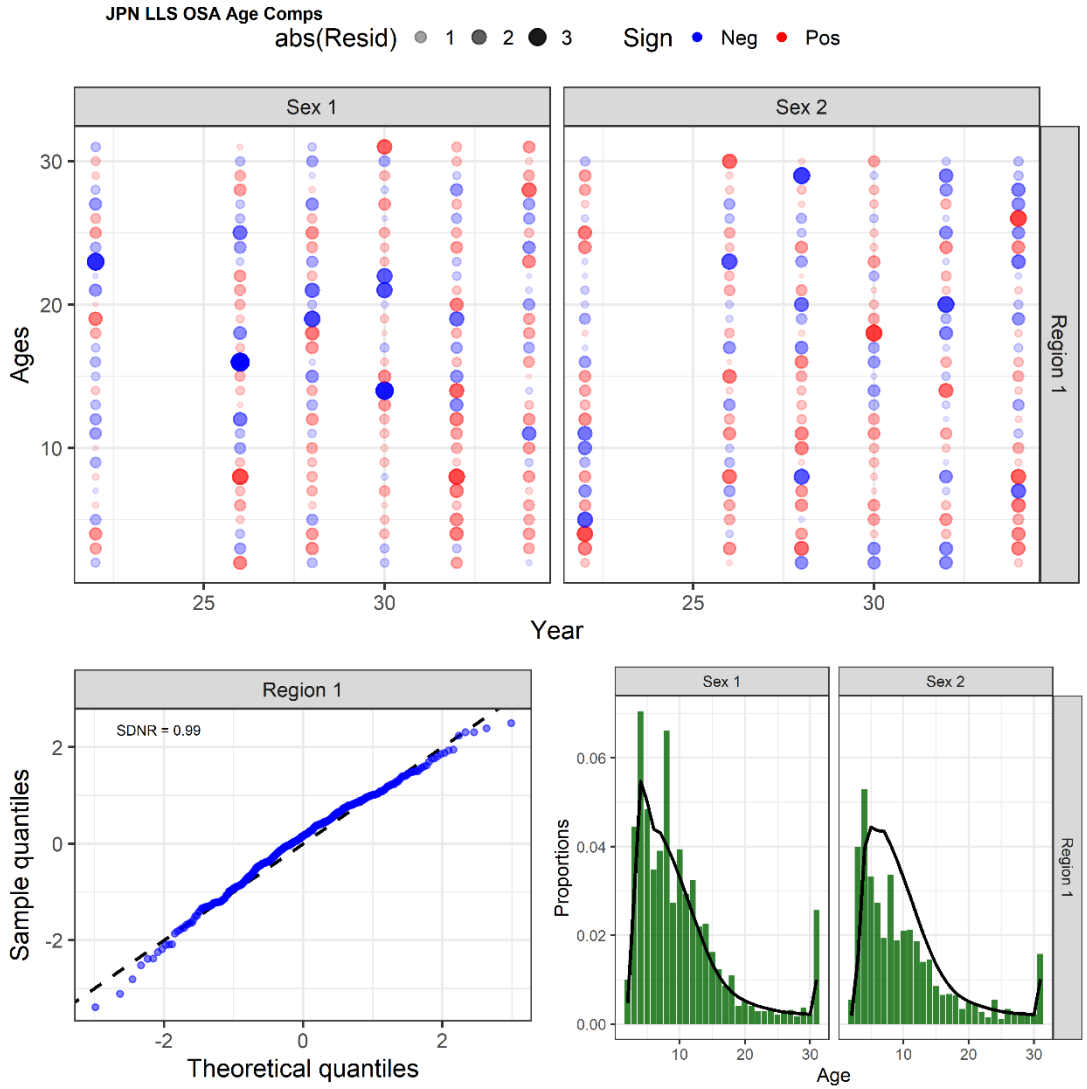

Figure SM3B.36. Fits to Japanese longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg*.

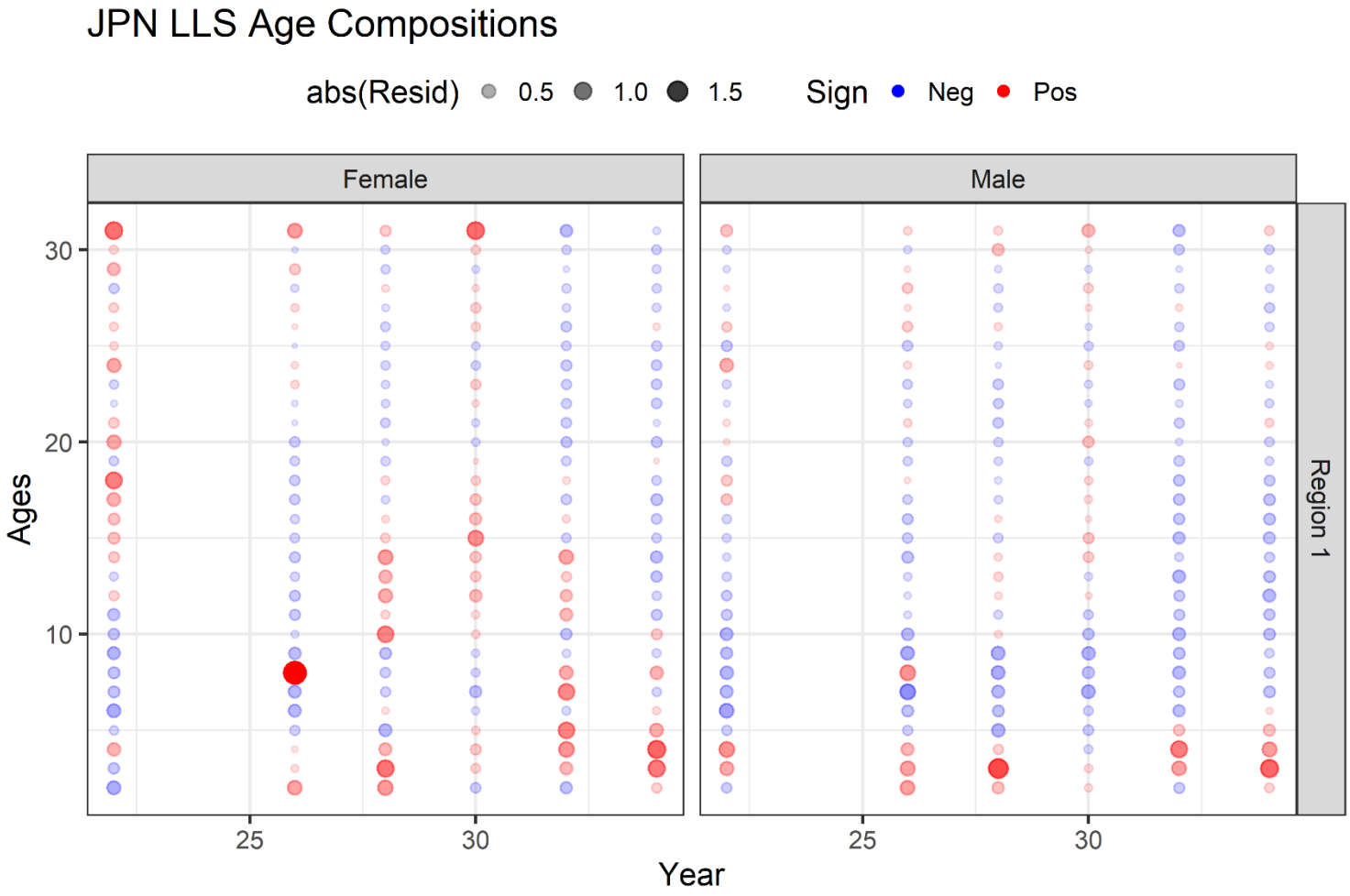

Figure SM3B.37. Fits to the Japanese Longline Survey age composition data based on Pearson residuals for model *25.12_1Reg_Cont*.

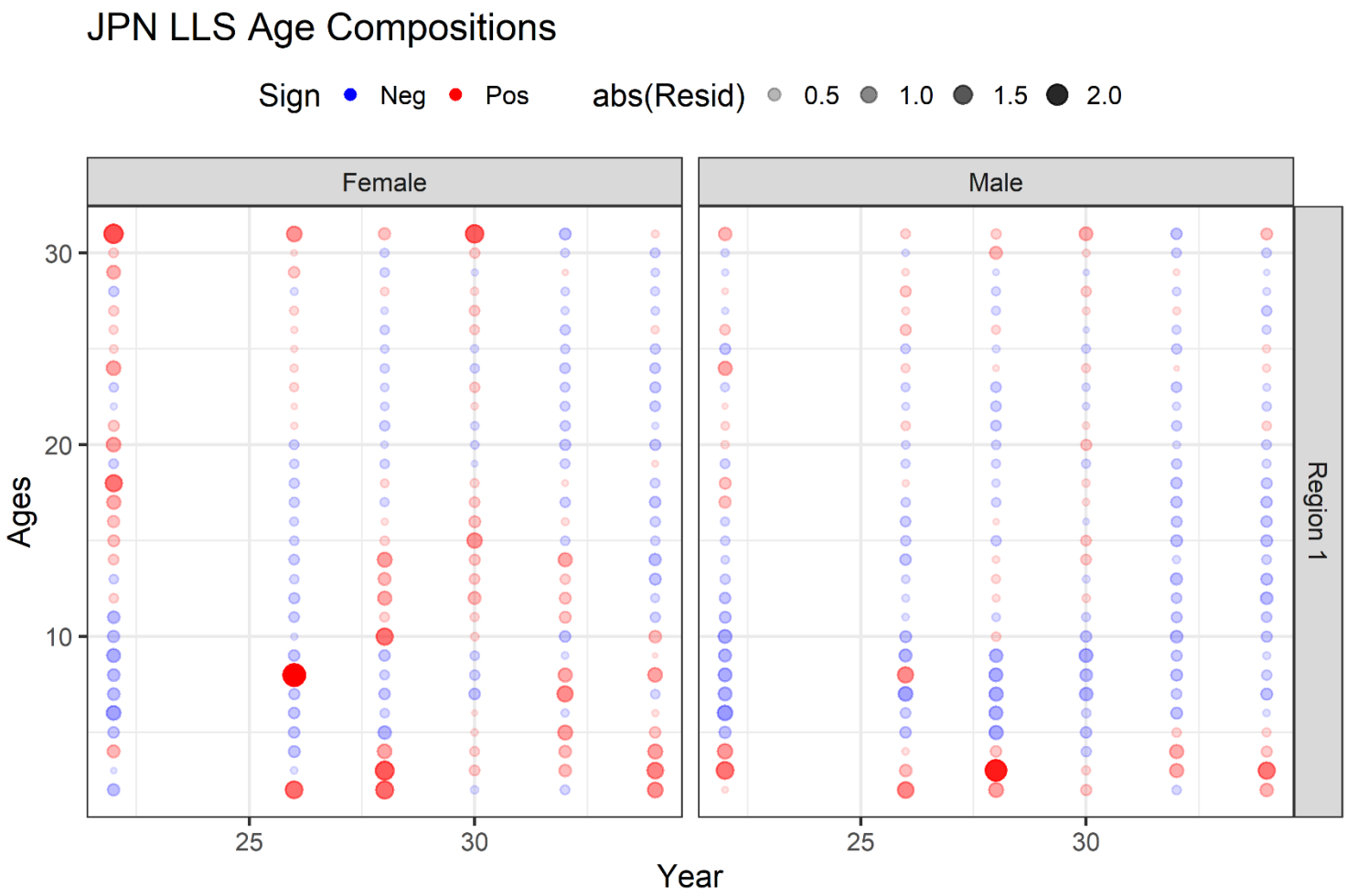

Figure SM3B.38. Fits to the Japanese Longline Survey age composition data based on Pearson residuals for model *26.1_1Reg*.

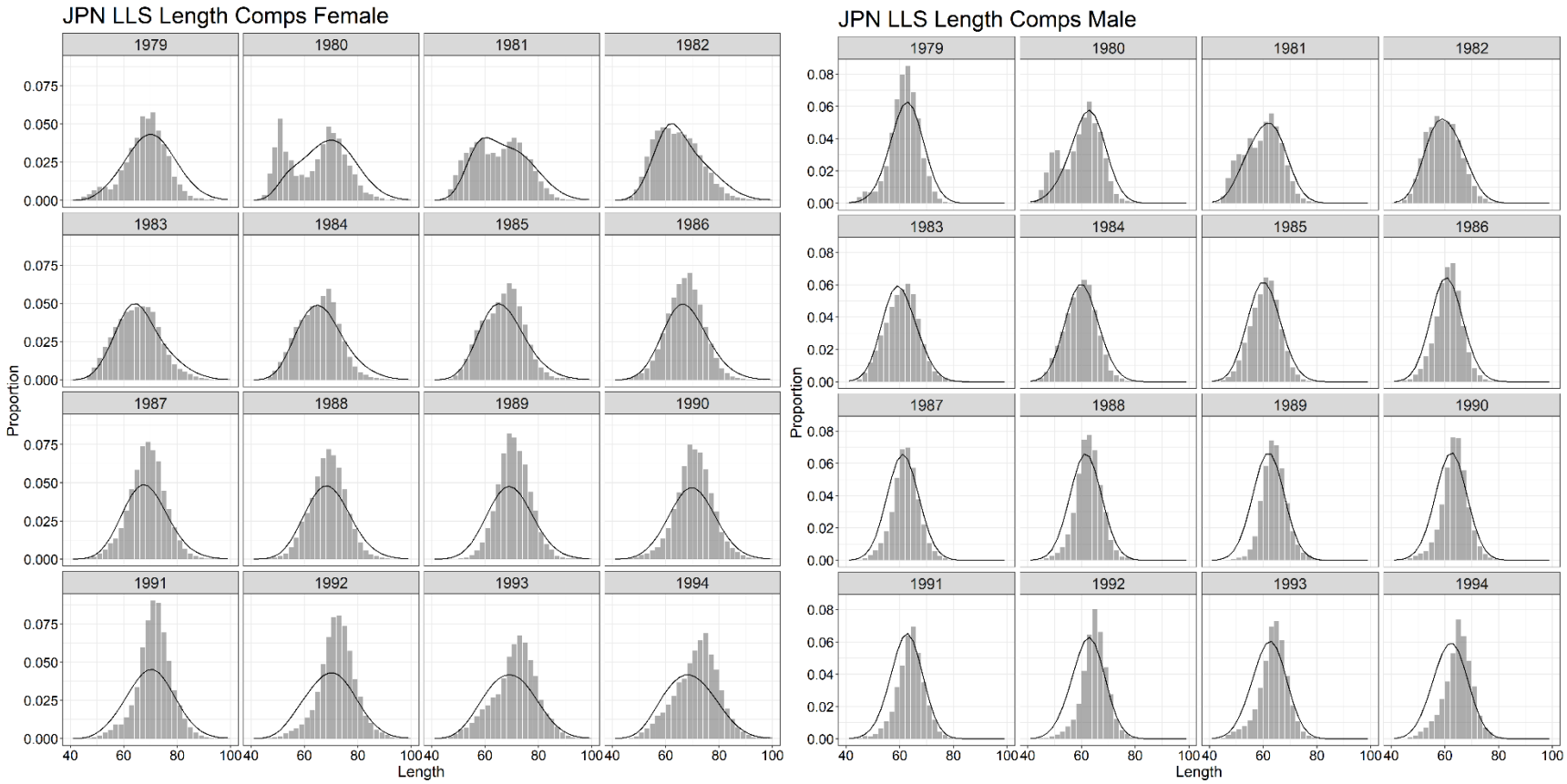

Figure SM3B.39. Fits to Japanese Longline Survey length composition data by year and sex (females left and males right) for Model *25.12_1Reg_Cont*.

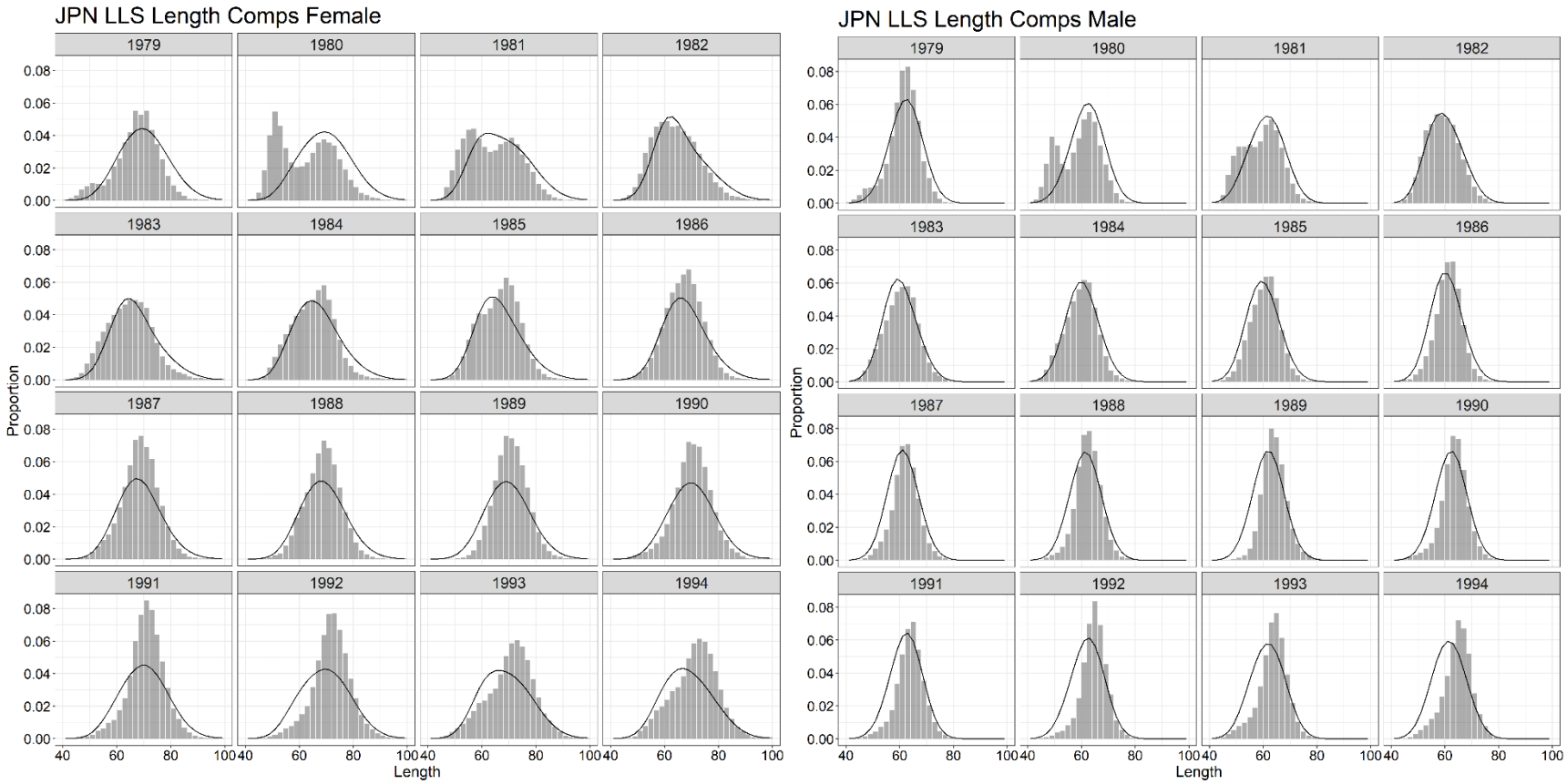

Figure SM3B.40. Fits to Japanese Longline Survey length composition data by year and sex (females left and males right) for Model *26.1_1Reg*.

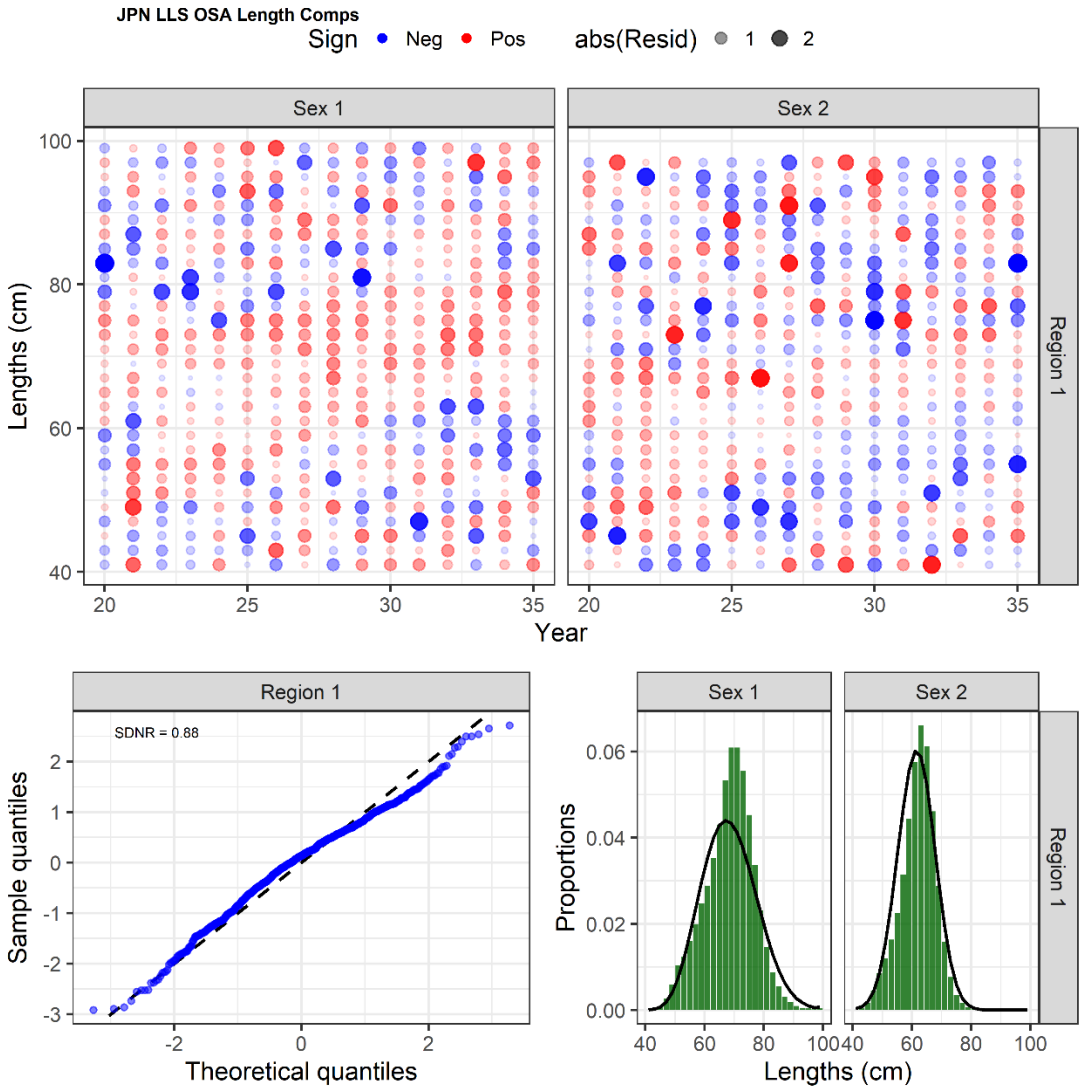

Figure SM3B.41. Fits to Japanese longline survey length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *25.12_1Reg_Cont*.

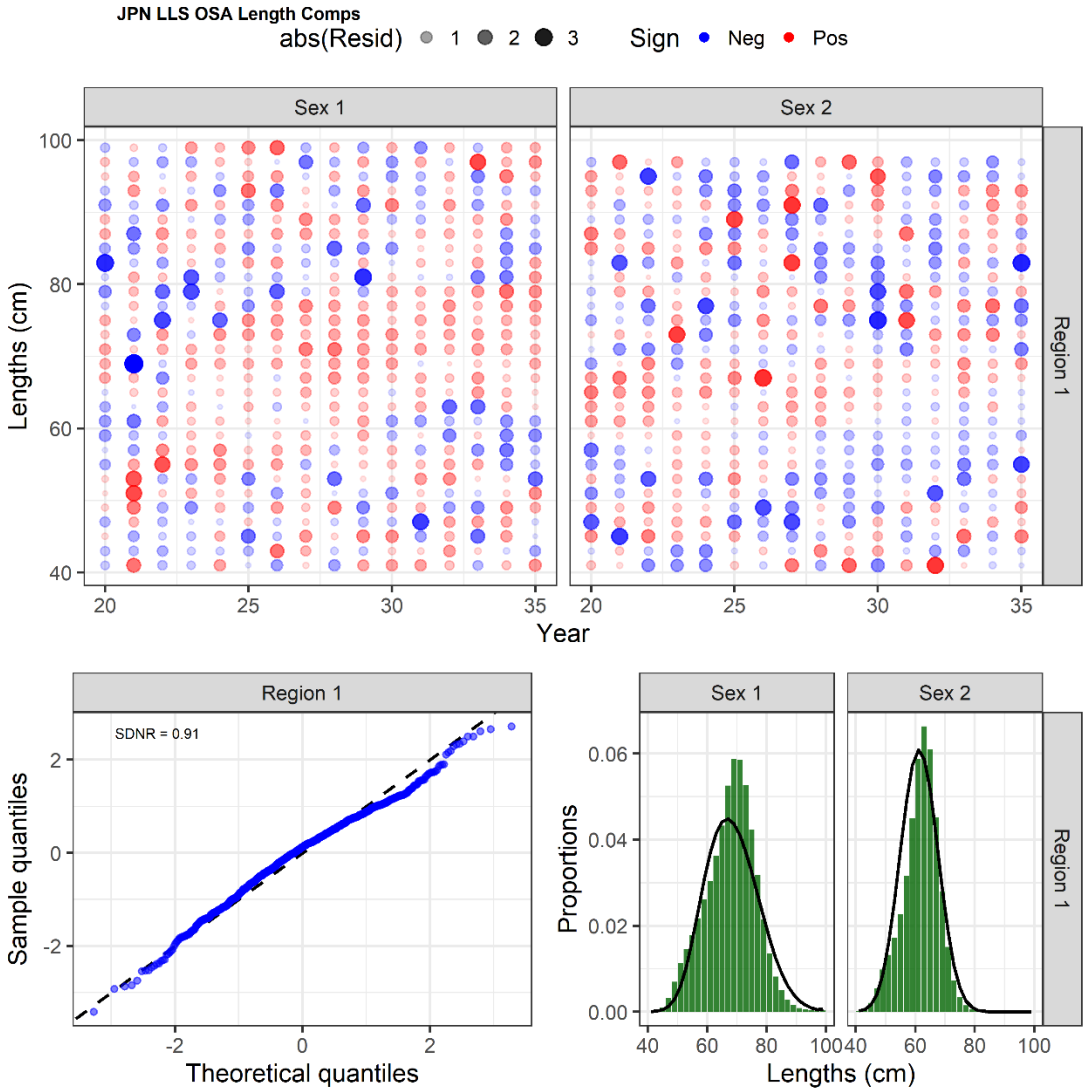

Figure SM3B.42. Fits to Japanese longline survey length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg*.

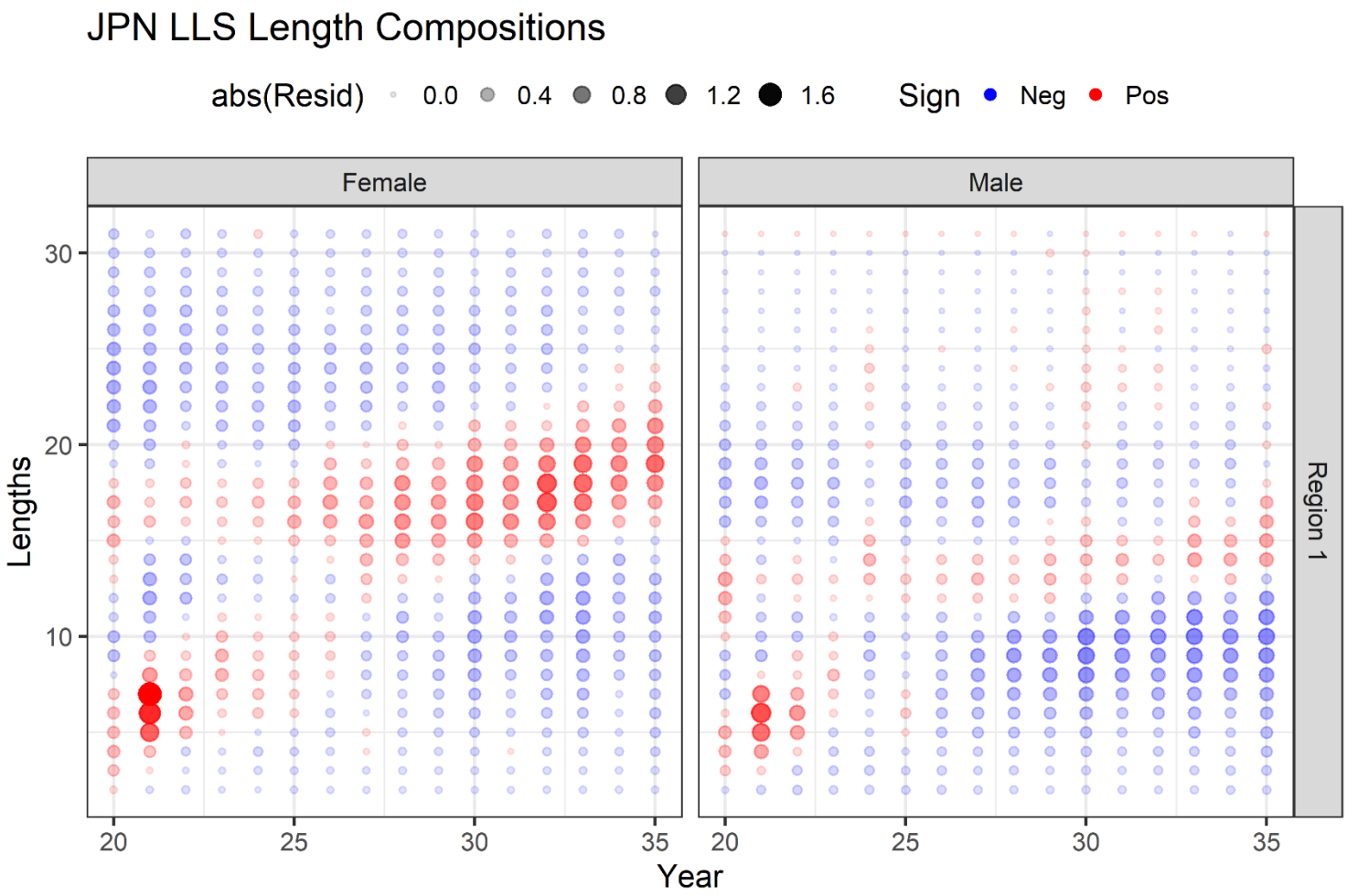

Figure SM3B.43. Fits to the Japanese Longline Survey length composition data based on Pearson residuals for model *25.12_1Reg_Cont*.

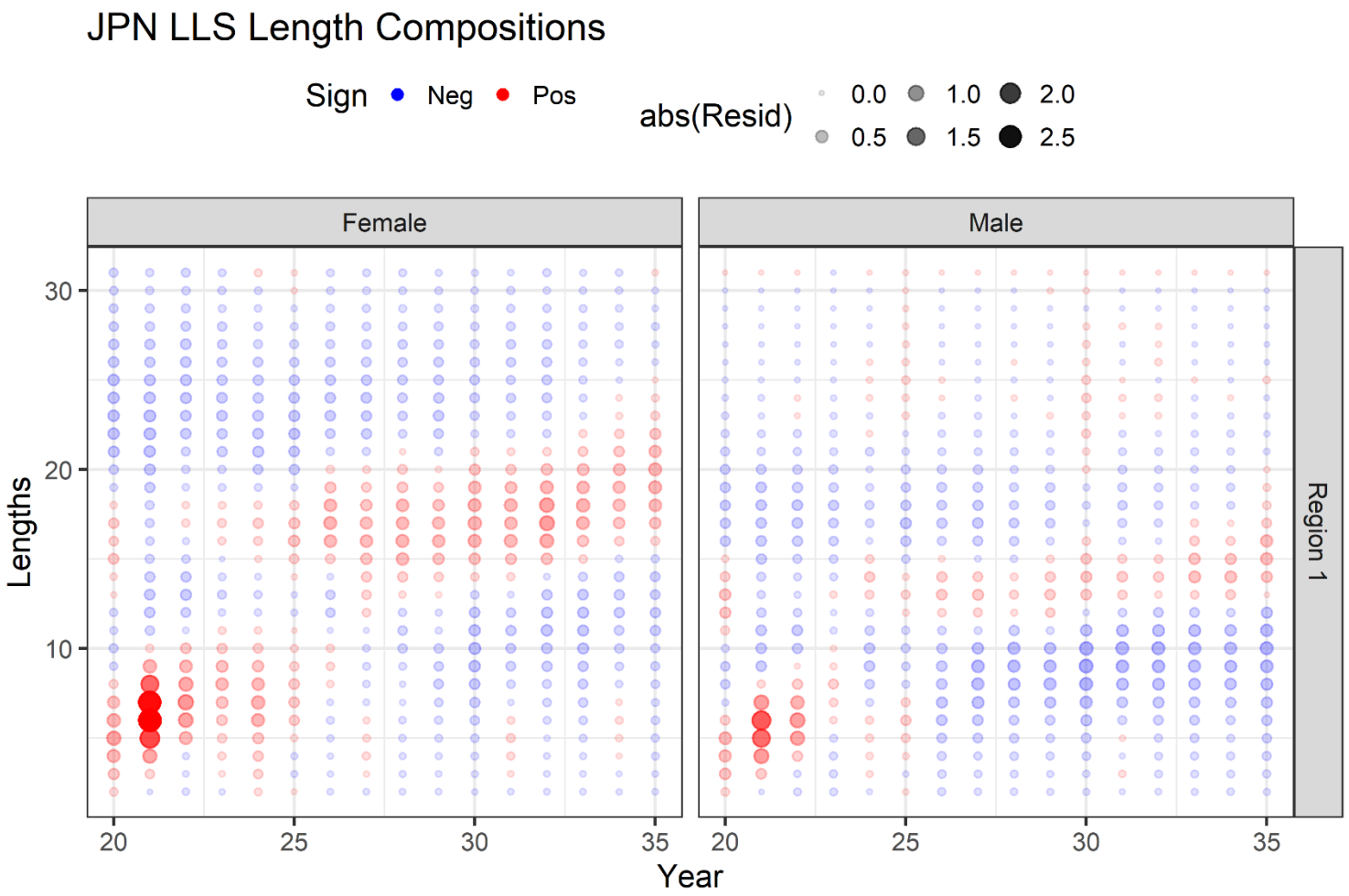

Figure SM3B.44. Fits to the Japanese Longline Survey length composition data based on Pearson residuals for Model *26.1_1Reg*.

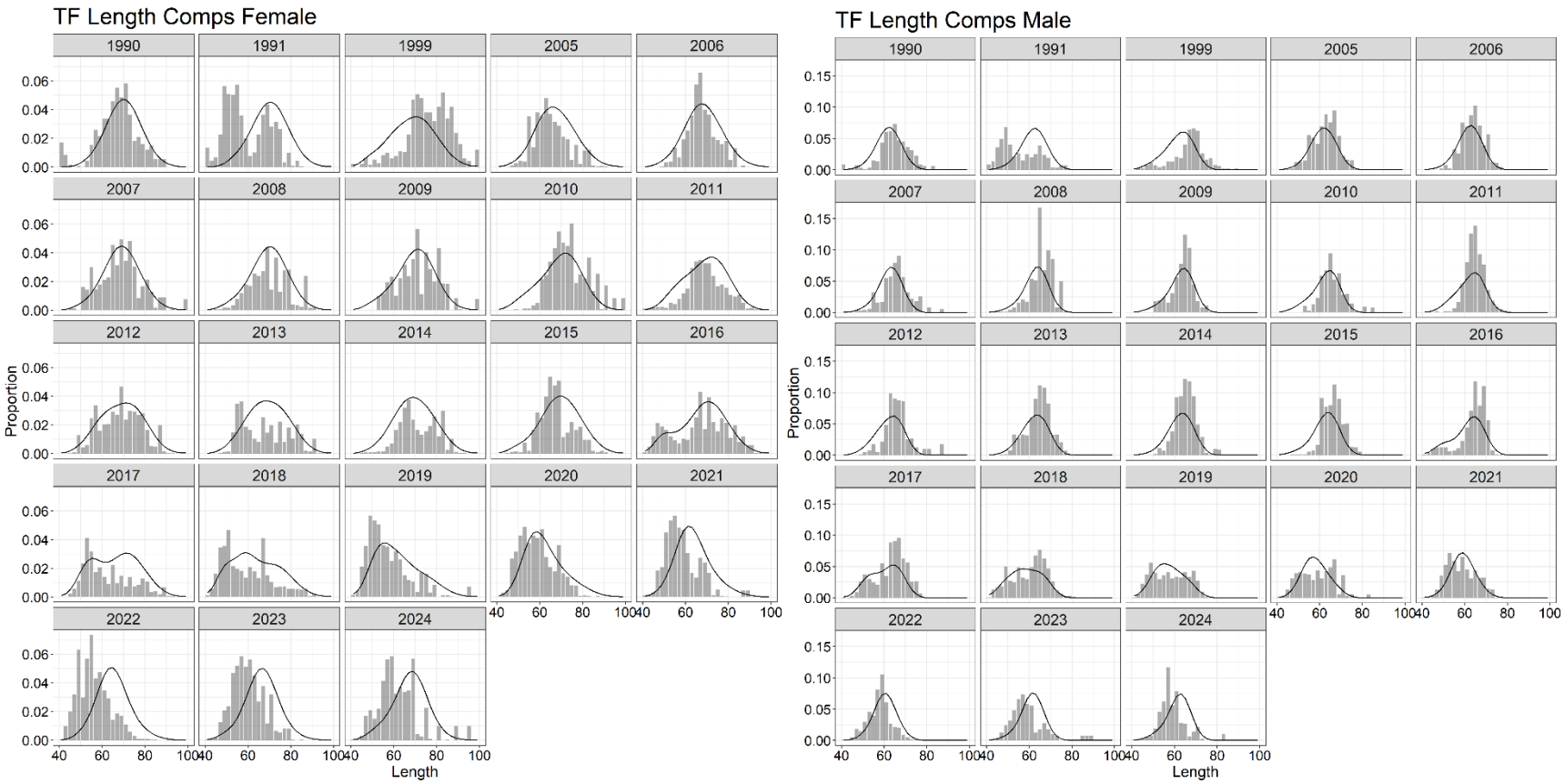

Figure SM3B.45. Fits to the trawl fishery length composition data by year and sex (females left and males right) for Model *25.12_1Reg_Cont*.

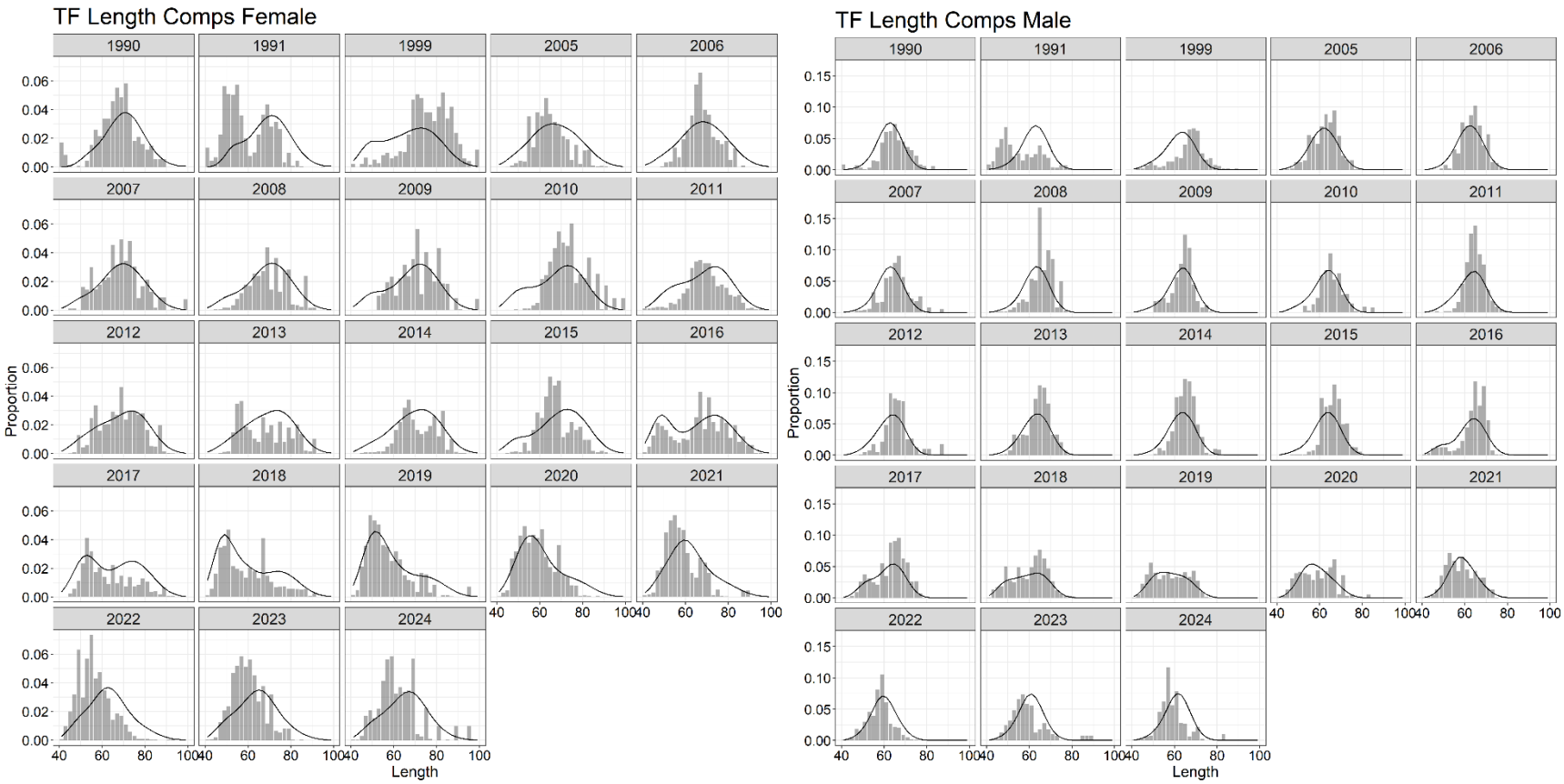

Figure SM3B.46. Fits to the trawl fishery length composition data by year and sex (females left and males right) for Model *26.1_1Reg*.

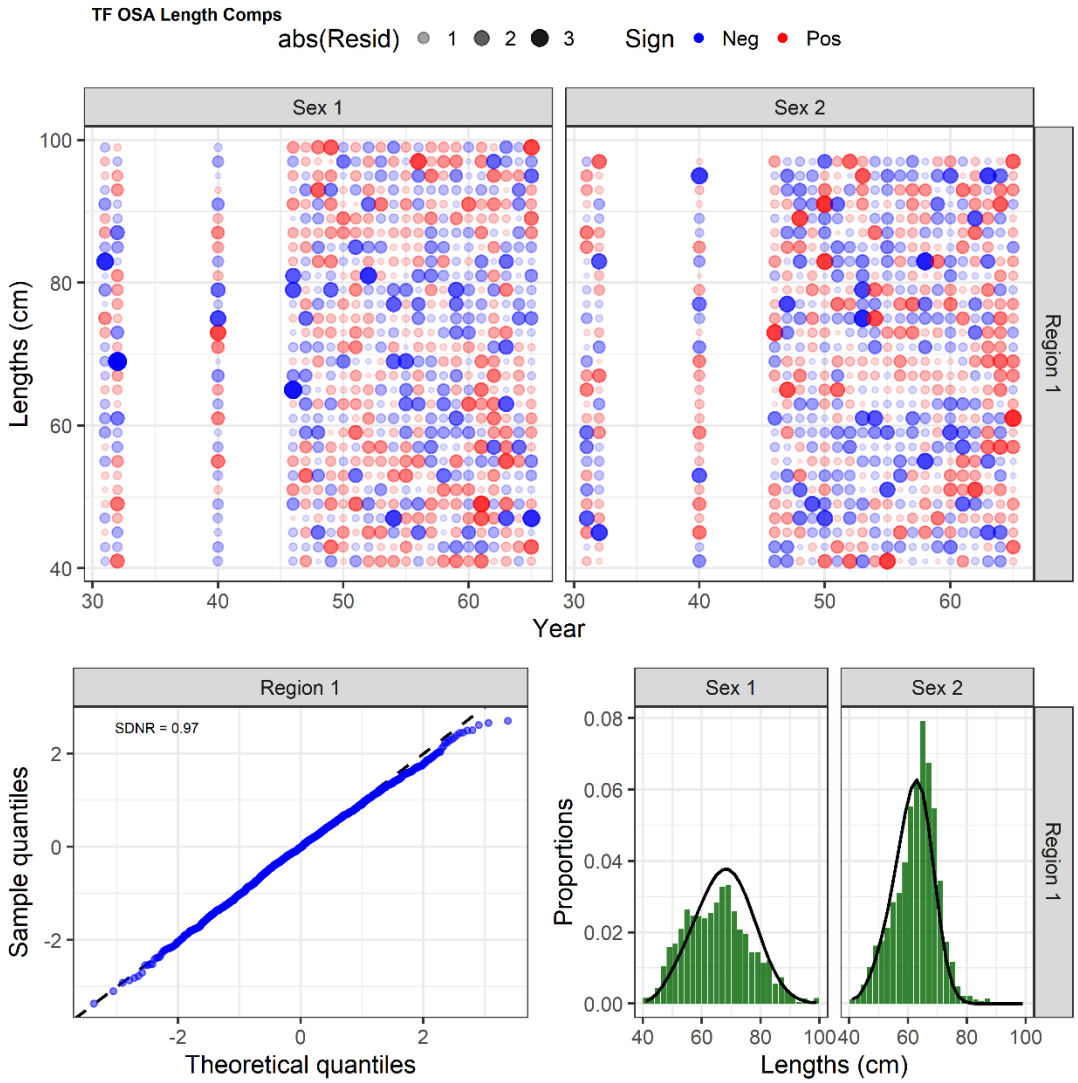

Figure SM3B.47. Fits to trawl fishery length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *25.12_1Reg_Cont*.

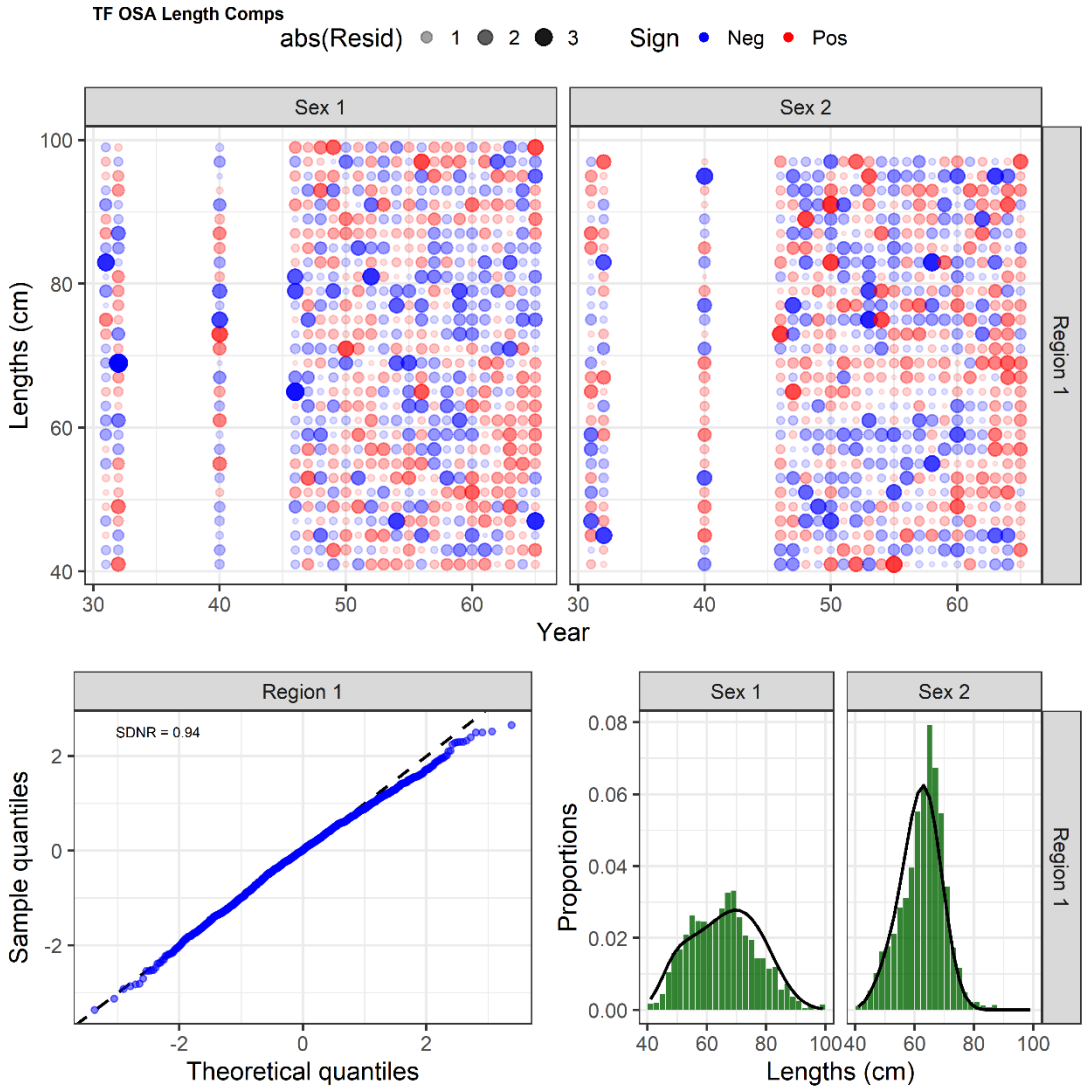

Figure SM3B.48. Fits to trawl fishery length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg*.

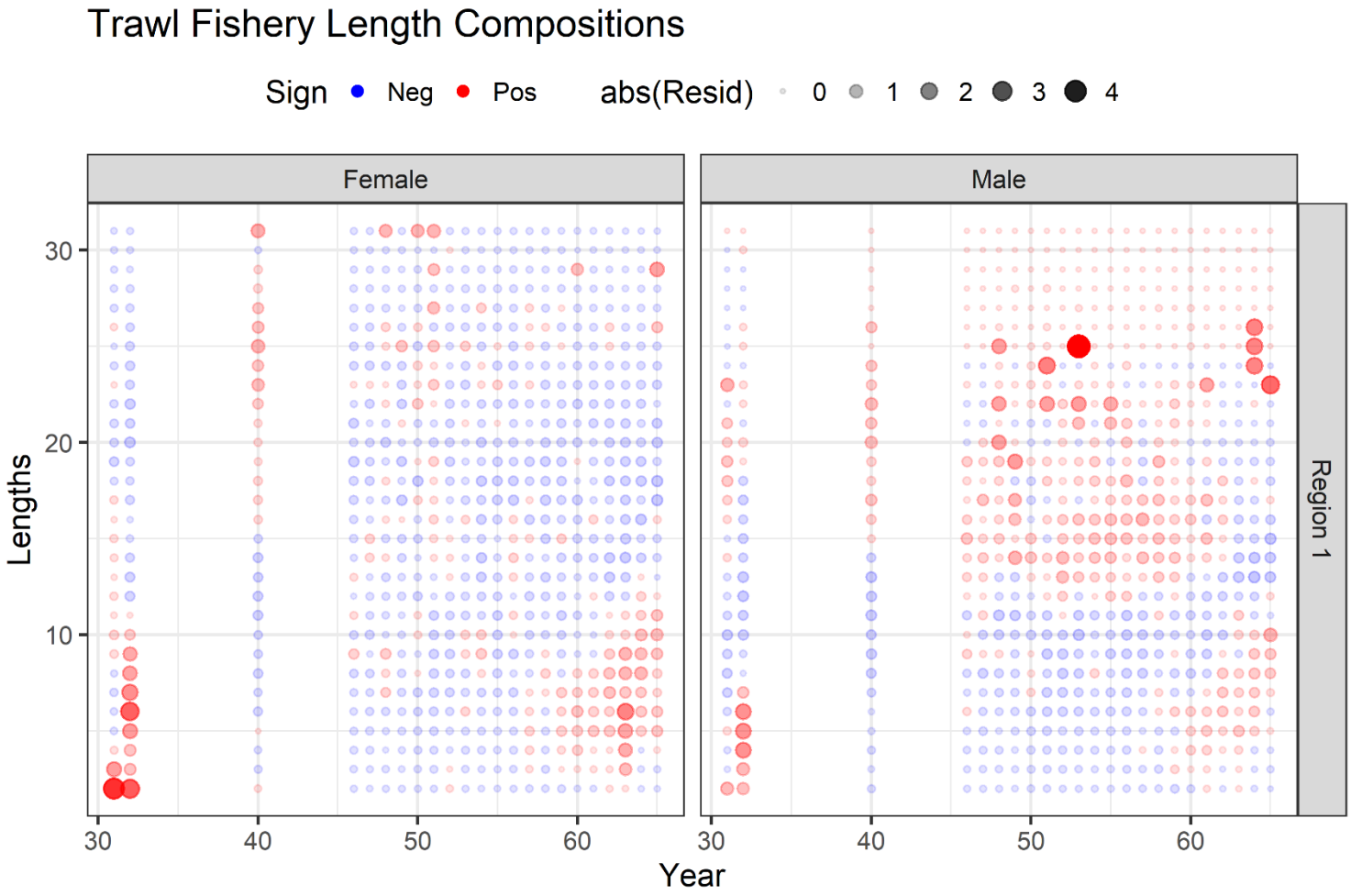

Figure SM3B.49. Fits to the trawl fishery length composition data based on Pearson residuals for model *25.12_1Reg_Cont*.

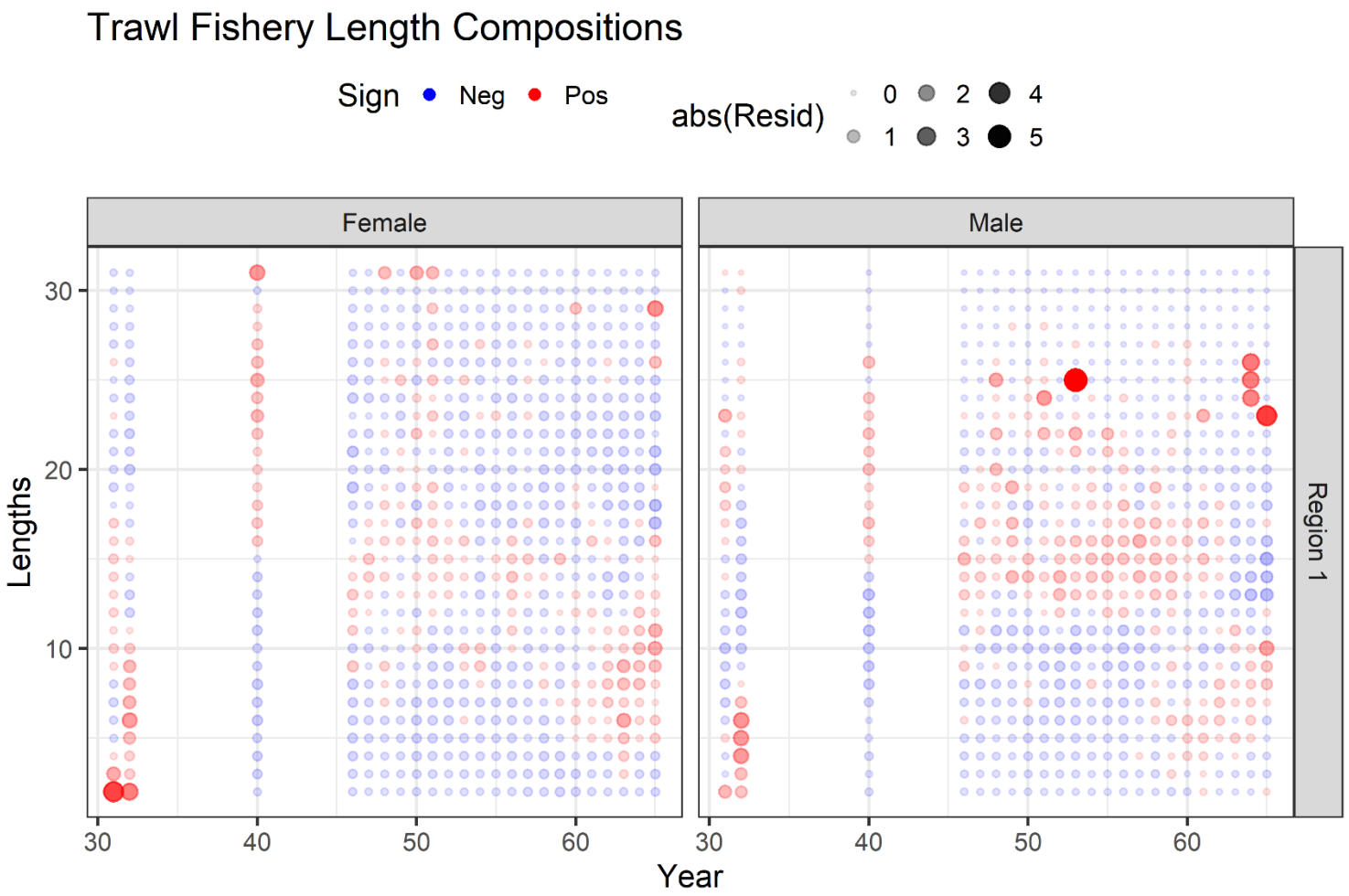

Figure SM3B.50. Fits to the trawl fishery length composition data based on Pearson residuals for Model *26.1_1Reg*.

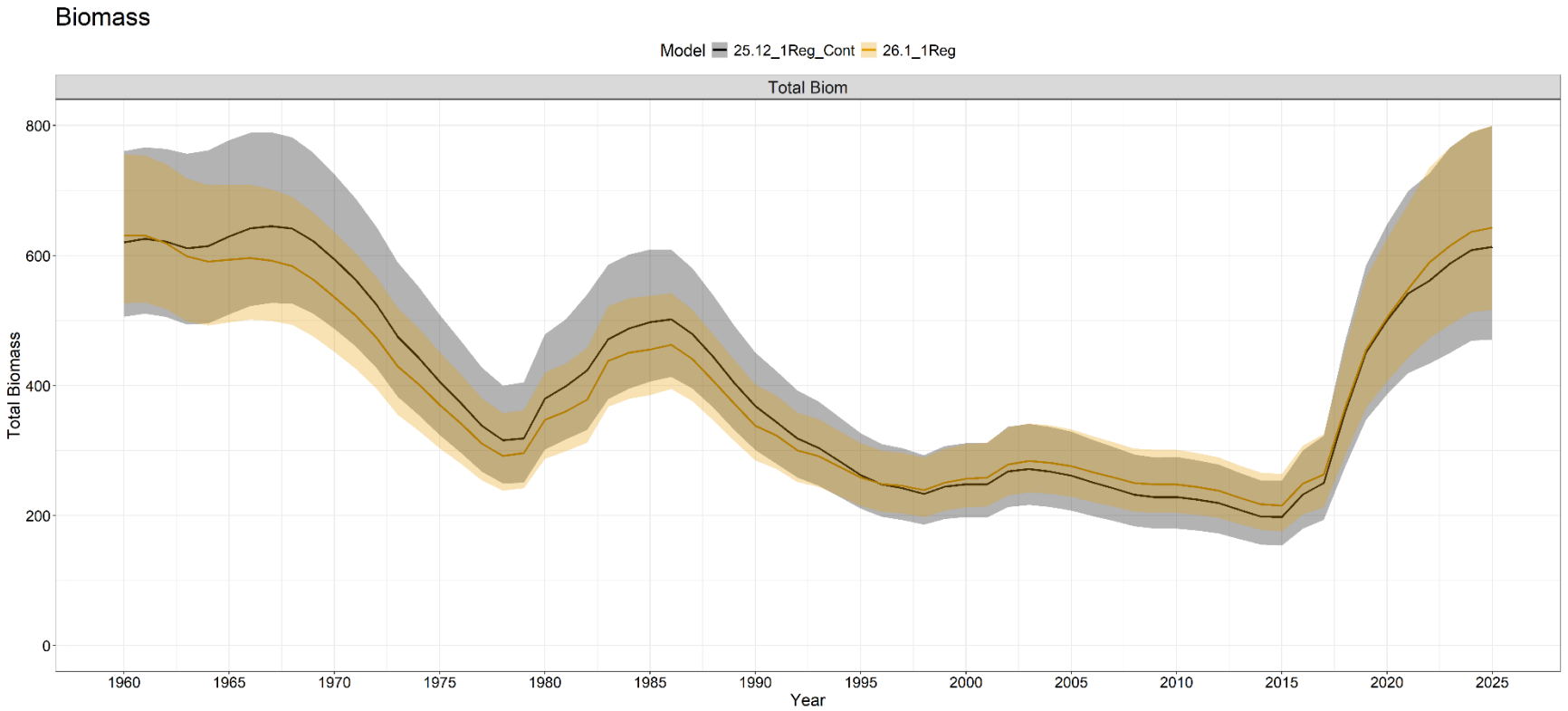

Figure SM3B.51. Time series estimates of total biomass (kt) for model *25.12_1Reg_Cont* and Model *26.1_1Reg*.

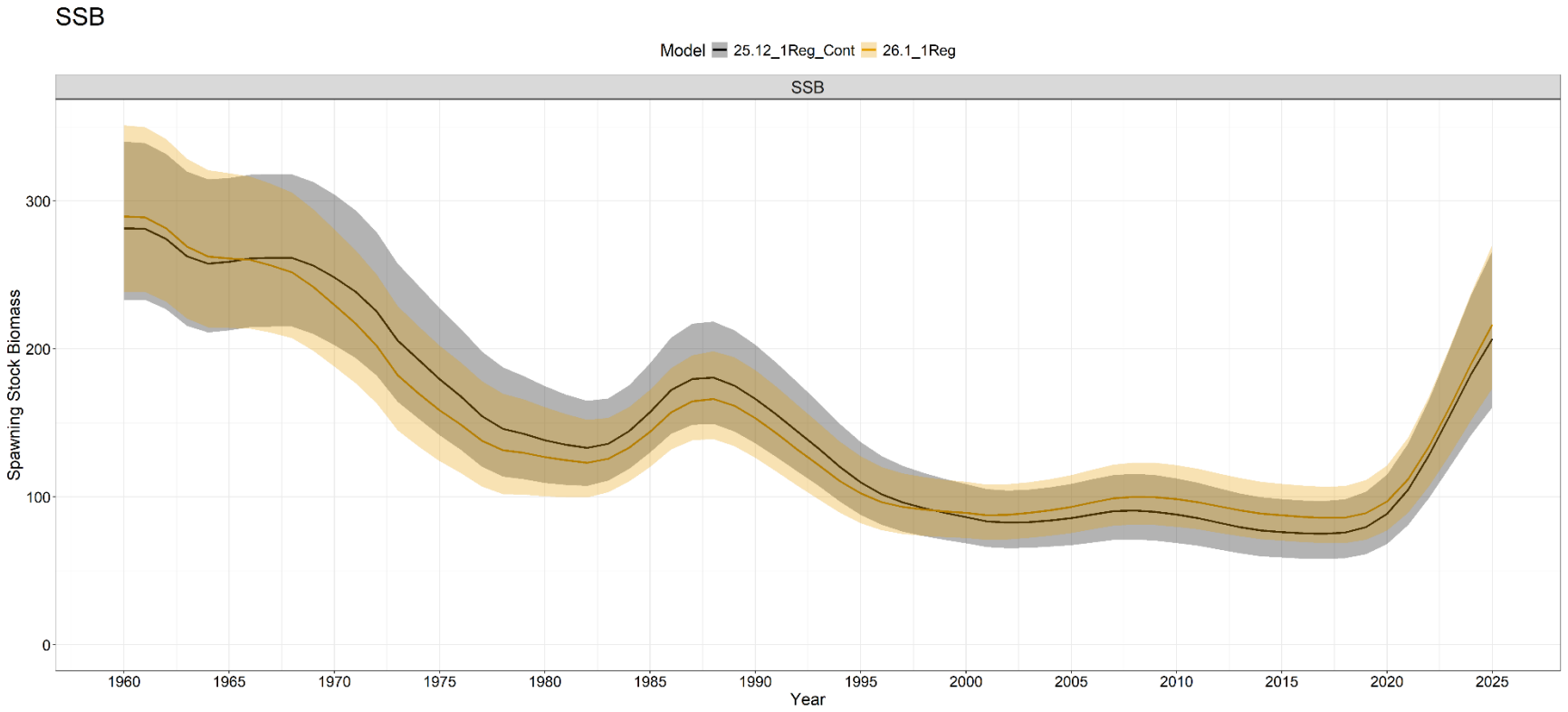

Figure SM3B.52. Time series estimates of spawning stock biomass (kt) for model *25.12_1Reg_Cont* and Model *26.1_1Reg*.

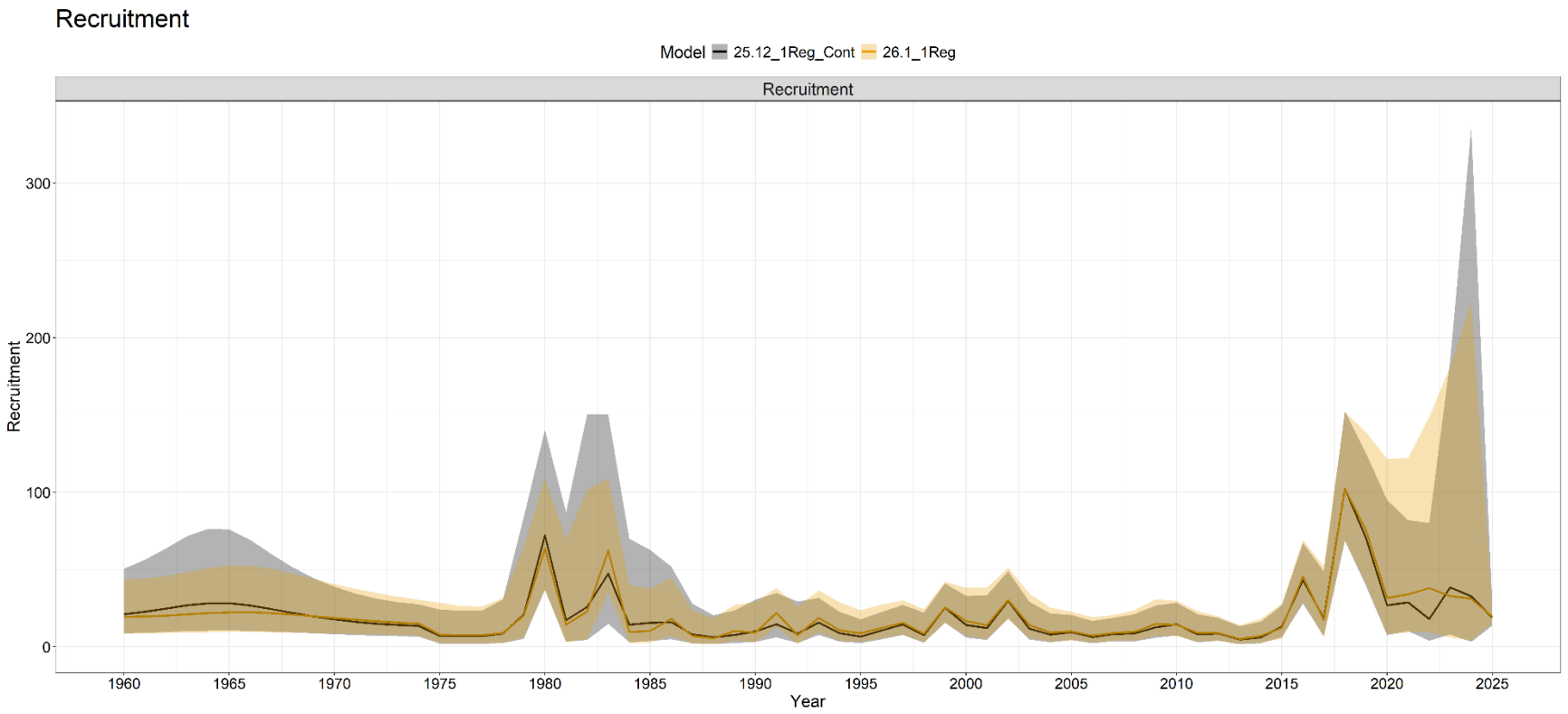

Figure SM3B.53. Time series estimates of recruitment (millions of fish) for model *25.12_1Reg_Cont* and Model *26.1_1Reg*.

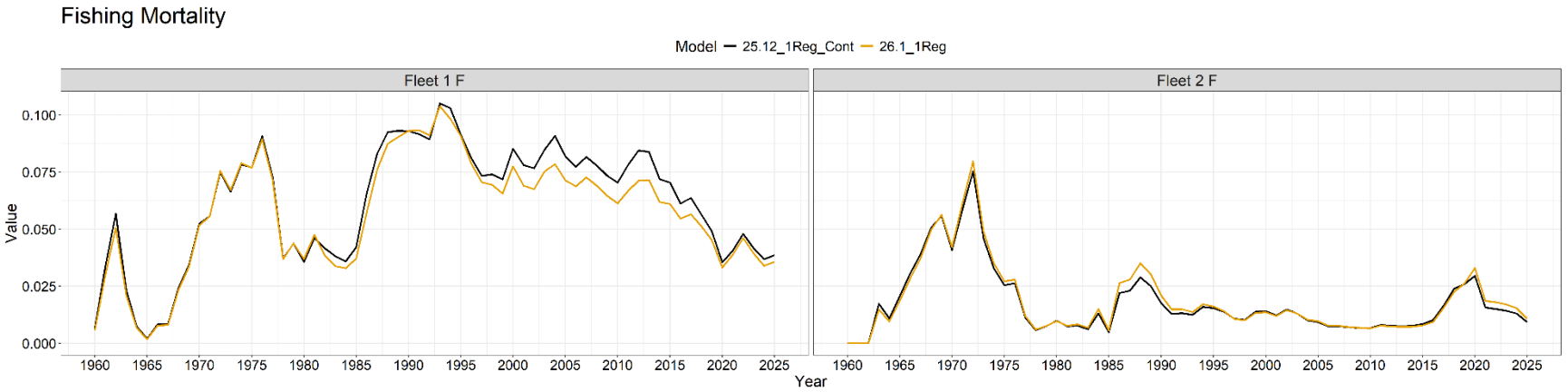

Figure SM3B.54. Time series estimates of fishing mortality (yr^{-1}) for model *25.12_1Reg_Cont* and Model *26.1_1Reg*. Fleet 1 is the fixed gear fishery and fleet 2 is the trawl fishery.

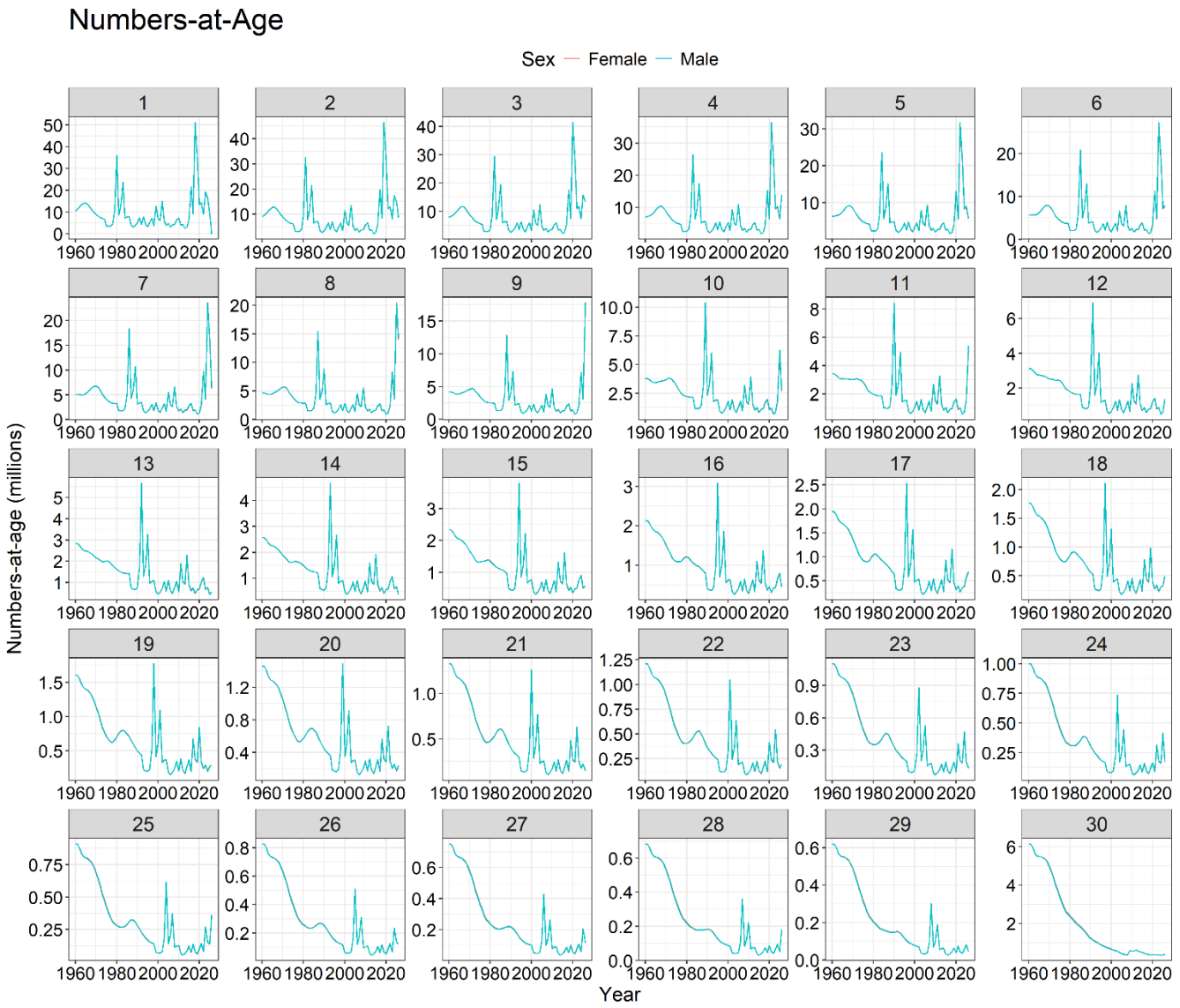

Figure SM3B.55. Time series estimates of numbers-at-age (millions of fish) by sex for model *25.12_1Reg_Cont*.

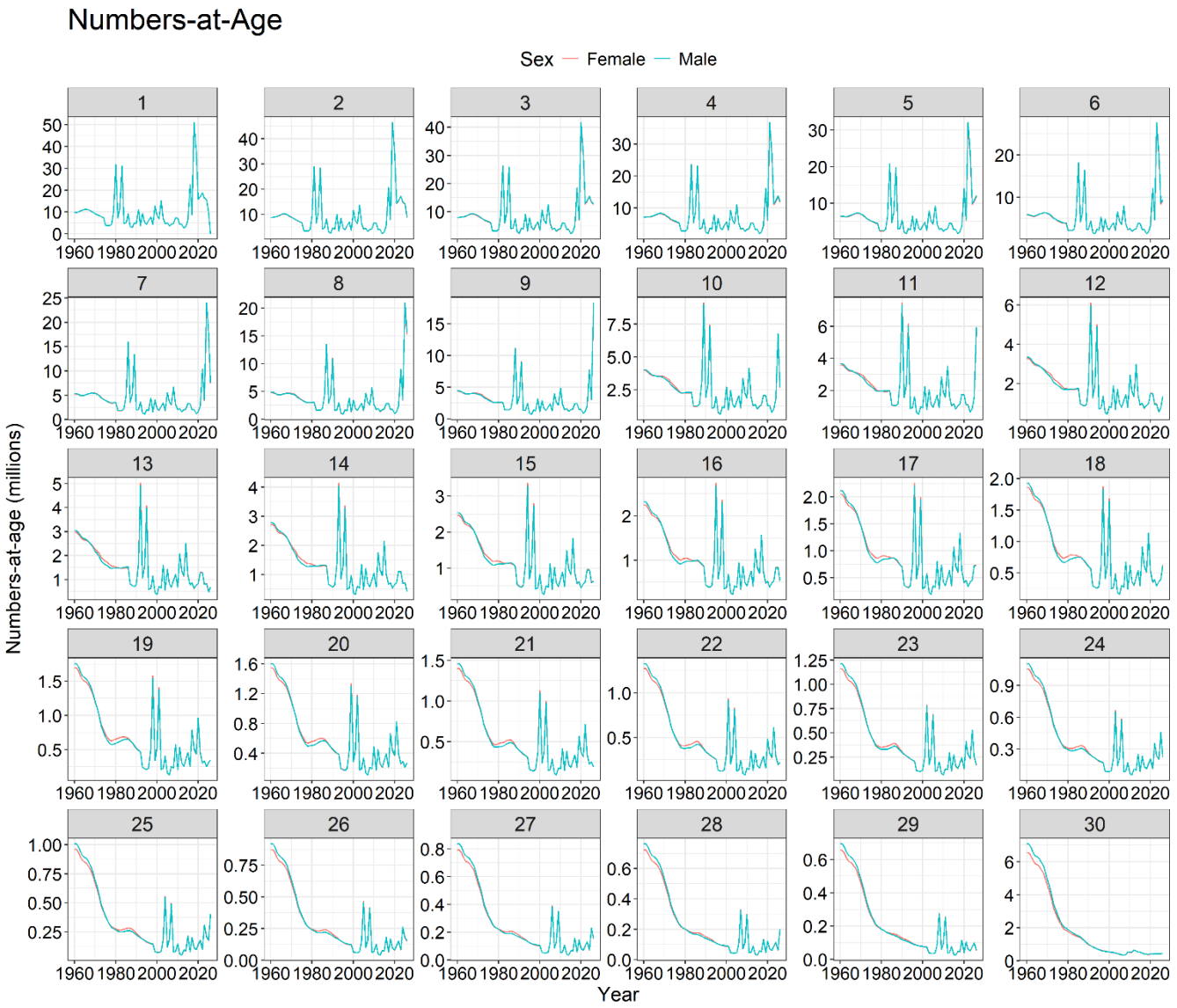

Figure SM3B.56. Time series estimates of numbers-at-age (millions of fish) by sex for model *25.12_1Reg_Cont*.

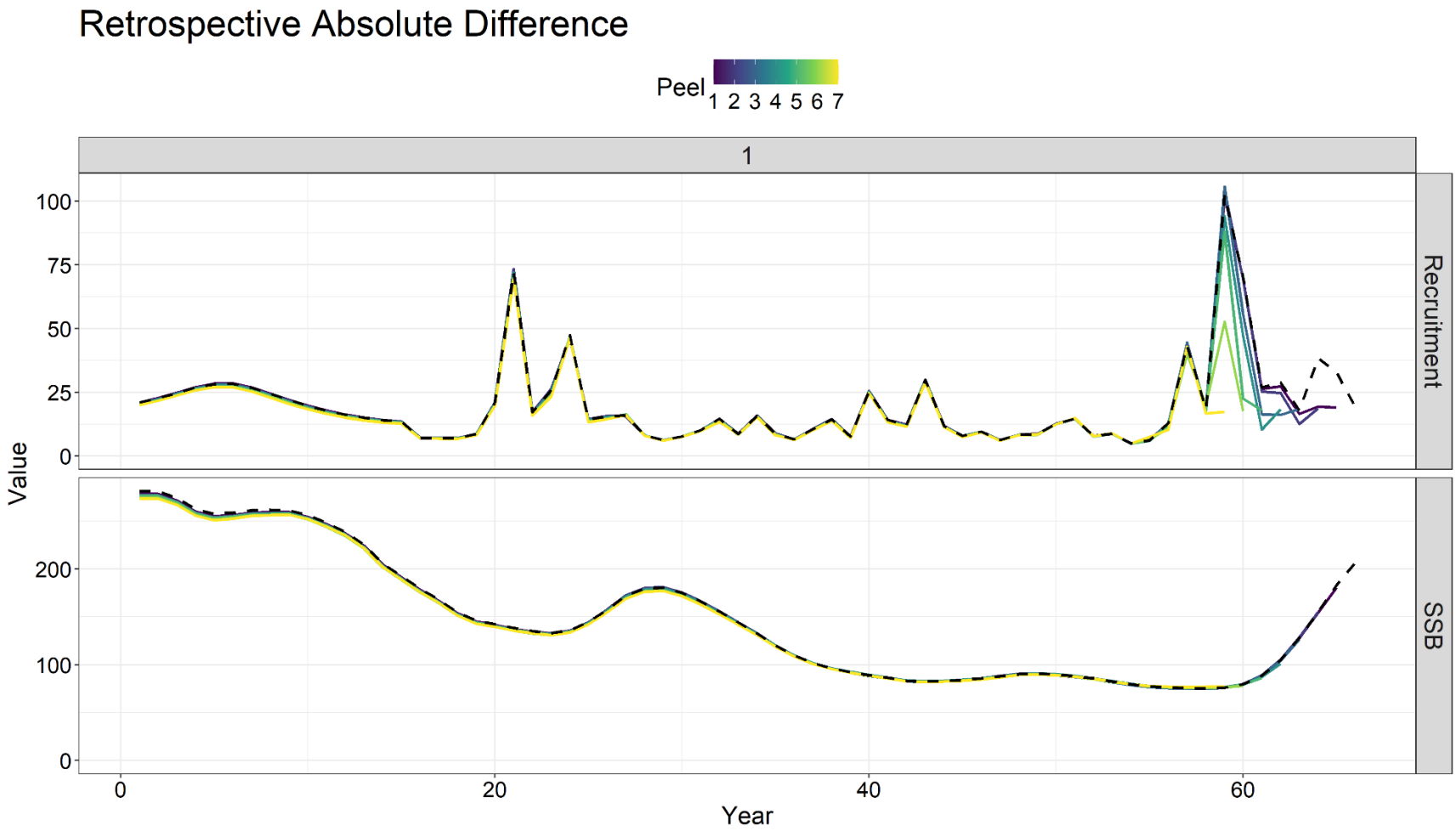

Figure SM3B.57. Comparison of recruitment (millions of fish; top panel) and spawning stock biomass (SSB; bottom panel) time series estimates from a seven year 'true' retrospective analysis for Model *25.12_1Reg_Cont*. Francis reweighting is undertaken for each peel to ensure consistency and to match how an assessment would be performed in a real-world situation. Mohn's rho for SSB was -0.014.

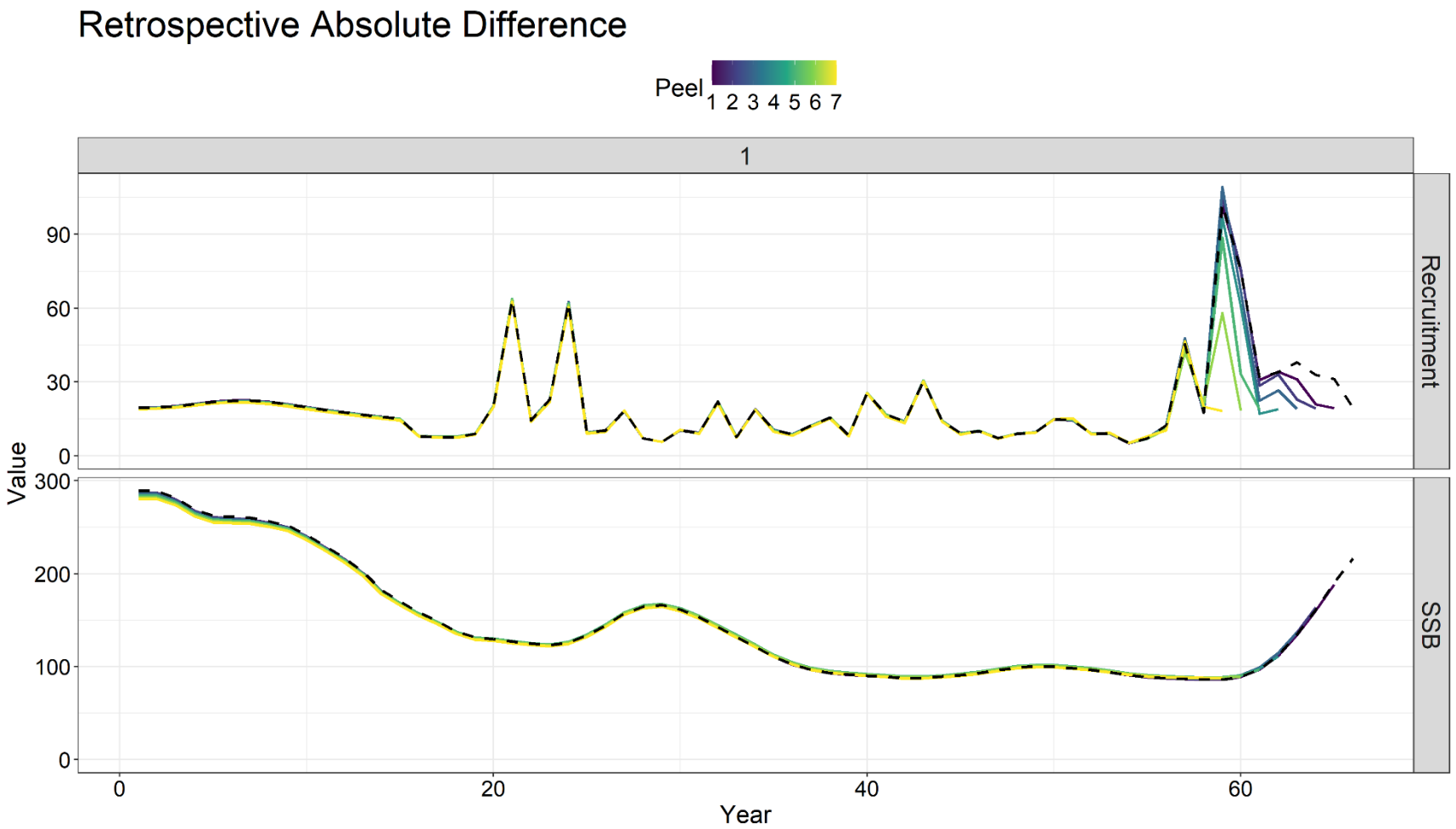

Figure SM3B.58. Comparison of recruitment (millions of fish; top panel) and spawning stock biomass (SSB; bottom panel) time series estimates from a seven year 'true' retrospective analysis for Model *26.1_1Reg*. Francis reweighting is undertaken for each peel to ensure consistency and to match how an assessment would be performed in a real-world situation. Mohn's rho for SSB was 0.0054.

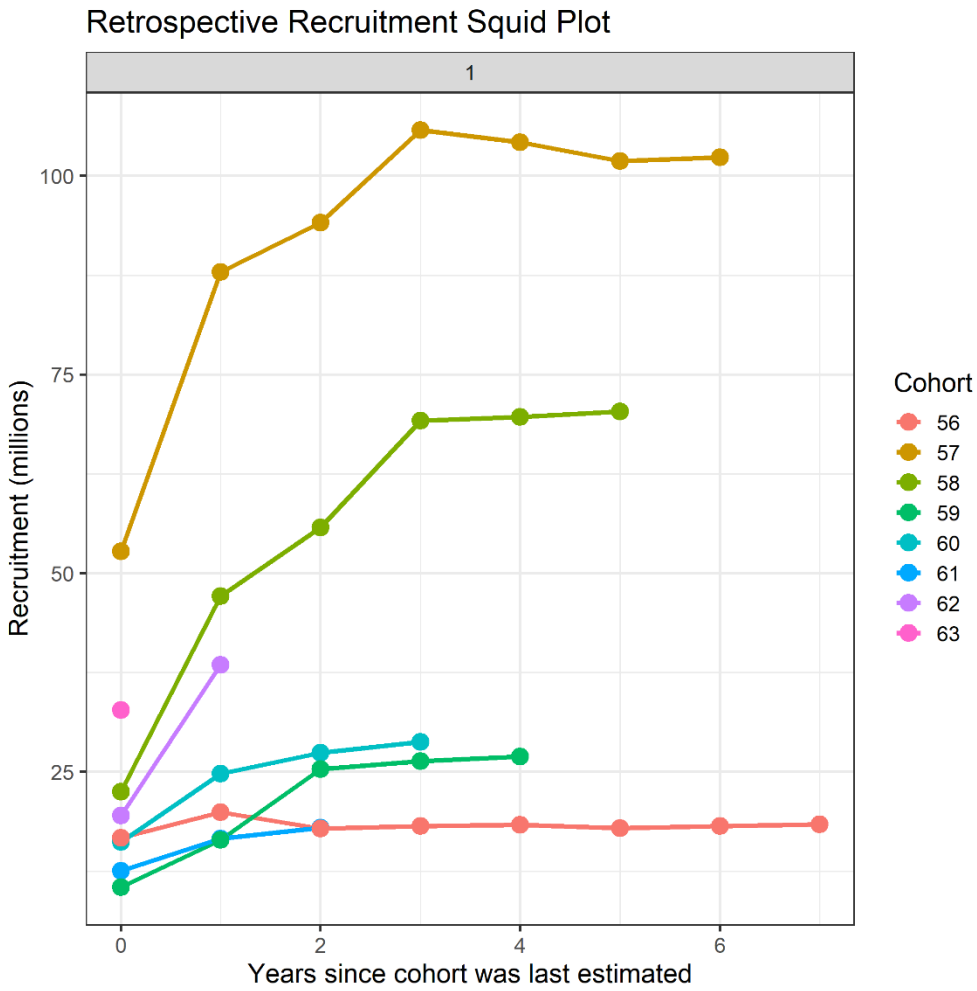

Figure SM3B.59. Comparison of a recruitment retrospective 'squid' plot representing how estimates of a given year class change as data are added in a seven year 'true' retrospective analysis for model *25.12_1Reg_Cont*. The 'true' retrospective analysis removes a year of data for each peel, but also accounts for the lag in availability of compositional data (i.e., fishery ages and lengths as well as longline survey ages become available on a one year lag from associated catch or index data). Francis reweighting is undertaken for each peel to ensure consistency and to match how an assessment would be performed in a real-world situation.

Figure SM3B.60. Comparison of a recruitment retrospective 'squid' plot representing how estimates of a given year class change as data are added in a seven year 'true' retrospective analysis for model *26.1_1Reg*. The 'true' retrospective analysis removes a year of data for each peel, but also accounts for the lag in availability of compositional data (i.e., fishery ages and lengths as well as longline survey ages become available on a one year lag from associated catch or index data). Francis reweighting is undertaken for each peel to ensure consistency and to match how an assessment would be performed in a real-world situation.

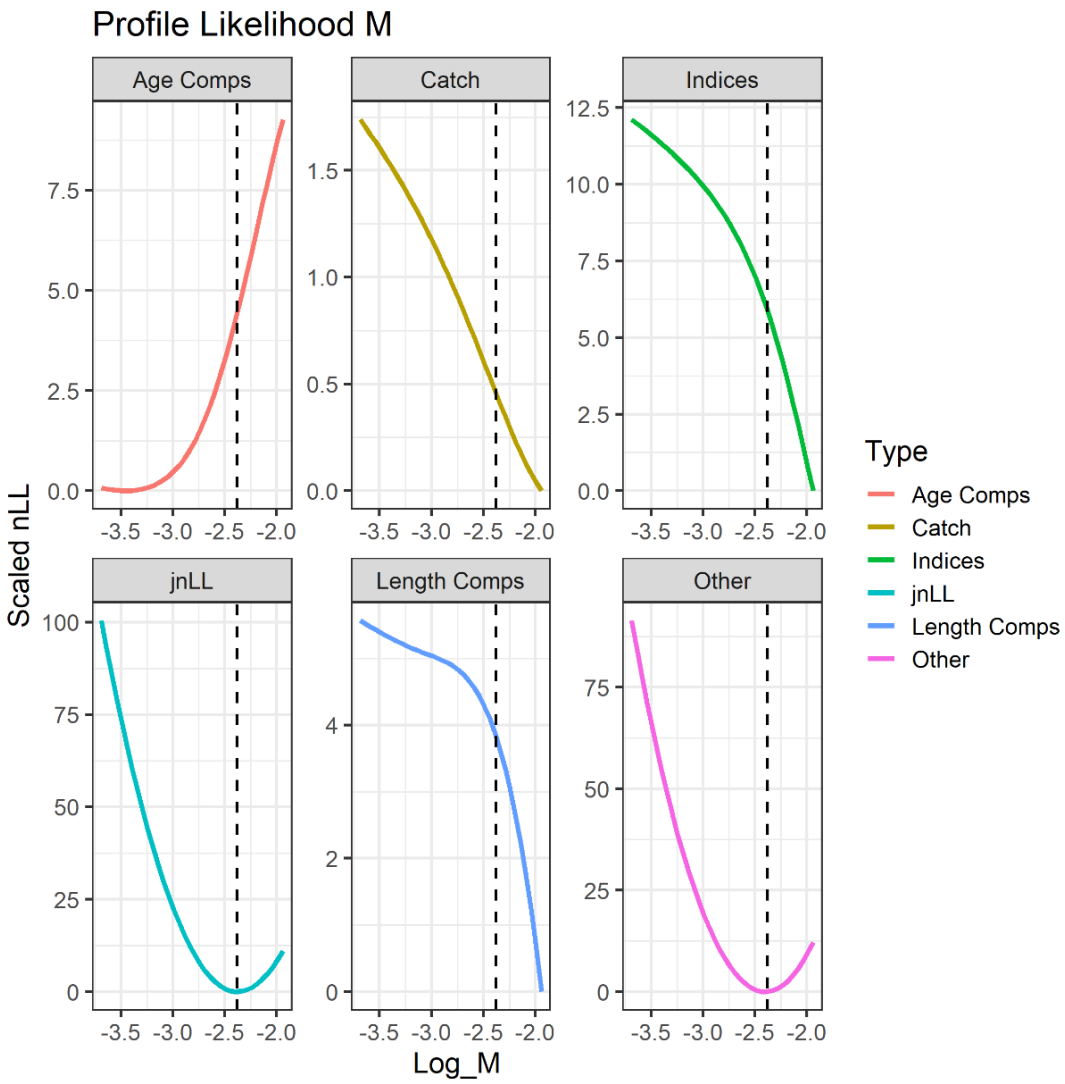

Figure SM3B.61. Likelihood profile for the natural mortality parameter (*M*), where panels represent the profile for each data type used in the model, for Model *25.12_1Reg_Cont*. The 'other' category represents the priors and penalties, which is particularly dominated by the *M* prior.

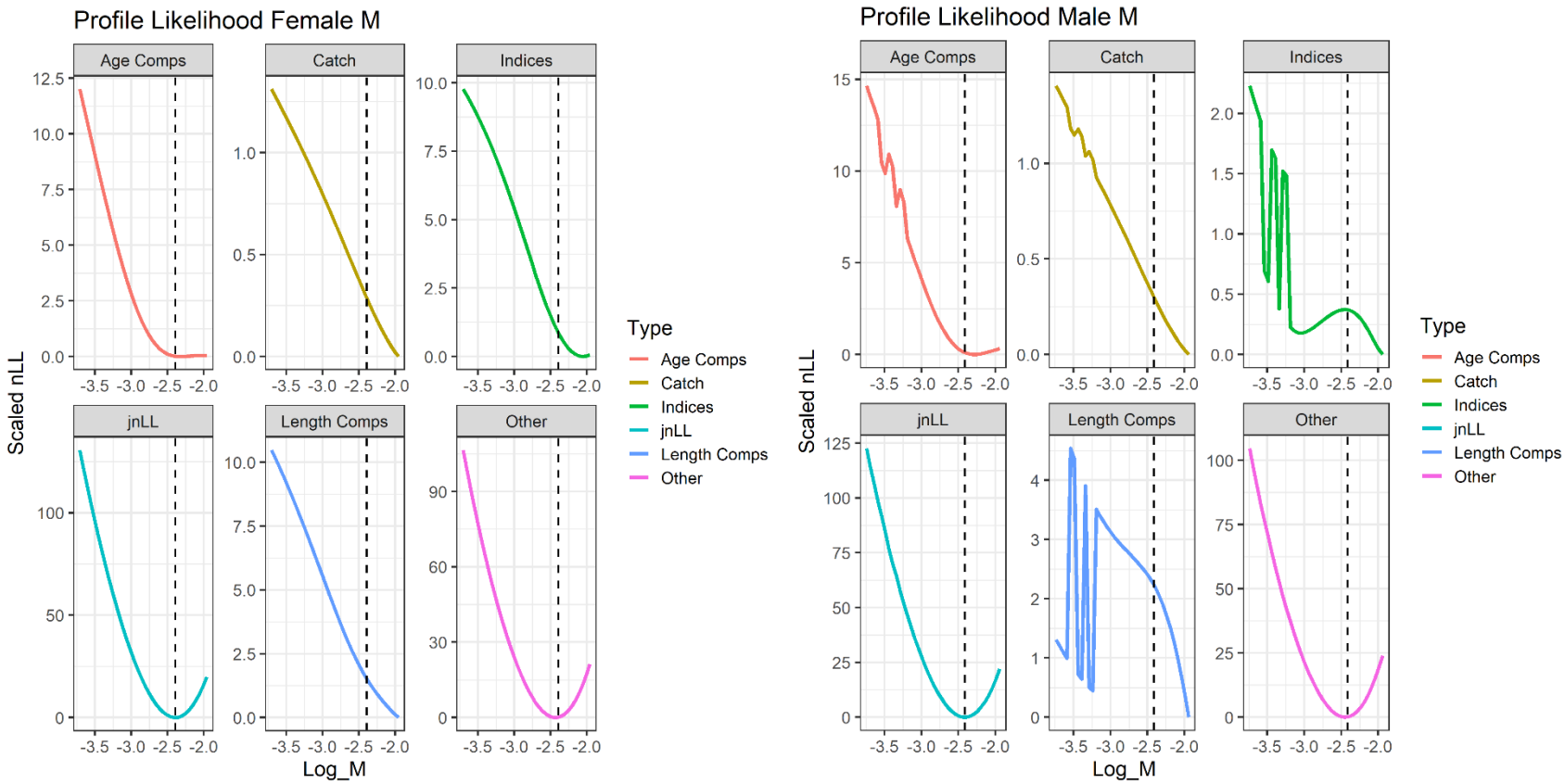

Figure SM3B.62. Likelihood profile for the natural mortality parameter (*M*) for females (left) and males (right), where panels represent the profile for each data type used in the model, for model *26.1_1Reg*. The 'other' category represents the priors and penalties, which is particularly dominated by the *M* prior.

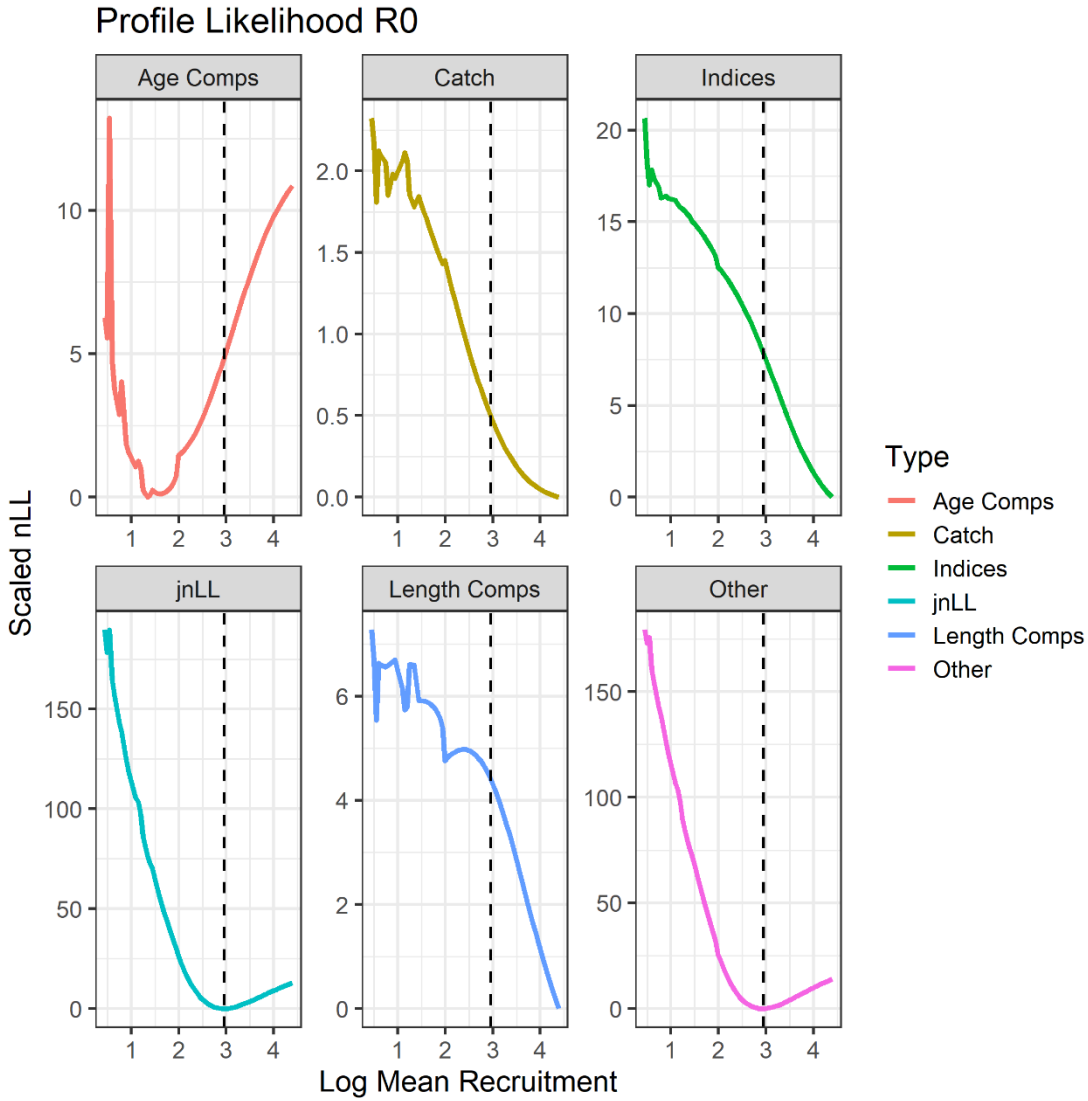

Figure SM3B.63. Likelihood profile for the mean recruitment (R_0), where panels represent the profile for each data type used in the model, for model *25.12_1Reg_Cont*.

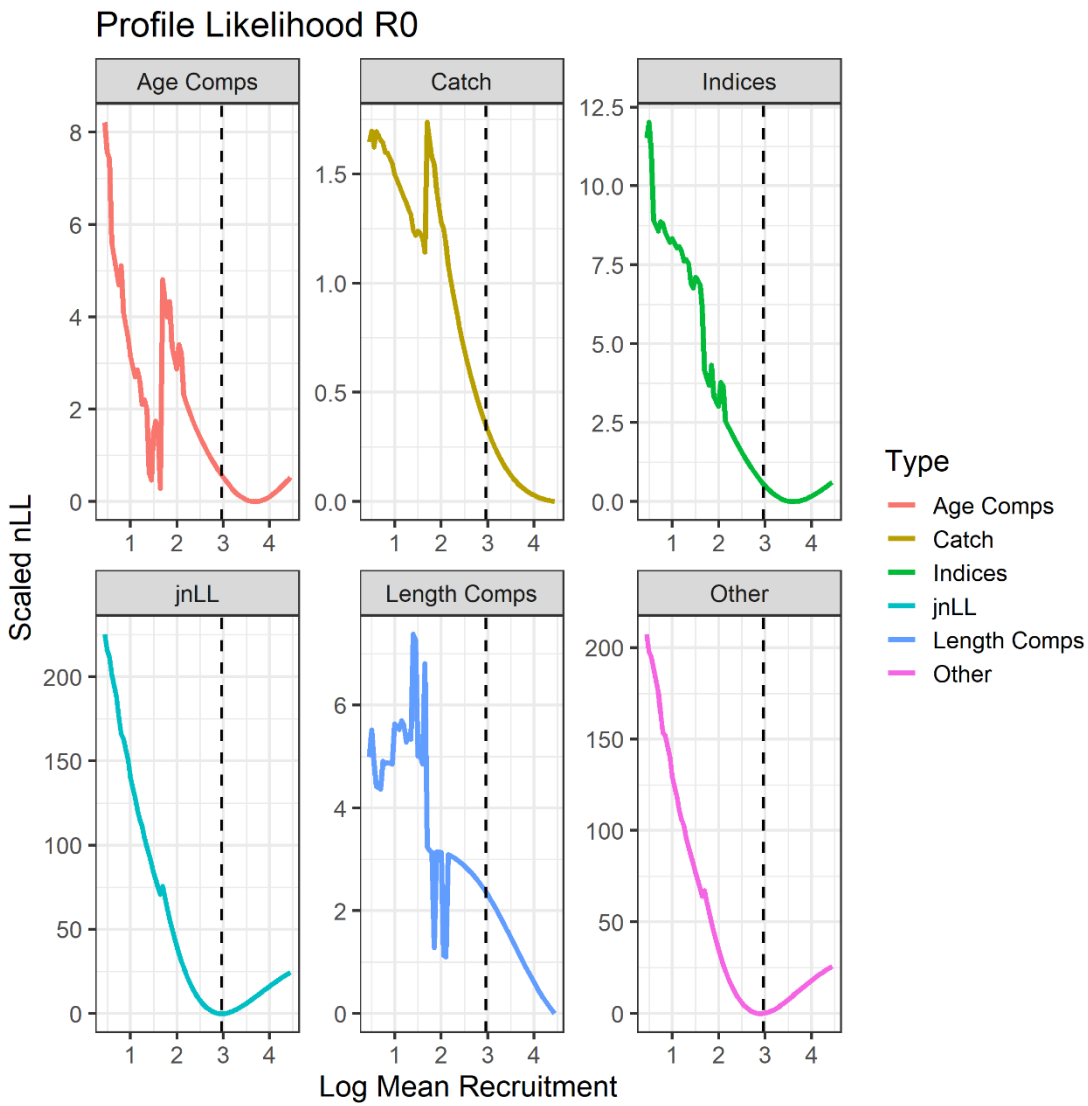

Figure SM3B.64. Likelihood profile for the mean recruitment (R_0), where panels represent the profile for each data type used in the model, for model *26.1_1Reg*.

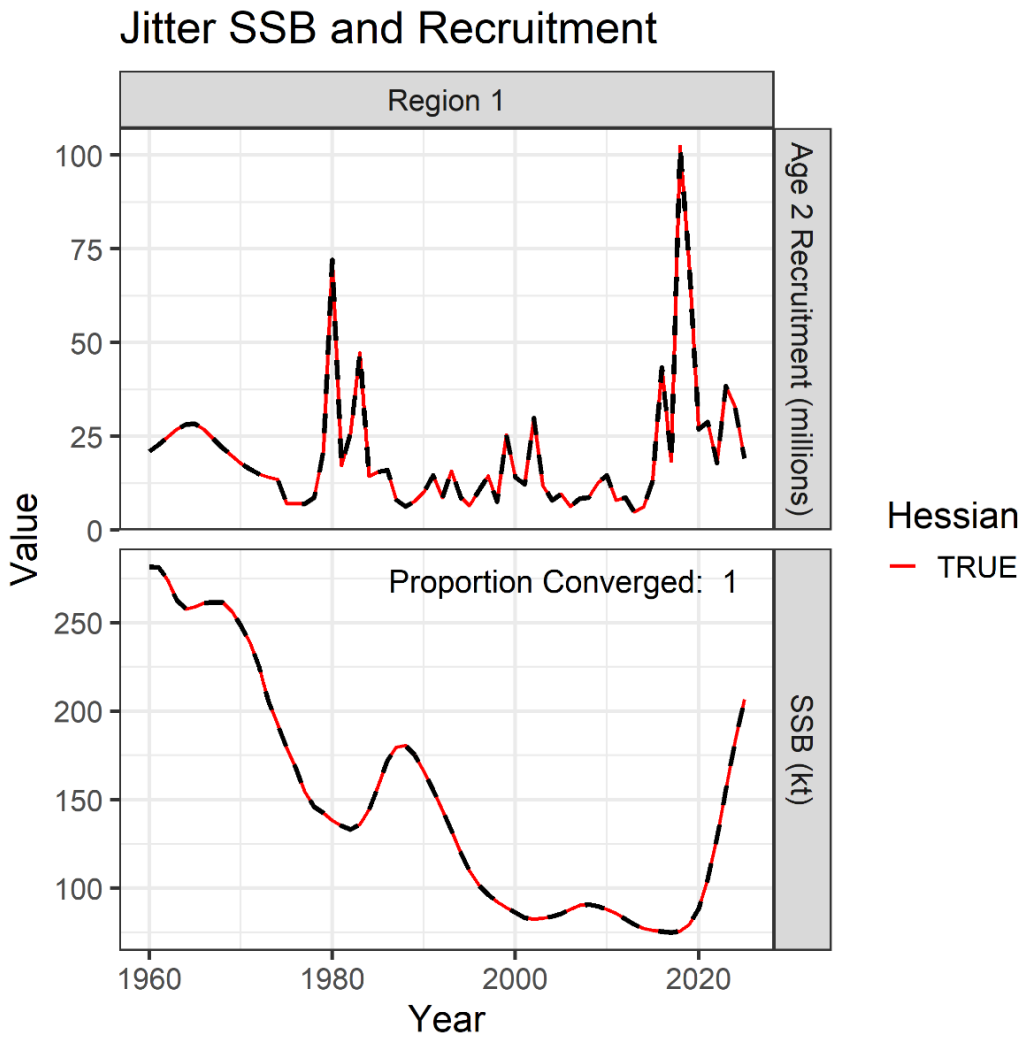

Figure SM3B.65. Results of a jitter analysis showing recruitment (top panel) and SSB (bottom panel) for each jitter iteration for model *25.12_1Reg_Cont* (left panel). The starting parameter values were randomly jittered assuming ~$N(0,0.1^2)$.

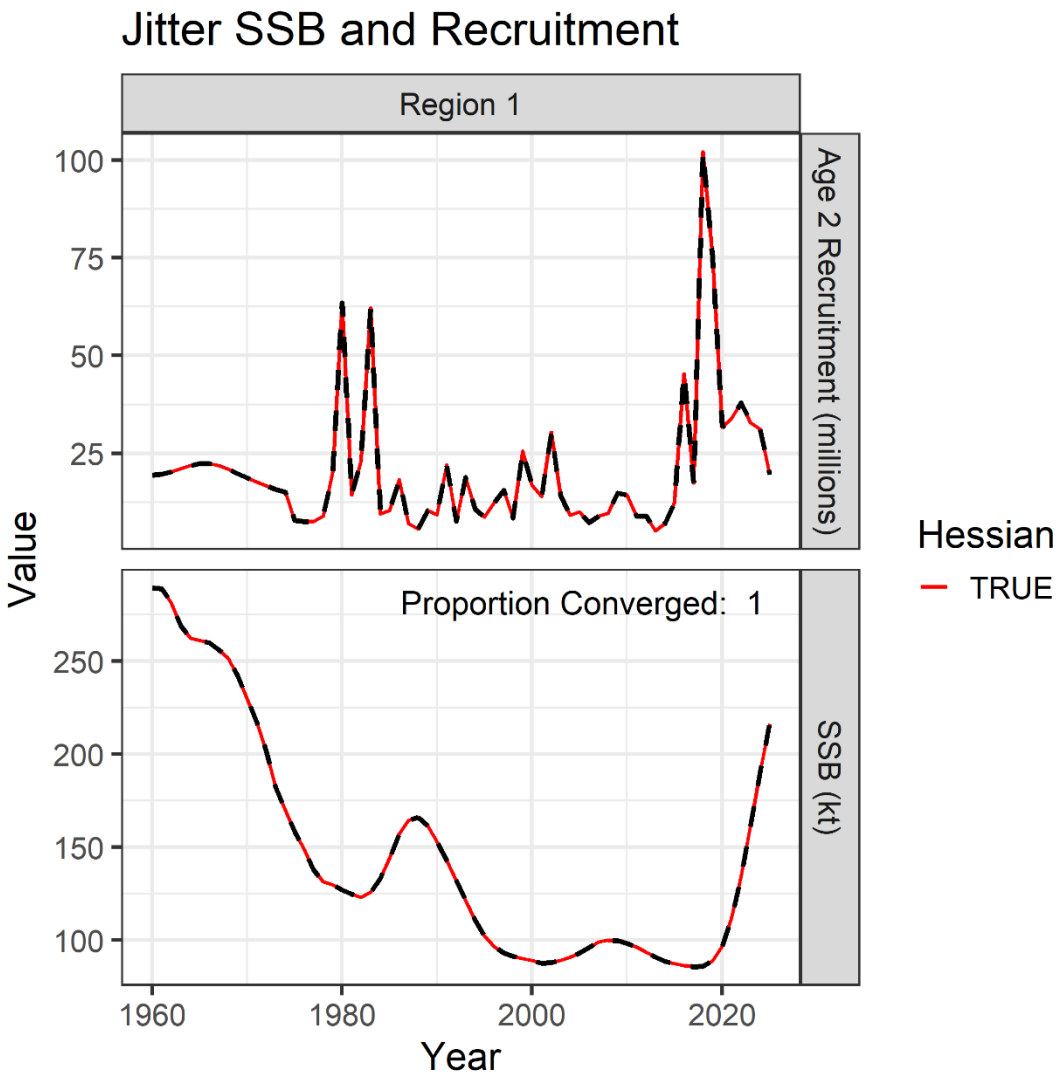

Figure SM3B.65. Results of a jitter analysis showing recruitment (top panel) and SSB (bottom panel) for each jitter iteration for Model *26.1_1Reg*. The starting parameter values were randomly jittered assuming ~$N(0,0.1^2)$.

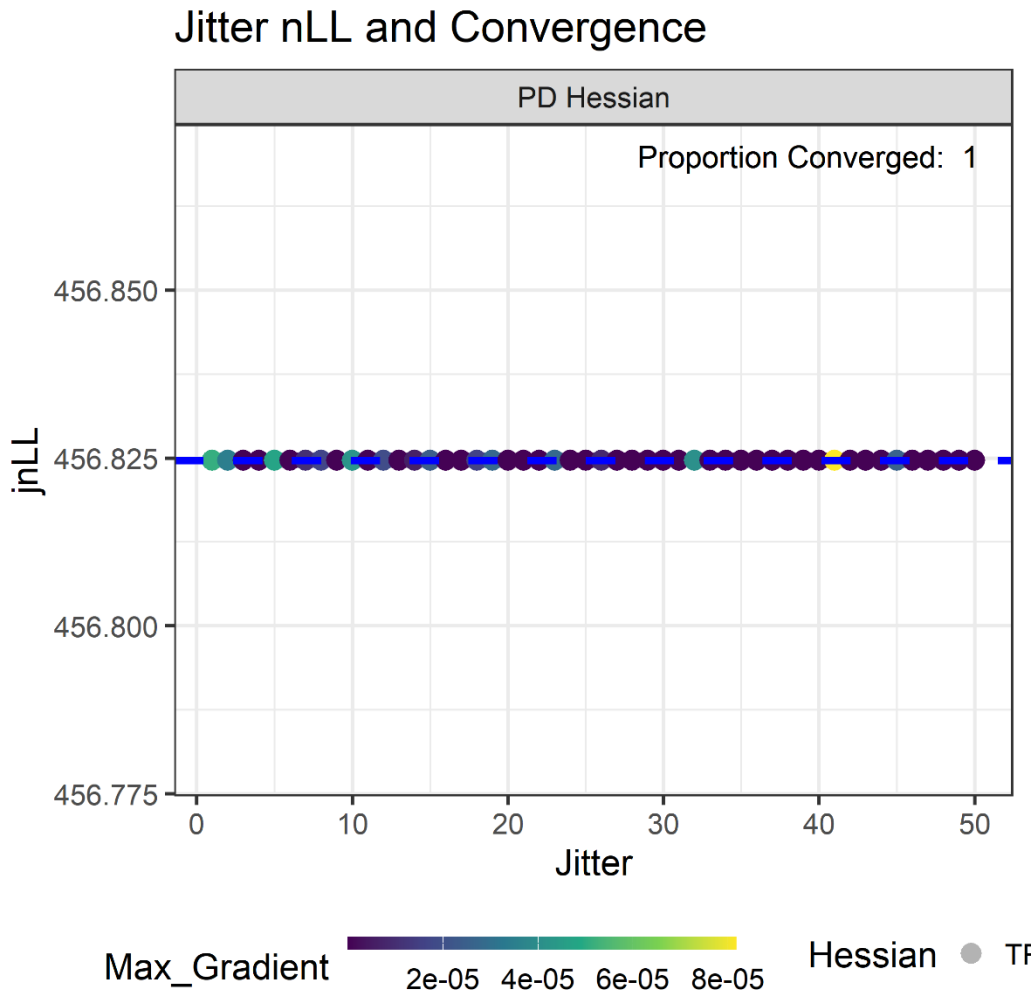

Figure SM3B.66. Results of a jitter analysis showing the joint negative log-likelihood values for converged (Positive Definite, PD, Hessian) and non-converged runs (non-Positive Definite Hessian) for model *25.12_1Reg_Cont*. The starting parameter values were randomly jittered assuming ~$N(0,0.1^2)$.

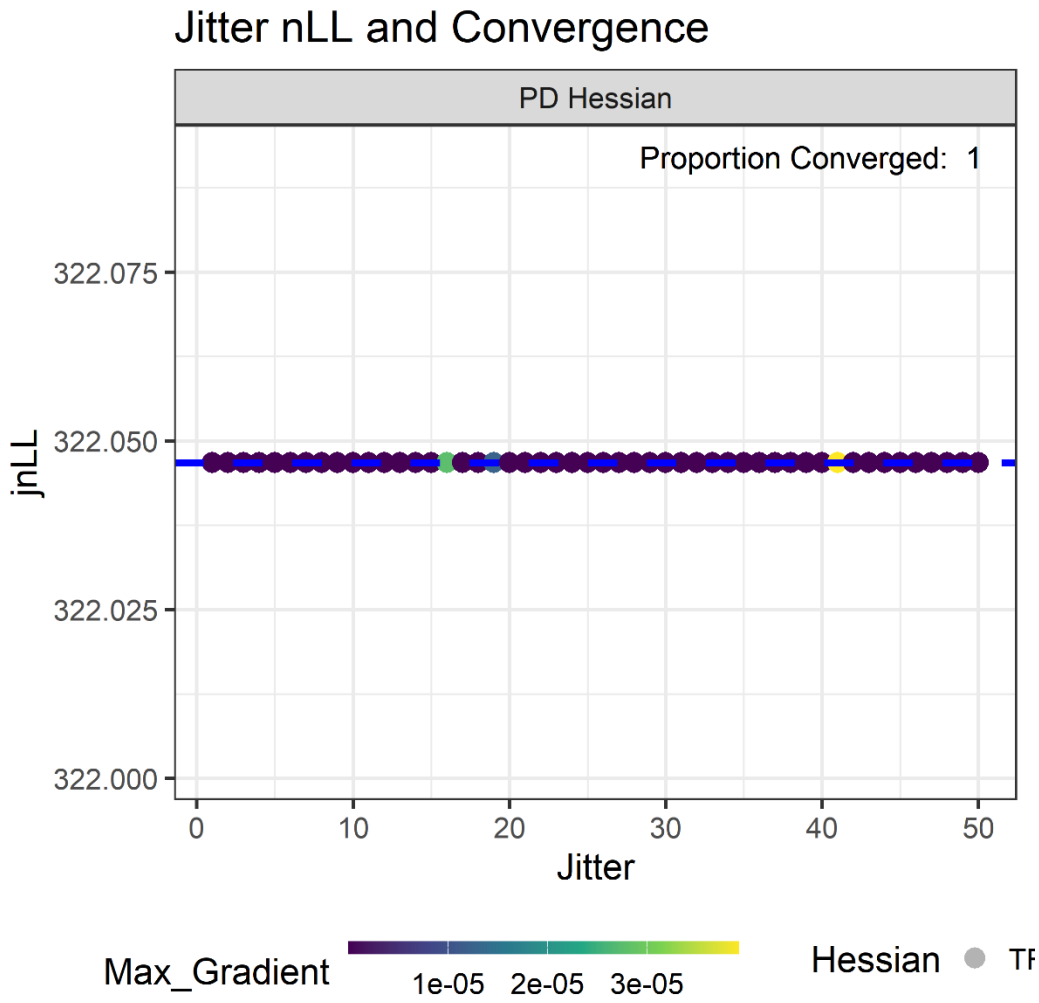

Figure SM3B.67. Results of a jitter analysis showing the joint negative log-likelihood values for converged (Positive Definite, PD, Hessian) and non-converged runs (non-Positive Definite Hessian) for model *26.1_1Reg*. The starting parameter values were randomly jittered assuming ~$N(0,0.1^2)$.